

Normalization by the Matching Number for Weighted Matching-Support Polynomials

Small ranks, unit and one-shore counterexamples, rank six, eventual scaling,
and three-tail boundaries

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Abstract

For a finite bipartite graph G with positive vertex activities, the matching-support polynomial $p_G(t) = \sum_k a_k t^k$ counts each pair of endpoint sets of a k -edge matching once, weighted by the product of its activities. Its degree is the matching number $\nu(G)$. This report collects batch-72 and batch-73 research notes on the question of *preorder-root-polytopes* (Parts XI–XIII) whether p_G is ultra-log-concave of order $\nu(G)$. Part I (sources 74, 72, 76 and 75) proves it for arbitrary positive activities on both shores whenever some minimum vertex cover meets one shore in at most two vertices, through an explicit transversal-matroid lift; in particular for every graph with $\nu(G) \leq 5$. With the rank-six Hall counterexample of *preorder-root-polytopes*, the least matching number of a weighted failure is exactly six, and there are failures at every larger matching number. Equality in any one normalized inequality occurs only for balanced star forests; graphs with a vertex cover of two vertices on each shore have real-stable basis polynomials and real-rooted support polynomials; and the full Hall family is ultra-log-concave for all activities exactly when one core has at most two vertices. Part II (sources 71, 73, 55, 47 and 51) answers the unit-weight and one-shore questions of *preorder-root-polytopes* negatively: a connected all-unit graph with 6916 vertices and matching number 3450 fails, and so does a 125-vertex graph whose left activities are all one and whose right activities are one except for activity 10000 on its 42 right-core vertices; for one common activity on the right core of a full Hall graph the least failing matching number is exactly 38. Part III (sources 60, 10, 23, 13, 05, 69 and 63) treats matching number six with all left activities one: the second and the last of the five inequalities hold for every graph and every positive right field, all five hold for several core families, and an all-right-core conditional cubic can fail its own Newton inequalities. Part IV (the embedded source 08s and sources 21, 43, 38, 35, 32, 29, 08, 03, 02 and 01) proves eventual ultra-log-concavity after a common scaling of the right part of a fixed minimum cover for every graph with $\nu(G) \leq 8$ and unit left activities, and locates the least matching number of a persistent failure under that scaling in [9, 38]. Part V (batch-73 sources 77–80) gives second routes to results of Parts I and III and four new results: at unit weights and matching number three a nonreal zero needs fifteen vertices, and the extremal graph is unique; a real-stable seed for star cores; a two-element Lorentzian deformation of every matroid basis polynomial; and a first-layer inequality for any number of left vertices, beside thirteen-vertex examples showing that the two-tail lifts of Part I do not extend to three tails. Nothing here is formalized in Lean or Rocq, and nothing is refereed.

Contents

1 Guide to this report

10

1.1	Sources, status and plan	10
1.2	What <i>preorder-root-polytopes</i> already proves	11
1.3	Notions and notation	12
1.4	Research questions of <i>preorder-root-polytopes</i> answered here	14
1.5	Labels and editorial marks	15
I	Weighted activities on both shores: normalization through matching number five	16
2	Definitions and main conclusions	17
3	Lorentzian input and singleton covers	18
4	The exact two-row formula	19
5	The full rank compression matroid lift	19
6	Exact failure at rank six and at every higher rank	21
6.1	A concrete rank-six witness	22
6.2	Permanent scaling failure at every rank at least six	22
6.3	The shapes recorded by source 74's verifier	23
7	An established geometric interpretation	23
8	A generic-line matroid model for the forced-left sector	24
9	The sharp all-core inequality at rank five	25
9.1	A finite exact coefficient certificate	26
9.2	The four-class sum-of-squares certificate	27
9.3	Passing to the right exterior and proving sharpness	28
10	Lorentzian transfer across a star core	28
10.1	A support-exact vertex-sum identity	29
10.2	Stars centered on the two-vertex shore	29
10.3	Stars centered on the opposite shore	30
11	Exact equality in the first Newton gap	31
12	Quantitative eventual strictness under common cover scaling	31
13	Exact verification and scope of the conclusions	32
14	Strictness and equality for weighted matching-support polynomials (source 72)	33
14.1	Definitions and conclusions	34
14.2	A weighted matroid equality criterion	34
14.3	The rank compression lift and its two bases	36
14.4	Contractions witnessing strictness	37

14.4.1	An ordinary column with two neighbors	37
14.4.2	A nonempty core	38
14.5	Completing the graph classification	38
14.6	The first inequality at every rank	39
14.7	Verification and scope	40
15	Stability of two-by-two covers and weighted rank four (source 76)	40
15.1	Definitions and the conclusions	41
15.2	Stability tools and the six cores	42
15.3	Independent columns and Boolean minor weights	43
15.3.1	The matching core	43
15.3.2	The three-edge core	43
15.3.3	Verification of all minor types	44
15.4	The complete core by two-block polarization	44
15.5	From stability to the rank-four inequalities	45
15.6	A sharp boundary for complete mixed cores	45
15.7	Scope and reproducibility	47
16	A sharp weighted classification of full Hall bottleneck graphs (source 75)	47
16.1	The family and the classification	48
16.2	A uniform formula for a two by s core	49
16.3	Twelve coefficient families	50
16.3.1	Pattern 10	51
16.3.2	Pattern 20	51
16.3.3	Pattern 11	51
16.3.4	Pattern 21	52
16.3.5	Pattern 22	52
16.3.6	The small index boundary and completeness	53
16.4	Singleton cores and deficient exterior populations	53
16.5	Explicit counterexamples when both cores are large	53
16.6	A positive theorem does not imply real rootedness	55
16.7	Verification and the remaining questions	55
II	Unit weights and one weighted shore: counterexamples and the rank-38 threshold	56
17	Unit weight failure of matching rank normalization (source 71)	56
17.1	The question and the counterexample	57
17.2	A graph family with pendant arms	57
17.3	Exact support counting	58
17.3.1	A short formula for the top three coefficients	58
17.4	The finite certificate	59
17.5	Failure when the smaller shore exceeds rank by one	59
17.5.1	Why deficiency one is not an isolated accident	60
17.6	Consequences for Ehrhart h^* -polynomials	60

17.7	Verification and remaining questions	61
18	One-shore activities can destroy matching-rank ULC (source 73)	61
18.1	The polynomial and two explicit counterexamples	61
18.2	Endpoint counting gives the complete certificate	62
18.3	A general construction criterion	64
18.4	What this settles and what remains open	64
18.5	Appendix: A pendant gadget that transfers fields between shores	65
18.5.1	A rank 310 example with activity 21	66
19	A sharp rank-38 boundary for the last Hall inequality, one common right-core activity (source 55)	67
19.1	The family and the sharp threshold	67
19.1.1	Deficient exterior populations	68
19.2	The exact endpoint quadratic	68
19.3	A finite field cannot create a hidden negative window	69
19.4	A cubic classification and the finite rank certificate	70
19.5	The two rank 38 shapes and the smaller witness	71
19.6	Verification and attribution	72
20	The sharp rank-38 ULC boundary for full Hall graphs, one common right-core activity (source 47)	72
20.1	The family and the conclusion	73
20.2	Counting and the finite population expansion	74
20.3	An exhaustive finite integer certificate	75
20.4	Closing the first and last gaps	75
20.4.1	The first gap	75
20.4.2	The endpoint classification	76
20.4.3	The remaining cover and population cases	77
20.5	The sharp witness and reproducibility	77
21	The last Hall gap does not control ultra-log-concavity, one common right-core activity (source 51)	77
III	Matching number six with one weighted shore at every finite activity	80
22	Rank-two matrix blocks and exterior conditioning, both shores weighted (source 60)	81
22.1	A rank two exterior column block	81
22.1.1	The rank two hypothesis cannot simply be increased	82
22.2	Exact conditioning when Hall slack is at most two	83
22.3	Forcing exterior vertices in a genuine cover	85
22.4	Eventual strictness and an explicit threshold	85
22.5	Verification and scope	86
23	Lorentzian marginals on three vertices and star cores, both shores weighted	

(source 10)	87
23.1 The selected-vertex marginal	87
23.2 Three Rayleigh inequalities	87
23.3 The remaining Hessian	88
23.4 Gluing at one left vertex	89
23.5 Exact verification and scope	90
24 Rooted Lorentzian Hall sectors and an unrooted cubic obstruction	
(source 23)	91
24.1 Matching supports and the rooted theorem	91
24.1.1 Exact inputs	92
24.1.2 Proof by transposed compression	92
24.2 Why the roots cannot be omitted	93
24.2.1 Exact conditional coefficients	94
24.2.2 The recombination obstruction	95
24.3 The original graph is safe under every common core field	95
24.4 Verification and remaining questions	95
25 The last Newton inequality at matching number six, one weighted shore	
(source 13)	96
25.1 Statement and support convention	96
25.2 The exact reciprocal activity quadratic	97
25.3 Two Schur bounds	98
25.4 The weighted plane incidence bound	99
25.5 The quantitative local lemma	99
25.6 Moments and exact finite identities	100
25.7 Summing the local bound	101
25.8 Reproduction and remaining scope	102
26 The second Newton inequality at matching number six, one weighted shore	
(source 05)	102
26.1 The inequality and its cubic strengthening	102
26.2 The homogenizing derivative	103
26.3 Exterior derivatives and rank two compression	104
26.4 Exact core profiles	105
26.5 The two dimensional Schur complement	106
26.6 The complete finite certificate	107
26.7 Scope and the next truncations	107
27 Matching number six with common exterior neighbourhoods, one weighted shore (source 69)	108
27.1 The theorem and its scope	108
27.2 Exact endpoint counting and the deleted-core baseline	109
27.3 The finite nonnegative coefficient certificate	110
27.4 Adjacent exterior classes and a precise barrier	111
27.5 Reproducibility and remaining question	112

28 Nested families and conditional barriers at matching number six, one weighted shore (source 63)	112
28.1 A nested exterior theorem	112
28.2 An obstruction to the full marginal approach	114
28.3 Every four right vertex sector is safe	115
28.3.1 Hall tight subfamilies	115
28.3.2 Two core vertices and two exterior vertices	116
28.3.3 A useful quadratic inequality	116
28.3.4 All three core vertices and a degree three exterior vertex	116
28.3.5 All three core vertices and a lower degree exterior vertex	117
 IV Eventual one-shore scaling through matching number eight	 119
29 Top conditional gaps and the rank-seven reduction, one weighted shore, right-cover scaling (source 08s)	120
29.1 Coefficients and conclusions	120
29.2 A two-plane incidence inequality	121
29.3 The exact top sectors	122
29.4 Quotient geometry and the unit counts	123
29.5 Scaling and the cubic consequences	124
29.6 The rank seven boundary	125
29.7 Appendix: The weighted local equality implication	126
29.8 Appendix: Exact checks and supplementary searches	126
 30 Rank-six conditioning for every core, one weighted shore, right-cover scaling (source 21)	 127
30.1 The conditional theorem	127
30.2 The ambient lower gap	128
30.3 A plane incidence inequality	128
30.4 Virtual vectors and exact support counts	129
30.5 Every core rank	130
30.6 The unit count and the higher gap	130
30.7 Eventual common scaling	131
30.8 Verification and scope	132
 31 Second route for the complete core: a rank-six cubic sector, one weighted shore (source 43)	 132
31.1 The conditional cubic	133
31.2 Three coupled inequalities from one matroid	134
31.3 The two shifted Newton gaps	135
31.3.1 The higher gap	135
31.3.2 The lower gap	136
31.4 Eventual strictness and a computable threshold	136
31.5 Verification and the remaining frontier	137

32 Second route for matching-hole cores, one weighted shore (source 38)	137
32.1 The graph and the conclusion	138
32.2 Unit left counts and exact plane defects	139
32.2.1 Right profiles	139
32.3 The Lorentzian flag polynomial and the higher gap	140
32.4 The lower gap by 25 neighborhood pairs	141
32.5 Eventual strictness and reproducibility	142
33 Second route for cores with two complete rows, one weighted shore (source 35)	143
33.1 The family and its conditional sector	143
33.1.1 Common notation	144
33.2 The exceptional row of degree one	144
33.2.1 A left flag and the higher gap	144
33.2.2 Coloops and unit flag constraints	146
33.2.3 The first gap	146
33.3 The exceptional row of degree zero	147
33.3.1 A direct first-gap proof	147
33.4 Eventual strictness and verification	148
34 Second route for cores with two complete columns, one weighted shore (source 32)	149
34.1 The family and exact support counts	149
34.1.1 The column of degree one	150
34.2 Plane and axis extensions	150
34.3 The lower conditional gap	152
34.3.1 Coloops and valid count constraints	152
34.3.2 Twenty-five exact coefficient identities	152
34.4 The zero-column case	153
34.5 Eventual scaling and verification	153
35 Second route for cores with at least six edges, one weighted shore (source 29)	154
35.1 The theorem and the finite frontier	154
35.1.1 The coloop branch	155
35.1.2 Exhaustive core classification	155
35.2 A common higher-gap criterion	156
35.3 The triangular core	156
35.4 A star plus a disjoint missing edge	158
35.5 The transposed star-edge core	158
35.6 The exact lower-gap certificates	159
35.6.1 Generator families and their hypotheses	160
35.7 From conditional gaps to eventual strictness	161
36 Eventual one-shore scaling through rank seven (source 08)	161
36.1 Main theorem and normalization	162
36.2 Two bounds for plane incidence	163

36.3	The auxiliary rank four matroid	164
36.4	Comparing the three middle coefficients	165
36.5	The middle gap and eventual scaling	166
36.6	Failure ranks and verification scope	166
37	Eventual scaling for five-plus-three covers, one weighted shore (source 03)	167
37.1	Main result and exact sectors	167
37.2	A graph generic additive plane bound	168
37.3	The top conditional gap	170
37.4	The strict next to top gap	170
37.5	A four dimensional unit counting lemma	171
37.6	The lower middle gap and eventual scaling	173
37.7	Scope and exact verification	173
38	The top conditional gap for four-plus-four covers, one weighted shore (source 02)	174
38.1	Graph normalization and the new top gap	174
38.2	Incidence quantities and a real obstruction	175
38.3	A coefficientwise graph generic upper bound	176
38.4	A coupled unit correction	177
38.5	The top conditional comparison	178
38.6	Status and reproducibility	179
39	Eventual one-shore scaling through rank eight (source 01)	179
39.1	Main theorem and the remaining cover split	179
39.2	The auxiliary quartic and a sufficient unit potential	180
39.3	A finite hyperplane transition bound	181
39.4	Two elementary minor count lemmas	182
39.5	The starting potential and its increments	183
39.6	Parallel classes and the strict middle gap	184
39.7	The lower population boundary and eventual scaling	185
39.8	Failure ranks and verification scope	186
V	Second routes and boundaries	188
40	Weighted rank four, real roots for two-by-two covers, and the rank-three minimum (source 77)	189
40.1	Support weights and the main theorem	190
40.2	The ambient theorem and the cover reduction	191
40.3	Two-center support data and path realizations	192
40.3.1	The local quadratic and rooted path model	192
40.3.2	No core edges or one core edge	193
40.3.3	Two adjacent core edges	193
40.4	Hermitian realizations for the next two core patterns	193
40.4.1	Cycle interference in a Gram determinant	194

40.4.2	Two disjoint core edges	194
40.4.3	Three core edges and two coupled phases	195
40.5	The complete core and a positive semidefinite block matrix	196
40.5.1	A side matrix and its variance margin	196
40.5.2	The full-core support identity	196
40.5.3	An aggregate coupling with a prescribed determinant	197
40.5.4	The block determinant identity	197
40.6	Completion and the real-rootedness boundary	198
40.6.1	Real roots can fail already at rank three	198
40.6.2	Why one weighted twin is not always enough	199
40.7	The smallest unit-weight counterexample at rank three	199
40.7.1	A direct lemma for one by two covers	200
40.7.2	An exact reduction to seven populations	200
40.8	A restricted rank-five family with four activity blocks	201
40.8.1	Support formula and removal of exterior activities	202
40.8.2	A finite positive-coefficient certificate for all parameters	202
40.9	Verification and further questions	203
41	Two-by-three cores with complete exteriors, and stable star cores (source 78)	204
41.1	Statement and normalization	204
41.2	Boolean support reconstruction	205
41.3	Exact middle gap certificates	206
41.4	The outer gaps and degree drops	207
41.5	Stable monomers for empty and star cores	208
41.6	Verification and boundaries	209
42	A two-element Lorentzian deformation of every matroid basis polynomial (source 79)	209
42.1	Statement and conventions	210
42.2	A direct all-matroid extension of the intake lift (source 80, Appendix A)	211
42.3	Two matroid constructions	212
42.3.1	A principal-line seed	212
42.3.2	Matroid union supplies the support	213
42.4	Contraction and the five small ranks	213
42.4.1	Rank zero	213
42.4.2	Rank one	214
42.4.3	Rank two	214
42.4.4	Ranks three and four	214
42.5	Removing the spanning hypothesis	215
42.6	Bipartite endpoint supports	215
42.7	Proof status and reproducibility	216
43	Three-tail obstructions and an all-rank first layer (source 80)	216
43.1	The sharp weighted rank threshold	217
43.2	Complete-block scalar counterexamples	218

43.2.1	The exact support formula	218
43.2.2	An explicit degree-six violation	218
43.2.3	Failure at every higher rank	219
43.3	A separate multivariate obstruction and its graph lift	219
43.3.1	Three parallel pairs	219
43.3.2	A thirteen-vertex graph with a full matching	220
43.4	The failure survives identifying selected-tail variables	221
43.5	An ordinary positive first-layer result for every rank	222
43.5.1	The full three-tail rank-zero base	223
43.6	Scope and exact reproducibility	224
44	Questions raised by Part V	225
A	Provenance of this report	225
A.1	Sources	225
A.2	What each source contributed	230
A.3	Where the merge chose	232
	References	233

1 Guide to this report

1.1 Sources, status and plan

This report is a merge of research manuscripts delivered to ProveIt’s incoming-report intake on 1 October 2026 (batch 72, arrival commit [1512ef835](#), placement commit [8bb543f0f](#)). The sources are numbered by their batch-72 manuscript numbers. Twenty-six archives and one manuscript delivered only inside archive 08 (called 08s here) are sources; a twenty-eighth archive (78) is an exact duplicate of the leaf-compression manuscript already printed as Part XIII of *preorder-root-polytopes* and is not a source. The full record, archive by archive, is Appendix A. Four manuscripts of batch 73 (arrival commit [f8c3a392a](#), placement commit [5e4f1263c](#)), written in parallel with batch 72, were added as Part V; they are numbered 77–80, continuing the batch-72 numbers, and five further batch-73 archives on the same object are duplicates and are not sources. The report has five Parts:

- Part I, weighted activities on both shores: sources 74 (the base), 72, 76 and 75;
- Part II, unit weights and one weighted shore: sources 71, 73, 55, 47 and 51;
- Part III, matching number six with one weighted shore: sources 60, 10, 23, 13, 05, 69 and 63;
- Part IV, eventual one-shore scaling: sources 08s, 21 (with 43, 38, 35, 32 and 29 as second routes), 08, 03, 02 and 01;
- Part V, second routes and boundaries (batch 73): sources 77, 78, 79 and 80.

Every source is printed in full: statements, proofs, examples, remarks, limitations and verification notes. A result proved by several sources is printed once, under the source named in the Part’s introduction, and the other proofs follow as marked second routes. A result that *preorder-root-polytopes* already proves is cited by its number there and is not presented as new;

where a source re-proves it, the source’s proof is either printed as a marked second route or replaced by a pointer, as recorded at the place and in Appendix A.

Status. All sources are AI-assisted research notes (their title pages read “Research note prepared for Vladimir Reshetnikov with OpenAI”, except source 72’s “AI-assisted mathematical research”; of the batch-73 sources, source 77’s reads “Research note prepared with OpenAI for Vladimir Reshetnikov”, source 78’s “Research manuscript prepared for Vladimir Reshetnikov”, with the PDF author field “Research manuscript prepared with OpenAI”, and those of sources 79 and 80 “Research note”). Their proofs are ordinary mathematical arguments supported by exact finite checks; they are not refereed, and nothing in them is formalized in Lean or Rocq. No source claims global priority. Text in square brackets marked *Merge note* was added in this merge; everything else is the sources’ text, with labels prefixed and section levels shifted (Section 1.5).

1.2 What *preorder-root-polytopes* already proves

The neighbouring report `SetTheory/Cardinals/docs/reports/enumerative-combinatorics/preorder-root-polytopes` ([8], cited here as *preorder-root-polytopes*) introduced the question and proves the following. They are cited by their printed numbers and labels in that report and are not reprinted here.

- Part V: Theorem 70.1 (`lor:thm:weightedULC`), the two-shore inequality: p_G is $\text{ULC}_{\min(|X|,|Y|)}$ for arbitrary positive activities (normalization by the smaller shore); its Research question 37 (`lor:q:first`) asks for normalization by the matching number.
- Part VI: the stability tools, Theorem 85.3 (`ade:mat:thm:stable-bridge`, the reciprocal bridge between the signed support polynomial and the basis polynomial), Theorem 91.2 (`ade:thm:expected`, independent-column expectations are real stable), Theorem 93.2 (`ade:mat:thm:vertex-stable`, stability under one-vertex sums), Theorem 94.1 (`ade:mat:thm:biclique`, complete bipartite blocks are stable) and Corollary 94.3 (`ade:mat:cor:block-family`).
- Part XI (Hall bottlenecks): the family $H_{s,n}$ and its exact coefficient formula, Theorem 175.3 (`hb:thm:coefficients`); the 30-vertex counterexample $H_{4,11}$ with weight 500 on both cores, Theorem 176.1 (`hb:thm:witness`); the complete classification of the core-three family $H_{3,n}$, Theorem 177.1 (`hb:thm:rank6phase`), whose last order-six margin fails at $n = 33$ (the rank-six counterexample $H_{3,33}$), with its asymptotic Corollary 177.2 (`hb:cor:asymptoticweight`); the endpoint threshold, Theorem 178.2 (`hb:thm:endpoint`); and the first-margin defect identity with its equality case, Theorem 180.2 (`hb:thm:firstidentity`) and Corollary 180.3 (`hb:cor:equality`): $(r - 1)a_1^2 \geq 2ra_2$ for every positively weighted graph of matching number $r \geq 2$, with equality exactly for r disjoint stars of equal edge totals.
- Part XII (rank-three matching supports): the first rank inequality, Theorem 190.2 (`mr:thm:first`), and its unweighted equality case, Theorem 190.3 (`mr:thm:equality`); normalization by the matching number at unit weights for $\nu \leq 3$, Theorem 191.1 (`mr:thm:rankthree`); the explicit rank-three bimatroid obstruction, Theorem 192.1 (`mr:thm:matrix`); and a connected rank-four family, Theorem 196.1 (`mr:thm:family`).
- Part XIII (leaf compression): the weighted smaller-shore inequality, Lemma 205.1 (`lc:lem:ambient`); weighted leaf compression, Lemma 206.1 (`lc:lem:compression`); normalization by the matching number for every minimum cover with a singleton shore or on one shore, Theorem 206.2 (`lc:thm:singleton`); and weighted rank three, Corollary 206.3 (`lc:cor:rankthree`).

Archive 78 of this batch is byte-identical to the batch-71 archive printed as Part XIII; it is not printed again.

1.3 Notions and notation

The sources share one setting but not one vocabulary. The following conventions hold throughout; where a source uses another word or letter, it is listed here and kept in the source’s text.

Notion	Meaning in this report, and the readings to avoid
Support polynomial $p_G(t) = \sum_k a_k t^k$	$a_k = \sum u_I v_J$ over pairs $I \subseteq X, J \subseteq Y$ with $ I = J = k$ for which $G[I, J]$ has a perfect matching. Each pair of endpoint sets is counted <i>once</i> , not once per matching: this is not the matching (rook) polynomial. At unit activities it is the perfectly matchable set polynomial of Davis–Kohl [5]. Source 75 writes $P_H(t) = \sum p_k t^k$; <i>preorder-root-polytopes</i> writes $p_G(t; u, v)$ (Part XIII) and $A_H(t; u, v)$ (Part V).
Matching number $\nu(G)$, also r	The maximum size of a matching. The sources call it the <i>matching rank</i> ; <i>preorder-root-polytopes</i> calls it the <i>matching number</i> . Both words are used here with one meaning. For positive activities $\deg p_G = \nu(G)$ and every a_k with $0 \leq k \leq \nu(G)$ is positive.
ULC _D and the gaps $\Delta_k^{(D)}$	p is ULC _D if $a_k / \binom{D}{k}$ (padded by zeros) is log-concave without internal zeros; for support polynomials this is $\Delta_k^{(D)} = k(D - k)a_k^2 - (k + 1)(D - k + 1)a_{k-1}a_{k+1} \geq 0$ for $1 \leq k \leq D - 1$ (display (1)). “Normalization by the matching number” means $D = \nu(G)$. The sources call Δ_k a <i>gap</i> , <i>margin</i> , <i>normalized gap</i> or <i>Newton gap</i> ; the <i>first</i> gap is $k = 1$, the <i>last</i> is $k = D - 1$. <i>Strict</i> ULC _D means all $D - 1$ gaps are positive.
Weightings	<i>Two-shore</i> (or simply <i>weighted</i>): arbitrary positive activities u on X and v on Y (Part I). <i>Unit</i> : $u \equiv v \equiv 1$ (Research question 84 of <i>preorder-root-polytopes</i> ; Part II). <i>One-shore</i> : $u \equiv 1$ on the left shore, v arbitrary positive on the right shore (Research question 85 of <i>preorder-root-polytopes</i> ; Parts II–IV). <i>One common right-core activity</i> : activity w on the right core of a full Hall graph and 1 elsewhere, a special one-shore weighting (sources 73, 55, 47, 51).
Displayed cover $A \sqcup B$	A vertex cover with $A \subseteq X$ on the left and $B \subseteq Y$ on the right; $L = X \setminus A$ and $R = Y \setminus B$ are the exterior sets and there are no L – R edges; $G[A, B]$ is the <i>core</i> . Source 74 writes a $2 + q$ cover for $ A = 2, B = q$; a cover is <i>genuine</i> when $\nu(G) = A + B $, that is, when it is minimum.
Full Hall graph $H_{a,b;n,m}$	Core A (a vertices) and B (b), exterior L (n) and R (m), edges $(A \times B) \cup (L \times B) \cup (A \times R)$ (sources 75, 73, 55, 47, 51; source 74 writes $H_{a,b;N,M}$, source 76 $H_{2,s;n,m}$). <i>preorder-root-polytopes</i> ’s $H_{s,n}$ is $H_{s,s;n,n}$ with weight w on <i>both</i> cores. The weighting differs by source: both cores (74, 75, <i>preorder-root-polytopes</i>), the right core only (73, 55, 47, 51).

Notion	Meaning in this report, and the readings to avoid
Scalings	Two different scalings occur. Source 74 (Section 6) multiplies the activities of a <i>whole</i> minimum cover, on both shores, by a common factor; then failures persist at every matching number ≥ 6 . The sources of Part IV multiply only the <i>right</i> part of a fixed minimum cover, with unit left activities; there no failure occurs through matching number eight. These statements do not conflict.
Real stability, real-rootedness, Lorentzian	A polynomial is real stable if it has no zero with all variables in the upper half-plane $\mathbb{H}$; a real-stable univariate polynomial is real-rooted. A homogeneous bivariate polynomial is Lorentzian exactly when its coefficients are ULC of its degree. Real-rootedness of p_G implies $\text{ULC}_{\deg p_G}$ by Newton's inequalities; the converse fails (source 75's rank-five example, Section 16).
Conditioning	<i>Forcing</i> prescribed vertices into the support (sources 60, 21, 23, 63): the conditional polynomial is the coefficient of the product of the forced activities. It is not probabilistic conditioning.
Shifted and unshifted Newton inequalities	For a conditional sector $t^j C(t)$ of a degree-six polynomial, the <i>shifted</i> inequalities are the degree-six gaps restricted to the sector; the <i>unshifted</i> ones are the degree-three Newton inequalities of the cubic C itself (constant three). Source 23's 25-vertex graph fails two unshifted inequalities while its shifted ones, and full ULC_6 , hold; it does not contradict source 21, which bounds shifted ones with smaller constants.
Letters local to a source	Letters are local to the source (often to the section) that defines them. In particular A, B are cover parts in source 74 but numerical gap factors in Section 3 of source 75 and <i>preorder-root-polytopes</i> 's core in Part XI; a, b, c are right-class activity sums in Section 5 but core sizes in Section 6; q is $ B $ in source 74 and $2^{-1/2}$ in source 76; $Q = ab + ac + bc + d$ in source 74; H is $G[L, B]$ in source 74, a full Hall graph in sources 75 and 76, and $H_{s,n}$ in <i>preorder-root-polytopes</i> ; h is a residual sector in source 74 and a population offset in source 75.
Part V vocabulary (sources 77–80)	<i>Tail</i> and <i>head</i> mean left and right vertex; <i>selected tails</i> are the left endpoints of a support, <i>unused</i> or <i>covered heads</i> the right cover vertices outside it. The <i>actual</i> or <i>surviving degree</i> is the matching number after zero-activity vertices are deleted, Part I's normalization order for nonnegative activities. $\Gamma(t) = \sum_k \gamma_k t^k$ (sources 78 and 80) is p_G , and source 77's p_n is a family of support polynomials indexed by a population size, not a coefficient.
Shores and cover parts in Part V	Sources 78–80 write the shores as $P \sqcup X$ and $Q \sqcup Y$, with P, Q the cover parts (Part I's A, B) and X, Y the exterior sets (Part I's L, R): there X, Y are <i>not</i> the whole shores, as they are elsewhere in this report. Source 77 writes $G = (L, R, E)$ with L, R the <i>whole</i> shores, not exterior sets. Source 79's A, B are two distinguished matroid elements, not cover parts; its μ is a population parameter, while source 78's μ is a signed monomer polynomial. Source 80's N is the matrix of Theorem 43.3, and an exterior population size only in Section 43.2; source 77's N is a population sum. Source 80's $H(t, x, z)$ is a polynomial, not a full Hall graph.

Renamed symbols. No printed symbol of a source was renamed. Source 69 defines the macro `\rank` as ν ; it is printed as ν here, which is how source 69 renders it, so that the macro can keep source 60's meaning (rank). In Part V, source 78's macro `\Ga` is printed as Γ , which is what it renders, and source 79's `\rk` keeps its rendering rk (this report's `\rk` prints rank).

1.4 Research questions of *preorder-root-polytopes* answered here

Question of <i>preorder-root-polytopes</i>	Status after this report
RQ 86 <code>hb:q:rank</code> (first possible weighted rank)	Answered: every weighted graph with $\nu \leq 5$ is ULC_ν (Corollary 2.2); Part XI's $H_{3,33}$ fails at $\nu = 6$. The least weighted failure rank is exactly six. Sources 77 and 80 (Part V) re-prove parts of this.
RQ 87 <code>hb:q:order</code> (a universal rank-only order)	In part: $\mathcal{D}(4) = 4$ and $\mathcal{D}(5) = 5$ (Corollary 2.2); the question beyond rank five is open.
RQ 93 <code>mr:q:rankfour</code> (finish rank four)	Answered affirmatively, for unit and arbitrary two-shore weights (Corollary 2.2; second route by real stability, Corollary 15.2; a further route, source 77 in Part V).
RQ 89 <code>hb:q:asymmetry</code> (asymmetric bottlenecks)	In large part: full Hall graphs are ULC for all activities exactly when one core has at most two vertices (Theorem 16.1); explicit failures for all unequal cores $a, b \geq 3$ (Sections 6 and 16.5). Unequal core activities: Part II (one core at activity one).
RQ 90 <code>hb:q:defects</code> (higher-index defect identities)	Bears on it: under the two-vertex-shore cover hypothesis, equality in any one gap holds only for balanced star forests (Theorem 14.1); no higher defect identity is given.
RQ 84 <code>hb:q:unit</code> (unit weights)	Answered negatively (source 71, Theorem 17.1): a connected all-unit graph with $\nu = 3450$ is not ULC_ν ; infinitely many deficiency-one examples. The least matching number of a unit counterexample is open.
RQ 85 <code>hb:q:oneshore</code> (one-shore fields)	Answered negatively (source 73: $\nu = 50$, 125 vertices; $\nu = 38$; source 55: for one common right-core activity on full Hall graphs the least failing matching number is exactly 38). Positive pieces: $\nu \leq 5$ (Part I); at $\nu = 6$ the first, second and last gaps for every graph (Part XI of <i>preorder-root-polytopes</i> , sources 05 and 13) and all gaps for several families (Part III). Eventually, after scaling the right part of a minimum cover, through $\nu = 8$ (Part IV).
RQ 88 <code>hb:q:phase</code> (Hall phase diagrams, $s \geq 4$)	Not answered (it concerns weight on both cores). Its one-shore analogue is answered by sources 47 and 51: the last gap controls all gaps through $\nu = 38$ but not at $\nu = 49$.
RQ 94 <code>mr:q:allrank</code> (a structural all-rank proof)	Answered negatively for unit weights (source 71): no all-rank proof can exist.
RQ 101 <code>mr:q:bimatroid</code> (the bimatroid boundary)	In part: rank-two exterior blocks with ordinary columns are Lorentzian (source 60, Theorem 22.1).
RQ 37 <code>lor:q:first</code> (Part V)	Weighted: holds through $\nu = 5$, fails from $\nu = 6$ (Part I). Unit and one-shore: fail (Part II).

1.5 Labels and editorial marks

Every label carries the prefix `mrn:.` Labels written in this merge are `mrn:sec:...`, `mrn:src:NN` (the start of source NN) and `mrn:w:tab:...`; a delivered label of source NN carries a sub-prefix naming its Part (`w` for Part I, `u` for Part II, `s` for Part III, `e` for Part IV, `v` for Part V) and the source, for example `mrn:w:rf:thm:main` for source 74's `thm:main`. The table of sub-prefixes is in Appendix A. Source 74, the base, keeps its sections as sections of Part I, and its appendices are ordinary sections there; every other source is one section, whose sections are subsections. Section and theorem numbers therefore differ from the delivered PDFs, which are not shipped. Part V also has the merge labels `mrn:part:five` and `mrn:v:sec:questions`, and seven labels added to unlabelled delivered sections so that literal section numbers could become references.

Part I

Weighted activities on both shores: normalization through matching number five

Contents of Part I. Part I prints four sources. Source 74, *Weighted Rank Five and Covers with a Two Vertex Shore*, is the base of the whole report; its sections are Sections 2–13, and its appendices are printed as ordinary sections (Sections 8–13). Source 72 (Section 14) classifies equality under the same cover hypothesis; source 76 (Section 15) proves real stability for covers with two vertices on each shore; source 75 (Section 16) classifies the full Hall family.

What is new here and what is cited. New relative to *preorder-root-polytopes*: the transversal lift of Theorem 2.1 and its consequences (normalization by the matching number for every minimum cover meeting one shore in at most two vertices, hence for $\nu \leq 5$, and the cover obstruction Corollary 5.1); the forced-sector and scaling refinements; the asymmetric negative shapes; the equality classification at every gap (source 72); real stability for $2 + 2$ covers, the $2 \times s$ stability threshold and the 17-vertex example (source 76); the full Hall classification, its strictness and real-rootedness refinements and the nonreal rank-five example (source 75). Cited, not new: the smaller-shore inequality and leaf compression (Parts V and XIII of *preorder-root-polytopes*; source 74’s Lemmas 3.1 and 3.2 and the corresponding steps of sources 72 and 76 are pointers); the first-gap inequality with its equality case (Part XI’s Theorem 180.2 and Corollary 180.3, Part XII’s Theorem 190.2; the Motzkin–Straus proofs of sources 74 and 72 are printed as second routes); the rank-six and rank-eight witnesses $H_{3,33}$ and $H_{4,11}$ (Part XI, restated by source 74); and the stability tools of source 76 (Part VI).

Same theorem in several sources. Source 74’s Corollary 2.2 contains source 76’s Corollary 15.2 ($\nu \leq 4$) and Part XIII’s Corollary 206.3 ($\nu \leq 3$); source 76’s real-stability route is printed as a second route. The positive half of source 75’s Theorem 16.1 is the full Hall case of source 74’s Theorem 2.1; source 75’s comparison-polynomial proof is printed as a second route. For equal cores and populations the negative half of Theorem 16.1 is Part XI’s endpoint theorem.

Non-claims of Part I. Source 74’s theorem asserts neither real-rootedness nor stability of the support polynomial (remark after Corollary 5.1); at higher rank it does not address the unrestricted unit-weight and one-shore problems (Section 13); finite computations are regression checks, not proofs (sources 72 and 74); source 76 does not assert real-rootedness for every graph of matching number four; source 75’s classification concerns the full Hall family only and its negative assertion says nothing about unit or one-shore weights.

Abstract of source 74, as delivered. Every bipartite matching-support polynomial with a minimum vertex cover meeting one shore in at most two vertices is ultra-log-concave after normalization by its matching rank, for arbitrary positive activities on both shores. The main construction replaces a displayed $2 + q$ cover by a rank- $(q + 2)$ represented matroid whose Lorentzian basis polynomial specializes exactly to the homogenized support polynomial. Consequently weighted rank normalization holds through rank five, and known rank-six Hall obstructions show that the first weighted failure rank is exactly six. We also prove a sharp all-core boundary inequality with constant three, give independent Lorentzian transfer proofs

for star cores, and derive quantitative eventual strictness under two natural activity scalings. Full proofs, finite exact coefficient certificates, sum-of-squares identities and independent verification scripts are included.

Scope and status. The rank-five weighted question posed in the inspected ProveIt draft is resolved here. At higher rank the theorem covers every minimum cover with a shore of size at most two; the unrestricted unit-weight and one-shore-weighted problems at higher rank are not addressed here. These are proofs in ordinary mathematics, not Lean-certified or independently refereed results. The work extends the inspected rank manuscript and the separate rank-four and full-Hall classification reports [10, 14, 13]; global priority is not claimed. [*Merge note:* The “inspected ProveIt draft” and “rank manuscript” [10] is the manuscript printed as Part XII of *preorder-root-polytopes* (pinned at 1b3960d8a); the companions [14] and [13] are sources 76 and 75, printed in Sections 15 and 16; [11] and [9] are Parts XIII and XI of *preorder-root-polytopes*. The unit-weight and one-shore problems that this paragraph leaves aside are the subject of Parts II–IV (batch-72 sources 71, 73 and later).]

2 Definitions and main conclusions

For a finite bipartite graph $G = (X, Y; E)$ and positive vertex activities u_x, v_y , write

$$p_G(t) = \sum_{k=0}^r a_k t^k, \quad a_k = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|=k \\ G[I,J] \text{ perfectly matchable}}} u_I v_J, \quad r = \nu(G).$$

Each endpoint pair is counted once. In particular $p_{K_{2,2}}(t) = 1 + 4t + t^2$ at unit activities. The polynomial is ULC_D , for $D \geq \deg p_G$, if its zero-padded sequence $a_k / \binom{D}{k}$ is log-concave and has no internal zeros. Equivalently, for these support polynomials,

$$\Delta_k^{(D)} := k(D-k)a_k^2 - (k+1)(D-k+1)a_{k-1}a_{k+1} \geq 0. \quad (1)$$

Strict ULC_D means all $D-1$ displayed gaps are positive.

A displayed $2+q$ cover is $A \sqcup B$, where $A = \{A_1, A_2\} \subseteq X$ and $|B| = q$, $B \subseteq Y$. Put $L = X \setminus A$ and $R = Y \setminus B$. There are no L – R edges. The induced graph $G[A, B]$ is called the core. We call the displayed cover *genuine* when $\nu(G) = q+2$.

Theorem 2.1 (A two-vertex cover shore). *If G has a displayed $2+q$ cover, fix positive activities v on its right shore. The homogeneous left marginal*

$$\sum_{\substack{I \subseteq X \\ |I| \leq q+2}} \left(\sum_{\substack{J \subseteq Y \\ (I,J) \text{ feasible}}} v_J \right) z_I s^{q+2-|I|}$$

is Lorentzian. In particular, the degree- $(q+2)$ homogenization of its weighted matching-support polynomial is Lorentzian. Thus p_G is ULC_{q+2} for every positive activity assignment. If the displayed cover is minimum, the normalization is by the actual matching rank.

[*Merge note:* Added 1 October 2026, batch 73: third and fourth routes to the graph statements, and a deformation of every matroid basis polynomial that contains this lift (source 79’s Theorem 42.1, with source 80’s union proof), are in Part V; the lift does not extend to three tails (Section 43).]

Corollary 2.2 (Weighted rank five and the first failure rank). *Every bipartite matching-support polynomial of matching rank $r \leq 5$ is ULC_r for arbitrary positive activities on both shores. The smallest rank admitting a weighted counterexample is exactly six. At every rank $r \geq 6$ there is a weighted counterexample.*

The positive conclusion follows from Theorem 2.1, the pure and singleton-side cover cases proved below, and König’s theorem. The negative construction is given in Section 6. For nonnegative activities, delete zero-activity vertices first; the same conclusion holds with r interpreted as the surviving degree. [Merge note: Corollary 2.2 contains Part XIII’s Corollary 206.3 of *preorder-root-polytopes* ($\nu \leq 3$, `lc:cor:rankthree`) and source 76’s Corollary 15.2 ($\nu \leq 4$), which is printed in Section 15 as a second route by real stability. The rank-six counterexample it relies on is Part XI’s $H_{3,33}$ (Theorem 177.1 of *preorder-root-polytopes*, restated in Section 6). It answers Research questions 86 and 93 of *preorder-root-polytopes* and gives $\mathcal{D}(4) = 4$, $\mathcal{D}(5) = 5$ in its Research question 87 (Section 1.4).] [Merge note: Added 1 October 2026, batch 73: in the report *preorder-gamma-rank-ulc*, the degree-four reduction for unit-activity preorders (its Section 3, from its source 16) treats connected bipartite components by a weighted bipartite rank-four theorem, which is source 77’s Theorem 40.1 here; a merge note there records that this corollary already covers them. That report also lists this corollary among the bipartite specializations of its Part II, and applies it in its Part III on the faces of a universal-sink family where no vertex is used both as a tail and as a head.]

Theorem 2.3 (Forced-sector and strict-scaling refinements). *For any displayed $2 + q$ cover with $q \geq 1$, the coefficient of the product of the two A activities in p_G is $t^2 C(t)$, where C is ULC_q . For a genuine cover, common scaling of only the two A activities makes p_G strictly ULC_{q+2} for all sufficiently large scale factors.*

For a rank-five $2 + 3$ cover whose core has matching rank two, common scaling of all five cover activities also makes all four gaps strictly positive eventually. An explicit coefficient bound gives a sufficient threshold for both scalings.

Every fixed minimum-cover scaling at rank at most five is therefore ULC at *every* positive scale factor. At each rank at least six, some fixed graph and minimum cover fail for every sufficiently large common cover scale, so the universal scaling threshold is sharp as well. [Merge note: Here the whole minimum cover is scaled, on both shores. Sources 08s, 08, 03, 02 and 01 (Part IV) scale only the right part of a minimum cover, with unit left activities, and there no failure occurs below matching number nine. See the entry “Scalings” in Section 1.3.]

3 Lorentzian input and singleton covers

We use the following published facts from Brändén–Huh [1]: matroid basis polynomials are Lorentzian (Theorem 3.10); partial derivatives, nonnegative linear substitutions, products and coefficientwise limits preserve this class (Corollary 2.11, Theorem 2.10, Corollary 2.32, and closedness); homogeneous stability preservers that preserve nonnegative coefficients preserve Lorentzian polynomials (Theorem 3.4); and multiaffine truncation and polarization preserve the class (Corollary 3.5 and Proposition 3.1). A homogeneous bivariate polynomial of degree D is Lorentzian exactly when its coefficients form a ULC_D sequence (Example 2.26). The zero polynomial is included by closure.

Lemma 3.1 (Weighted smaller-shore inequality). *Every bipartite support polynomial is $\text{ULC}_{\min(|X|, |Y|)}$.*

Proof (pointer). This is Theorem 70.1 of *preorder-root-polytopes* (Part V, `lor:thm:weighted ULC`), restated as Lemma 205.1 of its Part XIII (`lc:lem:ambient`). [Merge note: Source 74

proves it by the same specialization of the basis polynomial of the transversal matroid on the private copies $\bar{X} \sqcup Y$, at s/u_x and tv_y , followed by the bivariate Lorentzian characterization and transposition, and calls it the standard bimatroid specialization [7]. That proof is Part XIII's proof of Lemma 205.1 and is not reprinted.] $\square$

Lemma 3.2 (Singleton-side minimum covers). *If a rank- r graph has a minimum cover entirely on one shore or split as $1 + (r - 1)$, its support polynomial is ULC_r for all positive activities.*

Proof (pointer). This is Theorem 206.2 of *preorder-root-polytopes* (Part XIII, `lc:thm:singleton`), proved there by weighted leaf compression, Lemma 206.1 (`lc:lem:compression`). [Merge note: Source 74's proof is the same argument (delete isolated vertices for a pure cover; for a cover $\{x_0\} \sqcup B$ merge the leaves at x_0 into one leaf carrying their total activity, which leaves p_G unchanged, and apply Lemma 3.1); source 74 credits it as “the exact leaf-compression argument of” [11], which is Part XIII. It is not reprinted.] $\square$

4 The exact two-row formula

Let the two A activities be x, y , and let $H = G[L, B]$. For a virtual left vertex with neighborhood $S \subseteq B$ and activity one, define g_S by

$$p_H = f, \quad p_{H+S} = f + tg_S.$$

If S_1, S_2 are the actual core-row neighborhoods, define h by

$$p_{H+A} = f + t(xg_1 + yg_2) + xyt^2h, \quad g_i = g_{S_i}, \quad g_U = g_{S_1 \cup S_2}.$$

The feasible support set for g_U is the union of those for g_1, g_2 : in a matching the virtual row uses one edge, and that edge belongs to at least one of the original neighborhoods.

For R , let a, b, c be the sums of activities in classes with neighborhoods $\{A_1\}$, $\{A_2\}$ and A , respectively. If the individual common-class activities are $r_1, \dots, r_m$, put

$$d = \sum_{i < j} r_i r_j, \quad Q = ab + ac + bc + d, \quad 0 \leq d \leq c^2/2.$$

Then the exact full polynomial is

$$\begin{aligned} p_G = & f[1 + (x(a + c) + y(b + c))t + xyQt^2] + t(xg_1 + yg_2) \\ & + xyt^2(h + ag_2 + bg_1 + cg_U). \end{aligned} \tag{2}$$

To prove this, condition on the selected A vertices and then on the number of selected R vertices. With both A selected, two selected R contribute Qf ; one singleton-class vertex forces the corresponding A and gives ag_2 or bg_1 ; one common-class vertex leaves the union-neighborhood alternative and gives cg_U ; no selected R gives h . These cases have disjoint endpoint sets. In particular the common term is cg_U , not $c(g_1 + g_2)$.

5 The full rank compression matroid lift

We prove Theorem 2.1. First suppose $Q > 0$. The same construction includes $q = 0$, with no original B slots. Form a transversal matroid matched into the $q + 2$ slots

$$B \sqcup \{e_1, e_2\}.$$

Its elements and allowed slots are as follows:

- a private element $\bar{b}$ for each $b \in B$, allowed only in slot b ;
- an element for each $l \in L$, allowed in its original neighborhood in B ;
- two augmented core elements $\hat{A}_i$, allowed in $S_i \cup \{e_i\}$;
- two plane elements P_1, P_2 , allowed only in e_1, e_2 , respectively;
- one common element Z_j for each common right-exterior vertex, allowed in both e_1, e_2 .

The rank is $q + 2$, since all private B elements together with P_1, P_2 form a basis. Every basis is counted once, regardless of how many injections into the slots it admits. We call P_1, P_2, Z_j the plane elements, as they use only the two additional slots.

Give the private B variables the values s/v_b , and the exterior-left variables tu_l . Keep core variables X, Y , and specialize the plane variables to

$$P_1 : \frac{bs}{Q}, \quad P_2 : \frac{as}{Q}, \quad Z_j : \frac{r_js}{Q}. \quad (3)$$

Multiply this basis specialization by $Q \prod_{b \in B} v_b$. It is Lorentzian by the matroid and nonnegative-substitution theorems.

We now compute it exactly. Put

$$f_h = s^q f(t/s), \quad G_i = s^{q-1} g_i(t/s), \quad G_U = s^{q-1} g_U(t/s), \quad H_h = s^{q-2} h(t/s),$$

where $G_i = G_U = 0$ is defined directly for $q = 0$, and $H_h = 0$ is defined directly for $q < 2$; the displayed homogenizing formulas are used only for nonnegative exponents. Classify a basis by the number k of core elements and ℓ of plane elements. Other elements use only the q original slots, so $k + \ell \geq 2$ and $0 \leq k, \ell \leq 2$. There are six cases.

1. $(k, \ell) = (0, 2)$. Any pair of distinct plane elements can fill the two extra slots, and the other q elements fill the B slots. The sum of plane-pair activities is

$$\frac{s^2}{Q^2} \left(ab + (a + b)c + \sum_{i < j} r_i r_j \right) = \frac{s^2}{Q}.$$

After the exterior factor Q , the contribution is $s^2 f_h$.

2. $(k, \ell) = (1, 1)$. With core $\hat{A}_1$, the plane element must be able to occupy e_2 , hence is P_2 or a Z_j . Its total activity is $(a + c)s/Q$. The remaining q elements fill the B slots, giving $Xs(a + c)f_h$. With $\hat{A}_2$, the contribution is $Ys(b + c)f_h$.
3. $(k, \ell) = (1, 2)$. The plane pair fills both extra slots, forcing the core element into B . The remaining support uses row S_i and $q - 1$ ordinary elements, with profile G_i . The contribution is $Xs^2 G_1 + Ys^2 G_2$.
4. $(k, \ell) = (2, 0)$. The two core elements occupy their dedicated extra slots. The other q elements fill B , contributing $QXY f_h$.
5. $(k, \ell) = (2, 1)$. The element P_1 occupies e_1 , forcing $\hat{A}_2$ into e_2 and $\hat{A}_1$ into B ; hence its profile is G_1 and its activity coefficient is b . Similarly P_2 gives G_2 with coefficient a . A common Z_j can occupy either extra slot, so feasibility holds exactly when either core row completes the remaining B -slot support. This is the union profile G_U , counted once even if both alternatives work. Thus the contribution is $XYs(bG_1 + aG_2 + cG_U)$.

6. $(k, \ell) = (2, 2)$. The two plane elements fill the extra slots, forcing both core elements into B together with $q - 2$ other elements. The contribution is XYs^2H_h ; this case is absent for $q < 2$.

Consequently the constructed Lorentzian polynomial is

$$\begin{aligned}\mathcal{L}(s, t, X, Y) = & s^2f_h + sX[(a+c)f_h + sG_1] + sY[(b+c)f_h + sG_2] \\ & + XY[Qf_h + s(bG_1 + aG_2 + cG_U) + s^2H_h].\end{aligned}\tag{4}$$

Set $X = xt$ and $Y = yt$. Equation (2) identifies (4) with $s^{q+2}p_G(t/s)$, proving the bivariate assertion for $Q > 0$. More generally, keep each exterior-left element variable as its own z_l rather than tu_l , and set $X = z_{A_1}$, $Y = z_{A_2}$. The same six basis cases give exactly the homogeneous left marginal in Theorem 2.1. Every plane and private variable remains a nonnegative multiple of s , so this full marginal is Lorentzian. No analogous opposite-marginal assertion is inferred when $q > 2$.

The six cases are disjoint classes of matroid bases. In particular the union case does not count injections or matching allocations. The construction also has a generic matrix representation obtained by giving each allowed slot incidence its own indeterminate. Its six block-determinant expansions provide a second interpretation of the same identity, including dependent or zero original core-row vectors.

If $Q = 0$, add a singleton right vertex at each A_i with activity $\varepsilon > 0$. The perturbed graph still has a displayed $2 + q$ cover, its Q is positive, and its coefficients converge to those of the original graph as $\varepsilon \downarrow 0$. Closedness gives the result. Other zero activities are handled by continuity at fixed order $q + 2$; private reciprocals are used only before taking the positive-activity limit.

Corollary 5.1. *Any failure of weighted rank-normalized ultra-log-concavity must have every minimum vertex cover meet both shores in at least three vertices.*

Proof. A pure or singleton-side minimum cover is covered by Lemma 3.2; a cover meeting one shore in exactly two vertices is covered by Theorem 2.1. $\square$

Remark 5.2. The new rank- $(q + 2)$ transversal matroid is used through its Lorentzian basis polynomial. The theorem asserts neither real-rootedness of p_G nor stability of the original complemented-row polynomial. The support identity, and not the values of nonzero determinants, determines all coefficients.

6 Exact failure at rank six and at every higher rank

We include the elementary obstruction calculation to make the sharpness conclusion self-contained. It is the negative construction from the Hall reports [13, 9]. [*Merge note:* [13] is source 75 (Section 16.5) and [9] is Part XI of *preorder-root-polytopes*. For $a = b$ and $N = M$ the graph $H_{a,b;N,M}$ with weight w on both cores is Part XI's $H_{s,n}$, and formula (5) is Part XI's Theorem 175.3 (`hb:thm:coefficients`) extended to unequal cores and populations. The endpoint quadratic below is the asymmetric form of Part XI's endpoint analysis (Theorem 178.2, `hb:thm:endpoint`).]

Let $H_{a,b;N,M}$ have a complete $a \times b$ core, N exterior-left vertices adjacent to every right-core vertex, and M exterior-right vertices adjacent to every left-core vertex. Give core vertices activity w and exterior vertices activity one. If $N \geq b$ and $M \geq a$, the core is a minimum cover of size $r = a + b$: match its two shores independently into the opposite exterior sets.

A size- k endpoint support with i selected left-core and j selected right-core vertices is feasible exactly when $i + j \geq k$, subject to the cardinality bounds. Match all exterior vertices to the core first, then use the complete core for the remainder. Thus

$$a_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{b}{j} \binom{N}{k-i} \binom{M}{k-j} w^{i+j}. \quad (5)$$

Put $A_i = \binom{N}{b-i}$, $B_i = \binom{M}{a-i}$ for $i = 0, 1, 2$, and

$$X = aA_0B_1 + bA_1B_0, \quad Y = aA_1B_2 + bA_2B_1, \quad Z = \binom{a}{2}A_0B_2 + abA_1B_1 + \binom{b}{2}A_2B_0.$$

Counting at most two omitted core vertices gives

$$a_r = A_0B_0w^r, \quad a_{r-1} = A_1B_1w^r + Xw^{r-1}, \quad a_{r-2} = A_2B_2w^r + Yw^{r-1} + Zw^{r-2}.$$

Hence

$$\Delta_{r-1}^{(r)} = w^{2r-2}(\alpha w^2 + \beta w + \gamma), \quad (6)$$

where

$$\begin{aligned} \alpha &= (r-1)A_1^2B_1^2 - 2rA_0A_2B_0B_2, \\ \beta &= 2(r-1)A_1B_1X - 2rA_0B_0Y, \\ \gamma &= (r-1)X^2 - 2rA_0B_0Z. \end{aligned}$$

6.1 A concrete rank-six witness

Take $a = b = 3$ and $N = M = 33$. Here $A_0 = B_0 = 5456$, $A_1 = B_1 = 528$, $A_2 = B_2 = 33$, and exact simplification of (6) gives

$$\Delta_5^{(6)} = 404794368 w^{10}(-w^2 + 26784w + 522784).$$

At $w = 26804$, the parenthesis equals -13296 . This is an exact positive-integer-activity counterexample on 72 vertices and 207 edges, with minimum cover and matching rank six. Combined with Corollary 2.2, the first weighted failure rank is exactly six. [*Merge note*: This witness is Part XI's $H_{3,33}$: Theorem 177.1 of *preorder-root-polytopes* (`hb:thm:rank6phase`) gives the last order-six margin of $H_{3,n}$ at $n = 33$ as 16 times the same quadratic $-w^2 + 26784w + 522784$. Source 74 restates and verifies it (its bibliography says so); it is not new here.]

6.2 Permanent scaling failure at every rank at least six

Let $a, b \geq 3$. Write

$$K = (r-1)ab, \quad J = (r+1)ab - 2r(r-1).$$

Because $ab \geq 3(r-3)$ and $r \geq 6$,

$$J \geq r^2 - 4r - 9 > 0.$$

Choose $N = b + 30$, $M = a + 30$, so that

$$\frac{A_1^2B_1^2}{A_0A_2B_0B_2} = \frac{ab}{(a-1)(b-1)} \left(\frac{32}{31} \right)^2.$$

The inequality $\alpha < 0$ is equivalent to

$$K(2 \cdot 31 + 1) < J \cdot 31^2.$$

Indeed

$$15J - K = (14r + 16)ab - 30r(r - 1) \geq 12(r - 6)(r + 2) \geq 0,$$

and $63/961 < 1/15 \leq J/K$. Thus (6) is negative for every sufficiently large w . An explicit sufficient integer is

$$w \geq 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor,$$

since for $w \geq 1$ the bracket is at most $w(\alpha w + |\beta| + |\gamma|) < 0$. Taking $a = 3$, $b = r - 3$ supplies every $r \geq 6$. In contrast, every graph of rank at most five remains ULC at every positive minimum-cover scale by Corollary 2.2. This proves the sharp universal rank threshold for minimum-cover scaling.

6.3 The shapes recorded by source 74's verifier

[*Merge note:* This subsection was written in the merge from source 74's recorded output `data/74-weighted-rank-five-negative_shapes_verification.json`; its numbers were recomputed independently from (5). Source 74's text does not print them.] Source 74's negative-family verifier records seven exact witnesses, listed in Table 3. Each has exterior activity one and one common integer activity w on all $a + b$ core vertices; the populations are $N = b - 1 + h$ and $M = a - 1 + h$ with source 75's offset $h = 1 + \lceil 2K/H \rceil$ (Section 16.5), and w is the least integer at which the last gap $\Delta_{r-1}^{(r)}$ is negative for these populations: the primitive endpoint quadratic (the bracket of (6) divided by its content) is negative at w and positive at $w - 1$. The (4, 4) row is Part XI's 30-vertex counterexample (Theorem 176.1 of

(a, b)	r	(N, M)	vertices	edges	w	primitive quadratic	in <i>preorder-root-polytopes</i>
(3, 3)	6	(33, 33)	72	207	26804	$-w^2 + 26784w + 522784$	Part XI, $H_{3,33}$
(3, 4)	7	(16, 15)	38	121	2012	$-3w^2 + 6006w + 55601$	new
(3, 5)	8	(15, 13)	36	129	418	$-115w^2 + 46860w + 469964$	new
(4, 4)	8	(11, 11)	30	104	485	$-w^2 + 480w + 2304$	Part XI, $H_{4,11}$
(4, 5)	9	(11, 10)	30	115	203	$-215w^2 + 42560w + 192472$	new
(5, 5)	10	(10, 10)	30	125	131	$-11w^2 + 1400w + 4928$	new
(3, 8)	11	(16, 11)	38	185	235	$-158w^2 + 34860w + 518355$	new

Table 3: Exact last-gap failures of $H_{a,b,N,M}$ recorded by source 74's verifier (merge table).

preorder-root-polytopes, `hb:thm:witness`, which uses $w = 500$); the (3, 3) row is its rank-six witness. The other five rows are asymmetric or larger-core shapes, which bear on Research question 89 of *preorder-root-polytopes*; none has fewer than 30 vertices, and none has matching number below six, as Corollary 2.2 requires.

7 An established geometric interpretation

For unit activities, p_G is the *perfectly matchable set polynomial* of Davis–Kohl [5]. It is distinct from the ordinary matching or rook polynomial, which counts matchings with multiplicity. Their Theorem 3.10, restating results of Ohsugi–Tsuchiya, gives the following precise Ehrhart interpretation.

Given bipartite $G = (X, Y; E)$, form $\widehat{G}$ by adjoining x_0 on the left and y_0 on the right, with edges from x_0 to every old right vertex, from y_0 to every old left vertex, and x_0y_0 . For the edge polytope

$$Q_{\widehat{G}} = \text{conv}\{e_x + e_y : xy \in E(\widehat{G})\},$$

one has

$$h^*(Q_{\widehat{G}}; t) = p_G(t) = h^*(\text{STAB}(\overline{G}); t).$$

The second identity is their Lemma 3.11; $\text{STAB}(\overline{G})$ is the stable-set polytope of the complement.

Corollary 7.1. *If G has matching rank $r \leq 5$, or a minimum vertex cover meeting one shore in at most two vertices, these two h^* -polynomials are ULC_r , normalized by their degree $r = \nu(G)$.*

Only the unit-activity interpretation is used here. No assertion is made about arbitrary Ehrhart h^* -polynomials or about the h^* -polynomial of an unrelated preorder polytope. [*Merge note:* At unit activities without the cover or rank hypothesis the conclusion fails: source 71 (Part II) gives a connected graph with $\nu = 3450$ whose two h^* -polynomials are not ULC_{3450} .] [*Merge note:* Sections 8–13 were the appendices of source 74; they are printed here as ordinary sections of Part I.]

8 A generic-line matroid model for the forced-left sector

By (2), the polynomial in Theorem 2.3 is

$$C = Qf + ag_2 + bg_1 + cg_U + h. \quad (7)$$

Write $f_h = s^q f(t/s)$, $G_i = s^{q-1} g_i(t/s)$, $G_U = s^{q-1} g_U(t/s)$ and $H_h = s^{q-2} h(t/s)$. For $q = 1$ the last term is zero. The right-complement transversal matroid of $H + A$ gives a degree- q basis specialization

$$F = f_h + z_1 G_1 + z_2 G_2 + z_1 z_2 H_h.$$

The private B variables are s/v_b , the exterior-left variables are tu_l , the two distinguished elements have variables z_1, z_2 , and we multiply by $\prod_b v_b$.

Represent this matroid by a generic matrix: use an independent indeterminate for each allowed adjacency and private standard basis columns. A square determinant is nonzero exactly when its support admits a matching, since different matchings give distinct monomials. Denote the two distinguished columns by v_1, v_2 . Adjoin columns

$$U_j = \alpha_j v_1 + \beta_j v_2 \quad (1 \leq j \leq m),$$

with generic coefficients, algebraically independent over the original matrix field. For a basis containing just one U_j from this distinguished set, its determinant is $\alpha_j D_1 + \beta_j D_2$ and is nonzero iff at least one of D_1, D_2 is nonzero. Thus its remaining-element profile is exactly G_U . Any pair of distinct distinguished columns has determinant a nonzero scalar multiple of the determinant using v_1, v_2 , hence profile H_h . Three distinguished columns are dependent. These statements also hold when v_1, v_2 are dependent or zero: pair profiles then vanish, and the single profile remains the union.

The resulting matroid basis specialization is therefore

$$\begin{aligned} F_{\text{ext}} = & f_h + z_1 G_1 + z_2 G_2 + \left(\sum_j Z_j \right) G_U \\ & + \left[z_1 z_2 + (z_1 + z_2) \sum_j Z_j + \sum_{i < j} Z_i Z_j \right] H_h. \end{aligned} \quad (8)$$

It is Lorentzian. For $Q > 0$, substitute

$$z_1 = bs/Q, \quad z_2 = as/Q, \quad Z_j = r_js/Q.$$

The bracket becomes s^2/Q , and

$$QF_{\text{ext}} = Qf_h + s(aG_2 + bG_1 + cG_U) + s^2H_h = s^qC(t/s).$$

This proves the first part of Theorem 2.3. For $Q = 0$, add two singleton right vertices with positive activities tending to zero and use coefficientwise closedness.

Corollary 8.1 (Scaling only the two cover vertices). *For a genuine $2 + q$ cover with $q \geq 1$, scaling only the two A activities by a common sufficiently large factor makes every ULC_{q+2} gap strictly positive.*

Proof. Every matching of size $q + 2$ uses exactly one cover vertex per edge. Thus R matches both A vertices and L matches all B , so $Q > 0$ and $f_q > 0$. Consequently all coefficients $C_0, \dots, C_q$ are positive. Write the two activities as $w\alpha, w\beta$. With E the weighted edge sum incident with A at relative activities α, β ,

$$a_1(w) = wE + O(1), \quad a_k(w) = w^2\alpha\beta C_{k-2} + O(w) \quad (k \geq 2).$$

The graph induced by A and the opposite shore has matching rank two, so $E^2 \geq 4\alpha\beta C_0$. The first two leading gaps are respectively

$$(q+1)E^2 - 2(q+2)\alpha\beta C_0 \geq 2q\alpha\beta C_0 > 0, \quad 2q(\alpha\beta C_0)^2 > 0,$$

at powers w^2, w^4 . For $k = i + 2$, $1 \leq i < q$, the leading coefficient at w^4 is at least

$$\frac{2(q-i+1)}{i}(\alpha\beta)^2 C_{i-1} C_{i+1} > 0,$$

by substituting the ULC_q inequality for C_i^2 into (1). Positive leading coefficients prove the claim. $\square$

9 The sharp all-core inequality at rank five

The full lift gives a short general boundary theorem before the independent rank-five certificate below.

Theorem 9.1 (Sharp all-core constant for a two-vertex shore). *For a $2 + q$ cover with $q \geq 2$, factor the product of all core activities from the all-core sector and write it as $t^q(c_0 + c_1t + c_2t^2)$. Then*

$$c_1^2 \geq \frac{2q}{q-1} c_0 c_2,$$

and the constant is optimal for every q .

Proof. In the lifted matroid basis polynomial, set all private B variables to zero and differentiate once in each augmented core variable X, Y . These operations preserve the Lorentzian property, giving a degree- q polynomial. Under the remaining specializations its three plane-count sectors give, up to the fixed positive scaling factor,

$$c_0 s^2 t^{q-2} + c_1 s t^{q-1} + c_2 t^q.$$

The bivariate ULC_q inequality at index $q - 1$ is the desired bound. Zero activities and $Q = 0$ follow by continuity as before.

For the complete $2 \times q$ core with $N \geq q$ unit common-left vertices and $M \geq 2$ unit common-right vertices,

$$c_0 = \binom{N}{q-2}, \quad c_1 = M \binom{N}{q-1}, \quad c_2 = \binom{M}{2} \binom{N}{q}.$$

Their ratio is

$$\frac{c_1^2}{c_0 c_2} = \frac{2q}{q-1} \frac{M}{M-1} \frac{N-q+2}{N-q+1} \rightarrow \frac{2q}{q-1}.$$

□

The following argument independently proves the $q = 3$ case using only a finite quartic coefficient test and explicit sums of squares. It does not rely on the full lift.

Now fix a $2 + 3$ cover. Strip off the product of the five core activities from the sector in which all five core vertices are selected. Its polynomial is

$$t^3(c_0 + c_1 t + c_2 t^2).$$

Theorem 9.2 (Sharp constant three). *For arbitrary nonnegative exterior activities and every $2 + 3$ core,*

$$c_1^2 \geq 3c_0 c_2.$$

The constant 3 cannot be increased uniformly.

Let T be the weight of exterior-left triples matching all of B . Let C be the weight of single exterior-left vertices which, together with the two core rows, match B . Let P_i be the weight of exterior-left pairs completable with core row S_i , and let P_U use the union row. Set

$$O = P_1 + P_2 - P_U.$$

The union interpretation gives $0 \leq O \leq \min(P_1, P_2)$. The central left-side inequality is

$$4P_1 P_2 - 2O^2 \geq 3CT. \tag{9}$$

9.1 A finite exact coefficient certificate

Encode a nonempty neighborhood in B by its bit mask $1, \dots, 7$. For three nonempty masks, let $M(r, s, t)$ be one exactly when they admit a matching to all three columns. Hall's condition gives the elementary rule: their bitwise union is 7, and no singleton mask $1, 2, 4$ appears twice. For general tuples including an empty mask, use ordinary Boolean matchability.

Index left exterior vertices by i , with masks t_i and weights z_i . Put

$$\begin{aligned} c_i &= M(S_1, S_2, t_i), & p_{ij} &= M(S_1, t_i, t_j), & q_{ij} &= M(S_2, t_i, t_j), \\ o_{ij} &= p_{ij} q_{ij}, & \tau_{ijk} &= M(t_i, t_j, t_k). \end{aligned}$$

Then $C = \sum c_i z_i$, $P_1 = \sum_{i < j} p_{ij} z_i z_j$, and similarly for P_2, O, T . In the quartic

$$I = 4P_1 P_2 - 2O^2 - 3CT$$

every exponent is at most two. Its coefficient of $z_i^2 z_j^2$ is $2o_{ij} \geq 0$. Its coefficient of $z_i^2 z_j z_k$ for distinct indices is

$$4(p_{ij} q_{ik} + p_{ik} q_{ij}) - 4o_{ij} o_{ik} - 3c_i \tau_{ijk}. \tag{10}$$

Its coefficient of $z_1 z_2 z_3 z_4$ is

$$4 \sum_{ij|kl} (p_{ij} q_{kl} + p_{kl} q_{ij}) - 4 \sum_{ij|kl} o_{ij} o_{kl} - 3 \sum_i c_i \tau_{[4] \setminus \{i\}}, \quad (11)$$

where each of the three unordered pair partitions is used once.

If the core has matching rank at most one, $C = 0$ and (9) follows from $O^2 \leq P_1 P_2$. Otherwise row exchange and column permutation give exactly the eight representatives in Table 4. The table records the complete range of possible coefficients in (10) and (11). It is obtained by the stated three-row Boolean rule, over $7^3 = 343$ ordered triples and $\binom{10}{4} = 210$ neighborhood multisets per core. The included exact verifier implements an independent permutation matcher and checks all 4424 substitutions, as well as all 46 labeled matching-rank-two cores.

Core masks	Repeated-exponent coefficients	Squarefree coefficients
(1, 2)	0, 1, 4	0, 2, 3
(1, 3)	0, 1, 4	0, 2, 3, 6, 8, 12
(1, 6)	0, 1, 4	−4, 0, 2, 3
(1, 7)	0, 1, 4	−4, 0, 2, 3, 8, 12
(3, 3)	0, 1, 4	0, 2, 3, 6, 8, 12
(3, 5)	0, 1, 4	0, 2, 3, 8, 12
(3, 7)	0, 1, 4	0, 2, 3, 8, 12
(7, 7)	0, 1, 4	0, 2, 3, 8, 12

Table 4: The complete finite coefficient test. In each exceptional row the only four negative neighborhood multisets are listed in (12).

Only cores (1, 6) and (1, 7) have negative coefficients. In both cases the negative neighborhood multisets are exactly

$$(2, 3, 4, 5), \quad (2, 3, 5, 5), \quad (3, 3, 4, 5), \quad (3, 3, 5, 5), \quad (12)$$

and each coefficient is -4 . Thus every term involving another neighborhood class is nonnegative. It remains to treat the four classes 2, 3, 4, 5 in these two cores.

9.2 The four-class sum-of-squares certificate

For core (1, 6) let a, b, c, d now denote the activity sums in classes 2, 3, 4, 5, and let B_2, D_2 be the second elementary moments of classes 3, 5. Put $u = a + b$, $v = c + d$. Exact support counting yields

$$\begin{aligned} C &= u + v, & P_1 &= uv, \\ P_2 &= (a + c)(b + d) + bd + B_2 + D_2, & O &= ad + bc + bd, \\ T &= v(ab + B_2) + u(cd + D_2). \end{aligned}$$

Since $0 \leq B_2 \leq b^2/2$ and $0 \leq D_2 \leq d^2/2$, and I is affine in both moments, it suffices to check the four rectangle corners. Write $I_{00}, I_{01}, I_{10}, I_{11}$ for the lower/lower, lower/upper, upper/lower

and upper/upper choices. The following exact identities have only nonnegative summands:

$$\begin{aligned}
I_{00} &= 2(ad - bc)^2 + uv(ab + cd) + (ac + bd)(ad + bc) + 2b^2d^2, \\
I_{01} &= \frac{1}{2}d^2(a - b)^2 + \frac{1}{2}ad(b - d)^2 + ac(b - d)^2 + bc(a - d)^2 + R_{01}, \\
R_{01} &= a^2(bd + cd) + a\left(\frac{1}{2}b^2d + bc^2 + c^2d + \frac{1}{2}cd^2\right) \\
&\quad + b^2(2c^2 + cd) + b\left(c^2d + \frac{1}{2}cd^2 + \frac{1}{2}d^3\right), \\
I_{11} &= \frac{1}{2}bd(u - v)^2 + \frac{1}{2}(ad + bc + 3ac)(b - d)^2 \\
&\quad + (b + d)\left(ac(a + c) + \frac{1}{2}(a^2d + bc^2)\right).
\end{aligned}$$

The expression for I_{10} is obtained from I_{01} by interchanging (a, b) with (c, d) . For core $(1, 7)$, the restricted P_2, O both increase by ac while C, T, P_1 are unchanged. Since $P_1 - O = ac$, its quartic is exactly the preceding $I + 2a^2c^2$. This proves (9) in every case. The included symbolic checker verifies all five identities exactly.

9.3 Passing to the right exterior and proving sharpness

Return to the right-side notation a, b, c, d of Section 4 [*Merge note:* source 74 writes “Section 3”]. Selecting all core vertices gives

$$c_0 = C, \quad c_1 = aP_2 + bP_1 + cP_U, \quad c_2 = TQ.$$

Let $A = a + c$, $B = b + c$, $p = P_1$, $q = P_2$. Then

$$c_1 = qA + pB - Oc, \quad Q \leq AB - c^2/2.$$

For $pq > 0$, the exact quadratic identity

$$\begin{aligned}
&(qA + pB - Oc)^2 - (4pq - 2O^2)(AB - c^2/2) \\
&= \left(1 - \frac{O^2}{2pq}\right)(qA - pB)^2 + \frac{[O(qA + pB) - 2pqc]^2}{2pq} \geq 0
\end{aligned}$$

uses only $O^2 \leq pq$. Together with (9) it implies $c_1^2 \geq 3CTQ = 3c_0c_2$. If $pq = 0$, the same assertion follows by continuity, or directly from $O = 0$ and $CT = 0$.

For sharpness, take the complete 2×3 core, N common left-exterior vertices and M common right-exterior vertices, all with unit exterior activities. Then

$$c_0 = N, \quad c_1 = \binom{N}{2}M, \quad c_2 = \binom{N}{3}\binom{M}{2}, \quad \frac{c_1^2}{c_0c_2} = 3\frac{N-1}{N-2}\frac{M}{M-1} \rightarrow 3.$$

This completes Theorem 9.2.

10 Lorentzian transfer across a star core

Theorem 10.1 (Star-core transfer). *For a displayed $2+q$ cover with matching-rank-at-most-one core, both root orientations admit direct Lorentzian transfer proofs of ULC_{q+2} .*

This is now a special case of Theorem 2.1. We retain the independent proofs because the transfer operators and their interlacing certificates are useful beyond the full-lift construction.

10.1 A support-exact vertex-sum identity

Suppose G_1, G_2 intersect in just one vertex v of activity u , and write

$$p_1 = A(t) + utB(t), \quad p_2 = C(t) + utD(t).$$

Then

$$p_{G_1 \cup G_2} = AC + ut(BC + AD). \quad (13)$$

If v is absent, each component has balanced selected shores. If v is selected, exactly one component uses it, and that component is determined by its selected-vertex imbalance away from v : its opposite shore has one extra selected vertex. Thus the two root-using terms cannot count the same endpoint support, even if that support has several perfect matchings.

10.2 Stars centered on the two-vertex shore

Suppose every core edge is incident with $v = A_1$. Let G_1 have shores $L \cup \{v\}$ and B , and let G_2 have shores $\{v, A_2\}$ and R . Their intersection is just v . The right-complement basis construction for G_1 gives a Lorentzian polynomial

$$F(s, t, z) = A_h(s, t) + zB_h(s, t), \quad A_h = s^q A(t/s), \quad B_h = s^{q-1} B(t/s).$$

Here z is the root element variable; setting $z = ut$ gives the degree- q homogenization of p_1 .

For G_2 , write $C = 1 + dt$, $D = e + ft$. If A_2 has activity y , and the exterior class sums are a, b, c with common second moment δ , then

$$d = y(b + c), \quad e = a + c, \quad f = y(ab + ac + bc + \delta), \quad de - f = y(c^2 - \delta) \geq 0.$$

The homogeneous quadratic

$$K(s, t, z) = s^2 + dst + esz + ftz$$

is Lorentzian. Its Hessian is

$$\begin{pmatrix} 2 & d & e \\ d & 0 & f \\ e & f & 0 \end{pmatrix}, \quad \det = 2f(de - f) \geq 0.$$

For $f > 0$, its t, z principal submatrix has one positive and one negative eigenvalue; the determinant and eigenvalue interlacing exclude two positive eigenvalues of the full Hessian. For $f = 0$, $K = s(s + dt + ez)$ is a product of nonnegative linear forms.

The product FK is Lorentzian. Retain only its terms of degree at most one in z , then set $z = ut$. The output is

$$A_h(s^2 + dst) + ut[B_h(s^2 + dst) + A_h(es + ft)],$$

which is the degree- $(q + 2)$ homogenization of (13). Partial truncation preserves Lorentzian polynomials: polarize s, t , take the full multiaffine part (only powers z^2 are then removed), and identify the polarized variables again. Thus the output is Lorentzian and p_G is ULC_{q+2} .

10.3 Stars centered on the opposite shore

Suppose every core edge is incident with $v = B_1$. Let $G_1 = G[L, B]$ and let G_2 have shores A and $R \cup \{v\}$. For G_1 , the private-root variable now gives

$$F(s, t, z) = zA_h + tB_h, \quad A_h = s^{q-1}A(t/s), \quad B_h = s^{q-1}B(t/s).$$

The other private variables are s/v_b , the left variables are tu_l , and the basis polynomial is multiplied by $\prod_{b \neq v} v_b$. Hence F is Lorentzian and $uF(s, t, s/u) = s^q p_1(t/s)$.

For G_2 put $C_h = s^2 + c_1 st + c_2 t^2$ and $D_h = es + ft$. Give its two left vertices fixed activities α, β ; these are constants, distinct from the private-root indeterminate z . With exterior class sums a, b, c and second moment δ ,

$$c_1 = \alpha(a + c) + \beta(b + c), \quad c_2 = \alpha\beta(ab + ac + bc + \delta),$$

so

$$c_1^2 - 4c_2 = [\alpha(a + c) - \beta(b + c)]^2 + 4\alpha\beta(c^2 - \delta) \geq 0.$$

The rooted interlacing inequality is

$$c_1 ef - f^2 - c_2 e^2 \geq 0. \tag{14}$$

If v is adjacent only to the first left vertex, $e = \alpha$, $f = \alpha\beta(b + c)$ and the expression is $\alpha^3\beta(c^2 - \delta)$. The other singleton case gives $\alpha\beta^3(c^2 - \delta)$. If v is adjacent to both, $e = \alpha + \beta$, $f = \alpha\beta(a + b + c)$ and it is exactly

$$\alpha\beta \left[(a\alpha - b\beta + \frac{c}{2}(\alpha - \beta))^2 + \left(\frac{3c^2}{4} - \delta \right) (\alpha + \beta)^2 \right] \geq 0.$$

An isolated root has $e = f = 0$.

Factor $C_h = (s + \lambda_1 t)(s + \lambda_2 t)$, with $0 \leq \lambda_1 \leq \lambda_2$. If $e > 0$, (14) gives $\lambda_1 \leq f/e \leq \lambda_2$. Therefore

$$h(s, t) = \frac{stD_h}{C_h}$$

is a nonnegative linear combination of $st/(s + \lambda_i t)$ with positive total coefficient. Each such fraction maps $\mathbb{H}^2$ into $\mathbb{H}$, since its reciprocal is $1/t + \lambda_i/s$, of strictly negative imaginary part. Repeated λ_i give the same conclusion directly.

Consider the homogeneous degree-two operator

$$\mathcal{T}(P) = C_h P + stD_h \partial_z P.$$

It preserves real stability. At upper-half-plane values, a stable nonzero P has $\text{Im}(\partial_z P/P) \leq 0$, by its univariate factorization in z . A zero of $\mathcal{T}(P)$ with $D_h \neq 0$ would require this quotient to equal $-1/h$, which has strictly positive imaginary part. If $D_h = 0$, stable multiplication suffices.

The operator also preserves nonnegative coefficients. Restrict it to any finite coordinatewise degree box containing F ; its output lies in a larger degree box. Brändén–Huh’s Theorem 3.4 applies, so $\mathcal{T}(F)$ is Lorentzian. Finally

$$u\mathcal{T}(F)|_{z=s/u} = sA_h C_h + utB_h C_h + ustA_h D_h$$

is precisely the degree- $(q + 2)$ homogenization of (13). This proves the column-star case.

Every nonempty bipartite graph of matching rank one has all its edges incident with a common vertex. The two orientations therefore exhaust such cores. An empty core gives a product of ULC_q and ULC_2 polynomials, which is ULC_{q+2} by Lorentzian product closure. Theorem 10.1 follows.

11 Exact equality in the first Newton gap

[Merge note: The theorem of this section is *preorder-root-polytopes*'s Theorem 180.2 with Corollary 180.3 (Part XI, `hb:thm:firstidentity` and `hb:cor:equality`), proved there by an exact defect identity over a minimum vertex cover; the inequality is also Part XII's Theorem 190.2 (`mr:thm:first`), proved there by the same Motzkin–Straus route, and its unweighted equality case is Part XII's Theorem 190.3. Source 74's proof is printed as a second route; no novelty is claimed. Source 72 proves the same statement by the same route (Theorem 14.6).] The following gives a useful structural interpretation of non-strictness. Its ingredient is the classical Motzkin–Straus bound [6]; we include the short argument in the form needed here.

Theorem 11.1. *For a positive weighted bipartite graph of matching rank $r \geq 2$,*

$$(r - 1)a_1^2 - 2ra_2 = 0$$

if and only if every nontrivial connected component is a star and their weighted edge totals are all equal. In that case $p_G(t) = (1 + Et)^r$ for their common edge total E ; otherwise the first gap is strict.

Proof. Form the compatibility graph K whose vertices are the edges of G , with two vertices adjacent when the corresponding edges are disjoint. Give an edge xy weight $w_{xy} = u_x v_y$. Put $S = \sum w_e = a_1$ and

$$m_2 = \sum_{\{e,f\} \in E(K)} w_e w_f.$$

The clique number of K is r , and $m_2 \geq a_2$ because m_2 counts two-edge matchings rather than endpoint supports.

For completeness, the Motzkin–Straus bound is

$$m_2 \leq \frac{r-1}{2r} S^2.$$

To see it, merge the weights of two nonadjacent vertices onto whichever has the larger weighted neighbor sum. This preserves S and does not decrease m_2 . Continue until the positive support is a clique of size at most r , where the bound follows from Cauchy–Schwarz.

Equality in the asserted first gap forces equality in both preceding bounds. Since the original compatibility weights are all positive, they give an interior maximum of m_2 on the simplex of fixed sum. If K were not complete multipartite, it would contain an induced edge ik and a vertex j nonadjacent to both. The sum-preserving direction $e_i - 2e_j + e_k$ has strictly positive second derivative for m_2 , contradicting a local maximum. Hence K is complete multipartite. Its complement, the line graph of G , is a disjoint union of cliques. In a bipartite graph a nontrivial connected component whose edges are pairwise intersecting is a star. There are exactly r such components. The compatibility bound is then equality precisely when the total weights in the r parts are equal. Conversely, for stars support and matching counts coincide and the polynomial is the stated product. $\square$

12 Quantitative eventual strictness under common cover scaling

Fix a genuine $2 + 3$ cover whose core has matching rank two. Write every cover activity as w times its fixed positive relative activity. Let e_1, e_2 be the first two support coefficients of the

induced core at those relative activities, and let ζ be the product of all five relative activities. For the all-core coefficients of Theorem 9.2, set $\gamma_i = \zeta c_i$.

Every size-five matching uses exactly one cover vertex per edge; a core edge would consume two cover vertices and preclude five edges. Thus the exterior-left graph matches all three B vertices and the exterior-right graph matches both A vertices, giving $c_2 > 0$. A core two-edge matching leaves one B vertex unmatched; that vertex has an exterior-left neighbor, so adding it gives $c_0 > 0$. Also $e_1, e_2 > 0$ and $e_1^2 \geq 4e_2$ by Lemma 3.1 for the core.

Counting selected cover vertices gives

$$\begin{aligned} a_0 &= 1, & a_1 &= e_1 w^2 + O(w), & a_2 &= e_2 w^4 + O(w^3), \\ a_3 &= \gamma_0 w^5 + O(w^4), & a_4 &= \gamma_1 w^5 + O(w^4), & a_5 &= \gamma_2 w^5. \end{aligned}$$

Consequently the leading gap limits are

$$\begin{aligned} \Delta_1^{(5)}/w^4 &\longrightarrow 4e_1^2 - 10e_2 \geq 6e_2 > 0, \\ \Delta_2^{(5)}/w^8 &\longrightarrow 6e_2^2 > 0, \\ \Delta_3^{(5)}/w^{10} &\longrightarrow 6\gamma_0^2 > 0, \\ \Delta_4^{(5)}/w^{10} &\longrightarrow 4\gamma_1^2 - 10\gamma_0\gamma_2 \geq 2\gamma_0\gamma_2 > 0. \end{aligned}$$

The last inequality uses the sharp constant three from Theorem 9.2. All four gaps are therefore strictly positive eventually. This is stronger than the already proved non-strict ULC_5 property at every $w > 0$.

For either this scaling or Corollary 8.1, write each gap as

$$\Delta_k(w) = L_k w^{d_k} + \sum_{j < d_k} b_{kj} w^j, \quad L_k > 0.$$

A directly computable sufficient threshold is

$$W = 1 + \max_k \frac{\sum_{j < d_k} \max(-b_{kj}, 0)}{L_k}. \quad (15)$$

For $w \geq W$, every negative lower term is bounded in magnitude by its coefficient times w^{d_k-1} , so every gap is positive. This threshold depends on the graph and its fixed field; no uniform bound is asserted.

For a rank-five minimum cover of split $0 + 5$ or $1 + 4$, Lemma 3.2 already gives ULC_5 at every scale. For a $2 + 3$ cover, Theorem 2.1 does likewise; the calculation above adds strictness when its core has matching rank two. Thus all choices of minimum cover are covered, with the relative activities fixed before scaling.

13 Exact verification and scope of the conclusions

The proofs above are universal arguments. Their exact checks are independently implementable and do not use numerical root searches or numerical optimization as evidence for a theorem.

- The full transversal lift is checked by Boolean basis enumeration on 180 weighted graphs with $q = 0, 1, 2, 3, 4, 5$. It tests 140450 candidate basis subsets and compares the complete left-subset marginal, before diagonalizing the left variables.
- An independent finite-field representation check covers all 64 labeled 2×3 cores and 73441 basis determinants, comparing the full weighted support polynomial exactly. A separate generic-line check covers 192 graphs and 23553 determinants, including 54 rank-deficient cores.

- The union formula is checked on 1352 weighted graphs with $q = 1, 2, 3, 4$, including overlapping, nested and empty core neighborhoods.
- The rank-five quartic certificate checks all 4424 stated coefficient cases, all 46 labeled rank-two cores, the four moment-corner identities, the exceptional correction, and the right-side quadratic identity.
- The star-core and scaling checks cover 820 weighted star-core graphs for $q = 1, \dots, 5$ and 230 rank-five common-scaling instances. All displayed interlacing sums of squares, grading identities and computed strictness thresholds are verified exactly.
- The negative-family verifier checks complete endpoint support counts for seven core shapes and the universal exterior population choice for all $3 \leq a, b \leq 20$. The rank-six witness and its exact negative gap are included.

The package contains the Python sources, fixed seeds and JSON outputs. Only SymPy is needed beyond the Python standard library. Run `python3 checks/run_all.py`; the JSON outputs are regenerated in `data/`. The LaTeX sources compile with ordinary pdfLaTeX; `build_local.sh` also handles the local TeX configuration used to generate this PDF. [*Merge note:* In this report the scripts are shipped as `code/74-weighted-rank-five-*.py` and the outputs as `data/74-weighted-rank-five-*.json`; the delivered layout (`checks/`) is not kept, so rerun on a copy of the delivered layout as the README explains. `build_local.sh` is not shipped, and the eight section files are inlined into this report's `article.tex`.]

Röhrle–Ulirsch’s bimatroid theorem [7] normalizes by the smaller ambient shore size. The new lift replaces that order by $q + 2$, and hence by matching rank when the cover is minimum. The polynomial is the perfectly matchable set polynomial of [5], not a matching count with multiplicities. A bounded primary-literature comparison located no prior universal $2 + q$ -cover or weighted rank-five theorem; global priority remains unclaimed.

Higher-rank unrestricted unit-weight and one-shore-only questions are outside this report. [*Merge note:* Both are answered negatively by later sources of this batch (sources 71 and 73, Part II).] The proof uses Lorentzian matroid bases; it does not assume that transversal matroids have the half-plane property. The cover obstruction in Corollary 5.1 is sharp.

14 Strictness and equality for weighted matching-support polynomials (source 72)

Abstract of source 72, as delivered. For positively weighted bipartite graphs having a minimum vertex cover that meets one shore in at most two vertices, we characterize equality in every rank-normalized ultra-log-concavity inequality. Equality at any one index occurs precisely when all nontrivial connected components are stars with equal total edge activities. In that case the polynomial is $(1 + Et)^r$; otherwise every inequality is strict. In particular, this characterizes equality for every bipartite graph of matching rank at most five and gives strictness for connected graphs in that range with rank at least two. The proof combines the recently obtained rank-compression lift with a direct weighted version of the Chan–Pak contraction criterion and explicit rank-two contraction witnesses. We also give a short Motzkin–Straus proof of the same equality characterization for the first inequality at every rank, without a cover restriction.

14.1 Definitions and conclusions

Let $G = (X, Y; E)$ be a finite simple bipartite graph, with positive vertex activities u_x, v_y . Its matching-support polynomial is

$$p_G(t) = \sum_{k=0}^r a_k t^k, \quad a_k = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|=k \\ G[I, J] \text{ perfectly matchable}}} u_I v_J, \quad r = \nu(G).$$

An endpoint support is counted once, regardless of how many perfect matchings it has. Positive activities and a matching of size r imply $a_k > 0$ for every $0 \leq k \leq r$. The order- r normalized gap is

$$\Delta_k = k(r-k)a_k^2 - (k+1)(r-k+1)a_{k-1}a_{k+1}, \quad 1 \leq k \leq r-1. \quad (16)$$

The condition $\Delta_k \geq 0$ for all k is ULC_r . A star includes a single edge; isolated vertices are disregarded. The *edge total* of a star is the sum of $u_x v_y$ over its edges, equivalently its center activity times the sum of its leaf activities.

Theorem 14.1 (Equality classification). *Suppose $r \geq 2$ and G has a minimum vertex cover meeting one shore in at most two vertices. The following are equivalent:*

1. $\Delta_k = 0$ for at least one $k \in \{1, \dots, r-1\}$;
2. every nontrivial component of G is a star and all its edge totals are equal to one common $E > 0$;
3. $p_G(t) = (1 + Et)^r$ for some $E > 0$.

If these conditions fail, then $\Delta_k > 0$ for every $1 \leq k \leq r-1$.

Corollary 14.2. *Theorem 14.1 applies to every bipartite graph with $2 \leq r \leq 5$. Every connected graph in that range with rank at least two is strictly ULC_r , for every positive activity assignment.*

Indeed a minimum cover has size r by König's theorem, so its smaller shore contribution is at most two when $r \leq 5$.

This is a companion to the rank-compression theorem [12], which proves ULC_r under the cover hypothesis. [Merge note: [12] is source 74: Theorem 2.1 and Corollary 2.2.] The lift is recalled precisely below. The unweighted matroid equality characterization is due to Chan–Pak [3, Theorem 1.3]. Their Theorem 1.11 shows that single-index equality need not imply total equality in general matroids; our conclusion uses the special graph lift. We prove the needed real-weight version directly, rather than infer strictness by a limit from rational weights. The arguments are ordinary mathematical proofs, not externally refereed or proof-assistant formalized. No exhaustive publication-priority claim is made.

14.2 A weighted matroid equality criterion

Let M be a rank- r matroid on a ground set $E = \mathcal{R} \sqcup \mathcal{B}$, with positive element weights ω_e . Define

$$P_j = \binom{r}{j}^{-1} \sum_{\substack{S \text{ basis of } M \\ |S \cap \mathcal{R}|=j}} \prod_{e \in S} \omega_e.$$

Loops are allowed. Parallel classes below refer only to nonloops.

Theorem 14.3 (Weighted contraction criterion). *Let $1 \leq a \leq r - 1$ and $P_a > 0$. Then $P_a^2 = P_{a-1}P_{a+1}$ if and only if there is a common $\lambda > 0$ such that, for every independent I with*

$$|I| = r - 2, \quad |I \cap \mathcal{R}| = a - 1,$$

every nonloop parallel class J of M/I satisfies

$$\sum_{e \in J \cap \mathcal{R}} \omega_e = \lambda \sum_{e \in J \cap \mathcal{B}} \omega_e. \quad (17)$$

In particular, one such contraction with a monochromatic nonloop parallel class proves strictness at index a .

We use two established Lorentzian facts from Brändén–Huh [1]: a matroid basis polynomial is Lorentzian (Theorem 3.10), and every nonnegative directional derivative preserves the class (Corollary 2.11). A quadratic Lorentzian polynomial has a Hessian with at most one positive eigenvalue.

Lemma 14.4 (Kernel form of equality). *If a real symmetric matrix H has at most one positive eigenvalue and $y^T H y > 0$, then its quadratic form is negative semidefinite on $\{z : z^T H y = 0\}$. If also $z^T H z = 0$, then $H z = 0$.*

Proof. Otherwise the span of y and a positive vector in its H -orthogonal complement would have two positive directions. On that complement, zero quadratic value for a negative semidefinite form implies orthogonality to the whole complement. Together with orthogonality to y , it implies $H z = 0$. $\square$

Proof of Theorem 14.3: necessity. Let F be the unweighted basis generating polynomial. Let $f_e = \omega_e$ on $\mathcal{R}$ and zero on $\mathcal{B}$, and let $g_e = \omega_e$ on $\mathcal{B}$ and zero on $\mathcal{R}$. Write D_f, D_g for directional differentiation. The matrix

$$C = \text{Hess} (D_f^{a-1} D_g^{r-a-1} F) \quad (18)$$

has at most one positive eigenvalue. Direct differentiation gives

$$f^T C f = r! P_{a+1}, \quad f^T C g = r! P_a, \quad g^T C g = r! P_{a-1}. \quad (19)$$

At equality both neighbors are positive. Set

$$\lambda = P_a / P_{a-1} = P_{a+1} / P_a, \quad z = f - \lambda g.$$

Then $z^T C g = z^T C z = 0$, so Lemma 14.4 gives $C z = 0$.

Suppose first $r \geq 3$ and $a \geq 2$. For each red nonloop i , put

$$M_i = \text{Hess} (D_f^{a-2} D_g^{r-a-1} \partial_i F). \quad (20)$$

The exponents are nonnegative, and the differentiated polynomial has degree two. Commuting derivatives yields

$$C = \sum_{i \in \mathcal{R}} \omega_i M_i, \quad (C z)_i = z^T M_i f. \quad (21)$$

Loop terms are zero.

Every mixed value $f^T M_i g$ is strictly positive. Here is the required combinatorial verification. Equality supplies bases with red counts $a - 1$ and $a + 1$. Insert the nonloop i into each by one basis exchange, or keep it if already present. The resulting bases containing i have red counts at most a and at least $a + 1$, respectively. Bases of M/i are connected by exchanges, and each

exchange changes the red count by at most one. Consequently M/i has a basis with exactly $a - 1$ red elements. Its positive weight contributes to $f^T M_i g$.

Equations (21) and $Cz = 0$ now give

$$f^T M_i f = \lambda g^T M_i f > 0.$$

Each M_i has at most one positive eigenvalue, so Lemma 14.4 implies $z^T M_i z \leq 0$. Their positive-weighted sum is $z^T Cz = 0$; hence every term is zero. The same lemma gives $M_i z = 0$. By (19) applied to M/i , equality propagates to index $a - 1$ of that contraction with the same ratio λ .

Contract red elements until the index is one. Then reverse colors and contract blue elements until rank two. Color reversal replaces λ by $1/\lambda$, and reversing back restores λ . This covers $a = 1$ by starting with blue contractions and $a = r - 1$ by red contractions alone. Any independent I of the specified color counts can be ordered with red elements first and blue elements second; all prefixes remain independent, so each element contracted is a nonloop. The result therefore applies to every required M/I .

At rank two the Hessian has entry one on independent pairs and zero elsewhere. Its kernel equation for z says that the sum of z over the nonloops outside each parallel class is zero. Summing over the at least two classes makes the total over nonloops zero, and hence every class sum is zero. Loops have zero rows and impose no condition. This is precisely (17). $\square$

Proof of Theorem 14.3: sufficiency. For each specified independent I , let H_I be the Hessian of the basis polynomial of M/I . Condition (17) means $H_I z = 0$. Expansion of the directional derivatives gives

$$C = (a - 1)!(r - a - 1)! \sum_I \omega_I H_I.$$

Thus $Cz = 0$. Since $f^T Cg = r!P_a > 0$ and $\lambda > 0$, the three values in (19) are in geometric progression with ratio λ . This proves equality. $\square$

This proof establishes the criterion for arbitrary positive real weights. Parallel cloning proves its rational-weight version, but continuity alone would not justify the necessity or strictness conclusions for real weights.

14.3 The rank compression lift and its two bases

[Merge note: The lift recalled here is source 74's, Section 5; its six basis classes are the six cases there.] Suppose a minimum cover is $A \sqcup B$, where $A = \{A_1, A_2\}$, $|B| = q$ and $r = q + 2$. Put $L = X \setminus A$, $R = Y \setminus B$; there are no L - R edges. A size- r matching uses exactly one cover vertex per edge. Thus L matches all B and R matches both A vertices.

Partition R by neighborhoods $\{A_1\}, \{A_2\}, A$. Let their activity sums be a, b, c , and let the common-class individual activities be r_j . Put

$$d = \sum_{i < j} r_i r_j, \quad Q = ab + ac + bc + d > 0.$$

The lift of [12] is a rank- r transversal matroid with slots $B \sqcup \{e_1, e_2\}$. It has private columns for B , ordinary columns for L on their original neighborhoods, augmented core columns on $S_i \cup \{e_i\}$, private plane columns on e_1, e_2 , and common plane columns on both extra slots. Its red elements are the ordinary and augmented core columns; its blue elements are private and plane columns.

Give the red elements their original left activities, the private blue element at b weight $1/v_b$, and the plane elements weights

$$b/Q, \quad a/Q, \quad r_j/Q,$$

respectively. Delete zero-weight plane elements before applying Theorem 14.3. The positive value of Q guarantees a positive-weight independent pair of plane elements remains. The colored basis polynomial, multiplied by the harmless positive constant $Q \prod_b v_b$, is exactly

$$\sum_{k=0}^r a_k s^{r-k} t^k. \quad (22)$$

The common positive prefactor does not affect equality or strictness of the binomial-normalized coefficients. This identity counts bases once, not matching injections. It follows by classifying selected augmented core and plane columns: their six possible counts are $(0, 2), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)$. The one-plane/two-core case uses the union of the two possible core-row completion profiles. The independent plane-pair activity sum is $1/Q$, while the eligible single-plane sums for A_1, A_2 are $(a + c)/Q, (b + c)/Q$. These are exactly the support alternatives with zero, one or two selected vertices in R .

We use the following generic representation, which is the usual transversal representation after harmless column normalization. Let V have dimension q and let E_0 have basis e_1, e_2 . Private B columns are coordinate vectors of V . Ordinary L columns have independent generic entries on their neighborhoods in V and zero projection to E_0 . The core columns are

$$(v_1, e_1), \quad (v_2, e_2),$$

with independent generic top entries on S_1, S_2 . Plane columns are $(0, e_1), (0, e_2)$ and $(0, \alpha_j e_1 + \beta_j e_2)$, with generic independent coefficients. A permitted square submatrix has nonzero determinant exactly when it admits a matching; different matching monomials do not cancel.

Choose two positive-weight independent plane columns P, Q_0 . Together with the q private B columns they form a blue basis. Order them so that the coefficient of e_1 along P and of e_2 along Q_0 are nonzero. Such an ordering exists because the two-by-two change-of-basis matrix is invertible.

Choose q ordinary L columns matching all B ; together with the two core columns they form a red basis. Assign each ordinary column to its matching B coordinate and the two core columns to P, Q_0 . For every subset of assigned positions, the corresponding principal row-column submatrix is nonsingular over the generic field: the ordinary block contains its matching monomial, and the bottom block is empty, one of the selected nonzero diagonal entries, or the full invertible two-by-two block. The bottom-left block is zero.

14.4 Contractions witnessing strictness

We apply Theorem 14.3 at an arbitrary index $k \in \{1, \dots, r-1\}$. The following construction gives an independent I with $r-2$ elements and $k-1$ red elements. Leave two assigned positions i, j uncontracted. Select any $k-1$ of the other $r-2$ red basis elements, and contract them together with the blue basis elements at every remaining position except i, j . The principal-minor observation above shows that I is independent and the two uncontracted blue coordinate images are independent in M/I .

14.4.1 An ordinary column with two neighbors

Suppose some $u \in L$ has two B neighbors. Since the ordinary-column transversal matroid has rank q , its nonloop u extends to a basis. Choose a matching of this basis assigning u to i ,

and choose another neighbor j of u . Leave i, j uncontracted in the preceding construction, excluding both u and the ordinary red basis column assigned to j from I .

The contraction set and all blue columns are independent of every incidence parameter in column u . Its two entries at i, j are independent parameters multiplying independent quotient coordinate vectors. For any nonloop blue element b , the determinant of the images of u, b is a nonzero polynomial in those two parameters: its two coefficients cannot both vanish. Hence u is not parallel to any nonloop blue element in the generic matroid. Its parallel class is monochromatic red. Theorem 14.3 proves strictness at k . The construction works for every k .

14.4.2 A nonempty core

Suppose a core edge $A_1 B_i$ exists, and let $u = (v_1, e_1)$ be the augmented core column. Choose an ordinary basis matching all B . Leave the coordinate B_i and the bottom coordinate P uncontracted; exclude u and the ordinary column matched to B_i from the red contraction choices.

The contraction set and all blue elements are independent of the top entries of u . Its entry at B_i is an independent parameter multiplying the surviving B_i coordinate. Moreover the image of $u_0 = (0, e_1)$ is independent of that coordinate. Indeed the $q - 1$ contracted top columns together with B_i span V . The one contracted bottom projection is either Q_0 , not parallel to e_1 by the chosen ordering, or e_2 from the other core column. In either case adding e_1 gives a nonzero full determinant.

For any nonloop blue element b , if its determinant with the B_i coordinate is nonzero, the corresponding top parameter makes the determinant with u nonzero. Otherwise b is parallel to B_i , and its determinant with u_0 is nonzero. Thus the determinant of u, b is not identically zero in either case. The class of u is monochromatic red, again proving strictness at every k . A core edge incident with A_2 is handled by interchanging labels.

These arguments include empty singleton plane classes: only a positive-weight plane basis was used, not the presence of both private plane elements. They concern generic representation parameters, independent of the arbitrary positive activity values.

14.5 Completing the graph classification

Suppose neither witness case occurs. The core is empty, and every ordinary left vertex has at most one B neighbor. Since L matches all B , these components are stars with positive edge totals $E_1, \dots, E_q$. The remaining graph is on A_1, A_2 and R . If the two left activities are ξ, η , its support polynomial is

$$1 + Et + \xi\eta Qt^2, \quad E = \xi(a + c) + \eta(b + c).$$

Its discriminant equals

$$[\xi(a + c) - \eta(b + c)]^2 + 4\xi\eta(c^2 - d) \geq 0. \quad (23)$$

Thus it factors as $(1 + \lambda_1 t)(1 + \lambda_2 t)$ with positive parameters. If $c > 0$, then $c^2 - d \geq c^2/2 > 0$, so the two parameters differ. If $c = 0$, the whole graph is a disjoint union of stars, with additional edge totals $\xi a, \eta b$.

Lemma 14.5. *For $r \geq 2$ and $p(t) = \prod_{i=1}^r (1 + \lambda_i t)$ with all $\lambda_i > 0$, equality in any one order- r normalized gap occurs if and only if all λ_i are equal.*

Proof. Use a direct sum of r rank-one matroids, each with a red element of weight λ_i and a blue element of weight one. For any two classes and any index k , contract one element from

each other class, choosing exactly $k - 1$ red elements. Theorem 14.3 forces the two remaining class ratios to agree. Thus all λ_i agree. The converse follows from $(1 + \lambda t)^r$. $\square$

Equation (23) and Lemma 14.5 complete the classification for genuine $2 + q$ covers. They also handle $q = 0$ directly.

For a graph whose active right shore has size $r = \nu(G)$, use its ordinary right-complement transversal matroid with a blue coordinate basis. The first witness construction applies whenever a left vertex has two neighbors; otherwise the graph is a union of stars and Lemma 14.5 applies. This proves the same classification for pure minimum covers.

For a minimum cover split $1 + (r - 1)$, every nonisolated right vertex outside the cover is a leaf at its single left-cover vertex. Replace all these leaves by one with activity equal to their sum. [*Merge note:* This is the weighted leaf compression of *preorder-root-polytopes*, Lemma 206.1 (`lc:lem:compression`) of Part XIII.] There is at least one such leaf, since otherwise the rank would be at most $r - 1$. The support polynomial is unchanged and the compressed graph has an active right shore of size r . It is a union of stars exactly when the original graph is, and expansion of the compressed leaf preserves its component edge total. The pure-shore result therefore transfers. Shore reversal and König's theorem prove Theorem 14.1 and Corollary 14.2.

14.6 The first inequality at every rank

The following result requires no restriction on the cover and does not depend on the lift or on higher-gap ultra-log-concavity. [*Merge note:* Theorem 14.6 is *preorder-root-polytopes*'s Theorem 180.2 with Corollary 180.3 (Part XI, `hb:thm:firstidentity`, `hb:cor:equality`), and the inequality is Part XII's Theorem 190.2; it is also printed above as source 74's Section 11. Source 72's Motzkin–Straus proof is printed as a second route; as source 72 itself says below, it is not a new optimization theorem.]

Theorem 14.6. *For every positively weighted bipartite graph of matching rank $r \geq 2$,*

$$(r - 1)a_1^2 \geq 2ra_2.$$

Equality occurs exactly for disjoint stars with equal edge totals.

Proof. Give edge xy weight $w_{xy} = u_x v_y$ and put $S = \sum_e w_e = a_1$. Let m_2 count pairs of disjoint edges with their weight products. Every feasible two-edge endpoint support has at least one such matching, with the same product, so $a_2 \leq m_2$.

Form the edge-compatibility graph: its vertices are edges of G , and its adjacency means disjointness. Its clique number is r . The Motzkin–Straus theorem [6] gives $m_2 \leq (r - 1)S^2/(2r)$. The upper bound also has a short proof: merge the masses of a nonadjacent pair onto the endpoint of larger weighted neighbor sum, without decreasing the objective, until the positive support is a clique; then apply Cauchy–Schwarz to the masses on that clique.

If the support inequality is equality, the original strictly positive edge-weight vector is an interior global maximizer of the compatibility objective. If that graph were not complete multipartite, it would contain an edge ik and a vertex j nonadjacent to both. The fixed-sum direction $e_i - 2e_j + e_k$ has positive second variation, contradicting an interior maximum. Thus the complement, which is the line graph of G , is a disjoint union of cliques. Each nontrivial component of G has pairwise intersecting edges; in a bipartite graph such a component is a star. If their totals are $E_1, \dots, E_r$, equality in $m_2 = (S^2 - \sum E_i^2)/2$ requires all totals equal. The converse follows from $p_G(t) = \prod_i (1 + E_i t)$. $\square$

Theorem 14.6 is a graph-support consequence of the classical Motzkin–Straus theorem, not a new quadratic optimization theorem. It should not be used to infer higher-gap strictness by coefficient comparison.

14.7 Verification and scope

The accompanying exact script [*Merge note:* shipped as `code/72-strictness-verify.py` and `code/72-strictness-verify_witnesses.py`, with outputs `data/72-strictness-*`] enumerates all 512 simple 3×3 bipartite graphs under three positive integer activity assignments. Excluding ranks zero and one, it verifies 1434 weighted graph cases and 2175 individual gaps: 42 balanced-star cases have all gaps zero, and 1392 cases have all gaps positive. An independent Boolean-transversal test examines 375 genuine cover graphs with $q = 0, \dots, 5$ and verifies 1589 monochromatic contraction witnesses; four star cases have no such structural witness. These are finite regression tests, not proof of the universal statements.

The novelty boundary is the separate rank-compression report [12]: the present companion adds equality classification and strictness to its nonnegative-gap theorem. The unweighted matroid characterization is credited to [3]; the weighted criterion above is proved directly, and no broad bibliographic priority claim is made for that extension. The full graph classification uses the particular generic contraction witnesses, not a general claim about all transversal or represented matroids.

All activity assumptions in the strictness theorems are positive. If some activities vanish, delete those vertices and reconsider the surviving graph and its actual matching rank. A limit of strict inequalities need not be strict. In particular no unspecified continuity argument is used for equality cases.

15 Stability of two-by-two covers and weighted rank four (source 76)

Abstract of source 76, as delivered. Every bipartite graph with a vertex cover consisting of two vertices on each shore has a real stable complemented-row matching-support polynomial. In particular, its matching-support polynomial is real-rooted under arbitrary positive vertex activities. The proof reduces to six four-vertex core types. Two independent-column covariance constructions treat the matching and three-edge cores; a two-block polarization argument treats the complete core. Together with the smaller-shore inequality and exact leaf compression, this settles rank-normalized ultra-log-concavity for every bipartite matching-support polynomial of rank at most four, including arbitrary weights on both shores. All finite certificates are exact, and the universal argument is given in full.

Status and provenance. This note answers the incoming ProveIt manuscript’s question “Finish rank four” [10], strengthening that question to arbitrary positive vertex activities. It follows the separate leaf-compression note [11], which settled weighted rank three and all singleton-side minimum covers. The conclusions are proofs in ordinary mathematics, not Lean-certified results or independently refereed theorems. The stability machinery is classical [2]; novelty is asserted only relative to the inspected ProveIt drafts, not as a claim of worldwide priority. [*Merge note:* [10] is the manuscript printed as Part XII of *preorder-root-polytopes*, and “Finish rank four” is its Research question 93 (`mr:q:rankfour`); [11] is Part XIII. Source 74 (Corollary 2.2) answers the same question through matching number five, so Corollary 15.2 below is printed as a second route. The real-stability Theorem 15.1 is not implied by source 74, which asserts no real-rootedness. The tools of Section 15.2 are Part VI’s (see the notes there).]

15.1 Definitions and the conclusions

Let $G = (X, Y; E)$ be a finite simple bipartite graph. A pair (I, J) , with $I \subseteq X$, $J \subseteq Y$ and $|I| = |J|$, is *feasible* when $G[I, J]$ has a perfect matching. For positive activities u_x, v_y , define

$$p_G(t; u, v) = \sum_{(I, J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^r a_k t^k, \quad r = \nu(G).$$

Each pair is counted once, regardless of the number of perfect matchings. Thus $p_{K_{2,2}}(t; \mathbf{1}, \mathbf{1}) = 1 + 4t + t^2$.

A degree- r polynomial with nonnegative coefficients and no internal zeros is ULC_r when $a_k / \binom{r}{k}$ is log-concave. Equivalently,

$$k(r-k)a_k^2 \geq (k+1)(r-k+1)a_{k-1}a_{k+1} \quad (1 \leq k < r). \quad (24)$$

Write $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$. A nonzero real polynomial is *real stable* if it has no zero when every variable belongs to $\mathbb{H}$.

Introduce a private copy $\bar{x}$ of each $x \in X$ and put

$$F_G(z; y) = \sum_{(I, J) \text{ feasible}} z^{\bar{X} \setminus I} y^J. \quad (25)$$

This is the basis polynomial of the transversal matroid in which $\bar{x}$ is adjacent only to x and each $y \in Y$ is adjacent to its neighbours in G . It is homogeneous of degree $|X|$ and multiaffine.

Theorem 15.1 (Two vertices on each shore). *If G has a vertex cover $A \sqcup B$ with $A \subseteq X$, $B \subseteq Y$ and $|A| = |B| = 2$, then F_G is real stable. Consequently $p_G(t; u, v)$ has only real negative zeros for every positive activity assignment and is $\text{ULC}_{\deg p_G}$.*

[Merge note: Added 1 October 2026, batch 73: an elementary second route to the univariate conclusion, by weighted paths, Hermitian Gram determinants and a positive semidefinite block matrix, is source 77's Theorem 40.2 (Part V).] [Merge note: Added 1 October 2026, batch 73: the report *preorder-gamma-rank-ulc* generalizes the univariate conclusion to directed relations (its Theorem 5.10, from its source 20): under a role cover by at most two tails P and two heads Q , which may overlap, with arcs inside P and inside Q allowed and independent nonnegative tail and head activities, the signed physical monomer polynomial is real stable, so the support polynomial has only negative real roots and is ULC_d at its actual degree d . A bipartite graph oriented from one shore to the other, with two cover vertices on each shore, is the case of disjoint P and Q , so the univariate conclusion of this theorem is its bipartite, disjoint-cover case. The multivariate polynomials differ (F_G here, a monomer polynomial there), and that proof is a third route.]

Corollary 15.2 (Weighted rank four). *Every bipartite graph with matching rank $r \leq 4$ has ULC_r matching-support polynomial for arbitrary positive vertex activities. For nonnegative activities the same assertion holds with the actual surviving degree as normalization order. In rank four, the three inequalities are*

$$3a_1^2 \geq 8a_2, \quad 4a_2^2 \geq 9a_1a_3, \quad 3a_3^2 \geq 8a_2a_4.$$

The corollary does not assert real-rootedness for every rank-four graph. Its real-rootedness input concerns the two-plus-two cover class; other cover types use ultra-log-concavity directly. [Merge note: Corollary 15.2 is the case $\nu \leq 4$ of source 74's Corollary 2.2.]

15.2 Stability tools and the six cores

For a graph, also define its signed unmatched-support polynomial

$$\Phi_G(z) = \sum_{S \text{ perfectly matchable}} (-1)^{|S|/2} \prod_{v \notin S} z_v.$$

For bipartite G , the exact inversion identity is

$$\Phi_G(z) = \left(\prod_{y \in Y} z_y \right) F_G((z_x)_{x \in X}; (-1/z_y)_{y \in Y}). \quad (26)$$

The map $z \mapsto -1/z$ preserves $\mathbb{H}$, and the inverse substitution has the same form. Thus F_G is stable if and only if Φ_G is stable. Stability here concerns these polynomials, not the raw matched-support generating function. [*Merge note:* Identity (26) and this equivalence are *preorder-root-polytopes*'s Theorem 85.3 (Part VI, `ade:mat:thm:stable-bridge`).]

If F_G is stable, substitute $z_x = 1/u_x$ and $y_y = tv_y$ in (25). Boundary specialization and positive scaling preserve stability, possibly giving zero; here the constant term excludes zero. Therefore

$$p_G(t; u, v) = \left(\prod_{x \in X} u_x \right) F_G((1/u_x)_x; (tv_y)_y) \quad (27)$$

is a real stable univariate polynomial, so it is real-rooted. Its nonnegative coefficients and constant term one force all roots to be negative. Newton's inequalities give (24) with its actual degree.

Lemma 15.3 (Leaf attachment and vertex sums). *Attaching a leaf to a graph with stable Φ preserves stability. More generally, if two graphs with stable Φ intersect in one vertex and have no other connecting edges, their union has stable Φ .*

Proof (pointer). The vertex-sum statement is *preorder-root-polytopes*'s Theorem 93.2 (Part VI, `ade:mat:thm:vertex-stable`), proved there from the identity $\Phi_G = z_v A_1 A_2 + B_1 A_2 + A_1 B_2$ and $\text{Im}(B_i/A_i) \geq 0$. Attaching a leaf is the one-vertex sum with an edge, whose signed polynomial $z_v z_w - 1$ is stable. [*Merge note:* Source 76 gives the same two arguments (for the leaf, $\Phi_{G+w} = z_w \Phi_G - \partial_v \Phi_G$ and the sign of the logarithmic derivative); they are not reprinted.] $\square$

Delete isolated vertices and all exterior vertices whose neighbourhood in the opposite part of the cover is a singleton. These are pendant leaves and can be reattached by Lemma 15.3. The remaining exterior vertices form sets $L = \{L_1, \dots, L_n\} \subseteq X$ and $R = \{R_1, \dots, R_m\} \subseteq Y$, where every L_i is adjacent to both B_1, B_2 and every R_j to both A_1, A_2 . There are no edges between L and R .

Up to row and column permutations and transposition, the four-vertex core $G[A, B]$ has six types: empty, one edge, two adjacent edges, two disjoint edges, three edges, or four edges. For the first three types the graph is a vertex sum of complete bipartite graphs $K_{2,k}$, bridges and isolated vertices. The basis polynomial of $K_{2,k}$, with the two-vertex shore on the left, is the elementary symmetric polynomial e_2 on $k+2$ variables, hence is stable. Equation (26) removes dependence on orientation. Lemma 15.3 handles these three cases. [*Merge note:* Stability of $K_{2,k}$ is *preorder-root-polytopes*'s Theorem 94.1 (`ade:mat:thm:biclique`), and these three core types are covered by its Corollary 94.3 (`ade:mat:cor:block-family`). Only the matching, three-edge and complete cores treated below are new.]

15.3 Independent columns and Boolean minor weights

Lemma 15.4 (Expected determinant). *Let the columns K_j of a real random matrix be independent and have finite second moments. Then*

$$\mathbb{E} \det (\operatorname{diag}(z) + K \operatorname{diag}(y) K^T)$$

is real stable.

Proof (pointer). This is *preorder-root-polytopes*'s Theorem 91.2 (Part VI, `ade:thm:expected`). [Merge note: Source 76's proof is the one given there: stability of $\det(\operatorname{diag}(z) + \sum_j t_j C_j)$ with $C_j = \mathbb{E}(K_j K_j^T)$, the operators $1 + y_j \partial_{t_j}$, specialization $t_j = 0$, and independence of the columns. It is not reprinted.] $\square$

This is standard stability machinery, also used in the repository's even-multi-theta argument [8]; the covariance choices below are the particular constructions needed here.

Cauchy–Binet expands the expected determinant as

$$\sum_{|I|=|J|} \mathbb{E} \det(K[I, J])^2 z^{X \setminus I} y^J. \quad (28)$$

It therefore equals F_G if the squared-minor expectation is one on every feasible support and zero on every infeasible support.

Set

$$q = 2^{-1/2}, \quad \rho = \sqrt{q} = 2^{-1/4}.$$

Rows are indexed by $A_1, A_2, L_1, \dots, L_n$. Every R -column has covariance supported on the two A rows, with matrix

$$\begin{pmatrix} 1 & q \\ q & 1 \end{pmatrix}. \quad (29)$$

All columns are independent. Within either B -column, the covariance on the L rows is $(1 - q)I + qJ$, so diagonal entries are one and distinct-row entries are q .

15.3.1 The matching core

Use core edges $A_2 B_1, A_1 B_2$. In the B_1 column, the A_1 entry is zero, the A_2 variance is one, and its covariance with each L_i is ρ . In the B_2 column, the A_2 entry is zero, the A_1 variance is one, and its covariance with each L_i is $-\rho$.

These matrices are positive semidefinite in every dimension. For independent centred variance-one variables $Z, Z_1, \dots, Z_n$, take the nonzero A entry to be Z or $-Z$, respectively, and take

$$K_{L_i} = \rho Z + \sqrt{1 - q} Z_i.$$

Use independent copies for different columns. Matrix (29) is positive definite, so it too is a covariance matrix.

15.3.2 The three-edge core

Use edges $A_1 B_1, A_2 B_1, A_1 B_2$. In the B_1 column, the A -block is (29), and the covariance vector from (A_1, A_2) to any L_i is $(\rho/\sqrt{2}, 0)$. For independent centred variance-one Z, W, Z_i , realize it by

$$K_{A_1} = (Z + W)/\sqrt{2}, \quad K_{A_2} = W, \quad K_{L_i} = \rho Z + \sqrt{1 - q} Z_i.$$

In the B_2 column, take $K_{A_1} = Z$, $K_{A_2} = 0$, and the same formula for K_{L_i} , using a fresh independent family. This realizes variance one at A_1 and covariance ρ from A_1 to every L_i .

15.3.3 Verification of all minor types

Both constructions have the graph's zero pattern. Therefore unsupported minors vanish identically. Supported minors use at most two L rows and two R columns, and have size at most four.

Size-one supported minors have expectation one. A size-two supported minor has either one matching, giving one by column independence and unit entry variances, or two matchings around a four-cycle. In the latter case the expectation is $2 - 2q^2 = 1$, except for the path core's $\{A_1, L_i\} \times \{B_1, B_2\}$ minor, where it is

$$2 - 2(\rho/\sqrt{2})\rho = 1.$$

For size three, the only case without a forced row or column uses rows A_1, A_2, L_i and columns B_1, B_2, R_j . For the matching core, the two determinant terms have diagonal contributions one each, and their cross term is $-2\rho^2q$. Thus

$$\mathbb{E} \det(K[I, J])^2 = 2 - 2\rho^2q = 1.$$

For the path core there are three matching terms. Their three pairwise cross terms are

$$-2(\rho/\sqrt{2})\rho, \quad -2q^2, \quad 0,$$

so the expectation is again one. In every other supported size-three case a forced entry lies in a column independent of the remaining minor; its variance is one and the remaining size-two minor has squared expectation one.

Every supported size-four minor uses both A rows, both B columns, two L rows and two R columns. Up to sign its determinant factors as

$$\det K[\{L_i, L_j\}, \{B_1, B_2\}] \det K[\{A_1, A_2\}, \{R_s, R_t\}].$$

The factors depend on disjoint independent column families, and each has squared expectation one. This proves the Boolean identity required in (28) for every minor size. Lemma 15.4 proves stability for both core types.

15.4 The complete core by two-block polarization

Now suppose all four edges between A and B are present. Partition the ground set of the fundamental transversal matroid into

$$\mathcal{U} = \bar{A} \sqcup R, \quad \mathcal{V} = \bar{L} \sqcup B, \quad M = m + 2, \quad N = n + 2.$$

Its bases are exactly the N -element subsets containing at most two elements of $\mathcal{U}$. Indeed, let a support select i vertices of A , j of B , ℓ of L and s of R . Size balance is $i + \ell = j + s$. Feasibility requires $s \leq i$ and $\ell \leq j$, and these conditions are equivalent under balance. They suffice: match the exterior vertices to the selected opposite core, then match the remaining equally numerous core vertices across the complete core. The corresponding basis contains $2 - i + s$ elements of $\mathcal{U}$, so feasibility is exactly the bound of two.

Consequently the basis polynomial is

$$F_G = e_N(\mathcal{V}) + e_1(\mathcal{U})e_{N-1}(\mathcal{V}) + e_2(\mathcal{U})e_{N-2}(\mathcal{V}). \quad (30)$$

Its separate block diagonal, with $\mathcal{U}$ variables set to a and $\mathcal{V}$ variables to b , is

$$b^{N-2} \left(b^2 + MNab + \binom{M}{2} \binom{N}{2} a^2 \right).$$

The quadratic discriminant is

$$M^2N^2 - 4\binom{M}{2}\binom{N}{2} = MN(M + N - 1) > 0. \quad (31)$$

Since $M, N \geq 2$, its factors are $(b + \lambda_+ a)(b + \lambda_- a)$ with both $\lambda_{\pm} > 0$. The block diagonal is therefore stable.

Polynomial (30) is exactly its separate multiaffine polarization in M variables and N variables. The Grace–Walsh–Szegő theorem, in the convex upper half-plane, says that polarization preserves stability [2]. Thus F_G is stable. This completes all six core cases; leaf reattachment proves Theorem 15.1.

15.5 From stability to the rank-four inequalities

For completeness we recall the two inputs from the preceding note [11]. [*Merge note:* They are *preorder-root-polytopes*'s Theorem 70.1 (Part V) or Lemma 205.1 (Part XIII), and Lemma 206.1 with Theorem 206.2 (Part XIII); source 74 states them as Lemmas 3.1 and 3.2.] For arbitrary positive activities, p_G is $\text{ULC}_{\min(|X|, |Y|)}$: matroid basis polynomials are Lorentzian [1], and the nonnegative linear substitution

$$\left(\prod_{x \in X} u_x \right) F_G((s/u_x)_x; (tv_y)_y) = \sum_k a_k s^{|X|-k} t^k$$

preserves the Lorentzian property. The bivariate characterization gives $\text{ULC}_{|X|}$; transpose the graph for $\text{ULC}_{|Y|}$. This is the weighted form of the established smaller-shore input; see also [7].

If a minimum cover of size r lies wholly on one shore, remove isolated vertices and apply that input. If a minimum cover is $\{x_0\} \sqcup B$ with $|B| = r - 1$, every active right vertex outside B is a leaf at x_0 . A feasible support contains at most one such leaf, so replacing the class by one leaf of activity equal to the sum of its activities preserves the entire polynomial. The compressed right shore has size r , giving ULC_r . The transposed case is identical.

By König's theorem a rank-four graph has a cover of size four. Its split is $0 + 4$, $1 + 3$, $2 + 2$, or a transpose. The preceding paragraph treats the first two types, and Theorem 15.1 treats the third by Newton's inequalities. Covers of size at most three have a pure or singleton-side split, proving the smaller ranks too. For zero activities, delete their vertices and apply the result with the surviving degree. This proves Corollary 15.2.

15.6 A sharp boundary for complete mixed cores

For integers $s \geq 2$ and $n, m \geq 0$, let $H_{2,s;n,m}$ have left core A of size two, right core B of size s , n exterior left vertices L , and m exterior right vertices R . Its edges are all pairs in

$$(A \times B) \cup (L \times B) \cup (A \times R),$$

with no edges between L and R . When $n \geq s$ and $m \geq 2$, its matching rank is $s + 2$.

Theorem 15.5 (Sharp stability criterion). *The signed support polynomial of $H_{2,s;n,m}$ is real stable if and only if*

$$(m + 2)(n + 2)s \geq 2(m + 1)(n + 1)(s - 1). \quad (32)$$

In particular every two by two Hall core is stable. For a two by three core, stability is equivalent to

$$(n - 2)(m - 2) \leq 12. \quad (33)$$

Proof. In the extended transversal matroid put $F = \bar{A} \sqcup R$ and $O = \bar{L} \sqcup B$, with

$$M = |F| = m + 2, \quad N = |O| = n + s, \quad r = n + 2.$$

Exactly as in the complete two by two core, the bases are all r -element subsets containing at most two members of F . To check sufficiency directly, a selected set with $k \leq 2$ elements of F and j elements of B uses $r - k - j$ private L -elements. Match the F -elements into A , use $2 - k$ selected B -elements on the remaining positions of A , and use the other $k + j - 2$ selected B -elements on the vacant L -positions. All counts are nonnegative by the size of the selected subset.

Thus its basis polynomial is

$$\mathcal{B} = \sum_{k=0}^2 e_k(F) e_{r-k}(O).$$

Its two-block diagonal is

$$y^n \left[\binom{N}{r} y^2 + M \binom{N}{r-1} xy + \binom{M}{2} \binom{N}{r-2} x^2 \right].$$

All three bracketed coefficients are positive. The bracket is stable exactly when its discriminant is nonnegative, equivalently

$$M^2 \binom{N}{r-1}^2 \geq 4 \binom{N}{r} \binom{M}{2} \binom{N}{r-2}.$$

Using

$$\frac{\binom{N}{r-1}^2}{\binom{N}{r} \binom{N}{r-2}} = \frac{r}{r-1} \frac{N-r+2}{N-r+1}$$

reduces this inequality to (32). Necessity for full stability follows by identifying all variables within each block. Sufficiency follows by polarization, using M slots for x and N slots for y . The signed support polynomial is stable exactly when the extended basis polynomial is stable, by the invertible upper-half-plane substitution $z \mapsto -1/z$ on one shore. For $s = 3$, expansion of (32) gives (33). $\square$

A smallest unstable member of the rank-five family. Among members with $s = 3$, $n \geq 3$ and $m \geq 2$, instability requires $(n-2)(m-2) > 12$. If $(n-2) + (m-2) \leq 7$, the product is at most $\lfloor 7^2/4 \rfloor = 12$. Thus every unstable member has at least $n + m + 5 = 17$ vertices; $n = m = 6$ attains this bound. Its diagonal factor is

$$y^6(9y^2 + 288xy + 2352x^2),$$

whose quadratic discriminant is -1728 .

This does not refute positive-field real-rootedness or rank-normalized ULC. Indeed the same graph with unit vertex weights has

$$p(t) = 1 + 36t + 333t^2 + 974t^3 + 1005t^4 + 300t^5,$$

with five distinct negative roots. An exact certificate consists of its alternating signs at

t	-2	-1	$-\frac{1}{2}$	$-\frac{1}{5}$	$-\frac{1}{20}$	0
$p(t)$	-51	29	$-\frac{33}{16}$	$\frac{21}{25}$	$-\frac{1329}{16000}$	1

The intermediate value theorem supplies a root in each of the five disjoint intervals. Consequently the rank-four proof by universal signed stability has a genuine rank-five boundary, even though the rank-five ULC problem remains unresolved here. [*Merge note:* The rank-five problem is settled affirmatively by source 74 (Corollary 2.2); this graph has the $2 + 3$ minimum cover $A \sqcup B$, so Theorem 2.1 gives ULC_5 for every positive activity assignment.]

15.7 Scope and reproducibility

The rank-four question in [10] is therefore resolved for unit weights, one-shore weights and arbitrary two-shore weights. Combined with the rank-six weighted counterexamples in [9], the first possible failure rank for arbitrary two-shore rank normalization lies in $\{5, 6\}$. General unit-weight and one-shore rank normalization are not settled by this note. [*Merge note:* Superseded within the batch: the first failure rank is exactly six by source 74 (Corollary 2.2). The shipped `76-rank-four-PROOF_STATUS.md` says likewise that “the smallest failure rank is reduced to five or six”; it is kept as delivered. The unit-weight and one-shore questions are answered negatively in Part II (sources 71 and 73).]

The expected-determinant lemma, polarization theorem, uniform-matroid stability and gluing principles are established tools; they are not claimed as new. This note’s contributions relative to the inspected drafts are the two covariance realizations, the complete-core two-block application, the resulting two-plus-two cover theorem, and the weighted rank-four conclusion. A targeted search of the stability literature did not certify global priority, which is not claimed.

The accompanying scripts supplement the proof:

- [*Merge note:* The scripts are shipped as `code/76-rank-four-verify_*.py` and their outputs as `data/76-rank-four-*_verification.json`; the build wrapper is not shipped.]
- `verify_covariance.py` checks all 70 square minors of the four-by-four representative for each covariance construction, including infeasible and empty minors. All 140 exact identities pass in $\mathbb{Q}(2^{1/4})$. It also checks every principal covariance minor in those representatives.
- `verify_complete_core.py` independently checks the support-to-basis characterization and block counts for all 49 pairs $0 \leq n, m \leq 6$, together with the discriminant identity.
- `verify_boundary.py` checks 75 complete mixed cores against the two-block basis description, the exact stability threshold, and the rank-five unit-weight polynomial and rational root certificate.
- The six-core proof and the classification of all possible supported minor types establish the universal assertions; the finite scripts do not replace those arguments.

Run the scripts with Python 3 and SymPy, and compile this source twice with `pdflatex`. The package includes a build wrapper for the preparation environment and recorded verification outputs.

16 A sharp weighted classification of full Hall bottleneck graphs (source 75)

Abstract of source 75, as delivered. A full Hall-bottleneck graph has two complete core blocks, complete connections from each exterior block to the opposite core, and no exterior-to-exterior edges. We prove that this family satisfies matching-rank-normalized ultra-log-concavity for every positive vertex-activity assignment and every exterior population if and only if one core has size at most two. The new positive argument treats a two-by- s core uniformly for every s : deleting the core edges gives a real-rooted comparison polynomial, and restoring them increases every Newton margin. A twelve-case coefficient calculation proves this using only binomial ratios and Newton inequalities for exterior elementary symmetric functions. When both cores have size at least three, we give explicit finite populations and

an integer activity producing a failed last margin. The positive inequalities are strict in the genuine two-by- s rank regime. An exact weighted rank-five example has nonreal roots despite strict ultra-log-concavity. The classification concerns this full Hall family; arbitrary graphs with a two-plus- s minimum cover remain outside its scope.

Status. The results have ordinary mathematical proofs and reproducible exact algebraic checks. They are not formally verified in Lean or externally refereed. The contribution is compared with the inspected ProveIt Hall-bottleneck report; global priority is not asserted. [*Merge note:* The inspected Hall-bottleneck report [9] is Part XI of *preorder-root-polytopes* (pinned at 4a3b6991d). The positive half of Theorem 16.1 is the full Hall case of source 74's Theorem 2.1: for $n \geq b$, $m \geq a$ and $\min(a, b) \leq 2$ the core $A \sqcup B$ is a minimum cover with at most two vertices on one shore (for $\min(a, b) = 1$ it is also Part XIII's Theorem 206.2). Source 75's comparison-polynomial proof is printed as a second route, with what it adds: strictness when $\min(a, b) = 2$, real-rootedness when $\min(a, b) = 1$, and the nonreal rank-five example. For $a = b$ and $n = m$ the negative half is Part XI's endpoint theorem (Theorem 178.2 of *preorder-root-polytopes*, `hb:thm:endpoint`), and the population offset h below gives $n = 33$ and $n = 11$ at $s = 3, 4$, Part XI's minimal outer sizes; it is new for $a \neq b$ (Research question 89 of *preorder-root-polytopes*).]

16.1 The family and the classification

For positive integers a, b , define $H_{a,b;n,m}$ by the disjoint shores

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = a, \quad |B| = b, \quad |L| = n, \quad |R| = m,$$

and the exact edge set

$$(A \times B) \cup (L \times B) \cup (A \times R). \quad (34)$$

Thus every exterior-left vertex is adjacent to *all* of B , every exterior-right vertex to *all* of A , the core $A \times B$ is complete, and no exterior pair is adjacent. Assign an arbitrary positive activity to each vertex. Its support polynomial is

$$P_H(t) = \sum_{(I,J) \text{ feasible}} u_I v_J t^{|I|} = \sum_k p_k t^k,$$

where $I \subseteq X$, $J \subseteq Y$, $|I| = |J|$, and feasibility means that the induced bipartite graph admits a perfect matching. A pair of endpoint sets contributes once, regardless of how many matchings realize it.

For a nonnegative degree- d coefficient sequence with no internal zeros, write ULC_d for log-concavity of $p_k / \binom{d}{k}$. Equivalently,

$$\Delta_k^{(d)}(P) := k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1} \geq 0 \quad (1 \leq k < d). \quad (35)$$

For positive activities the degree equals the maximum matching size. If $n \geq b$ and $m \geq a$, this rank is $d = a + b$: the core is a cover of that size, attained by matching each core into the opposite exterior block.

Theorem 16.1 (Sharp full-Hall classification). *Fix positive core sizes a, b . The following are equivalent.*

1. For all $n \geq b$, $m \geq a$ and every positive activity assignment, $P_{H_{a,b;n,m}}$ is ULC_{a+b} .
2. $\min(a, b) \leq 2$.

If $\min(a, b) = 2$, every internal inequality is strict. If $a, b \geq 3$, the failure of the last inequality can be realized with activity one at every exterior vertex and one common positive integer activity at every core vertex; explicit parameters are given in Section 16.5.

[Merge note: Added 1 October 2026, batch 73: at $s = 3$ the positive direction, with strictness, is also proved by the batch-73 archive `complete-two-by-three-result.zip`, which is not printed; source 78 (Part V) extends it, by certificates, to every 2×3 core with complete exterior blocks.] The positive assertion also extends to smaller exterior populations with normalization by the actual matching rank, as explained in Section 16.4. Zero activities are handled there separately. Nothing in the negative assertion concerns unit weights or activities on just one shore. [Merge note: Those cases are treated by sources 71, 73, 55, 47 and 51 in Part II.]

16.2 A uniform formula for a two by s core

Assume $a = 2$, $b = s$, $n \geq s$ and $m \geq 2$, and put $d = s + 2$. Normalize both exterior activity sums to one by dividing every left activity by $e_1(L)$ and every right activity by $e_1(R)$. This only replaces $P_H(t)$ by $P_H(ct)$ for a positive c , and multiplies its k th margin by c^{2k} .

Let u_1, u_2 be the normalized left-core activities, $v_1, \dots, v_s$ the normalized right-core activities, and define

$$U_1 = u_1 + u_2, \quad U_2 = u_1 u_2, \quad V_j = e_j(v_1, \dots, v_s), \quad E_j = e_j(L), \quad z = e_2(R).$$

Here $E_0 = E_1 = 1$, $0 \leq z \leq 1/2$, and every out-of-range elementary symmetric function is zero.

A size- k support selecting i left-core and j right-core vertices is feasible exactly when $i + j \geq k$. Necessity follows because its $k - i$ exterior-left vertices must be matched into its j selected right-core vertices. Conversely match those exterior vertices first, then the selected exterior-right vertices into the left core, and finally the remaining core vertices to one another. Therefore

$$p_k = E_k V_k + U_1 E_{k-1} (V_k + V_{k-1}) + U_2 E_{k-2} (V_k + V_{k-1} + z V_{k-2}). \quad (36)$$

Deleting the $2s$ core edges gives two disjoint complete bipartite graphs. Their support polynomial Q has coefficients

$$q_k = E_k V_k + U_1 E_{k-1} V_{k-1} + U_2 E_{k-2} z V_{k-2}. \quad (37)$$

The proof will establish the stronger comparison

$$\Delta_k^{(d)}(P_H) - \Delta_k^{(d)}(Q) \geq 0 \quad (1 \leq k < d). \quad (38)$$

Lemma 16.2 (The comparison polynomial). *The polynomial Q has degree d and only negative real zeros. Its order- d Newton inequalities are strict.*

Proof. For a complete bipartite graph with left activities $\alpha_1, \dots, \alpha_p$ and right activities $\beta_1, \dots, \beta_q$, its support polynomial is

$$\left(\prod_i \alpha_i \right) e_p(\alpha_1^{-1}, \dots, \alpha_p^{-1}, t\beta_1, \dots, t\beta_q).$$

Elementary symmetric polynomials are real stable: differentiate $\prod_i (w + x_i)$ in w the required number of times and specialize $w = 0$. Differentiation, positive scaling, and real specialization preserve stability or give zero [4]. The displayed univariate polynomial is nonzero with positive constant term and nonnegative coefficients, so its zeros are negative and real.

Apply this to $K_{2,m}$ and $K_{n,s}$. Their product has degree $2 + s = d$. The first factor is

$$1 + U_1 t + z U_2 t^2.$$

Its discriminant obeys

$$U_1^2 - 4zU_2 \geq (u_1 - u_2)^2 + 2u_1u_2 > 0,$$

so Q has at least two distinct negative zeros. Newton's inequalities are strict at every internal index unless all roots coincide. This proves the lemma. $\square$

16.3 Twelve coefficient families

Every core variable has exponent at most two in the difference in (38). The possible nonzero left exponent patterns, up to swapping u_1, u_2 , are

$$10, \quad 20, \quad 11, \quad 21, \quad 22.$$

Fix one representative monomial with left pattern λ . Suppose its right-core exponents have p twos and q ones, and put

$$a = k - p, \quad j = 2a - q, \quad \ell = d - k, \quad A = k\ell, \quad B = (k + 1)(\ell + 1).$$

The letters A, B in this section are numerical gap factors, not vertex blocks. A nonzero increment coefficient has

$$0 \leq j < |\lambda|, \quad k \geq a, \quad a \geq \lceil j/2 \rceil, \quad \ell \geq a - j + 2. \quad (39)$$

Indeed the total right-core degree is $2k - j$; a core-edge contribution raises total core degree above $2k$, giving $j < |\lambda|$. Also the number of right-core variables occurring is

$$p + q = k + a - j \leq s = d - 2,$$

which gives the last inequality. These restrictions leave exactly twelve pattern- j families.

Divide the coefficient by the following positive central binomial factor:

$$C_0 = \binom{2a}{a}, \quad C_1 = \binom{2a-1}{a-1}, \quad C_2 = \binom{2a-2}{a-1}, \quad C_3 = \binom{2a-3}{a-2}.$$

Each is used only when its indices are allowed by (39). To derive the formulas below, the coefficient of a monomial with p twos and q ones in $V_i V_h$ is

$$\binom{q}{i-p} \quad (i + h = 2p + q), \quad (40)$$

with the usual zero convention. The repeated variables occur in both subsets; one chooses which of the q single-occurrence variables belong to the first subset. Thus all twelve formulas follow by direct substitution of (36) and (37) into (35), followed by elementary ratios of binomial coefficients. No matching multiplicities are introduced.

We use these consequences of Newton's inequalities:

$$\begin{aligned} E_{k-2}E_{k+1} &\leq \frac{k-1}{k+1}E_{k-1}E_k, \\ E_{k-2}E_k &\leq \frac{k-1}{k}E_{k-1}^2, & E_{k-3}E_{k+1} &\leq \frac{k-2}{k+1}E_{k-2}E_k, \\ E_{k-3}E_k &\leq \frac{k-2}{k}E_{k-2}E_{k-1}, & E_{k-3}E_{k-1} &\leq \frac{k-2}{k-1}E_{k-2}^2. \end{aligned} \quad (41)$$

For $k \geq 2$ these also cover boundary cases by the zero convention. They follow, for example, from log-concavity of $j!E_j$. The case $k = 1$ is checked at the end of the argument.

16.3.1 Pattern 10

Only $j = 0$ occurs. The normalized coefficient is

$$2AE_kE_{k-1} - B\frac{a}{a+1}(E_kE_{k-1} + E_{k-2}E_{k+1}).$$

Using the first inequality in (41), it is at least

$$\frac{2k(\ell - a)}{a+1}E_kE_{k-1} \geq 0,$$

since $\ell \geq a+2$.

16.3.2 Pattern 20

Put $X = E_{k-1}^2$ and $Y = E_{k-2}E_k$. For $j = 0$ the normalized coefficient is

$$AX - B\frac{a}{a+1}Y;$$

for $j = 1$ it is twice this expression. The bound $Y \leq (k-1)X/k$ reduces its sign to

$$k(a+1)\left(A - B\frac{a}{a+1}\frac{k-1}{k}\right) = k^2(\ell - a) + a(\ell + 1) \geq 0.$$

Here $\ell \geq a+1$ in both cases.

16.3.3 Pattern 11

Keep X, Y and put $Z = E_{k-3}E_{k+1}$. The two normalized coefficients are

$$\begin{aligned} j = 0 : & \quad 2A(X + Y) - B\frac{a}{a+1}(X + 2Y + Z), \\ j = 1 : & \quad 4AX + 2AY - B\left(X + \frac{4a}{a+1}Y + \frac{a-1}{a+1}Z\right). \end{aligned}$$

Use $Z \leq (k-2)Y/(k+1)$, and write the resulting lower bound as

$$\alpha\left(X - \frac{k}{k-1}Y\right) + \beta Y. \tag{42}$$

For $j = 0$ the two factors are

$$\alpha = 2A - \frac{Ba}{a+1} \geq 0, \quad \beta = \frac{2k(2k-1)(\ell-a)}{(k-1)(a+1)} \geq 0.$$

The sign of α is immediate for $a = 0$; for $a \geq 1$, use $k \geq a$ and $\ell \geq a+2$ in $A/B = k\ell/((k+1)(\ell+1))$.

For $j = 1$ one has $a \geq 1$, $\ell \geq a+1$, and

$$\alpha = 4A - B \geq 0,$$

while

$$\begin{aligned} \frac{(k-1)(a+1)}{2}\beta &= \ell(3k^2 - 3k + 1 + a) \\ &\quad - a(3k^2 - k - 1) - 2k + 1. \end{aligned}$$

The right side increases with ℓ . At $\ell = a+1$ it becomes

$$(a-k+1)(a-k+2) + 2k(k-1) \geq 0,$$

because the first product consists of consecutive integers. Equation (42) and Newton's bound now prove both cases.

16.3.4 Pattern 21

Put $M = E_{k-2}E_{k-1}$ and $N = E_{k-3}E_k$. The normalized coefficient is

$$AMS_j - B(MH_j + NJ_j),$$

where

j	S_j	H_j	J_j
0	2	$\frac{a+1}{2a}$	$\frac{a+1}{2a}$
1	4	$\frac{a+1}{2a}$	$\frac{a+1}{2a}$
2	$2 + \frac{2z(a-1)}{a}$	$z + \frac{a-1}{a}$	$\frac{a-1}{a} + \frac{z(a-1)(a-2)}{a(a+1)}$

Newton gives $N \leq \nu M$, where $\nu = (k-2)/k$. For $j = 0, 1$,

$$\frac{H_j + \nu J_j}{S_j} = \frac{a}{a+1} \frac{k-1}{k} \leq \frac{a}{a+1} \frac{k}{k+1} \leq \frac{A}{B},$$

using $\ell \geq a$. These two coefficients are nonnegative.

For $j = 2$, $a \geq 1$ and $\ell \geq a$. Define

$$D(\ell, z) = k\ell S_2 - (k+1)(\ell+1)(H_2 + \nu J_2).$$

It is affine in each of ℓ, z . Direct simplification gives

$$D(a, 0) = \frac{2(a^2 + k^2 - 1)}{ak} \geq 0, \quad D(a, \frac{1}{2}) = \frac{(k-a)^2 + (k-a) + 2a(a-1)}{ak} \geq 0.$$

Thus $D(a, z) \geq 0$ on $0 \leq z \leq 1/2$. Since $H_2 + \nu J_2 \geq 0$,

$$\frac{\partial D}{\partial \ell} = \frac{D(a, z) + (k+1)(H_2 + \nu J_2)}{a} \geq 0.$$

Therefore $D(\ell, z) \geq 0$ for all $\ell \geq a$, proving the last case.

16.3.5 Pattern 22

Put $U = E_{k-2}^2$ and $W = E_{k-3}E_{k-1}$. The normalized coefficient is

$$AUS_j - BWT_j,$$

with

j	S_j	T_j
0	1	$\frac{a+1}{2a}$
1	2	$\frac{a+1}{2a}$
2	$1 + \frac{2z(a-1)}{a}$	$\frac{a-1}{a} + \frac{2z(a^2 - a + 1)}{a(a+1)}$
3	$2z$	$\frac{2z(a-1)}{a}$

For $j \leq 2$, $T_j/S_j \leq a/(a+1)$. In the $j = 2$ row this follows by checking $z = 0, 1/2$ in the affine cross-multiplied inequality; the endpoint ratios are $(a-1)/a$ and $a/(a+1)$. For $j = 3$ and $z > 0$ the ratio is $(a-1)/a$; at $z = 0$ the row is zero.

Now

$$W \leq \frac{k-2}{k-1}U, \quad \frac{k-2}{k-1} \leq \frac{k}{k+1}.$$

When $j \leq 2$, feasibility gives $\ell \geq a$, so A/B dominates the product of the moment ratio and $a/(a+1)$. When $j = 3$, use $\ell \geq a-1$ and the ratio $(a-1)/a$ instead. All four cases follow.

16.3.6 The small index boundary and completeness

At $k = 2$, all displayed arguments remain valid with $E_{-1} = 0$. At $k = 1$, patterns 21 and 22 vanish. For patterns 10 and 11 with $j = 0$ the normalized coefficient is $2A - Ba/(a + 1)$; here $a = 0$ or 1 and $\ell \geq a + 2$, making it nonnegative. Pattern 20 gives A or $2A$, and pattern 11 with $j = 1$ gives $4A - B \geq 0$. Thus the boundary is covered without division by $k - 1$.

The twelve cases exhaust every nonzero monomial coefficient of the increment in the independent positive core activities. Every coefficient is nonnegative, proving (38). Lemma 16.2 then proves strict ULC_{s+2} for every full two-by- s Hall graph in the genuine-rank regime.

16.4 Singleton cores and deficient exterior populations

A singleton-core case has an elementary stable-matroid description. Suppose $a = 1$. In the extended transversal matroid, let

$$F = \bar{A} \sqcup R, \quad O = \bar{L} \sqcup B.$$

Its rank is $n + 1$, and its bases are exactly the $(n + 1)$ -element subsets containing at most one element of F . Indeed F can only be matched to the one core vertex, and every selected element of B is available for any remaining left position. Its basis polynomial is

$$\mathcal{B} = e_{n+1}(O) + e_1(F)e_n(O).$$

Its two-block diagonal is $y^n(c_0y + c_1x)$ for positive constants c_0, c_1 . This is real stable, and polarization separately in the two blocks recovers $\mathcal{B}$, preserving stability [4]. Setting each private variable to the inverse left activity and each right variable to t times its activity recovers P_H , up to a positive factor. Thus P_H is real-rooted and satisfies Newton's inequalities. Exchanging shores handles $b = 1$.

For $\min(a, b) \leq 2$ but insufficient exterior populations, the matching rank is

$$\min\{a + b, n + a, m + b\}.$$

The support criterion proves this formula directly: for every k below all three bounds, choose $i = \min(a, k)$ and $j = \min(b, k)$ selected core vertices. If $n < b$ or $m < a$, the formula equals the smaller shore size. The established weighted smaller-shore ULC theorem therefore applies [7, 1]. [*Merge note:* This is *preorder-root-polytopes*'s Theorem 70.1 (Part V), source 74's Lemma 3.1.] Its weighted form follows by specializing the basis polynomial of the extended transversal matroid at private variables s/u_x and right variables tv_y , then using the bivariate Lorentzian criterion. Hence Theorem 16.1's positive direction holds for every exterior population when the actual rank is used.

For nonnegative activities, delete the zero-activity vertices. The graph remains a full Hall graph with cores of size at most their original sizes. An empty core leaves a complete bipartite graph and isolates; singleton cores and all other cases are covered above. Thus the positive direction remains true with normalization by the surviving degree, not merely by its original value.

16.5 Explicit counterexamples when both cores are large

[*Merge note:* Source 74 (Section 6) prints the same family with the populations $N = b + 30$, $M = a + 30$; source 75's offset h is a second choice, and Table 3 records its first failing integer activities for seven shapes.] Now let $a, b \geq 3$, and set

$$r = a + b, \quad K = (r - 1)ab, \quad H = (r + 1)ab - 2r(r - 1).$$

Since $ab \geq 3(r-3)$ and $r \geq 6$,

$$H \geq r^2 - 4r - 9 > 0.$$

Choose the explicit integer

$$h = 1 + \left\lceil \frac{2K}{H} \right\rceil, \quad n = b - 1 + h, \quad m = a - 1 + h.$$

Give every exterior vertex activity one and every core vertex activity $w > 0$. Put

$$A_i = \binom{n}{b-i}, \quad B_i = \binom{m}{a-i} \quad (i = 0, 1, 2),$$

and define

$$\begin{aligned} X &= aA_0B_1 + bA_1B_0, \\ Y &= aA_1B_2 + bA_2B_1, \\ Z &= \binom{a}{2}A_0B_2 + abA_1B_1 + \binom{b}{2}A_2B_0. \end{aligned}$$

The top three support coefficients are exactly

$$\begin{aligned} p_r &= A_0B_0w^r, \\ p_{r-1} &= A_1B_1w^r + Xw^{r-1}, \\ p_{r-2} &= A_2B_2w^r + Yw^{r-1} + Zw^{r-2}. \end{aligned}$$

They are obtained by omitting zero, one or two core vertices in the general support formula. Consequently

$$\Delta_{r-1}^{(r)}(P_H) = w^{2r-2}(\alpha w^2 + \beta w + \gamma), \quad (43)$$

where

$$\begin{aligned} \alpha &= (r-1)A_1^2B_1^2 - 2rA_0A_2B_0B_2, \\ \beta &= 2(r-1)A_1B_1X - 2rA_0B_0Y, \\ \gamma &= (r-1)X^2 - 2rA_0B_0Z. \end{aligned}$$

The leading coefficient is strictly negative. Indeed

$$\frac{A_1^2B_1^2}{A_0A_2B_0B_2} = \frac{ab}{(a-1)(b-1)} \left(1 + \frac{1}{h}\right)^2,$$

while the choice of h gives

$$\frac{H}{K} \geq \frac{2}{h-1} > \frac{2}{h} + \frac{1}{h^2}.$$

Therefore this coefficient ratio is strictly below $2r/(r-1)$, exactly the threshold for $\alpha < 0$.

For a completely finite activity choice, take

$$w = 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor.$$

Then $w \geq 1$ and

$$\alpha w^2 + \beta w + \gamma \leq \alpha w^2 + (|\beta| + |\gamma|)w < 0.$$

This gives an explicit positive-integer two-shore activity counterexample for every $a, b \geq 3$, completing the negative direction of Theorem 16.1. The activity bound is conservative and is not claimed minimal.

16.6 A positive theorem does not imply real rootedness

The two-by- s conclusion is genuinely an ultra-log-concavity theorem. For $s = 3$, $n = m = 9$, activity 100 at all five core vertices and activity one at every exterior vertex, replacing t by $t/100$ gives

$$F(t) = 1 + 645t + 38730t^2 + 124500t^3 + 37800t^4 + 3024t^5.$$

Its discriminant is

$$-4431992565218959594687680000000 < 0,$$

so not all roots are real. Exact root counting gives three real roots. Its four order-five margins are nevertheless

$$1276800, \quad 8036447400, \quad 75433572000, \quad 1950480000,$$

all strictly positive. The program `verify_separation.py` [*Merge note*: shipped as `code/75-full-hall-verify_separation.py`] regenerates these integers from the graph parameters.

16.7 Verification and the remaining questions

The accompanying exact programs generate the twelve normalized coefficient families directly from (36)–(37). A separate check proves their signs by substituting the Newton bounds, reducing to corners of multilinear boxes, and verifying nonnegative polynomial coefficients in the free integer slacks. Denominator positivity and the $k = 1, 2$ boundary are checked explicitly. The generic extraction also reproduces all 49 orbit coefficients of the two-by-three case and all 80 of the two-by-four case exactly.

The readable argument above is uniform in s ; it is not an inference from finitely many core sizes. Exact scripts audit algebraic identities and boundary bookkeeping. They do not constitute a Lean formalization, and numerical root approximations are not used in any proof.

The existing Hall-bottleneck report [9] already disproves universal weighted rank normalization using a three-by-three core and studies symmetric higher cores. Theorem 16.1 gives a sharp boundary for the *full* Hall-core shapes, including all unequal cores and arbitrary independent positive activities in the positive range. Arbitrary graphs with a two-plus-three minimum cover are still not classified here: missing core edges and proper exterior neighbourhoods need different arguments. In particular, this report does not settle whether the first weighted counterexample among all bipartite graphs has rank five or six, and gives no negative result for unit weights or weights confined to one shore. [*Merge note*: Settled within the batch: the first weighted counterexample rank is exactly six (source 74, Corollary 2.2); unit-weight and one-shore counterexamples are sources 71 and 73 (Part II). Source 75’s delivered README, which is not shipped, likewise says that general weighted rank five remains open “here”. Arbitrary graphs with a $2 + 3$ minimum cover, which source 75 leaves unclassified, are covered by source 74’s Theorem 2.1.]

Part II

Unit weights and one weighted shore: counterexamples and the rank-38 threshold

Contents of Part II. Part II prints five sources. Source 71 (Section 17) gives a connected all-unit counterexample, answering Research question 84 of *preorder-root-polytopes* negatively. Source 73 (Section 18) gives one-shore counterexamples (all left activities one), answering Research question 85 there negatively. Sources 55, 47 and 51 (Sections 19, 20 and 21) study the full Hall graphs with *one common activity on the right core* and one elsewhere: the last gap first fails at matching number 38 (55), so does full ULC (47), and at matching number 49 the last gap no longer controls the others (51).

Which weighting. Every section title of this Part states its weighting. Source 71 is unit weighted; sources 73, 55, 47 and 51 are one-shore weighted, and the last three only with one common activity on the right core. None of them answers Research question 88 of *preorder-root-polytopes*, which concerns the two-shore family $H_{s,n}$ with weight on both cores; sources 47 and 51 answer its one-shore analogue.

Same theorem in several sources. Source 71's unit pendant-arm identity is the case $T = 1$ of source 73's weighted identity; it is printed once, in the weighted form, in source 73's appendix. Source 47 recalls source 55's endpoint theorem without proof (kept as a recall, with pointers) and re-proves the first-gap inequality by the Motzkin–Straus route (replaced by a pointer to Part XI of *preorder-root-polytopes*; its strictness step is kept). Source 55 extends source 73: the top-three formula and the rank-38 construction of 73 reappear in 55 with an exact threshold. The smaller-shore steps of sources 55 and 47 restate *preorder-root-polytopes*'s Theorem 70.1.

Non-claims of Part II. Source 71 does not determine the least matching number, vertex count or edge count of a unit counterexample, and keeps the smaller-shore bound and ordinary log-concavity; source 73 makes no minimal-rank claim; sources 55, 47 and 51 concern one common right-core activity on full Hall graphs only, and not arbitrary one-shore activities; source 55 concerns the last gap only.

17 Unit weight failure of matching rank normalization (source 71)

Abstract of source 71, as delivered. We give a connected bipartite graph whose unweighted matching-support polynomial fails ultra-log-concavity normalized by its maximum matching size. The explicit graph has 6916 vertices, 174228 edges and matching rank 3450. Its last Newton margin is a strictly negative integer, certified by a short exact formula for its top three coefficients. Every vertex activity is one. A second example has matching rank exactly one below the size of the smaller shore, showing that even this smallest possible discrepancy can obstruct rank normalization. The construction uses pendant length-two arms and admits exact support counting without enumerating the large graphs' matchings.

Status and scope. The counting identities and the counterexamples below are established in ordinary mathematics and independently checked in exact arithmetic. This is an unrefereed

research note, not a Lean-certified result. Its claim of resolving a question is relative to the inspected ProveIt manuscript *Hall Bottlenecks and the Limits of Rank Normalization* [9]; worldwide priority is not asserted. The examples do not determine the smallest possible matching rank or number of vertices of a counterexample.

17.1 The question and the counterexample

For a finite simple bipartite graph $G = (X, Y; E)$, define

$$p_G(t) = \sum_{k \geq 0} p_k t^k,$$

where p_k is the number of pairs $I \subseteq X$, $J \subseteq Y$ with $|I| = |J| = k$ for which $G[I, J]$ has a perfect matching. A feasible pair is counted once, regardless of its number of perfect matchings. This is the bipartite form of the perfectly matchable set polynomial; see [5] for earlier work on this polynomial. In particular $p_{K_{2,2}}(t) = 1 + 4t + t^2$.

For a polynomial of degree R with positive coefficients, rank-normalized ultra-log-concavity means that $p_k / \binom{R}{k}$ is log-concave. Equivalently,

$$k(R - k)p_k^2 \geq (k + 1)(R - k + 1)p_{k-1}p_{k+1} \quad (1 \leq k < R). \quad (44)$$

Here the proposed normalization is $R = \nu(G)$, the maximum matching size. The smaller-shore normalization is known to hold [7, Theorem A]; replacing the smaller shore by $\nu(G)$ is a genuinely stronger assertion. The unit-weight version was explicitly left open in [9, question “Unit weights”]. [*Merge note:* That question is Research question 84 of *preorder-root-polytopes* (`hb:q:unit`); the smaller-shore normalization is its Theorem 70.1 (Part V). The positive unit-weight theorems there, Theorem 191.1 ($\nu \leq 3$), Theorem 196.1 (a rank-four family) and Theorem 206.2 (singleton covers), and Theorem 2.1 of Part I, do not apply to the graph below.]

Theorem 17.1. *There is a connected simple bipartite graph with all vertex activities equal to one, 6916 vertices, 174228 edges and matching rank $R = 3450$, for which (44) fails at $k = 3449$.*

The complete graph definition, the rank certificate, and the exact negative margin are given below. No external field or limiting activity is used in this theorem. [*Merge note:* Theorem 17.1 answers Research question 84 of *preorder-root-polytopes* negatively, and with it the unit-weight half of its Research question 37 and its Research question 94 (no all-rank proof at unit weights can exist). By Corollary 5.1, every minimum cover of this graph meets both shores in at least three vertices; the cover of Lemma 17.2 has $12 + 3040 = 3052$ vertices on the left and 398 on the right.]

17.2 A graph family with pendant arms

Let $a, b, N, M \geq 1$, with $N \geq b$ and $M \geq a$, and let $W \geq 2$ be an integer. Start with left classes A, L of sizes a, N and right classes B, C of sizes b, M . Include all edges in

$$A \times B, \quad A \times C, \quad L \times B,$$

and no edges in $L \times C$. At each $c_j \in C$, attach $W - 1$ new length-two arms

$$c_j - z_{j,h} - \ell_{j,h} \quad (1 \leq h < W),$$

where $z_{j,h}$ is a new left vertex and $\ell_{j,h}$ is a new right leaf. There are no further edges. Write

$$K = M(W - 1), \quad r = a + b, \quad R = r + K.$$

Denote the resulting graph by $G(a, b, N, M, W)$. *Every activity is one; W specifies an arm count and is not a vertex weight.*

Lemma 17.2 (Size and rank). *The two shore sizes of $G(a, b, N, M, W)$ are $a + N + K$ and $b + M + K$. Its edge count is $ab + aM + bN + 2K$, and its matching rank is $R = a + b + K$.*

Proof. Match every vertex of A to a distinct vertex of C , every vertex of B to a distinct vertex of L , and every new center $z_{j,h}$ to its leaf $\ell_{j,h}$. This matching has $a + b + K$ edges. The set consisting of A , B , and all new centers is a vertex cover of that same size. The size and edge formulas follow directly from the construction. Connectivity holds because the original three complete blocks form a connected graph, and all new vertices lie on attached arms. $\square$

17.3 Exact support counting

Lemma 17.3 (The unit pendant-arm identity). *Let a vertex x of a bipartite graph have $k \geq 1$ new pendant length-two arms attached to it, all with unit activities. Write $p_G = f + g$, where f counts original supports omitting x and g counts original supports containing x . Then the augmented support polynomial is*

$$(1+t)^{k-1}[(1+(k+1)t)f + (1+t)g]. \quad (45)$$

Proof (pointer). This is the case $T = 1$ of source 73's weighted pendant-arm identity, Lemma 18.6 (Section 18.5), with the two shores exchanged: there the arms hang at a left vertex and their centers carry activity T . [Merge note: The same theorem in two sources: source 71's proof is source 73's proof at $T = 1$ (supports containing x contribute $(1+t)^k g$; for supports omitting x , choosing j centers gives $j+1$ distinct endpoint sets, whence $\sum_j \binom{k}{j} (j+1)t^j f = (1+t)^{k-1}(1+(k+1)t)f$; no matching multiplicities are counted). It is printed once, in the weighted form.] $\square$

For the original three-block Hall graph, mark the number ρ of selected vertices of C . Its corresponding support polynomial is

$$P_\rho(t) = \binom{M}{\rho} \sum_{\substack{0 \leq i \leq a, \\ 0 \leq j \leq b, \\ i \geq \rho}} \binom{a}{i} \binom{b}{j} \binom{N}{j+\rho-i} t^{j+\rho}, \quad (46)$$

where out-of-range binomial coefficients are zero. Indeed, i and j count selected core vertices in A and B ; balance requires $j + \rho - i$ selected vertices in L . Feasibility is precisely $i \geq \rho$: match the exterior vertices first, then match the remaining core vertices across the complete core. Applying Lemma 17.3 at all M vertices of C gives

$$p_{G(a,b,N,M,W)}(t) = \sum_{\rho=0}^a P_\rho(t) (1+t)^{K-M+\rho} (1+Wt)^{M-\rho}. \quad (47)$$

The factor $1 + Wt$ records W different endpoint sets; it does not assign an activity to a vertex.

17.3.1 A short formula for the top three coefficients

For $a, b \geq 2$ and $\delta \in \{0, 1, 2\}$, let e be the deficit of the original support size from $r = a + b$. If u left-core and v right-core vertices are omitted, then

$$u + v \leq e, \quad \ell = b - e + u, \quad \rho = a - e + v$$

are the exterior-left and exterior-right support sizes. Put $A_\rho = K - M + \rho$ and $B_\rho = M - \rho$. Extracting the top coefficients in (47) yields the entirely integer formula

$$p_{R-\delta} = \sum_{e=0}^{\delta} \sum_{u=0}^e \sum_{v=0}^{e-u} \binom{a}{u} \binom{b}{v} \binom{N}{b-e+u} \binom{M}{a-e+v} \cdot \sum_{z=0}^{\min(B_\rho, \delta-e)} \binom{A_\rho}{\delta-e-z} \binom{B_\rho}{z} W^{B_\rho-z}. \quad (48)$$

To see the last line, the pendant factors have degree K . Selecting their coefficient of degree $K - (\delta - e)$ means omitting $\delta - e$ of their linear terms, with z omissions from the B_ρ factors of slope W . Thus (48) counts supports exactly and requires only a few summands.

17.4 The finite certificate

Take

$$(a, b, N, M, W) = (12, 398, 410, 16, 191).$$

Thus $K = 3040$, the shore sizes are 3462 and 3454, and Lemma 17.2 gives $R = 3450$ and 174228 edges. The graph has 6916 vertices in total.

Define the positive integer

$$D = 160160547610973088925200.$$

Formula (48) gives the compact exact certificate

$$\begin{aligned} p_{3450} &= 455D \, 191^4, \\ p_{3449} &= 3020455956D \, 191^3, \\ p_{3448} &= 10022638278784267D \, 191^2. \end{aligned} \quad (49)$$

The required endpoint inequality is $3449p_{3449}^2 \geq 6900p_{3448}p_{3450}$. Its margin is

$$\begin{aligned} &3449p_{3449}^2 - 6900p_{3448}p_{3450} \\ &= D^2 \, 191^6 (3449 \cdot 3020455956^2 - 6900 \cdot 455 \cdot 10022638278784267) \\ &= -314102056577041236 \, D^2 \, 191^6 < 0. \end{aligned}$$

This proves Theorem 17.1. The exact coefficient formula, together with the displayed finite integer substitution, is the proof of the counterexample; the search that led to the parameters is not used as a premise.

17.5 Failure when the smaller shore exceeds rank by one

The phenomenon does not require a large gap between the smaller shore and the matching rank.

Theorem 17.4. *There is a connected unit-weight bipartite graph whose smaller shore has size 125509, whose matching rank is 125508, and whose support polynomial is not ULC_{125508} .*

Proof. Take $G(8, 100, 101, 30, 4181)$. Then $K = 125400$, $R = 125508$, the shore sizes are 125509 and 125530, and the graph has 251039 vertices and 261940 edges. Put $D_1 = 8567325$. The same formula (48) gives

$$\begin{aligned} p_R &= 69D_1 \, 4181^{22}, \\ p_{R-1} &= 61945839672D_1 \, 4181^{21}, \\ p_{R-2} &= 27806230344578726426D_1 \, 4181^{20}. \end{aligned}$$

Consequently

$$(R-1)p_{R-1}^2 - 2Rp_{R-2}p_R = -415296714996000603216 D_1^2 4181^{42} < 0.$$

The rank and the two shore sizes follow from Lemma 17.2. $\square$

The smaller-shore theorem [7] still guarantees ULC_{125509} for this graph. The conclusion is failure of the stronger order 125508, not failure of that published bound.

17.5.1 Why deficiency one is not an isolated accident

Fix a, b, N, M and let W grow, still keeping every vertex activity one. Write

$$h = N - b + 1, \quad \theta = \frac{b}{h}, \quad \zeta = \frac{b(b-1)}{h(h+1)}, \quad \mu = \frac{a}{M-a+1}, \quad \eta = \frac{a(a-1)}{(M-a+1)(M-a+2)}.$$

Expanding (48), with $R = a + b + M(W-1)$, gives

$$\frac{(R-1)p_{R-1}^2 - 2Rp_{R-2}p_R}{p_R^2 W^3} \longrightarrow M \left[\mu^2(a+\theta)^2 - 2\eta \left(\binom{a}{2} + a\theta + \zeta \right) \right]. \quad (50)$$

For example, the leading terms of p_{R-1}/p_R and p_{R-2}/p_R are respectively

$$W(M + \mu(a+\theta)), \quad W^2 \left[\frac{M^2}{2} + M\mu(a+\theta) + \eta \left(\binom{a}{2} + a\theta + \zeta \right) \right],$$

which imply (50) directly. Set $h = 2$, so $N = b + 1$. After division by μ^2 , the bracket in (50) tends as $M \rightarrow \infty$ to

$$2a - 1 + \frac{4a-1}{3a}b - \frac{a-4}{12a}b^2.$$

For any fixed $a \geq 5$, choose b sufficiently large, then choose a finite $M \geq a + 2$ sufficiently large. The bracket is negative, so all sufficiently large integers W give counterexamples. Their smaller shore is always $R + 1$. This proves existence of infinitely many deficiency-one examples; Theorem 17.4 supplies one explicit finite instance.

17.6 Consequences for Ehrhart h^* -polynomials

For either constructed graph $G = (X, Y; E)$, form $\widehat{G}$ by adding a new left vertex v , a new right vertex w , and the edges vy for $y \in Y$ and xw for $x \in X \cup \{v\}$. Let

$$Q_{\widehat{G}} = \text{conv}\{e_i + e_j : ij \in E(\widehat{G})\},$$

and let $\text{STAB}(\overline{G})$ be the stable set polytope of the complement of G . Davis–Kohl [5, Theorem 3.10 and Lemma 3.11] give

$$h^*(Q_{\widehat{G}}; t) = p_G(t) = h^*(\text{STAB}(\overline{G}); t).$$

Thus these two specific Ehrhart h^* -polynomials have degree $R = \nu(G)$ and fail ULC_R . [*Merge note:* Compare source 74's corollary in Section 7, which gives ULC_r for these polynomials when $\nu \leq 5$ or a minimum cover meets one shore in at most two vertices.] They also have nonreal zeros: otherwise their nonnegative coefficients would satisfy the degree-normalized Newton inequalities. Their coefficients still satisfy ordinary log-concavity and the smaller-shore ULC bound by [7].

17.7 Verification and remaining questions

The package contains standard-library Python scripts [*Merge note:* shipped as `code/71-unit-weight-verify*.py`, with outputs `data/71-unit-weight-*.json`] checking the pendant identity by independent endpoint-set enumeration on 150 small graphs, including nonuniform activities as an extra regression test. For both displayed examples, they evaluate (48) by integer binomial sums and independently recompute the tails from normalized reciprocal sums of the pendant factors. They also check the exact negative margins, explicit matching and cover constructions, edge counts and coverage. All checks pass with fixed seeds. They count supports without matching multiplicity and supplement the written proof.

The minimum matching rank, vertex count and edge count of an all-unit counterexample remain undetermined. [*Merge note:* By Part I (Corollary 2.2) such a counterexample has matching number at least six, so the least unit failure rank lies in $[6, 3450]$.] Rank 3450 is the best found in a bounded parameter search, not a proved minimum. Smaller deficiency-one examples and structural classes retaining rank-normalized ultra-log-concavity remain natural questions. The published smaller-shore normalization and ordinary log-concavity still hold.

18 One-shore activities can destroy matching-rank ULC (source 73)

Abstract of source 73, as delivered. The restriction to activities on one shore does not restore matching-rank-normalized ultra-log-concavity of perfectly matchable endpoint supports. We give an explicit full Hall-bottleneck graph on 125 vertices, with matching rank 50, whose left activities are all one and whose right activities take only the values one and 10000. Its last Newton margin is a positive factor times the negative integer -39998827 . A second example has matching rank 38. Both certificates follow from a closed formula for the top three coefficients and a quadratic in the common right-core activity. Thus no numerical root calculation or limiting argument is needed to certify either finite counterexample. We also give a general criterion for constructing such examples and record a separate pendant-arm transfer construction.

Status and scope. These are ordinary mathematical proofs with reproducible exact-integer checks, not formal Lean verification or external peer review. The result resolves the one-shore-activity question in the inspected ProveIt Hall-bottleneck report negatively. It does not produce a unit-weight counterexample, identify the smallest possible rank, or assert exhaustive literature priority. [*Merge note:* The question is Research question 85 of *preorder-root-polytopes* (`hb:q:oneshore`), with exactly these hypotheses: all left activities one, right activities arbitrary positive. The weighting here, activity w on the right core B and one elsewhere, is the “one common right-core activity” of Section 1.3. The unit-weight counterexample is source 71’s (Section 17).]

18.1 The polynomial and two explicit counterexamples

For a bipartite graph $G = (X, Y; E)$ with positive vertex activities u_x, v_y , define

$$P_G(t) = \sum_{(I, J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^d p_k t^k.$$

Here $I \subseteq X$, $J \subseteq Y$, $|I| = |J|$, and feasibility means that $G[I, J]$ admits a perfect matching. Each pair of endpoint sets is counted once, regardless of the number of realizing matchings.

The degree d is the maximum matching size. This is the vertex-weighted perfectly matchable set polynomial, with t recording half the support size.

The property ULC_d requires log-concavity of $p_k/\binom{d}{k}$ and no internal zeros. For these positive-activity support polynomials there are no internal zeros, so the inequalities are

$$\Delta_k^{(d)} = k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1} \geq 0. \quad (51)$$

In particular, the last inequality is $(d-1)p_{d-1}^2 \geq 2dp_{d-2}p_d$.

Write $H_{a,b;n,m}$ for the graph with shores $X = A \sqcup L$, $Y = B \sqcup R$, block sizes a, n, b, m , and exactly the edges

$$(A \times B) \cup (L \times B) \cup (A \times R). \quad (52)$$

All exterior-left vertices are adjacent to all of B , all exterior-right vertices to all of A , and there are no exterior-to-exterior edges. Give every vertex activity one except the b vertices of B , each of which has activity $w > 0$. Thus the weighting is confined to one shore. If $n \geq b$ and $m \geq a$, the matching rank is $d = a + b$: the core is a cover of size d , attained by disjoint matchings $A \rightarrow R$ and $L \rightarrow B$.

Theorem 18.1 (A counterexample on 125 vertices). *Let $G = H_{8,42;44,31}$, with activity w on B and one elsewhere. Its matching rank is 50, and*

$$\Delta_{49}^{(50)} = 98 \left[\binom{44}{42} \binom{31}{7} \right]^2 w^{82} (-w^2 + 5958w + 421173). \quad (53)$$

At $w = 10000$ the parenthesized integer is $-39998827 < 0$. The first positive integer w for which this particular margin fails is 6028, where the integer is -787 .

This graph has 125 vertices and 2432 edges. Only 42 activities differ from one. The theorem is an exact certificate against universal one-shore ULC_d .

Theorem 18.2 (A counterexample of matching rank 38). *Let $G = H_{8,30;32,1000}$ with the same one-shore weighting. Put*

$$Q(w) = -2404w^2 + 4878807600w + 10888224221475.$$

Then

$$\Delta_{37}^{(38)} = \frac{3}{2272} \left[\binom{32}{30} \binom{1000}{7} \right]^2 w^{58} Q(w). \quad (54)$$

At $w = 3000000$, one has $Q(w) = -6988688975778525 < 0$. The first positive integer giving a negative value of this particular margin is 2031684.

No minimal-rank claim is made. The rank-38 example has 1070 vertices; the rank-50 example is a substantially smaller graph. [*Merge note:* Within full Hall graphs with one common right-core activity, source 55 (Section 19) proves that 38 is the least matching number at which the last gap can fail, with 864 vertices the fewest at that rank, and source 47 (Section 20) extends this to every gap.]

18.2 Endpoint counting gives the complete certificate

A size- k support selecting i vertices of A and j vertices of B is feasible exactly when $i + j \geq k$. Necessity follows because the $k - i$ selected exterior-left vertices must match into the j selected

vertices of B . For sufficiency, first match the exterior-left vertices into B , then the exterior-right vertices into A , and finally the remaining core vertices to each other. Consequently

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j, \quad (55)$$

with out-of-range binomial coefficients set to zero.

Lemma 18.3 (The top three coefficients). *Let $a, b \geq 2$, $d = a + b$, $n \geq b$, $m \geq a$, and set*

$$X_i = \binom{n}{b-i}, \quad Y_i = \binom{m}{a-i} \quad (i = 0, 1, 2).$$

Then

$$p_d = X_0 Y_0 w^b, \quad (56)$$

$$p_{d-1} = (X_1 + aX_0)Y_1 w^b + bX_1 Y_0 w^{b-1}, \quad (57)$$

$$\begin{aligned} p_{d-2} = & \left(X_2 + aX_1 + \binom{a}{2} X_0 \right) Y_2 w^b \\ & + b(X_2 + aX_1)Y_1 w^{b-1} + \binom{b}{2} X_2 Y_0 w^{b-2}. \end{aligned} \quad (58)$$

Proof. At $k = d$, the condition $i + j \geq d$ forces $(i, j) = (a, b)$. At $k = d - 1$, the possible deficits $(a - i, b - j)$ are $(0, 0), (1, 0), (0, 1)$. At $k = d - 2$, these and $(2, 0), (1, 1), (0, 2)$ are possible. Substitution in (55) gives the formulas. $\square$

For a compact parametrization write

$$h = n - b + 1, \quad t = \frac{b}{h}, \quad z = \frac{b(b-1)}{h(h+1)}, \quad R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2}, \quad J = \binom{a}{2} + at + z. \quad (59)$$

Thus $t = X_1/X_0$, $z = X_2/X_0$, $R = Y_0/Y_1$ and $\rho = Y_2/Y_1$. Put $D = X_0 Y_1 > 0$.

Proposition 18.4 (Quadratic endpoint criterion). *For $a, b \geq 2$ and the above parameters,*

$$\Delta_{d-1}^{(d)} = D^2 w^{2b-2} (\alpha w^2 + \beta w + \gamma), \quad (60)$$

where

$$\alpha = (d-1)(a+t)^2 - 2dR\rho J, \quad (61)$$

$$\beta = 2bR((d-1)t(a+t) - d(at+z)), \quad (62)$$

$$\gamma = bR^2(b(d-1)t^2 - d(b-1)z). \quad (63)$$

Proof. Divide the three identities in Lemma 18.3 by D and substitute them into $(d-1)p_{d-1}^2 - 2dp_{d-2}p_d$. Grouping the three powers $w^{2b}, w^{2b-1}, w^{2b-2}$ yields the displayed coefficients. $\square$

For Theorem 18.1, formulas (61)–(63) give

$$(\alpha, \beta, \gamma) = 98(-1, 5958, 421173).$$

For Theorem 18.2, they give

$$(\alpha, \beta, \gamma) = \frac{3}{2272}(-2404, 4878807600, 10888224221475).$$

This proves both exact factorizations. Each primitive quadratic has a negative leading coefficient and a positive constant term, so it has precisely one positive zero. Direct substitution at the stated first integer and the preceding integer proves the threshold claims; the preceding values are respectively 5310 and 1792188719.

18.3 A general construction criterion

For fixed a, b, h the leading coefficient (61) tends, as $m \rightarrow \infty$, to

$$\alpha_\infty = (d-1)(a+t)^2 - \frac{2d(a-1)}{a} \left(\binom{a}{2} + at + z \right). \quad (64)$$

This is only a device for finding examples; the preceding theorems use finite exact identities.

Proposition 18.5 (From a negative limiting coefficient to a finite graph). *If $\alpha_\infty < 0$, set*

$$C = \frac{2d(a-1)}{a} \left(\binom{a}{2} + at + z \right), \quad m = a - 1 + \left\lfloor \frac{C}{-\alpha_\infty} \right\rfloor.$$

Then $m \geq a$ and $\alpha < 0$. With β, γ defined by the finite parameters, the positive integer

$$w = 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor \quad (65)$$

necessarily makes the last margin negative.

Proof. One has $\alpha = \alpha_\infty + C/(m - a + 2)$. The chosen denominator is $1 + \lfloor C/(-\alpha_\infty) \rfloor > C/(-\alpha_\infty)$, so $\alpha < 0$. Also $C > -\alpha_\infty$, since their difference is $(d-1)(a+t)^2 > 0$, proving $m \geq a$. For (65), $(-\alpha)w > |\beta| + |\gamma|$ and $w \geq 1$, hence

$$\alpha w^2 + \beta w + \gamma \leq -(-\alpha)w^2 + |\beta|w + |\gamma| < 0.$$

□

For $a = 8, b = 30, h = 3$ (thus $n = 32$),

$$\alpha_\infty = -\frac{61}{4}, \quad \alpha = \frac{48379 - 61m}{4(m-6)}.$$

The leading coefficient is negative exactly when $m \geq 794$. Choosing $m = 1000$ gives Theorem 18.2. The alternative $a = 7, b = 31, h = 3$ also has rank 38 and gives $\alpha_\infty = -764/63$. These are construction examples, not a proof of any global optimum.

18.4 What this settles and what remains open

The inspected ProveIt Hall-bottleneck report [9] distinguishes the arbitrary-two-shore weighted failure from its unit-weight and one-shore variants. Theorems 18.1 and 18.2 supply a negative answer to the one-shore variant, even inside the full two-step Ferrers family (52). Since a degree- d polynomial with positive coefficients and only real zeros satisfies Newton's inequalities with normalization d , these examples also fail positive-field real-rootedness.

There is no conflict with the ambient-shore normalization in the bimatroid theorem of Röhrle and Ulirsch [7]. Nor does the current non-nesting-rook result of Alexandersson and Jal settle the stronger normalization used here: Corollary 4.1 of version 3 explicitly normalizes by the board's row count [15]. Our two examples have smaller shore sizes 52 and 40, respectively, while their matching ranks are 50 and 38. The older version's statement should not be substituted for that current row-normalized theorem.

The full-Hall constructions here have nonunit activities. Further concrete questions include the smallest rank admitting one-shore failure, the smallest graph admitting such failure, and which fixed full-Hall core shapes remain universally rank-ULC under one-shore activities. [Merge note: Re-scoped by the batch: for one common right-core activity on full Hall graphs the smallest failing rank is exactly 38 (sources 55 and 47); among all one-shore graphs it lies in [6, 38], since Part I gives ULC_ν for $\nu \leq 5$.] The present calculation also suggests studying the sign of (64) systematically; its negativity is sufficient for a counterexample, not asserted necessary for every one-shore failure.

Reproducibility. The accompanying script `verify_full_hall.py` [*Merge note:* shipped as `code/73-one-shore-verify_full_hall.py`, output `data/73-one-shore-full_hall_verification.json`] evaluates the complete feasibility sum (55), checks it independently against Boolean matching on six small graphs, and compares the top-three formula on 100 deterministic parameter samples. It then verifies both factorizations at four stated integer activities using exact integers and rational numbers. For all four instances, the only negative order- d Newton margin is the last one. These finite checks supplement the proof; they are not its logical basis.

18.5 Appendix: A pendant gadget that transfers fields between shores

The direct Hall counterexamples above give much smaller matching rank. The following independent construction gives a one-shore counterexample whose only nonunit activity is the modest integer 21. It also explains how a weighted obstruction can survive a transfer of fields between shores.

[*Merge note:* Lemma 18.6 contains source 71’s Lemma 17.3 as the case $T = 1$ (shores exchanged); it is printed here once for both sources.]

Lemma 18.6 (Exact pendant-arm identity). *Let x be a left vertex of activity one in a bipartite graph, and write $p_G(t) = f(t) + g(t)$, where f counts supports omitting x and g counts supports containing x . Attach $k \geq 1$ disjoint length-two arms $x - z_i - \ell_i$. Give the new right vertices z_i activity $T > 0$ and the new left leaves ℓ_i activity one. Then*

$$p_{G'}(t) = (1 + Tt)^{k-1}[(1 + (k + 1)Tt)f(t) + (1 + Tt)g(t)]. \quad (66)$$

Moreover, $\nu(G') = \nu(G) + k$.

Proof. Fix a support counted by g , so x must match within the original graph. Each selected new center then forces its own leaf, contributing $(1 + Tt)^k g$. For a support counted by f , choose j new centers. Either all their leaves are selected and x is absent, or x replaces exactly one of those j leaves. The latter support forces x to that leaf’s center, so it cannot also use x in the original matching. These $j + 1$ endpoint sets are distinct. Their total contribution is

$$\sum_{j=0}^k \binom{k}{j} (j + 1)(Tt)^j f = (1 + Tt)^{k-1} (1 + (k + 1)Tt)f.$$

This counts supports, not allocations or perfect matchings. An original maximum matching together with all k center–leaf edges has size $\nu(G) + k$. An original minimum vertex cover together with the k centers covers the augmented graph, giving the reverse inequality. $\square$

The top power of T in (66) is

$$T^k t^k ((k + 1)f + g).$$

Thus the forced-center limit simulates activity $1/(k + 1)$ at x , up to a positive scalar and the degree shift k . For gadgets at several left vertices, these factors multiply. The rank shift must be accounted for: a polynomial $t^K p(t)$ of degree $r + K$ has last normalized inequality

$$(r + K - 1)p_{r-1}^2 \geq 2(r + K)p_{r-2}p_r.$$

In particular, a top-three ratio strictly below 2 survives every such shift, and continuity then gives a finite positive center activity. This observation is a sufficient criterion; the finite example below is checked by its exact endpoint polynomial.

18.5.1 A rank 310 example with activity 21

[*Merge note:* This example is one-shore weighted, not unit weighted: its ten right-core vertices have activity 21. Its rank is superseded by the rank-50 and rank-38 examples above; it is kept for the transfer construction.] Start with a full 20×10 Hall core, with exterior populations $N = 14$ on the left and $M = 30$ on the right. Attach 20 pendant arms to each exterior-left vertex. Give all left vertices activity one. Give the ten original right-core vertices activity 21, and every other right vertex activity one, including every pendant center. There are $K = 280$ arms, so the graph has

$$634 \text{ vertices,} \quad 1500 \text{ edges,} \quad \nu(G) = 30 + 280 = 310.$$

For completeness, the original 30-edge matching pairs the twenty left-core vertices to distinct exterior-right vertices and the ten right-core vertices to distinct exterior-left vertices. Add all 280 center–leaf edges. The thirty original core vertices together with the 280 centers form a cover of the same size.

Here is an exact count of the endpoint coefficients. More generally let a, b, N, M be Hall parameters, put $W = k + 1$, $K = N(W - 1)$, and let $P_\ell(t)$ count original supports using exactly ℓ exterior-left vertices when only the right core has activity W . Then

$$P_\ell(t) = \binom{N}{\ell} \sum_{\substack{0 \leq i \leq a, \\ j \geq \ell}} \binom{a}{i} \binom{b}{j} \binom{M}{i + \ell - j} W^j t^{i + \ell}, \quad (67)$$

where out-of-range binomial coefficients are zero. The condition $j \geq \ell$ is exactly the Hall-support feasibility condition after balancing the shores. Repeated application of Lemma 18.6 gives the full polynomial

$$P(t; T) = \sum_{\ell=0}^{\min(N, b)} P_\ell(t) (1 + Tt)^{K - N + \ell} (1 + W T t)^{N - \ell}. \quad (68)$$

Every factor and coefficient in this expression is explicit.

For $(a, b, N, M, W, T) = (20, 10, 14, 30, 21, 1)$, coefficient extraction gives

$$\begin{aligned} p_{308} &= 8131749153344342901404784768563550, \\ p_{309} &= 39894459936702682423206366171900, \\ p_{310} &= 97561086654439123859418490215. \end{aligned}$$

Consequently the endpoint margin is

$$309p_{309}^2 - 620p_{308}p_{310} = -726235 \cdot 4485169759863727935000 \cdot 21^{26} < 0.$$

This is an exact one-shore-weighted counterexample. Its full negative integer margin, full coefficient vector, edge list, matching, and cover are included in the verification data. [*Merge note:* Shipped as `data/73-one-shore-gadget_verification-one_shore_rank310_witness.json`, beside `code/73-one-shore-gadget_verification-*.py`; the gadget identity record is byte-identical to source 71's `data/71-unit-weight-gadget_identity_verification.json`.]

Keeping the same graph and allowing each center activity to be T , the stronger exact identity is

$$\begin{aligned} 309p_{309}(T)^2 - 620p_{308}(T)p_{310}(T) &= 4485169759863727935000 \cdot 21^{26} T^{558} \\ &\quad \cdot (-91777T^2 - 1459920T + 825462). \end{aligned}$$

At $T = 1$ the bracket is -726235 . This identity makes the finite activity explicit; no limiting value is used in the counterexample.

Verification. The gadget identity was checked against independent weighted endpoint-set enumeration on 150 small bipartite graphs, with a fixed seed. A separate standard-library script expands (68) in full, checks the three displayed coefficients and exact margin, and verifies the explicit rank certificates. An independent audit recomputed the three endpoint coefficients directly from Hall subset feasibility and the top coefficients of each gadget factor. The finite checks supplement the counting proofs above.

19 A sharp rank-38 boundary for the last Hall inequality, one common right-core activity (source 55)

Abstract of source 55, as delivered. For full Hall graphs with one common activity on the right core and unit activities elsewhere, the first possible failure of the last matching-rank-normalized Newton inequality occurs at rank 38. We prove this for all finite exterior populations and every positive core activity, rather than only in a limiting regime. An explicit positive identity rules out a negative finite-field window whenever the leading quadratic coefficient is nonnegative. A cubic then gives an exact parameter criterion, and a finite integer certificate proves the rank threshold. Precisely two core shapes occur at rank 38. The smaller of their minimal populations gives an 864-vertex example with an exact integer certificate.

Scope. The activity is common across one specified core block. The theorem concerns the *last* Newton inequality. It does not determine the first failure among arbitrary individual one-shore activities or among all gaps. General finite-activity rank-six one-shore ultra-log-concavity remains open. [*Merge note:* At rank six, the first gap holds for every positively weighted graph (*preorder-root-polytopes*, Theorem 180.2), and the second and last gaps for every one-shore graph (sources 05 and 13, Part III); the third and fourth stay open in general.] These are ordinary mathematical proofs with exact certificates, not formal kernel verification or external peer review; no global priority claim is made.

19.1 The family and the sharp threshold

Let $H_{a,b;n,m}$ have shores $A \sqcup L$ and $B \sqcup R$, with respective block sizes a, n, b, m , and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Here $a, b \geq 1$ and $n, m \geq 0$ are integers. Every vertex has activity one except the vertices of B , whose common activity is $w > 0$. Write

$$P(t) = \sum_{k=0}^d p_k t^k,$$

where p_k is the total activity of pairs of k -element endpoint sets admitting a perfect matching. Each pair is counted once, regardless of its number of matchings. The degree d is the matching rank. The last rank-normalized Newton inequality is

$$\Delta_{d-1} := (d-1)p_{d-1}^2 - 2dp_{d-2}p_d \geq 0 \quad (d \geq 2). \quad (69)$$

Theorem 19.1 (Sharp common-core-field endpoint boundary). *Within this family, the smallest matching rank for which (69) fails at some $w > 0$ is exactly 38. At rank 38, the only triples*

$$(a, b, h), \quad h = n - b + 1,$$

permitting failure for some finite m, w are

$$(7, 31, 3) \quad \text{and} \quad (8, 30, 3).$$

Their least possible m are 923 and 794, respectively. Thus the fewest vertices at rank 38 in this family with a failing last gap is 864, attained by $H_{8,30;32,794}$. For that graph the first failing positive integer activity is $w = 292602646$.

[Merge note: Source 47 (Theorem 20.1) shows that in this family no gap at all fails below matching number 38, so the least failure rank of full ULC is also 38. This is the one-shore analogue of Research question 88 of *preorder-root-polytopes*; that question itself, with activity on both cores, is not answered here.]

The qualifier “at rank 38” in the vertex statement matters: a previously constructed rank-50 example has only 125 vertices [16]. The new conclusion is a sharp rank threshold for the stated family and gap.

19.1.1 Deficient exterior populations

The actual matching rank is

$$d = \min\{a + b, a + n, b + m\}. \quad (70)$$

For a direct construction, first match $\min(a, m)$ pairs from A to R and $\min(b, n)$ pairs from L to B , then match the remaining core vertices to each other. This attains (70); the three displayed quantities are upper bounds from the core cover and the two shores.

If $n < b$ or $m < a$, this rank is the smaller shore size, and the established weighted smaller-shore theorem gives the entire actual-rank ULC sequence. [Merge note: That theorem is *preorder-root-polytopes*’s Theorem 70.1 (Part V), Lemma 3.1 here; the specialization recalled next is its proof.] Here is the weighted specialization explicitly. For any bipartite graph $G = (X, Y)$, introduce a private element $\bar{x}$ for each $x \in X$. Match these private elements and Y into X , with $\bar{x}$ adjacent only to x . The resulting transversal matroid has bases

$$\{\bar{x} : x \notin I\} \sqcup J$$

for exactly the feasible support pairs (I, J) . Its Lorentzian basis polynomial $\mathcal{B}$, specialized by

$$\left(\prod_{x \in X} u_x \right) \mathcal{B}((s/u_x)_{x \in X}, (tv_y)_{y \in Y}) = \sum_k p_k s^{|X|-k} t^k,$$

gives $\text{ULC}_{|X|}$ by the bivariate characterization and closure under nonnegative linear substitutions [1]. Transpose to obtain the smaller order. This is the weighted form of the relation/bimatroid argument in [7]. Consequently deficient populations cannot fail (69). We henceforth assume $n \geq b$ and $m \geq a$, and write $r = a + b = d$.

19.2 The exact endpoint quadratic

A size- k support selecting i core-left and j core-right vertices is feasible exactly when $i + j \geq k$. Indeed, the selected exterior-left vertices must fit into the selected B vertices; after matching the exterior vertices, the remaining core vertices can be matched because the core is complete. Thus

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j, \quad (71)$$

where out-of-range binomial coefficients are zero.

Set

$$h = n - b + 1, \quad x = \frac{b}{h}, \quad y = \frac{b(b-1)}{h(h+1)},$$

$$R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2}, \quad J = \binom{a}{2} + ax + y, \quad D = \binom{n}{b} \binom{m}{a-1} > 0.$$

Enumerating total core deficits zero, one and two in (71) gives

$$p_r = DRw^b,$$

$$p_{r-1} = D((a+x)w^b + bxRw^{b-1}),$$

$$p_{r-2} = D \left(J\rho w^b + b(ax+y)w^{b-1} + \binom{b}{2}yRw^{b-2} \right).$$

When $b = 1$, the last term is zero. Substitution in (69) yields

$$\Delta_{r-1} = D^2 w^{2b-2} (\alpha_m w^2 + 2bRBw + R^2 C), \quad (72)$$

where

$$\alpha_m = (r-1)(a+x)^2 - 2rR\rho J, \quad (73)$$

$$B = (r-1)x(a+x) - r(ax+y) = \frac{b[b(r-1) - ah^2]}{h^2(h+1)}, \quad (74)$$

$$C = b(b(r-1)x^2 - r(b-1)y)$$

$$= \frac{b^2\{b^2(r-1) + h[r(2b-1) - b^2]\}}{h^2(h+1)} > 0. \quad (75)$$

The last positivity uses $r(2b-1) - b^2 = a(2b-1) + b(b-1) > 0$. Define

$$K = \frac{2r(a-1)}{a}J, \quad \alpha_\infty = (r-1)(a+x)^2 - K. \quad (76)$$

Then

$$\alpha_m = \alpha_\infty + \frac{K}{m-a+2} \geq \alpha_\infty. \quad (77)$$

19.3 A finite field cannot create a hidden negative window

Lemma 19.2 (Positive discriminant margin). *Put $H = ah^2 - b(r-1)$ and define*

$$Q_0 = a^2(b^2 + 2b - 1) + a(b-1)(2b^2 + 3b - 1) + b(b-1)^2,$$

$$Q_1 = (r-1)(2ab - a - b),$$

$$R_0 = a(b^2 + 2b - 1) + (b-1)(3b - 1),$$

$$R_1 = a(3b - 1) + (b-1).$$

For all positive integers a, b ,

$$\frac{ah^3(h+1)^2}{b^2r}(\alpha_\infty C - b^2 B^2) = H(Q_0 + Q_1 h) + br(r-1)(R_0 + R_1 h). \quad (78)$$

If $H \geq 0$, the right side is strictly positive.

Proof. Substitution of (74), (75), and (76), followed by collecting powers of h , proves the identity. Each displayed term in Q_0, R_0, R_1 is nonnegative and their first terms are positive. Also $2ab - a - b = a(b-1) + b(a-1) \geq 0$, so $Q_1 \geq 0$. Finally $br(r-1) > 0$ and $R_0 + R_1 h > 0$. $\square$

Theorem 19.3 (Exact finite-field and population criterion). *Fix a, b, n, m with $n \geq b, m \geq a$. The last inequality fails for some $w > 0$ if and only if $\alpha_m < 0$. If $\alpha_m \geq 0$, it is strictly satisfied for every $w > 0$. If $\alpha_m < 0$, failure occurs precisely above the unique positive root of the quadratic in (72).*

For fixed a, b, h , there exist finite m and $w > 0$ giving failure if and only if $\alpha_\infty < 0$. In that case the least eligible integer population is

$$m_{\min} = a - 1 + \left\lfloor \frac{K}{-\alpha_\infty} \right\rfloor. \quad (79)$$

Proof. If $B \geq 0$ and $\alpha_m \geq 0$, every term in the quadratic is nonnegative and its constant term is positive. If $B < 0$, then $H > 0$ by (74). Lemma 19.2 and (77) imply

$$\alpha_m C - b^2 B^2 \geq \alpha_\infty C - b^2 B^2 > 0.$$

In particular $\alpha_m > 0$, and the quadratic has negative discriminant and is positive on the entire real line. This rules out every putative negative finite- w window with nonnegative leading coefficient.

If $\alpha_m < 0$, the constant term is positive, so the two real roots have opposite signs and failure is exactly above the positive root. Now (77) shows that some finite m has $\alpha_m < 0$ exactly when $\alpha_\infty < 0$. In that case the necessary and sufficient condition is

$$m - a + 2 > \frac{K}{-\alpha_\infty},$$

whose least integer solution is (79). Since $K + \alpha_\infty = (r-1)(a+x)^2 > 0$, the quotient in (79) exceeds one, so $m_{\min} \geq a$. $\square$

19.4 A cubic classification and the finite rank certificate

Clearing the positive denominator in (76) gives

$$ah^2(h+1)\alpha_\infty = F_{a,b}(h) = A_3 h^3 + A_2 h^2 + A_1 h + A_0, \quad (80)$$

where

$$\begin{aligned} A_0 &= ab^2(r-1), \\ A_1 &= -b(r-1)(ab - 2a - 2b), \\ A_2 &= a(a^2 + 2ab - a + 2b^2 - b), \\ A_3 &= a(a^2 + 2ab - a - b). \end{aligned}$$

The coefficients A_0, A_2, A_3 are strictly positive. Only A_1 can be negative. In particular $a \leq 2$ or $b \leq 2$ forces $F_{a,b}(h) > 0$ for all $h \geq 1$.

This gives a complete test for any fixed core shape, at arbitrary rank. If $A_1 \geq 0$, failure is impossible. Otherwise the unique positive critical point is

$$h_* = \frac{-A_2 + \sqrt{A_2^2 - 3A_3A_1}}{3A_3}.$$

Since $F''(h) = 6A_3h + 2A_2 > 0$ on $h \geq 0$, the integer minimum for $h \geq 1$ is attained at

$$\max\{1, \lfloor h_* \rfloor\}, \quad \max\{1, \lceil h_* \rceil\}.$$

Some exterior populations and a common core field fail the last gap exactly when at least one of the two corresponding cubic values is negative.

For the sharp finite-rank certificate we use an integer-only tail bound. Set

$$H_0 = \max \left\{ 1, \left\lceil \frac{-A_1}{A_2} \right\rceil \right\}. \quad (81)$$

For $h \geq H_0$, the sum $A_2h^2 + A_1h = h(A_2h + A_1)$ is nonnegative, so $F(h) > 0$. It therefore suffices to evaluate $1 \leq h \leq H_0$, with the boundary included redundantly.

The following complete integer procedure covers every rank from 4 through 38:

$$\begin{aligned} r &= 4, \dots, 38; & a &= 2, \dots, r-2; & b &= r-a; \\ h &= 1, \dots, H_0(a, b); & & & & \text{evaluate } F_{a,b}(h). \end{aligned}$$

It contains 630 core pairs and exactly 2237 integer evaluations; the largest cutoff is 10. Every value at rank at most 37 is positive. At rank 38, the only negative values are

$$F_{7,31}(3) = -3056, \quad F_{8,30}(3) = -4392. \quad (82)$$

No zeros occur. All omitted $a = 1$ or $b = 1$ cases have $A_1 \geq 0$, and (81) proves positivity for all omitted h . The integer verifier and complete evaluation list accompany this report. For ease of inspection, Table 5 records the minimum evaluated value at each rank. The tail argument, not a numerical extrapolation, makes the finite calculation exhaustive.

Table 5: Minimum of the cubic over the complete finite certificate at each rank.

r	Minimum	r	Minimum	r	Minimum
4	96	16	7680	28	30936
5	200	17	9248	29	31248
6	360	18	11016	30	30921
7	588	19	12996	31	30312
8	896	20	15200	32	29184
9	1296	21	17640	33	26976
10	1800	22	20328	34	24304
11	2420	23	23276	35	21088
12	3168	24	26337	36	16281
13	4056	25	28000	37	8424
14	5096	26	29200	38	-4392
15	6300	27	30132		

19.5 The two rank 38 shapes and the smaller witness

For $(a, b, h) = (8, 30, 3)$ one has

$$\alpha_\infty = -\frac{61}{4}, \quad \alpha_m = \frac{48379 - 61m}{4(m-6)}.$$

Thus $\alpha_m < 0$ exactly when $m \geq 794$. For $(7, 31, 3)$, the limiting coefficient is $-764/63$, and (79) gives $m_{\min} = 923$. These are the complete possibilities at rank 38 by (82).

For the first graph at its minimal population, put

$$D = \binom{32}{30} \binom{794}{7}, \quad q(w) = -22w^2 + 6437219280w + 11385894392295.$$

The exact endpoint factorization is

$$\Delta_{37} = \frac{5}{6304} D^2 w^{58} q(w). \quad (83)$$

Direct integer substitution gives

$$q(292602645) = 725475345 > 0, \quad q(292602646) = -5711821777 < 0.$$

The unique positive root proves the stated first failing integer. This graph has 864 vertices and 7552 edges. The other rank-38 shape has at least $38 + 33 + 923 = 994$ vertices, establishing the rank-38 vertex minimum in Theorem 19.1.

For completeness, at that second minimal population its primitive endpoint quadratic is

$$-2w^2 + 3479879022w + 8332557477297,$$

whose first failing positive integer is 1739941906. The preceding and failing values are 1715636157 and -1764252443 .

19.6 Verification and attribution

The top-three formula and the earlier rank-38 construction appeared in the companion [16]. [*Merge note:* Source 73: Lemma 18.3, Proposition 18.4 (whose α, β, γ are the α_m , $2bRB$, R^2C above) and Proposition 18.5.] The additional conclusions here are the exact finite-field criterion, exhaustive rank-38 endpoint classification, and minimal populations within that rank. The smaller-shore input is established Lorentzian theory [1, 7].

The standard-library script `verify_boundary.py` [*Merge note:* shipped as `code/55-rank38-endpoint-verify_boundary.py` and `code/55-rank38-endpoint-verify_symbolic.py`, outputs `data/55-rank38-endpoint-*.json`] regenerates all 2237 exact cubic values, verifies both rank-38 population thresholds, and checks the endpoint formula against the complete support sum on 160 deterministic rational-activity samples. It also checks the full support coefficients of both new minimal-population witnesses; in each, the last inequality is the only failing gap. The separate script `verify_symbolic.py` expands (78) and (80) exactly over $\mathbb{Q}(a, b, h)$. The complete certificate, scripts, and outputs are included. The general identities and the proven finite tail bound establish the result; sample regressions are additional checks.

The exact first-failure question for arbitrary individual one-shore activities is broader. Earlier Newton gaps in the common-core-field family are also outside this report's classification. [*Merge note:* The earlier gaps are settled by source 47 (Section 20).] Neither is settled by the last-gap criterion proved here.

20 The sharp rank-38 ULC boundary for full Hall graphs, one common right-core activity (source 47)

Abstract of source 47, as delivered. For full Hall graphs with a common positive activity on the right core and unit activities elsewhere, the first possible failure of matching-rank-normalized ultra-log-concavity occurs at rank 38. We close the earlier-gap question left open by the endpoint classification. Every intermediate Newton gap through rank 38

has positive coefficients in the common field, for every eligible pair of exterior populations. The proof expands each gap in two binomial population bases and checks a finite array of 22,452,529 positive integer coefficients. This is an exact all-population certificate, not a bounded population or activity scan. Together with the first-gap inequality and the previously established endpoint theorem, it gives the complete failure classification at rank 38.

Scope. The field is common across one specified core block. The result does not determine the first failure for arbitrary individual one-shore activities or for arbitrary bipartite graphs. General finite-activity rank-six one-shore ultra-log-concavity remains open. [*Merge note:* See the note in the scope paragraph of source 55 (Section 19).] This is a computer-assisted mathematical proof with reproducible integer certificates, not formal kernel verification or an external peer-review claim.

20.1 The family and the conclusion

Let $H_{a,b;n,m}$ have shores $A \sqcup L$ and $B \sqcup R$, with block sizes a, n, b, m , respectively, and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Here $a, b \geq 1$ and $n, m \geq 0$ are integers. Every activity is one except those of B , which all equal $w > 0$. The support polynomial is

$$P(t) = \sum_{k=0}^d p_k(w) t^k.$$

Each pair of equally sized endpoint sets admitting a perfect matching is counted once, regardless of its number of matchings. Its degree is

$$d = \min\{a + b, a + n, b + m\}.$$

For $1 \leq k < d$, put

$$\Delta_k = k(d - k)p_k^2 - (k + 1)(d - k + 1)p_{k-1}p_{k+1}.$$

The sequence is ULC_d precisely when every $\Delta_k \geq 0$, since all coefficients up to the matching rank are positive.

Theorem 20.1 (Sharp boundary for all gaps). *Every graph in this common-core-field family of matching rank at most 37 is ULC_d for every $w > 0$ and all exterior populations.*

At matching rank 38, every gap except possibly the last is strictly positive whenever $a, b \geq 3$ and $n \geq b, m \geq a$. A failure at some finite m, w is possible only for

$$(a, b, n - b + 1) = (7, 31, 3) \quad \text{or} \quad (8, 30, 3).$$

The least corresponding m are 923 and 794. The smallest graph at rank 38 in this family admitting any failed gap therefore has 864 vertices, namely $H_{8,30;32,794}$. Its first failing positive integer core activity is 292602646.

The new ingredient is the following stronger statement about intermediate gaps.

Theorem 20.2 (Positive common-field coefficients). *Suppose $a, b \geq 3$, $r = a + b \leq 38$, $n \geq b$, and $m \geq a$. For every $2 \leq k \leq r - 2$, all coefficients of*

$$w^{-\max(0, 2k-2a)} \Delta_k$$

from degree zero through degree $\min(2b, 2k) - \max(0, 2k - 2a)$ are strictly positive. In particular $\Delta_k > 0$ for every $w > 0$.

At larger rank the last gap does not control the earlier ones: the separate rank-49 example $H_{14,35;38,624}$ has its last gap positive for every $w > 0$ but a failing penultimate gap at $w = 40000000$ [18]. The rank restriction in Theorem 20.2 is therefore essential to the present certificate.

20.2 Counting and the finite population expansion

Write $\binom{x}{y} = 0$ when $y < 0$ or $y > x$, for nonnegative integer x . For $j, q \geq 0$, define

$$T_{j,q}(n) = \sum_{\ell=0}^j \binom{a}{q+\ell} \binom{n}{j-\ell}; \quad (84)$$

set $T_{j,q} = 0$ when either subscript is negative. A support selecting j vertices of B and q of R must select at least q vertices of A . If it selects $q + \ell$ of A , it selects $j - \ell$ of L . The complete core makes this necessary condition sufficient. Hence

$$p_k(w) = \sum_{j=0}^b \binom{b}{j} \binom{m}{k-j} T_{j,k-j}(n) w^j. \quad (85)$$

This is endpoint-support counting throughout.

Fix $2 \leq k \leq r - 2$. For a coefficient of w^s , write $L = 2k - s$. Its possible range is

$$\max(0, 2k - 2a) \leq s \leq \min(2b, 2k). \quad (86)$$

For $0 \leq p \leq \lfloor L/2 \rfloor$, define

$$\begin{aligned} G_{k,s,p}(n) &= \binom{L-p}{p} \sum_{j=\max(0, s-b)}^{\min(b, s)} \binom{b}{j} \binom{b}{s-j} (U_j - V_j), \\ U_j &= k(r-k) \binom{L-2p}{k-j-p} T_{j,k-j}(n) T_{s-j, k-s+j}(n), \\ V_j &= (k+1)(r-k+1) \binom{L-2p}{k-1-j-p} T_{j, k-1-j}(n) T_{s-j, k+1-s+j}(n). \end{aligned} \quad (87)$$

Lemma 20.3 (Exact finite population transfer). *For every nonnegative integer m ,*

$$[w^s] \Delta_k = \sum_{p=0}^{\lfloor L/2 \rfloor} \binom{m}{L-p} G_{k,s,p}(n). \quad (88)$$

Proof. Two subsets of an m -element set, of sizes q_1, q_2 and intersection size p , can be specified by their union, their intersection, and the first disjoint remainder. Thus

$$\binom{m}{q_1} \binom{m}{q_2} = \sum_p \binom{m}{q_1 + q_2 - p} \binom{q_1 + q_2 - p}{p} \binom{q_1 + q_2 - 2p}{q_1 - p}.$$

Both products in the coefficient of w^s have $q_1 + q_2 = L$. Applying this identity to (85) gives exactly (87)–(88). $\square$

This transfer keeps all finite-population terms, including intersections. The diffuse limit alone would retain only $p = 0$ and would not justify the finite- m conclusion.

20.3 An exhaustive finite integer certificate

The polynomial $T_{j,q}(n)$ has degree at most j , so $G_{k,s,p}(n)$ has degree at most s . Its Newton finite-difference expansion about $n = b$ is

$$G_{k,s,p}(b+h) = \sum_{e=0}^s c_{a,b,k,s,p,e} \binom{h}{e}, \quad (89)$$

$$c_{a,b,k,s,p,e} = \sum_{i=0}^e (-1)^{e-i} \binom{e}{i} G_{k,s,p}(b+i). \quad (90)$$

These are integer identities. Combining them with Lemma 20.3 gives

$$\Delta_k = \sum_s \sum_{p=0}^{\lfloor (2k-s)/2 \rfloor} \sum_{e=0}^s c_{a,b,k,s,p,e} w^s \binom{m}{2k-s-p} \binom{n-b}{e}. \quad (91)$$

Every basis factor is nonnegative at the populations and activities in question.

Proposition 20.4 (Verified coefficient array). *Every integer $c_{a,b,k,s,p,e}$ is strictly positive over the finite range*

$$\begin{aligned} 6 \leq r \leq 38, \quad 3 \leq a \leq r-3, \quad b = r-a, \quad 2 \leq k \leq r-2, \\ \max(0, 2k-2a) \leq s \leq \min(2b, 2k), \quad 0 \leq p \leq \lfloor (2k-s)/2 \rfloor, \quad 0 \leq e \leq s. \end{aligned}$$

There are 561 core pairs, 13,651 intermediate gaps, 1,583,703 polynomials G , and 22,452,529 checked coefficients.

Exact verification. The accompanying standard-library verifier evaluates (87) at $n = b, \dots, b+s$, forms successive difference rows, and checks their first entries against zero. Every arithmetic operation is an integer addition, multiplication, binomial evaluation, or comparison. It visits precisely the displayed finite range; there are no numerical tolerances, sampled fields, or omitted population tails. Per-gap coefficient counts and SHA-256 records are included, allowing all entries to be regenerated and compared. A separate implementation constructs the support counts by adding exterior vertices using Pascal recurrence and computes the coefficients by the explicit inverse Pascal matrix. It agrees with every per-gap hash and count. The smallest checked coefficient is 9; there are no zero entries. $\square$

Proof of Theorem 20.2. Equation (91) and Proposition 20.4 give coefficientwise nonnegativity in w for all $n \geq b$ and $m \geq a$. For each s in (86), take $p = \lfloor L/2 \rfloor$ and $e = 0$. Then

$$L-p = \lceil L/2 \rceil \leq a \leq m,$$

so the corresponding population binomial is positive. Its coefficient is strictly positive by Proposition 20.4. Thus every stated common-field coefficient is positive. $\square$

20.4 Closing the first and last gaps

20.4.1 The first gap

[Merge note: Source 47 proves here, by the Motzkin–Straus route, that $\Delta_1 \geq 0$ for every positively weighted bipartite graph; this is *preorder-root-polytopes*’s Theorem 180.2 (Part XI, `hb:thm:firstidentity`) and the inequality printed with this route in Sections 11 and 14. The argument is not reprinted; in it $M_2 \geq p_2$ is the weighted number of two-edge matchings and the Motzkin–Straus theorem gives $M_2 \leq \frac{d-1}{2d} p_1^2$. Source 47’s strictness step is kept verbatim:]

In the genuine full Hall cases with $a, b \geq 3$, two vertices from each core form a $K_{2,2}$. Its support contributes two positive matchings to M_2 but only once to p_2 . Hence $M_2 > p_2$ and $\Delta_1 \geq 2d(M_2 - p_2) > 0$. [*Merge note:* In *preorder-root-polytopes*'s identity this is the positive term B_4 .]

20.4.2 The endpoint classification

We use the exact endpoint theorem of [17]. [*Merge note:* That theorem is source 55, printed in Section 19: the quadratic (72), Lemma 19.2, Theorem 19.3, the cubic (80) and the two negative values (82). Source 47 recalls its formulas without proof; the recall is kept below, in source 47's notation, which agrees with source 55's.] Its essential formulas are recalled to make the scope and resulting boundary explicit. Assume $n \geq b, m \geq a$, and put

$$h = n - b + 1, \quad x = \frac{b}{h}, \quad y = \frac{b(b-1)}{h(h+1)}, \quad R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2},$$

$$J = \binom{a}{2} + ax + y, \quad D = \binom{n}{b} \binom{m}{a-1}.$$

Then

$$\Delta_{r-1} = D^2 w^{2b-2} (\alpha_m w^2 + 2bRBw + R^2 C),$$

where

$$\alpha_m = (r-1)(a+x)^2 - 2rR\rho J, \quad B = \frac{b[b(r-1) - ah^2]}{h^2(h+1)},$$

$$C = \frac{b^2\{b^2(r-1) + h[r(2b-1) - b^2]\}}{h^2(h+1)} > 0.$$

The endpoint theorem proves that this quadratic is positive for every $w > 0$ if $\alpha_m \geq 0$. If $\alpha_m < 0$, it has one positive root and is negative precisely above it. Thus finite fields introduce no hidden negative window.

Writing

$$K = \frac{2r(a-1)}{a} J, \quad \alpha_\infty = (r-1)(a+x)^2 - K$$

gives $\alpha_m = \alpha_\infty + K/(m-a+2)$. Some finite m, w fail exactly when $\alpha_\infty < 0$, and then

$$m_{\min} = a - 1 + \left\lfloor \frac{K}{-\alpha_\infty} \right\rfloor.$$

The sign of α_∞ is the sign of the cubic

$$\begin{aligned} F_{a,b}(h) = & a(a^2 + 2ab - a - b)h^3 + a(a^2 + 2ab - a + 2b^2 - b)h^2 \\ & - b(r-1)(ab - 2a - 2b)h + ab^2(r-1). \end{aligned}$$

Only its linear coefficient can be negative. The endpoint certificate uses the finite tail cutoff $\max(1, \lceil -A_1/A_2 \rceil)$ for these cubic coefficients. Its 2237 exact evaluations show positivity through rank 37, and exactly

$$F_{7,31}(3) = -3056, \quad F_{8,30}(3) = -4392$$

as the rank-38 negative cases. The tail bound makes this exhaustive in h .

20.4.3 The remaining cover and population cases

If $n < b$ or $m < a$, the actual rank is the smaller shore size. The standard weighted transversal-matroid specialization then gives actual-rank ULC [1, 7]. [*Merge note: preorder-root-polytopes's* Theorem 70.1; Lemma 3.1 here.] Explicitly, with a private element for each left vertex, the extended transversal basis polynomial specializes to

$$\sum_k p_k s^{|X|-k} t^k$$

by assigning a private variable s/u_x and a right variable tv_y , then multiplying by $\prod_x u_x$. Matroid basis polynomials are Lorentzian, so this is $\text{ULC}_{|X|}$; transpose for the smaller shore.

When $n \geq b, m \geq a$ and $\min(a, b) \leq 2$, the previously proved minimum-cover theorem [12] gives actual-rank ULC for arbitrary positive activities. [*Merge note: Theorem 2.1; also source 75's Theorem 16.1.*] All remaining cases are precisely those in Proposition 20.4. The first gap and the endpoint classification therefore complete the proof of Theorem 20.1.

20.5 The sharp witness and reproducibility

For $H_{8,30;32,794}$, let $D = \binom{32}{30} \binom{794}{7}$. The endpoint factorization is

$$\Delta_{37} = \frac{5}{6304} D^2 w^{58} (-22w^2 + 6437219280w + 11385894392295).$$

The quadratic takes the values 725475345 and -5711821777 at 292602645 and 292602646, respectively. The graph has 864 vertices and 7552 edges. The other rank-38 shape requires at least 994 vertices. [*Merge note: These are source 55's numbers, (83); restated by source 47.*] The new intermediate-gap theorem shows that these endpoint possibilities also exhaust every-gap failure at rank 38.

The new certificate and the endpoint certificate have different purposes. The former covers every intermediate gap and both unbounded exterior populations through binomial-basis identities. The latter covers the endpoint through a cubic tail bound and an exact quadratic criterion. The source package contains the generators, compact per-gap coefficient records, regression checks of the population identity, and instructions for replay. [*Merge note: Shipped as `code/47-rank38-ulc-*.py` and `data/47-rank38-ulc-*.json` (33 per-rank certificate files, ranks 6–38); the replay instructions were in the delivered README, which is not shipped (this report's README gives them).*] Numerical scans used during exploration are not part of the proof.

The new conclusion is restricted to a common core field on full Hall graphs. The minimum rank for failure with arbitrary individual one-shore activities remains between 6 and 38. [*Merge note: The lower end six rests on source 74's Corollary 2.2 (Part I); the upper end on sources 73 and 55.*] The rank-49 separation [18] prevents extrapolating endpoint control to all higher ranks.

21 The last Hall gap does not control ultra-log-concavity, one common right-core activity (source 51)

Abstract of source 51, as delivered. We give a full Hall graph of rank 49 with a common activity on one right-core block for which the last rank-normalized Newton inequality holds strictly at every positive activity, but the preceding inequality fails. The example has 711 vertices. Four explicit coefficient formulas give an exact quartic certificate. This separates the

last-gap criterion from full ultra-log-concavity without changing the previously proved sharp rank-38 endpoint threshold.

Let $H_{a,b;n,m}$ have left blocks A, L of sizes a, n , right blocks B, R of sizes b, m , and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Give every vertex activity one except the vertices of B , each of which has activity $w > 0$. The coefficient p_k of its support polynomial counts each pair of k -element endpoint sets admitting a perfect matching once, with the product of its vertex activities. For matching rank r , write

$$\Delta_k = k(r - k)p_k^2 - (k + 1)(r - k + 1)p_{k-1}p_{k+1}.$$

Rank-normalized ultra-log-concavity requires $\Delta_k \geq 0$ at every $1 \leq k < r$.

Theorem 21.1. *The graph $H_{14,35;38,624}$ has matching rank 49, 711 vertices and 10556 edges. Its last gap Δ_{48} is strictly positive for every $w > 0$. At $w = 40000000$, its preceding gap Δ_{47} is negative. The first positive integer w for which $\Delta_{47} < 0$ is 38189138.*

Proof. The core is a cover of size 49, attained by disjoint matchings $A \rightarrow R$ and $L \rightarrow B$. This proves the rank. The vertex and edge counts are

$$14 + 35 + 38 + 624 = 711, \quad 14 \cdot 35 + 38 \cdot 35 + 14 \cdot 624 = 10556.$$

For general full Hall parameters, a size- k support selecting i vertices of A and j of B is feasible exactly when $i + j \geq k$. Hence, with out-of-range binomial coefficients zero,

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j. \quad (92)$$

Set $D = \binom{38}{35} \binom{624}{12} > 0$. The four coefficients needed here are

$$\begin{aligned} \frac{p_{46}}{Dw^{32}} &= \frac{8801172765}{2} + 32499495w + \frac{136955}{2}w^2 + \frac{27846}{613}w^3, \\ \frac{p_{47}}{Dw^{33}} &= 72736965 + 299880w + 273w^2, \\ \frac{p_{48}}{Dw^{34}} &= \frac{1258425}{2} + 1071w, \\ \frac{p_{49}}{Dw^{35}} &= \frac{14382}{7}. \end{aligned} \quad (93)$$

These follow directly from (92): the core-deficit pairs $(14 - i, 35 - j)$ have total at most 3, 2, 1, 0, respectively. Thus the formulas require only ten, six, three and one terms.

Substituting (93) gives

$$\Delta_{47} = \frac{882}{613} D^2 w^{66} q(w), \quad (94)$$

$$\Delta_{48} = 89964 D^2 w^{68} (w^2 + 47940w + 48443370), \quad (95)$$

where

$$\begin{aligned} q(w) &= -13w^4 + 496441140w^3 + 673931488920w^2 \\ &\quad + 331783888454280w + 68528424519495225. \end{aligned}$$

Every factor in (95) is positive for $w > 0$. In contrast,

$$q(40000000) = -1506688736346303933404280504775 < 0,$$

proving the finite counterexample. The quartic has one sign change in descending coefficient order; Descartes' rule, together with $q(0) > 0$ and $q(w) \rightarrow -\infty$, gives exactly one positive root. Finally,

$$\begin{aligned} q(38189137) &= 358201675329747057329792 > 0, \\ q(38189138) &= -365864633482267706996743 < 0. \end{aligned}$$

This proves the first-integer claim. □

Scope and reproducibility. The sharp rank-38 theorem in [17] concerns the last gap in this common-core-activity family. It remains valid. The present example shows that its criterion does not characterize the full ULC sequence. It does not decide whether an earlier gap can fail below rank 38, or the first failure for arbitrary individual one-shore activities. [*Merge note:* The first is decided by source 47 (Theorem 20.1): no gap fails below rank 38 in this family. The second stays open; it lies in [6, 38].]

The accompanying standard-library script [*Merge note:* shipped as `code/51-last-gap-verify_counterexample.py`, output `data/51-last-gap-verification.json`] independently sums (92), verifies all four coefficient identities and both polynomial factorizations exactly, and checks the stated integer signs. At $w = 40000000$, it also finds that $k = 47$ is the only negative gap. No numerical root approximation is used. These are ordinary proofs with reproducible exact arithmetic, not formal kernel verification; global priority is not claimed.

Part III

Matching number six with one weighted shore at every finite activity

Contents of Part III. Part III prints seven sources on graphs of matching number six with *one weighted shore*: all left activities one, right activities arbitrary positive (Research question 85 of *preorder-root-polytopes*). By source 74 one may assume a genuine $3 + 3$ minimum cover. Source 60 (Section 22) lifts rank-two exterior column blocks of matrices and gives exact conditioning reductions; source 10 (Section 23) proves a Lorentzian three-vertex marginal and all five inequalities for star cores; source 23 (Section 24) proves rooted sector theorems and gives a 25-vertex graph whose all-right-core conditional cubic fails its unshifted Newton inequalities; source 13 (Section 25) proves the *last* inequality and source 05 (Section 26) the *second*, for every graph and every activity; source 69 (Section 27) proves all five for every core with complete exterior neighbourhoods, and source 63 (Section 28) for a complete core with nested exterior neighbourhoods, with a barrier and a conditional theorem. Sources 60 and 10 also allow weights on both shores in their main theorems; their section titles say so.

Status at matching number six, one weighted shore. Gap one holds for every positively weighted graph (*preorder-root-polytopes*, Theorem 180.2); gap two holds by source 05 and gap five by source 13, for every graph and every finite activity. Gaps three and four are open in general; they hold for star cores (source 10), for every core with complete exterior neighbourhoods (source 69) and for the nested complete-core family (source 63). Several sources, written before others, call more gaps open; their sentences are kept with notes.

Same results in several sources. Source 60's 9×9 rank-three example is *preorder-root-polytopes*'s Theorem 192.1 (Part XII, `mr:thm:matrix`); it is replaced by a pointer. Source 05's Lemma 26.3 is source 60's Theorem 22.1 at $q = 0$ (a second route, printed), and source 05's one-vertex compression is the case $h = 1$ of source 60's slack reductions. Source 05's quintic example is source 23's 25-vertex graph. Source 13's incidence lemma and sector identity are source 21's (Part IV), and the lemma is the case $b = 3$ of source 08s's two-plane lemma; it is printed here with source 13's proof. The first-gap arguments of sources 69 and 63 are *preorder-root-polytopes*'s Theorem 180.2 and become pointers. Gaps two and five of sources 69 and 63 also follow from sources 05 and 13, and *preorder-root-polytopes*'s unit $H_{3,n}$ result (Corollary 177.2) is a special case of source 69's theorem.

Non-claims of Part III. No source proves ULC_6 at every finite one-shore activity; source 60's conditioned and eventual conclusions do not assert ULC of the original polynomial at every finite activity; source 23's obstruction does not refute rank-six ULC or an eventual-scaling theorem; source 13's local lemma is graph-generic only; source 05's quartic route is a research direction and its quintic route is refuted, not scalar ULC_6 ; source 63's barrier rules out the full-marginal approach, not ULC_6 ; source 69's barrier coefficient obstructs a proof method, not ULC.

22 Rank-two matrix blocks and exterior conditioning, both shores weighted (source 60)

Abstract of source 60, as delivered. An explicit matroid lift proves ultra-log-concavity of order $q + 2$ for weighted regular-minor support polynomials of matrices whose columns consist of q ordinary columns and a block of rank two. The normalization is the actual matrix rank when that rank is $q + 2$. Exact conditioning reductions with Hall slack at most two then show that, in a genuine $a + q$ bipartite cover, forcing any matchable $a - 2$ exterior-right vertices leaves an actual-degree- $(q + 2)$ ULC polynomial after the forced factor is removed. Scaling those activities sufficiently far also makes the full original polynomial strictly rank-ULC, with an explicit field-dependent threshold. The compressed object may be a matrix rather than a graph; a positive rank-three matrix example shows why this distinction matters. The general finite-activity rank-six one-shore problem remains open.

22.1 A rank two exterior column block

Let M be a matrix over a field, with finite row set X and column set Y . For positive real activities u_i, v_j , define its regular-minor support polynomial by

$$P_M(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, \det M[I, J] \neq 0}} u_I v_J t^{|I|}, \quad u_I = \prod_{i \in I} u_i, \quad v_J = \prod_{j \in J} v_j.$$

The empty minor is one. Each row-column pair is counted once, and determinant magnitudes are not used. The degree is $r = \text{rank } M$. For $p(t) = \sum p_k t^k$, write

$$\Delta_k^{(D)}(p) = k(D - k)p_k^2 - (k + 1)(D - k + 1)p_{k-1}p_{k+1}.$$

The property ULC_D means these gaps are nonnegative and the coefficient sequence has no internal zeros. This is equivalently log-concavity after division by $\binom{D}{k}$.

Theorem 22.1. *Partition the columns as $Y = B \sqcup R$, where $|B| = q$ and $\text{rank } M[X, R] = 2$. For every positive right activity vector, the polynomial*

$$F_M(s, z) = \sum_{\substack{I, J: |I|=|J| \\ \det M[I, J] \neq 0}} z_I v_J s^{q+2-|I|} \tag{96}$$

is Lorentzian. Consequently P_M is ULC_{q+2} for all positive activities on both sides. This is actual-rank normalization if $\text{rank } M = q + 2$.

If the rank is smaller, the theorem asserts order $q + 2$, not the stronger normalization by that smaller rank. The activity parameters are positive real numbers, independently of the field containing the matrix entries.

Proof. Factor the exterior block and write

$$M = [C, DE], \quad D \text{ has two columns, } E \text{ has two rows and rank two.}$$

For the column E_j indexed by $j \in R$, put

$$Q = \sum_{\substack{i < j \text{ in } R \\ \det(E_i, E_j) \neq 0}} v_i v_j > 0, \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Construct a representable matroid in dimension $q + 2$ with three kinds of elements:

$$a_i = (C_i^\top, D_i^\top) \ (i \in X), \quad e_b \ (b \in B), \quad p_j = (0, J_0 E_j) \ (j \in R).$$

The private elements e_b are the first q coordinate vectors. Together with any independent plane pair p_i, p_j , they form a basis, so the matroid has rank $q + 2$.

In its basis polynomial substitute

$$a_i \mapsto z_i, \quad e_b \mapsto s/v_b, \quad p_j \mapsto sv_j/Q, \quad (97)$$

and multiply by $Q \prod_{b \in B} v_b$. We claim the result is (96).

Fix a selected original row set I and write the private set as $B \setminus J_B$. The basis-size equation forces the number of plane elements to be

$$\ell = 2 + |J_B| - |I|.$$

Only $\ell = 0, 1, 2$ are possible. Eliminate the private coordinates and consider the remaining determinant.

If $\ell = 2$, it factors as the ordinary minor $C[I, J_B]$ times the plane-pair determinant, up to sign. The total plane-pair activity is Q/Q^2 , which cancels the outside factor Q . This records selection of no exterior column.

If $\ell = 1$, the plane element p_j yields the nonzero-minor criterion for $M[I, J_B \cup \{j\}]$. The elementary identity behind this is

$$\det(d, J_0 e) = d^\top e.$$

Its activity v_j/Q cancels the outside factor Q , recording that one exterior column.

If $\ell = 0$, the augmented minor uses the two columns of D . An original minor with exterior columns i, j equals this augmented minor times $\det(E_i, E_j)$, up to sign. Thus either no exterior pair works or exactly all independent pairs work; their total activity is Q , as supplied by the outside factor. Larger exterior selections are dependent.

In all three cases the private activities supply v_{J_B} , and the degree in s is $q + 2 - |I|$. This proves the claimed polynomial identity without a determinant multiplicity. Matroid basis polynomials are Lorentzian, and nonnegative linear substitutions preserve the property [1]. Setting $z_i = u_i t$ proves the ULC conclusion. $\square$

[Merge note: Theorem 22.1 answers in part Research question 101 of *preorder-root-polytopes* (`mr:q:bimatroid`, “Which representable bimatroids satisfy actual-rank ULC despite the dense rank-three obstruction?”): every regular-minor matrix whose columns are q ordinary columns and a block of rank two, when its rank is $q + 2$. At $q = 0$ it is source 05’s Lemma 26.3, proved there by a different route.]

The construction generalizes the two-vertex-shore graph lift: there the exterior columns are supported on two rows. [Merge note: Source 74’s lift, Section 5.] Theorem 22.1 requires only a rank-two exterior column space, not a particular zero pattern. The normalization can be smaller than the ambient-shore normalization in the general bimatroid theorem [7].

22.1.1 The rank two hypothesis cannot simply be increased

[Merge note: Source 60 considers here the positive 9×9 matrix $M_{ij} = 1 + ij + (ij)^2$ of rank three, all of whose minors of orders one to three are positive, so that at unit activities $P_M(t) = 1 + 81t + 1296t^2 + 7056t^3$ and $\Delta_2^{(3)}(P_M) = -69984 < 0$. This is *preorder-root-polytopes*’s Theorem 192.1 (Part XII, `mr:thm:matrix`) at $m = n = 9$, with the same Cauchy–Binet and

Vandermonde proof; source 60 does not cite it. Its value -69984 is twice Part XII's defect $1296^2 - 3 \cdot 81 \cdot 7056 = -34992$, because $\Delta_2^{(3)} = 2 \cdot 1296^2 - 6 \cdot 81 \cdot 7056$ carries the factor $k(D-k) = 2$. The statement and proof are not reprinted; source 60's comments on the example follow.] The rank-two case is covered by Theorem 22.1 with $q = 0$, whereas rank one is trivial. Thus arbitrary regular-minor matrices can first fail at rank three. This elementary example is used as a scope check, not as a novelty claim. Its entry-support graph is complete bipartite and has matching rank nine; that graph's endpoint polynomial is a different polynomial.

22.2 Exact conditioning when Hall slack is at most two

For a weighted bipartite graph $G = (X, Y)$ and a right set F , let

$$P_G^{(F)}(t) = \sum_{\substack{(I,J) \text{ feasible} \\ F \subseteq J}} u_I v_J t^{|I|}.$$

This is the coefficient of $\lambda^{|F|}$ when the activities in F are multiplied by λ . Suppose F is matchable, $|F| = h \geq 1$, and let $N = N_G(F)$. The Hall slack here means $|N| - h$.

Proposition 22.2 (Slack zero and one). *If $|N| = h$, then*

$$P_G^{(F)} = v_F u_N t^h P_{G-N-F}.$$

If $|N| = h + 1$, put

$$E_h = \sum_{\substack{I \subseteq N, |I|=h \\ G[I,F] \text{ matchable}}} u_I > 0.$$

Delete N, F and add one left vertex of activity u_N/E_h . Its neighborhood is the union of the unforced-right neighborhoods of those $i \in N$ for which $N \setminus \{i\}$ matches F . Call the resulting graph H . Then

$$P_G^{(F)} = v_F E_h t^h P_H. \quad (98)$$

The remaining graph has matching rank $\nu(G) - h$ in either case.

Proof. In slack zero all of N is consumed by F . In slack one, supports selecting exactly h members of N likewise consume them all; summing their weights gives E_h , with the virtual vertex absent. A support selecting all $h + 1$ neighbors has the fixed product u_N . Its one available vertex can be matched outside F exactly through the stated union neighborhood. This is a Boolean existence test, so different successful choices do not multiply a support's weight. The virtual activity makes $E_h(u_N/E_h) = u_N$.

A matchable right set extends to a maximum matchable right set. Thus $P_G^{(F)}$ has degree $\nu(G)$; the exact identities prove the rank statements. $\square$

For example, conditioning on a right vertex of degree two merges its neighbors a, b into a union-neighborhood vertex of activity $u_a u_b / (u_a + u_b)$, with outside factor $v_F(u_a + u_b)t$.

Theorem 22.3 (Slack two kernel reduction). *If $|N| = h + 2$, the conditioned polynomial has an exact representation*

$$P_G^{(F)} = v_F E_h t^h P_{M'}, \quad (99)$$

where M' is an explicit regular-minor matrix with positive row activities and rank $\nu(G) - h$. In general M' need not represent a graph endpoint polynomial.

Proof. Represent G by its generic matrix M , putting an algebraically independent indeterminate on every edge and zero elsewhere. Work over the rational-function field. A square minor is nonzero exactly when its endpoint graph has a perfect matching, since distinct matchings give distinct determinant monomials.

Let $\Gamma = M[N, F]$, of rank h , and define E_h by summing u_I over its nonsingular h -row submatrices. The coefficient space

$$K = \ker(\Gamma^\top)$$

has dimension two. For every $i \in N$ such that $\Gamma[N \setminus \{i\}, F]$ has rank h , choose a nonzero vector k_i spanning the line

$$K \cap \{x_i = 0\}.$$

If this intersection has dimension two, omit the index i : the corresponding $h + 1$ rows cannot cover all the forced columns. Define the virtual row and its activity by

$$r'_i = k_i^\top M[N, Y \setminus F], \quad w_i = u_{N \setminus \{i\}} / E_h. \quad (100)$$

Retain the original rows outside N , their activities, and the columns outside F . Together with the virtual rows these form M' . Nonzero rescaling of a virtual row does not change any minor's zero/nonzero status.

For retained i, j , the kernel lines k_i, k_j are independent exactly when

$$\det \Gamma[N \setminus \{i, j\}, F] \neq 0. \quad (101)$$

Indeed, their intersection is the kernel supported on that h -row complement. It is zero exactly when the complement is nonsingular. A nonsingular h -row set exists, so the retained lines span K .

Partition supports by their selected neighbors in N . Exactly h selected neighbors contribute E_h times the unchanged outside support polynomial. Exactly $h + 1$ neighbors, $N \setminus \{i\}$, reduce after eliminating the forced columns to the one virtual row r'_i . Their activity is recovered because $E_h w_i = u_{N \setminus \{i\}}$.

When all $h + 2$ neighbors are selected, any independent kernel pair is a basis of K . Changing that basis multiplies a residual determinant by a nonzero scalar. All such pairs therefore have the same residual nonzero-minor profile, including the possibility that this profile is identically zero. Their total assigned activity is

$$\begin{aligned} \sum_{\substack{i < j \\ k_i, k_j \text{ independent}}} w_i w_j &= \frac{u_N}{E_h^2} \sum_{\substack{i < j \\ \det \Gamma[N \setminus \{i, j\}, F] \neq 0}} u_{N \setminus \{i, j\}} \\ &= \frac{u_N}{E_h}. \end{aligned} \quad (102)$$

Thus the prefactor E_h recovers u_N exactly once. Dependent kernel pairs and all triples of virtual rows contribute zero. This proves (99).

The independence in (102) is independence of the kernel vectors, not necessarily of their image rows. Independent kernel vectors can have dependent images in a rank-deficient residual matrix; then their common determinant profile is zero, and the identity remains valid.

Finally, projecting the original row space onto the F coordinates has rank h . Its kernel is spanned by the outside rows and the combinations from K . The retained virtual lines span K , so deleting the zero F coordinates leaves rank $\text{rank } M - h = \nu(G) - h$. $\square$

For a single degree-three forced vertex, with forced-column entries $\gamma_1, \gamma_2, \gamma_3$ at its neighbors, the virtual rows can be written as

$$\gamma_2 r_1 - \gamma_1 r_2, \quad \gamma_3 r_1 - \gamma_1 r_3, \quad \gamma_3 r_2 - \gamma_2 r_3.$$

Their respective activities are $u_1u_2/E_1, u_1u_3/E_1, u_2u_3/E_1$, with $E_1 = u_1 + u_2 + u_3$. Their pair-activity sum is $u_1u_2u_3/E_1$, giving a concrete instance of (102).

22.3 Forcing exterior vertices in a genuine cover

Write a bipartite graph as $X = A \sqcup L, Y = B \sqcup R$, with $A \cup B$ a minimum vertex cover, $|A| = a$ and $|B| = q$. Call the displayed cover genuine when $\nu(G) = a + q$. There are no L – R edges. Any maximum matching then consists of an a -matching $A \rightarrow R$ and a disjoint q -matching $L \rightarrow B$: an edge inside the cover would consume two cover vertices and prevent attaining its size.

Theorem 22.4. *Suppose $a \geq 3$ and the displayed $a + q$ cover is genuine. Let $F \subseteq R$ be any matchable set of $h = a - 2$ vertices. For arbitrary positive activities on both shores,*

$$P_G^{(F)}(t) = C_F t^h Q_F(t), \quad (103)$$

where $C_F > 0$ and Q_F has actual degree $q + 2$ and is ULC_{q+2} . Consequently the entire conditioned sector is ULC_{a+q} .

Proof. The neighborhood $N(F)$ is contained in A and has size $h, h + 1$, or $h + 2$. Slack zero deletes h core-left vertices and leaves a genuine $2 + q$ cover. Slack one leaves one unchanged core-left vertex and one virtual union vertex, again giving a genuine $2 + q$ cover. A generic matrix of either graph has a rank-two exterior-column block and total rank $q + 2$, so Theorem 22.1 applies.

In slack two, $N(F) = A$. The generic exterior block originally has rank a , and its h forced columns are independent. Eliminating those columns reduces its rank to $a - h = 2$. The total compressed rank is $a + q - h = q + 2$. Thus Theorems 22.3 and 22.1 give (103), with the compressed object interpreted as a regular-minor matrix.

Multiplication by t^h weakens the required Newton ratios. At coefficient index j of a degree- d polynomial, $d = q + 2$, the required ratio changes from

$$\frac{(j+1)(d-j+1)}{j(d-j)} \quad \text{to} \quad \frac{(j+h+1)(d-j+1)}{(j+h)(d-j)}.$$

The latter is smaller. This proves ULC at order $d + h = a + q$ for the shifted sector. $\square$

[Merge note: Compare source 23's Theorem 24.1, which forces all of B and exterior-left vertices instead.] Further forcing is a coefficientwise limit at the same fixed normalization. In particular, for a genuine $3 + 3$ cover, conditioning on any exterior-right vertex leaves t times a degree-five ULC polynomial, for arbitrary fields on both shores. Every further conditioned sector containing such a vertex is safe. Among three-right forced sets, only the all- B set escapes this reduction. This does not prove the original polynomial is ULC at arbitrary finite activities.

22.4 Eventual strictness and an explicit threshold

Theorem 22.5. *Under the assumptions of Theorem 22.4, multiply just the activities in F by a common $\lambda > 0$, keeping all other activities fixed. The full, unconditioned polynomial is strictly ULC_{a+q} for all sufficiently large λ . The threshold depends on the fixed graph and activities.*

Proof. Put $h = |F| \geq 1$, $d = q + 2$, and $r = h + d$. Write $p_k(\lambda)$ for the full coefficient. For $0 \leq k \leq h$,

$$p_k(\lambda) = b_k \lambda^k + O(\lambda^{k-1}),$$

where (b_k) is the coefficient sequence of the graph restricted to right shore F . It has rank h and is ULC_h . One can see this directly from the usual basis matroid on its original row vectors and private coordinate elements in dimension h ; positive specialization gives its degree- h homogenization. For $0 \leq j \leq d$,

$$p_{h+j}(\lambda) = c_j \lambda^h + O(\lambda^{h-1}),$$

where (c_j) is a positive scalar multiple of the ULC_d sequence in Theorem 22.4. All b_k, c_j are positive, and $b_h = c_0$.

For $k < h$, the coefficient of λ^{2k} in the order- r gap is at least

$$\frac{(k+1)d}{h-k} b_{k-1} b_{k+1} > 0,$$

by the stronger ULC_h inequality. At $k = h$, the square has order λ^{2h} while the adjacent product has at most order λ^{2h-1} ; the leading coefficient is $hdc_0^2 > 0$. Above it, at $k = h + j$ with $1 \leq j < d$, ULC_d gives a leading coefficient at least

$$\frac{h(d-j+1)}{j} c_{j-1} c_{j+1} > 0$$

at order λ^{2h} . Every gap polynomial therefore has a positive leading coefficient. There are finitely many gaps, proving eventual strictness. $\square$

For an explicit coefficient bound, write each gap as

$$G_k(\lambda) = L_k \lambda^{m_k} + \sum_{j < m_k} g_{kj} \lambda^j, \quad L_k > 0,$$

and set

$$\Lambda = 1 + \max_k \frac{\sum_{j < m_k} \max\{-g_{kj}, 0\}}{L_k}. \quad (104)$$

For every $\lambda \geq \Lambda$, the leading term dominates all negative terms, since $\lambda \geq 1$ and

$$G_k(\lambda) \geq \lambda^{m_k-1} \left(L_k \lambda - \sum_{j < m_k} \max\{-g_{kj}, 0\} \right) > 0.$$

This is a field-dependent bound, not a uniform numerical threshold. With exact numerical activities, it is directly computable from the finitely many coefficient polynomials.

22.5 Verification and scope

Run `python3 checks/run_all.py`. [*Merge note:* Shipped as `code/60-rank-two-blocks-run_all.py` and `code/60-rank-two-blocks-check_*.py`, outputs `data/60-rank-two-blocks-*`; rerun on a copy of the delivered layout (README).] The package independently checks 32 rank-two-block basis identities, 32 degree-three contractions, 50 general kernel compressions with $h = 1, 2, 3, 4$, and 200 graph conditioning identities. The kernel cases include omitted and parallel kernel lines. Further exact checks verify all 8,433 positive minors in the rank-three scope example and the leading coefficients and explicit threshold on 36 genuine covers with positive integer activities. The universal proofs are the determinant-sector, kernel, and leading-coefficient arguments above; finite examples are regression checks.

The regular-minor matrix after slack-two compression must not be replaced by its entry-support graph. Its linear dependencies are part of the construction. Likewise, order $q + 2$

is actual-rank ULC only when the total matrix rank is $q + 2$. The conditioned and eventual conclusions do not assert ULC at every finite choice of positive activities on the original graph. In particular, the general rank-six one-shore question, including its all-core-right cubic limit, remains unresolved here. [*Merge note:* Within the batch, at matching number six with one weighted shore, gap one holds by *preorder-root-polytopes*'s Theorem 180.2, gap two by source 05 (Section 26) and gap five by source 13 (Section 25), at every activity; gaps three and four are open in general. The all-core-right cubic limit is source 21's theorem (Section 30).]

The note uses the matroid Lorentzian theorem and the regular-minor viewpoint of bimatroids [1, 7]; it makes no exhaustive literature-priority claim. The arguments and exact verifiers received independent mathematical review. They are not formal theorem-prover verification or external peer review.

23 Lorentzian marginals on three vertices and star cores, both shores weighted (source 10)

Abstract of source 10, as delivered. For a bipartite graph with three vertices on one shore, fix arbitrary positive activities on the other shore. The degree-three homogenization of its selected-vertex marginal is Lorentzian. The proof combines rank-three matroid Rayleigh inequalities with an exact positive-coefficient certificate for one Hessian determinant. An endpoint-set vertex-sum identity then proves weighted rank-normalized ultra log-concavity for every genuine $3 + q$ cover whose core is a star centered on the three-vertex shore. In particular, both star orientations of a $3 + 3$ cover satisfy all five rank-six inequalities for arbitrary positive activities on both shores. No general rank-three-block recombination theorem is asserted.

23.1 The selected-vertex marginal

Let $G = (X, Y)$ be bipartite, with $X = \{1, 2, 3\}$, and give each $y \in Y$ an activity $v_y > 0$. For $I \subseteq X$, define

$$c_I = \sum_{\substack{J \subseteq Y, |J|=|I| \\ G[I, J] \text{ perfectly matchable}}} \prod_{y \in J} v_y, \quad c_\emptyset = 1.$$

Each pair of endpoint sets is counted once, regardless of the number of perfect matchings realizing it. Write c_i, c_{ij}, c_{123} for the corresponding coefficients.

Theorem 23.1. *The homogeneous cubic*

$$H(s, z_1, z_2, z_3) = s^3 + \sum_i c_i z_i s^2 + \sum_{i < j} c_{ij} z_i z_j s + c_{123} z_1 z_2 z_3 \quad (105)$$

is Lorentzian.

For a homogeneous polynomial with nonnegative coefficients and M -convex support, the Lorentzian criterion is that each quadratic coordinate derivative has a Hessian with at most one positive eigenvalue [1, Theorem 2.25]. For a cubic these are its first coordinate derivatives. We verify them directly. Positive left activities can subsequently be introduced by nonnegative linear substitution.

23.2 Three Rayleigh inequalities

First suppose that every c_I is positive. Represent the right vertices by columns whose entries at graph edges are algebraically independent, and adjoin the three private coordinate columns

e_1, e_2, e_3 . This gives a rank-three transversal matroid. A generic minor is nonzero exactly when its endpoints are matchable, since distinct matchings give distinct determinant monomials.

Every rank-three matroid is Rayleigh [19, Theorem 1.1]. Set the private variable at i to zero, retain the other two private variables, and specialize all right variables to their activities. For $\{i, j, k\} = \{1, 2, 3\}$, its basis polynomial is

$$c_{123} + x_j c_{ik} + x_k c_{ij} + x_j x_k c_i.$$

The Rayleigh inequality, extended to zero specializations by continuity, gives

$$c_{ij} c_{ik} \geq c_i c_{123}. \quad (106)$$

The nonzero Hessian block of $\partial_{z_i} H$ is

$$\begin{pmatrix} 2c_i & c_{ij} & c_{ik} \\ c_{ij} & 0 & c_{123} \\ c_{ik} & c_{123} & 0 \end{pmatrix}.$$

Its determinant is $2c_{123}(c_{ij}c_{ik} - c_i c_{123}) \geq 0$. Its lower principal block has one positive and one negative eigenvalue. Interlacing and the determinant therefore imply that the full block has at most one positive eigenvalue.

23.3 The remaining Hessian

The Hessian of $\partial_s H$ is

$$\begin{pmatrix} 6 & 2c_1 & 2c_2 & 2c_3 \\ 2c_1 & 0 & c_{12} & c_{13} \\ 2c_2 & c_{12} & 0 & c_{23} \\ 2c_3 & c_{13} & c_{23} & 0 \end{pmatrix}.$$

Eliminating its positive first pivot reduces the assertion to $Q \succeq 0$, where

$$Q_{ii} = \frac{2}{3}c_i^2, \quad Q_{ij} = \frac{2}{3}c_i c_j - c_{ij} \quad (i \neq j). \quad (107)$$

The one- and two-dimensional principal minors are nonnegative: $0 \leq c_{ij} \leq c_i c_j$ gives $-\frac{1}{3}c_i c_j \leq Q_{ij} \leq \frac{2}{3}c_i c_j$. We prove the remaining determinant inequality by an exact finite identity.

Group right vertices by their nonempty neighborhoods $S \subseteq \{1, 2, 3\}$. Let

$$m_S = \sum_{y: N(y)=S} v_y, \quad h_S = \sum_{y: N(y)=S} v_y^2.$$

For distinct i, j , put

$$s_{ij} = \sum_{S \supseteq \{i, j\}} m_S, \quad h_{ij} = \sum_{S \supseteq \{i, j\}} h_S, \quad c_i = \sum_{S \ni i} m_S.$$

Exact endpoint counting yields

$$c_{ij} = c_i c_j - \frac{s_{ij}^2 + h_{ij}}{2}. \quad (108)$$

Indeed, $c_i c_j$ includes the forbidden repeated use of each common neighbor, and counts each distinct pair of common neighbors twice. Its excess is the sum of their squared activities plus their second elementary symmetric sum.

Encode neighborhoods by binary masks $1, \dots, 7$. For the four nonsingleton classes set

$$y_S = \sqrt{h_S}, \quad z_S = m_S - y_S \geq 0. \quad (109)$$

Then $m_S = y_S + z_S$ and $h_S = y_S^2$ exactly. The singleton classes enter only through their masses x_1, x_2, x_4 . Thus $M = 6Q$ has entries

$$M_{ii} = 4c_i^2, \quad M_{ij} = -2c_i c_j + 3(s_{ij}^2 + h_{ij}), \quad (110)$$

as polynomials in the eleven nonnegative variables

$$(x_1, x_2, x_4, y_3, z_3, y_5, z_5, y_6, z_6, y_7, z_7).$$

Lemma 23.2. *After the substitutions above, $\det M$ has exactly 5339 nonzero monomials. Every coefficient is a positive integer; the least is 36 and the greatest is 10656.*

Proof. This is the exact coefficient identity in the accompanying `data/determinant.json`. The generator constructs (110) from the neighborhood sums and expands the six permutation terms using sparse integer arithmetic. An independently written checker uses explicit neighborhood forms and, for a symmetric matrix $\begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}$, expands

$$abc + 2def - af^2 - be^2 - cd^2.$$

Both reconstructions agree coefficient by coefficient and check that every listed coefficient is positive and every exponent vector has total degree six. The complete identity, rather than a numerical evaluation or residual tolerance, is the certificate. Section 23.5 specifies its replay. $\square$

Consequently $\det Q \geq 0$. All principal minors of Q are nonnegative, so $Q \succeq 0$. This proves the required Hessian condition. [*Merge note:* Added 1 October 2026, batch 73: an ordinary proof of $Q \succeq 0$, for every number of left vertices, is source 80's Theorem 43.3 (Part V); with it source 80 reproves Theorem 23.1 as its Corollary 43.4, without Lemma 23.2.] If all $c_I > 0$, the support of (105) is the degree-three homogenization of the independent sets of a free matroid on three elements; it is M -convex. The cubic criterion proves Theorem 23.1 in this case. For an arbitrary graph, add at each left vertex a private right leaf of activity $\varepsilon > 0$. All eight coefficients become positive. Let $\varepsilon \downarrow 0$ and use closure of the Lorentzian cone to obtain the theorem, including isolated vertices and deficient matching rank.

23.4 Gluing at one left vertex

For a polynomial $p(t) = \sum_{k=0}^D p_k t^k$, write $p \in \text{ULC}_D$ when it has no internal zeros and

$$k(D-k)p_k^2 \geq (k+1)(D-k+1)p_{k-1}p_{k+1}, \quad 1 \leq k < D.$$

Equivalently, its degree- D bivariate homogenization is Lorentzian [1, Example 2.26].

Theorem 23.3. *Suppose G has a vertex cover $A \sqcup B$ with $|A| = 3$, $|B| = q$, and all edges of $G[A, B]$ are incident to one vertex $v \in A$. For arbitrary positive activities on both shores, $P_G \in \text{ULC}_{q+3}$. If the cover is minimum, this is ULC at the actual matching rank $q + 3$.*

Proof. Write the shores as $A \sqcup L$ and $B \sqcup R$. Since $A \sqcup B$ is a cover, there are no L – R edges. Split G into G_1 on $(L \sqcup \{v\}, B)$ and G_2 on (A, R) ; these meet only at v . Write

$$p_1 = A(t) + u_v t B(t), \quad p_2 = C(t) + u_v t D(t).$$

Then

$$P_G = AC + u_v t(BC + AD). \quad (111)$$

If v is omitted, both selected components are balanced. If v is selected, the component using it is determined uniquely by its imbalance away from v : its opposite shore has one extra selected vertex. Thus (111) counts each feasible endpoint support once.

For $q \geq 1$, the right-complement transversal basis polynomial of G_1 gives the degree- q Lorentzian polynomial

$$F(s, t, x) = A_h(s, t) + xB_h(s, t), \quad A_h = s^q A(t/s), \quad B_h = s^{q-1} B(t/s).$$

Explicitly, take the transversal matroid on the left vertices and one private element for each $b \in B$, with the B vertices as its q slots. Substitute s/v_b for the private element, tu_ℓ for each $\ell \in L$, and x for v , and multiply by $\prod_{b \in B} v_b$. This is Lorentzian because every matroid basis polynomial is Lorentzian, and nonnegative substitutions preserve the class [1, Theorems 3.10 and 2.10].

[Merge note: Identity (111) is the vertex-sum identity of source 74 (display (13)) and of *preorder-root-polytopes*'s Part VI; source 74's star transfer (Section 10) treats $2 + q$ covers in the same way.] Theorem 23.1, with the other two left variables specialized to tu_i and the root variable to x , gives

$$K(s, t, x) = C_h(s, t) + xD_h(s, t), \quad C_h = s^3 C(t/s), \quad D_h = s^2 D(t/s).$$

The product FK is Lorentzian [1, Corollary 2.32]. Retain only terms of x -degree at most one, then set $x = u_v t$. The result is

$$A_h C_h + u_v t (B_h C_h + A_h D_h) = s^{q+3} P_G(t/s),$$

which is Lorentzian. The case $q = 0$ is Theorem 23.1 itself. □

For completeness, the partial truncation used here preserves the Lorentzian property directly. Its support is the intersection of an M -convex set with the coordinate bound $\deg_x \leq 1$, which is again M -convex when nonempty. In a quadratic coordinate derivative, a differentiation using no x simply removes the nonnegative x^2 coefficient from the original quadratic; this subtracts a positive semidefinite diagonal Hessian entry and cannot increase the positive index. A derivative using x once is the corresponding derivative of $\partial_x(FK)$ specialized at $x = 0$. A derivative using x twice or more vanishes. The Hessian criterion gives the assertion.

Corollary 23.4. *Every genuine $3 + 3$ -cover graph whose core has matching rank at most one has $P_G \in \text{ULC}_6$ for arbitrary positive activities on both shores.*

Proof. A nonempty bipartite graph of matching rank one is a star. A row star is Theorem 23.3; a column star follows after interchanging the shores. For an empty core, the polynomial is a product of two smaller-shore ULC_3 polynomials, so product closure applies. □

The proof also retains every non-root left activity as its own variable: the selected-left marginal of the row-star cover, homogenized to degree $q + 3$, is Lorentzian. The scalar conclusion then allows arbitrary left activities. This vertex-sum result does not provide a rank-three exterior-block lift for general cores.

23.5 Exact verification and scope

The source package contains the manuscript, the complete determinant coefficient list and three standard-library Python checks. [Merge note: Shipped as `code/10-three-vertex-*.py`

and `data/10-three-vertex-*.json`; the determinant list is `data/10-three-vertex-determinant.json`.] From its root run

```
python -O code/replay_all.py.
```

The two independent polynomial reconstructions produce the same lexicographically ordered coefficient list. Its canonical SHA-256 is

```
0f6b0733bd999e34e058223139f524830
bcdcd6a1436a8fae89983cba49c3522.
```

The third checker independently enumerates endpoint sets in 240 weighted three-left graphs and 160 weighted star-core graphs, including 38 genuine minimum covers. It verifies the overlap identity, Rayleigh specialization, rooted gluing and all fixed-cover Newton inequalities. These finite regressions supplement the proof; only the complete determinant expansion is used as a finite certificate for a universal assertion.

The result concerns all fields in the stated star-core class. The general finite-activity one-shore rank-six problem remains open at its three middle Newton gaps. [*Merge note:* After source 05 only two middle gaps remain: at matching number six with one weighted shore, gap one holds by *preorder-root-polytopes*'s Theorem 180.2, gap two by source 05 (Section 26) and gap five by source 13 (Section 25), at every activity; gaps three and four are open in general.] No universal real-rootedness or arbitrary-core Lorentzian recombination is claimed. The standard Lorentzian and Rayleigh theorems are attributed below; no exhaustive priority claim is made.

24 Rooted Lorentzian Hall sectors and an unrooted cubic obstruction (source 23)

Abstract of source 23, as delivered. For a genuine $a + q$ bipartite cover, condition on all q right-core vertices and on any matchable $q - 2$ exterior-left vertices. The remaining degree- a exterior-right marginal is Lorentzian, for arbitrary positive activities on both shores. This follows from exact weighted Hall-slack compression and the rank-two exterior-block matroid lift. The roots cannot simply be omitted: a connected 25-vertex rank-six graph, with unit left activities and a unique $3 + 3$ minimum cover, has an all-right-core conditional cubic failing both degree-three Newton inequalities. Its shifted degree-six inequalities remain strict, and its full polynomial is strictly ULC₆ for every common right-core activity. This identifies a normalization and recombination barrier without refuting rank-six ULC.

24.1 Matching supports and the rooted theorem

For a bipartite graph with left and right activities $u_x, v_y > 0$, its matching-support polynomial is

$$P_G(t) = \sum_{\substack{|I|=|J| \\ G[I,J] \text{ has a perfect matching}}} u_I v_J t^{|I|}.$$

Each feasible endpoint pair is counted once. Call a displayed cover $A \sqcup B$ genuine when

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = a, \quad |B| = q, \quad \nu(G) = a + q.$$

There are no L – R edges. Every maximum matching consists of disjoint exterior matchings of sizes a and q .

Theorem 24.1. *Let $a \geq 1$, $q \geq 2$, and let $F \subseteq L$ be any matchable set of $q - 2$ vertices. Condition simultaneously on selecting all B and all F . After removing the size factor t^q and fixed positive activity factors, the remaining polynomial has degree a and is ULC_a .*

More strongly, keep the exterior-right activities as separate variables. The degree- a homogenization

$$H_F(s, z_R) = \sum_{J \subseteq R} c_F(J) z_J s^{a-|J|}$$

is Lorentzian, where $c_F(J)$ sums the fixed left activities of feasible endpoint sets containing F and matched to $B \cup J$.

The theorem permits arbitrary positive activities on both shores. The fixed positive normalization does not affect the Lorentzian or Newton properties. In particular, in a genuine 3+3 cover, forcing all three B vertices and one specified exterior-left vertex leaves a degree-three Lorentzian marginal.

24.1.1 Exact inputs

We use the following previously proved weighted results [21] [*Merge note:* source 60: Theorem 22.1, Proposition 22.2 and Theorem 22.3; identity (112) is source 60's (102). They are restated here as source 23 states them]; the compression is a regular-minor construction and need not produce an ordinary graph.

First, if a matrix consists of a ordinary columns and an exterior column block of rank two, with total rank $a + 2$, its degree- $(a + 2)$ left marginal of nonzero square minors is Lorentzian after fixing positive column activities.

Second, a matchable forced set of size h with h , $h + 1$ or $h + 2$ neighbors admits exact weighted compression. Slack zero deletes the forced set and its neighbors. Slack one replaces the neighbors by a weighted union-profile element. Slack two replaces them by a two-dimensional kernel-line family. The compressed matrix has rank reduced by h , and its support polynomial recovers the conditioned polynomial after a positive factor and t^h .

For precision, in the slack-two case let Γ be the forced-column matrix on its $h + 2$ neighboring rows, write α_i for their activities, and put

$$E_h = \sum_{\substack{|I|=h \\ \det \Gamma[I] \neq 0}} \alpha_I.$$

Each eligible omitted index i gives a kernel line in $\ker(\Gamma^\top) \cap \{x_i = 0\}$, with virtual activity $\alpha_{N \setminus \{i\}}/E_h$. Independent kernel pairs have a common residual determinant profile, and their total activity is

$$\sum_{\text{independent kernel pairs } i < j} \frac{\alpha_{N \setminus \{i\}} \alpha_{N \setminus \{j\}}}{E_h^2} = \frac{\alpha_N}{E_h}. \quad (112)$$

This identity counts the full neighbor set once. Independence is in the kernel, not necessarily in the image rows; zero image profiles contribute zero. These exact weighted identities are essential.

24.1.2 Proof by transposed compression

Proof of Theorem 24.1. For $q = 2$, use the rank-two-block marginal theorem directly after transposition, with the a original A elements as ordinary columns. Extracting both B rows gives degree a .

For $q \geq 3$, transpose the graph and force the right set F , of size $h = q - 2$. Its neighbor set N lies in the original B , so $h \leq |N| \leq h + 2$. Apply the exact compression above. The residual matrix has rank $a + 2$.

In the transposed residual matrix, the original A elements remain exactly a ordinary columns. All other columns come from $L \setminus F$. Their span originally had rank q and was supported only on B . Eliminating the h independent forced directions reduces this exterior-column rank to $q - h = 2$. The kernel family spans the residual directions. Thus the rank-two-block theorem makes the residual degree- $(a + 2)$ left marginal Lorentzian. Its left variables are the retained or virtual original-right variables.

Selecting all original B now has a uniform description:

- If $|N| = h$, exactly two original B rows remain, and both must be selected.
 - If $|N| = h + 1$, one original B row and one union-profile virtual row must be selected.
 - If $|N| = h + 2 = q$, an independent pair from the kernel-line family must be selected.
- Equation (112) recovers the original full- N activity exactly.

Call these distinguished original or virtual rows D . Give each its assigned positive activity times a common variable y , while keeping all z_R separate. Nonnegative linear substitution preserves the Lorentzian property [1]. The y -degree is at most two: there are two distinguished elements in the first cases, and the virtual family has rank two in the last.

Apply $\frac{1}{2}\partial_y^2$ and set $y = 0$. Derivatives and nonnegative boundary specialization preserve Lorentzianity. The result is the coefficient of y^2 , homogeneous of degree a , and the exact compression identities identify it, up to a positive factor, with H_F . The total size shift is $h + 2 = q$, and the power of s is $(a + 2) - 2 - |J| = a - |J|$.

Finally, F extends to a size- q exterior-left basis. Together with an a -matching from A to R , this gives a maximum matching containing F and all B . The top coefficient is positive; the same left basis gives a positive constant coefficient. Thus the actual residual degree is a , as claimed. $\square$

24.2 Why the roots cannot be omitted

[Merge note: This 25-vertex graph is also the example of source 05's last section (Section 26), which uses it, without citing source 23, against a quintic Lorentzian route; it is printed once, here.] We construct a connected graph whose unrooted all- B cubic fails ULC_3 , although Theorem 24.1 applies after any one exterior-left vertex is additionally selected.

Take $A = \{A_1, A_2, A_3\}$ and $B = \{B_1, B_2, B_3\}$. The core edges are

$$A_1B_2, \quad A_2B_1.$$

There are eight exterior-left vertices, each adjacent to all of B . All left activities and the three B activities are one. The exterior-right vertices are:

neighborhood in A	population	activity per vertex
$\{A_1\}$	1	15
$\{A_2\}$	1	15
$\{A_1, A_2\}$	7	1
$\{A_3\}$	1	15
$\{A_1, A_3\}$	1	1/10

The graph has 25 vertices and 45 edges. Three common exterior-left vertices match all B , and the three private exterior-right vertices match all A , so its matching rank is six.

Its minimum cover is uniquely $A \cup B$. Omitting any B vertex would require all eight common left neighbors, impossible in a six-vertex cover. Once all B are selected, omitting either A_1 or A_2 would require all seven common right neighbors. The remaining one-vertex budget must include A_3 , because its private neighbor and the hybrid neighbor are distinct. The hybrid joins the former A_3 star to the other component, so the graph is connected.

24.2.1 Exact conditional coefficients

First give the hybrid vertex activity ε . Removing it leaves an isolated A_3 star and a quadratic factor. Direct support counting gives

$$C_0(t) = (1 + 15t)(120 + 3500t + 25536t^2). \quad (113)$$

Indeed, with $n = 8$, $p = \binom{8}{2} = 28$, $q = \binom{8}{3} = 56$, the constant quadratic coefficient is $q + 2p + n = 120$. Its linear coefficient is $2(q + p)15 + (2q + p)7 = 3500$, and its quadratic coefficient is $q(15^2 + 30 \cdot 7 + \binom{7}{2}) = 25536$.

The hybrid contributes

$$C_\varepsilon(t) = C_0(t) + \varepsilon(204t + 5992t^2 + 44016t^3). \quad (114)$$

For the first increment, $2q + 3p + n = 204$. Pairing the hybrid with the four old right classes contributes

$$15(q + p) + 15(2q + p) + 7(3q + p) + 15(q + p) = 5992.$$

The top increment is $q(2 \cdot 15^2 + 3 \cdot 15 \cdot 7 + \binom{7}{2}) = 44016$. These are endpoint-support counts, not counts of matching allocations.

At $\varepsilon = 1/10$, the conditioned polynomial is $t^3 C(t)$, with

$$C(t) = 120 + \frac{26602}{5}t + \frac{393176}{5}t^2 + \frac{1937208}{5}t^3. \quad (115)$$

Multiply by the harmless positive factor five, and write

$$\hat{C}(t) = 600 + 26602t + 393176t^2 + 1937208t^3.$$

Its degree-three Newton gaps are

$$\begin{aligned} 26602^2 - 3 \cdot 600 \cdot 393176 &= -50396, \\ 393176^2 - 3 \cdot 26602 \cdot 1937208 &= -13454672. \end{aligned}$$

Both fail strictly. Its cubic discriminant is -9199780352 , so scalar real-rootedness fails as well.

By contrast, the two nontrivial shifted degree-six gaps are

$$\begin{aligned} 8 \cdot 26602^2 - 15 \cdot 600 \cdot 393176 &= 2122747232 > 0, \\ 5 \cdot 393176^2 - 12 \cdot 26602 \cdot 1937208 &= 154533548288 > 0. \end{aligned}$$

Thus this is not a counterexample to the required rank-six normalization. [*Merge note:* The tempting false reading: this cubic does not contradict source 21's all-core bounds (Theorem 30.1). They are bounds for the *shifted* degree-six inequalities, with constants $2n/(n-1)$ and $21/8$, both below three; on this graph the two ratios are $2.99979 \geq 16/7$ and $2.99974 \geq 21/8$ (recomputed at placement). Only the *unshifted* degree-three inequalities, with constant three, fail.]

24.2.2 The recombination obstruction

Forcing a specified common exterior-left vertex replaces each factor $\binom{8}{j}$ counting selected common left vertices by $\binom{7}{j-1}$. Summing over the eight possible roots therefore counts a support j times when it uses j exterior-left vertices:

$$8 \binom{7}{j-1} = j \binom{8}{j}.$$

All eight rooted marginals are Lorentzian by Theorem 24.1; they do not recover the unrooted marginal with its required multiplicity one. This example shows that the normalization loss cannot be repaired by simply dropping the roots.

24.3 The original graph is safe under every common core field

There is a stronger check for this particular graph. Keep the exterior-right activities as above, and give every B vertex a common activity $w > 0$. Direct endpoint enumeration gives the full support coefficients

k	$p_k(w)$
0	1
1	$296/5 + 26w$
2	$11241/10 + (14951/10)w + 117w^2$
3	$34593/5 + (276443/10)w + (61207/10)w^2 + 120w^3$
4	$(830232/5)w + (521166/5)w^2 + (26602/5)w^3$
5	$(2905812/5)w^2 + (393176/5)w^3$
6	$(1937208/5)w^3$

Put $G_k = k(6-k)p_k^2 - (k+1)(7-k)p_{k-1}p_{k+1}$. The third and fourth gaps have nonnegative coefficients and are nonzero. The remaining gaps have the exact forms

$$\begin{aligned} 5G_1 &= 20170 - 12746w + 9880w^2, \\ 25G_2 &= 99127242 - 8903046w + 94261217w^2 + 7629855w^3 + 1567800w^4, \\ 25G_5 &= w^4(22918732029648 - 690327945216w + 154533548288w^2). \end{aligned}$$

The displayed quadratic parts have positive leading coefficients and respective discriminants

$$-634657884, \quad -37296153647015940, \quad -13690299259267221125283840.$$

They are positive for every real w . Hence every $G_k(w) > 0$ for $w > 0$. The original graph is strictly ULC_6 throughout this common-core-field family, while its all- B cubic fails ULC_3 . No statement for arbitrary independent B fields is inferred from this check.

24.4 Verification and remaining questions

The package [*Merge note*: shipped as `code/23-rooted-sectors-*.py` and `data/23-rooted-sectors-*.json`] includes a Hall-subset enumeration of the conditional coefficients, all four hybrid increments, the full common-field coefficient table and its exact gap certificates. A separate endpoint-state dynamic enumeration verifies the original 24-vertex construction. An exhaustive enumeration of all 2048 left-cover masks confirms the connected graph's unique minimum cover and rank.

The Lorentzian theorem depends on the exact weighted matrix/kernel compression inputs [21]. The residual object must not be replaced by its entry-support graph, since its linear

dependencies matter. The counterexample uses unit left activities, positive right activities, a genuine $3 + 3$ cover and no degenerating zero field.

The general finite-activity rank-six one-shore ULC problem remains open. The present obstruction also does not refute an all-core eventual-scaling theorem at the weaker rank-six normalization. [*Merge note:* Such a theorem is source 21’s Corollary 30.4. On the open problem: at matching number six with one weighted shore, gap one holds by *preorder-root-polytopes*’s Theorem 180.2, gap two by source 05 (Section 26) and gap five by source 13 (Section 25), at every activity; gaps three and four are open in general.] It identifies why the blanket stronger scalar- ULC_3 and real-rootedness routes cannot prove that result. No global minimal-vertex or priority claim is made.

25 The last Newton inequality at matching number six, one weighted shore (source 13)

Abstract of source 13, as delivered. For every bipartite graph of matching rank six, with unit activities on one shore and arbitrary positive activities on the other, we prove the last matching-rank-normalized Newton inequality. The three core activities reduce the gap to a positive-definite quadratic in their reciprocals. Two matroid Hessians and a weighted plane-incidence inequality reduce its Schur complement to a quantitative two-point Rayleigh bound. In the graph-generic setting, that bound has ten support profiles; exact moment aggregation and 858 rational weighted-square terms certify all ten. The conclusion holds at every finite positive activity. The full rank-six ULC question remains open at the three middle gaps.

25.1 Statement and support convention

For a bipartite graph $G = (X, Y)$, define

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|} = \sum_k p_k t^k.$$

Each feasible pair of endpoint sets is counted once, regardless of how many perfect matchings realize it. Its degree is the matching rank.

Theorem 25.1. *If G has matching rank six, every $u_x = 1$, and every $v_y > 0$, then*

$$5p_5^2 \geq 12p_4p_6. \tag{116}$$

This is the last ULC_6 inequality. It concerns arbitrary finite positive activities. The first gap is already universally nonnegative by the edge-compatibility Motzkin–Straus argument [6]; the present result leaves only gaps 2, 3, 4 as possible failures at rank six. We do not claim full ULC_6 . [*Merge note:* The first gap is *preorder-root-polytopes*’s Theorem 180.2 (Part XI), printed with the Motzkin–Straus route in Section 11. Gap two is settled by source 05 (Section 26); only gaps three and four remain open in general.]

We use the following precise earlier result: if a bipartite graph has a minimum vertex cover with at most two vertices on either shore, its support polynomial is ULC_r at its actual matching rank r , for arbitrary positive activities on both shores. The two-vertex case is [12, Theorem 1.1]; the pure and singleton cases are proved in Section 2 of that report. [*Merge note:* Theorem 2.1 and Section 3 here; the pure and singleton cases are *preorder-root-polytopes*’s

Theorem 206.2.] Thus, after choosing a minimum vertex cover, it suffices to treat a genuine $3 + 3$ cover

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = |B| = 3, \quad \nu(G) = 6.$$

There are no L – R edges. Each maximum matching avoids core edges, so both exterior blocks have rank three. Isolated vertices may be removed.

We use two established inputs: matroid basis polynomials are Lorentzian, with the usual derivative and nonnegative-specialization closures [1]; and every rank-three matroid is Rayleigh [19]. The new finite input is the ten exact identities in Section 25.6.

25.2 The exact reciprocal activity quadratic

Represent the graph by a matrix whose nonzero edge entries are algebraically independent. A square minor is nonzero exactly when its endpoint sets are matchable: distinct matchings give distinct determinant monomials. Write K for the 3×3 core, ℓ_i for the exterior-left rows, and r_j for the exterior-right columns. Matrix indeterminates are separate from vertex activities.

Let q count independent triples in L , let p_i count pairs matching $B \setminus \{i\}$, and let n_i count left neighbors of B_i . All are positive. In the right rank-three matroid on R and the private coordinate basis $E = \{e_1, e_2, e_3\}$, let U, a, T be the weighted basis counts with one, two and three R elements, respectively. Its degree-three Newton inequality gives

$$a^2 \geq 3TU. \quad (117)$$

For each independent left pair P , choose a nonzero annihilator column v_P and introduce a virtual right vector $w_P = Kv_P$. Retain every pair label, including zero or parallel virtual vectors. Let P_B count weighted bases consisting of one virtual vector and two R columns, and let P_V count bases with one virtual vector, one R column and one private coordinate vector.

For a left row ℓ , let β_ℓ be the total activity of those r_j for which

$$\begin{pmatrix} K & r_j \\ \ell & 0 \end{pmatrix} \quad (118)$$

is nonsingular, and put $N = \sum_{\ell \in L} \beta_\ell$. Forcing all B and removing their activity product and t^3 gives a residual cubic with

$$C_1 = qU + P_V + N, \quad C_2 = qa + P_B, \quad C_3 = qT. \quad (119)$$

The support partition uses three, two or one selected left-exterior vertices. [*Merge note:* The same sector decomposition is source 21's (180).] Eliminating a selected independent pair gives its virtual column, and complementary private coordinate vectors record the selected A rows once.

For each independent right pair J , choose a row normal b_J and introduce a virtual left row $z_J = b_J K$, with activity v_J . Let B_i count weighted bases (ℓ, e_i, z_J) . If exactly $B \setminus \{i\}$ is selected, its residual t^2 coefficient is

$$E_i = p_i a + B_i. \quad (120)$$

This is the same determinant elimination after transposition. Let D count weighted bases (ℓ, z_J, z_H) for distinct pair labels J, H . Their activity products may share original R factors.

Write $w_i > 0$ for the three B activities, $W = w_1 w_2 w_3$ and $\tau_i = 1/w_i$. Exact endpoint counting gives

$$\begin{aligned} p_6/W &= qT, \\ p_5/W &= C_2 + T \sum_i p_i \tau_i, \\ p_4/W &= C_1 + \sum_i E_i \tau_i + T \sum_{i < j} n_k \tau_i \tau_j, \quad \{i, j, k\} = \{1, 2, 3\}. \end{aligned} \tag{121}$$

For example, the top coefficient with exactly two B vertices uses all A and two L vertices, giving $p_i T$. With exactly one B vertex it gives $n_i T$.

Set $x_i = T \tau_i$. The last gap is

$$\frac{5p_5^2 - 12p_4 p_6}{W^2} = A_0 + 2c^\top x + x^\top Q x, \tag{122}$$

where

$$A_0 = 5C_2^2 - 12qTC_1, \quad c_i = 5C_2 p_i - 6qE_i,$$

and

$$Q_{ii} = 5p_i^2, \quad Q_{ij} = 5p_i p_j - 6qn_k \quad (i \neq j). \tag{123}$$

25.3 Two Schur bounds

Let $\mathcal{N}$ have zero diagonal and $\mathcal{N}_{ij} = n_k/2$. The rank-three left matroid on L and the private coordinates has basis polynomial

$$qt^3 + \sum_i p_i x_i t^2 + \sum_{i < j} n_k x_i x_j t + x_1 x_2 x_3.$$

Its derivative Hessian, with positive pivot $6q$, gives

$$H = pp^\top - 3q\mathcal{N} \succeq 0. \tag{124}$$

Thus $Q = pp^\top + 4H$ is positive definite. Indeed, a vector in its kernel is orthogonal to p and killed by H , hence by $\mathcal{N}$, whereas

$$\det \mathcal{N} = \frac{n_1 n_2 n_3}{4} > 0.$$

No simplicity, coloop or nonsingular-core assumption is needed.

Adjoin all virtual left rows z_J to this matroid, with a common variable multiplying their activities. The derivative Hessian in the original-left direction gives

$$\begin{pmatrix} H & d \\ d^\top & P_B^2 - 3qD \end{pmatrix} \succeq 0, \quad d = pP_B - \frac{3q}{2}B. \tag{125}$$

To check the factors, before taking its Schur complement the entries at the original-left pivot are $6q$, $2p_i$ and $2P_B$. The private block is $2\mathcal{N}$, its mixed virtual entries are B_i , and the virtual diagonal is $2D$. Eliminate $6q$ and multiply the resulting negative matrix by $-3q/2$.

From (120),

$$c = (P_B - qa)p + 4d.$$

The Cauchy–Schwarz inequality applied to (125) yields

$$c^\top Q^{-1} c \leq (P_B - qa)^2 + 4P_B^2 - 12qD. \tag{126}$$

For a direct verification, use $|d^\top y|^2 \leq (P_B^2 - 3qD)y^\top Hy$, then apply the two-term Cauchy–Schwarz inequality to $c^\top y$. The denominator is $y^\top Qy = (p^\top y)^2 + 4y^\top Hy$.

Put $\mathcal{R} = aP_B - TP_V$. Subtraction in (126) and (119) gives

$$A_0 - c^\top Q^{-1}c \geq 4q[q(a^2 - 3TU) + 3(\mathcal{R} - TN + D)]. \quad (127)$$

It remains to establish

$$D \geq \frac{3}{4}TN, \quad \mathcal{R} \geq \frac{1}{4}TN. \quad (128)$$

25.4 The weighted plane incidence bound

We recall the rank-three incidence lemma from the all-core conditional argument [22], including its short proof. [*Merge note:* Source 21’s Lemma 30.2; it is the case $b = 3$ of source 08s’s two-plane incidence lemma (Lemma 29.4). It is printed with its proof here, where it is first used; Part IV refers back.] For a weighted rank-three configuration and a plane H_0 , let β be the mass outside H_0 , q_0 the weighted basis sum, and h_δ the pair mass whose span meets H_0 in the line δ . Put $h = \sum_\delta h_\delta$. Then

$$h^2 - \sum_\delta h_\delta^2 \geq \frac{3}{2}q_0\beta. \quad (129)$$

Adjoin diffuse generic plane points of total activity y and one point x_δ on each line. The limiting matroid basis cubic is

$$q_0t^3 + hyt^2 + \frac{\beta}{2}y^2t + \sum_\delta x_\delta((h - h_\delta)t^2 + \beta yt) + \beta t \sum_{\delta < \eta} x_\delta x_\eta.$$

Its ∂_t Hessian has at most one positive eigenvalue. Eliminating its positive yy pivot β leaves line block $-\beta I$, cross entries $-2h_\delta$ and tt entry $6q_0 - 4h^2/\beta$. The remaining Schur complement is nonpositive, which is (129). Parallel elements and arbitrary positive weights are allowed.

Apply (129) to the weighted R configuration for each fixed ℓ . If K is invertible, the effective plane is $\ker(\ell K^{-1})$. If $\text{rank } K = 2$ and ℓ is nonzero on $\ker K$, it is $\text{im } K$. In the remaining rank-two cases, and when $\text{rank } K \leq 1$, $\beta_\ell = 0$.

In each nonzero case, independent mapped pairs of right-pair normals correspond exactly to distinct intersection lines with this plane. This follows by restricting the normals to the plane: their projective restrictions agree exactly when their kernels there agree. At rank two the quotient by the kernel of the transpose identifies with the image plane. Consequently the ordered virtual-left pair count at ℓ is at least $3T\beta_\ell/2$. The definition of D uses unordered pairs. Summing over the unit left rows proves the first inequality in (128).

25.5 The quantitative local lemma

Lemma 25.2. *Let E be the coordinate basis in dimension three. Let R consist of columns generic independently on specified nonempty coordinate supports, with positive activities. Let v_1, v_2 be two independent vectors over the field of the other graph entries, so they are independent of the R indeterminates.*

Let β be the total R activity outside $\text{span}(v_1, v_2)$. Let P count weighted bases with two R columns and one of v_1, v_2 , and let B count bases with one R column, one coordinate element and one of v_1, v_2 . Then

$$aP - TB \geq \frac{1}{2}T\beta. \quad (130)$$

The graph-generic hypothesis is explicit. The analogous quantitative assertion for arbitrary numerical configurations is not needed and is not proved here.

Write $S_1 = \text{supp}(v_1)$, $S_2 = \text{supp}(v_2)$ and $S_0 = \text{supp}(v_1 \times v_2)$. A determinant involving one virtual vector and one or two generic R columns is nonzero exactly when the coordinate masks permit a matching. Distinct monomials in the independent R entries cannot cancel. The outside-plane indicator for a column of type S is $\mathbf{1}_{S \cap S_0 \neq \emptyset}$. Thus the actual normal support is retained; the plane is not replaced by a generic plane.

Up to coordinate permutations and swapping the two virtual vectors, there are ten profiles, listed in Table 6. We use masks 1, 2, 4 for the three coordinates. If $|S_0| = 1$, the plane is a coordinate plane and gives the first three types. If $|S_0| = 2$, it is spanned by the remaining coordinate axis and a line supported on the other two coordinates, giving four types. If $|S_0| = 3$, every nonzero point has support of size at least two; equal size-two supports would give dependent vectors, leaving three types.

Equivalently, on common support coordinates, a zero minor means equality of the ratios $(v_1)_i/(v_2)_i$. Set partitions of these ratios, with the one-product minors forced nonzero, give exactly 55 labeled nonparallel profiles and ten orbits. This is an exhaustive classification, rather than a vector sample.

25.6 Moments and exact finite identities

In one coordinate class, with positive activities u_j , set

$$s = \sum_j u_j, \quad y = \sqrt{\sum_j u_j^2}, \quad z = s - y \geq 0.$$

Its elementary symmetric moments satisfy

$$e_1 = y + z, \quad e_2 = yz + \frac{z^2}{2}, \quad e_3 \leq \frac{yz^2}{2} + \frac{z^3}{6}. \quad (131)$$

The last inequality follows from $\sum_j u_j^3 \leq (\sum_j u_j^2)^{3/2}$. Only e_1 matters for a singleton coordinate class, only e_1, e_2 for a size-two class, and only the full-support class can contribute e_3 to T .

Replace the full-support e_3 by its upper bound. This lowers $2aP - T(2B + \beta)$, since its multiplier is nonnegative. The other used moments are exact. Denote the resulting quartic by F_{S_1, S_2, S_0} , in the eleven nonnegative variables

$$X = (x_1, x_2, x_4, y_3, z_3, y_5, z_5, y_6, z_6, y_7, z_7).$$

The first three are singleton-class totals.

For complete specification, let $\chi(M_1, \dots, M_k)$ be the indicator that the masks admit distinct coordinate representatives. For a multiset I of right types, put $e(I) = \prod_S e_{\text{mult}_I(S)}(S)$, using (131). Then

$$\begin{aligned} a &= \sum_{|I|=2} e(I) \sum_{e \in \{1,2,4\}} \chi(I, e), \\ P &= \sum_{|I|=2} e(I) (\chi(I, S_1) + \chi(I, S_2)), \\ T &= \sum_{|I|=3} e(I) \chi(I), \\ B &= \sum_S e_1(S) \sum_{e \in \{1,2,4\}} (\chi(S, e, S_1) + \chi(S, e, S_2)), \\ \beta &= \sum_{S \cap S_0 \neq \emptyset} e_1(S), \quad F = 2aP - T(2B + \beta). \end{aligned}$$

Multisets and elementary symmetric moments count distinct original column labels exactly once.

For every profile the supplied rational data give the exact identity

$$F = \sum_j \frac{\lambda_j X^{\mu_j}}{2r_j} (r_j X^{m_j} - X^{n_j})^2 + \sum_{\alpha} d_{\alpha} X^{\alpha}. \quad (132)$$

All exponents are nonnegative integers; all $\lambda_j, r_j, d_{\alpha}$ are positive rationals. Every multiplier X^{μ_j} is therefore nonnegative on the entire moment domain. There are no residual tolerances in (132). This proves $F \geq 0$, hence Lemma 25.2.

Table 6: The exhaustive virtual profiles and exact certificate sizes.

S_1	S_2	S_0	Weighted squares	Positive monomials
1	2	4	64	716
1	3	4	61	718
3	3	4	54	728
4	3	3	74	839
4	7	3	122	792
3	7	3	90	821
7	7	3	93	798
3	5	7	95	807
3	7	7	107	807
7	7	7	98	775
Total			858	7801

The accompanying standard-library checker independently reconstructs all ten targets from the matching rule and expands every rational term of (132). It checks 8680 target monomials. A numerical linear program found the decompositions; exact rational reconstruction and exact coefficient equality are the proof-relevant steps. Two active-basis solutions were solved again over the rationals, and every final residual is exactly zero.

25.7 Summing the local bound

For each ℓ with $\beta_{\ell} > 0$, choose two other left rows completing it to a basis. The normals of the two pairs containing ℓ span $\ker \ell$. Their images under K are independent and span the effective plane whose outside R activity is β_{ℓ} . Indeed, nonsingularity of (118) for some r_j forces the image of $\ker \ell$ to have dimension two; the rank cases were identified above. Apply Lemma 25.2 to these two virtual points.

For any individual virtual point w_P , the ordinary rank-three Rayleigh theorem gives

$$aP_B(w_P) - TP_V(w_P) \geq 0.$$

This follows by summing the Rayleigh inequalities between w_P and the three private coordinate elements, and setting their external activities to zero by continuity. It includes zero and parallel virtual points.

Each labeled left pair can be chosen at most twice, once at each endpoint. Summing the local inequalities and using nonnegativity of every unused individual slack gives

$$2(aP_B - TP_V) \geq \frac{1}{2}T \sum_{\ell} \beta_{\ell}.$$

This is the second bound in (128). Now (117), (127) and (128) imply $A_0 - c^\top Q^{-1}c \geq 0$. Finally,

$$A_0 + 2c^\top x + x^\top Qx = (x + Q^{-1}c)^\top Q(x + Q^{-1}c) + A_0 - c^\top Q^{-1}c \geq 0.$$

Equations (122) and (116) prove Theorem 25.1. In fact the reciprocal-core quadratic is nonnegative without restricting x to the positive orthant.

25.8 Reproduction and remaining scope

The package [*Merge note:* shipped as `code/13-last-newton-*.py` and `data/13-last-newton-*.json`] includes the complete ten-profile rational certificate, an independent standard-library replay, the exhaustive ratio-partition orbit checker, and the endpoint-set dynamic program. The latter checks the top-three coefficient formulas, positive definiteness and 1536 full quadratic identities on 512 deterministic graph shapes. These finite graph checks are supplementary; the universal proof is the reduction and exact identities above.

The theorem concerns the last gap at every finite positive one-shore activity. It strengthens the earlier eventual-scaling conclusion in this particular gap. The three middle rank-six inequalities remain open. [*Merge note:* The earlier eventual-scaling conclusion is source 21's (Corollary 30.4). Of the three middle inequalities, the second is settled by source 05; the third and fourth remain open.] The unrestricted-vector quantitative local lemma, optimal constants, equality classification for the new last-gap theorem and global novelty priority are not claimed. These are ordinary mathematical proofs with independently checked exact certificates, not a formal-kernel proof or an externally refereed publication.

26 The second Newton inequality at matching number six, one weighted shore (source 05)

Abstract of source 05, as delivered. For every bipartite graph of matching rank six, with unit activities on one shore and arbitrary positive activities on the other, we prove the second matching-rank-normalized Newton inequality. For a genuine $3 + 3$ minimum cover, we prove the stronger Lorentzian property of a degree-three truncation in the individual right variables. A neighbor-feature Gram identity and exact kernel compression settle its homogenizing and exterior derivatives. Each remaining core derivative reduces to a two-dimensional Schur complement. The empty core column has a direct proof; the other columns have an exact certificate on 61 rooted cores and 51 exterior-basis cones, comprising 50 037 694 nonnegative coefficient checks. The third and fourth rank-six gaps remain open. An explicit quadratic derivative rules out the analogous universal quintic Lorentzian approach to the fourth gap.

26.1 The inequality and its cubic strengthening

For a bipartite graph $G = (X, Y)$ with positive activities, let

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|} = \sum_k p_k t^k.$$

Every feasible endpoint pair is counted once. We never count its matching allocations. The degree is the matching rank.

Theorem 26.1. *If G has matching rank six, $u_x = 1$ for every $x \in X$, and $v_y > 0$ for every $y \in Y$, then*

$$8p_2^2 \geq 15p_1p_3. \tag{133}$$

This is the second ULC_6 inequality. Together with the universal first gap and the earlier last-gap theorem, it leaves only gaps 3, 4 open at rank six. [*Merge note:* The universal first gap is *preorder-root-polytopes*'s Theorem 180.2; the earlier last-gap theorem is source 13, Theorem 25.1, which source 05 does not cite.] It holds at every finite positive activity.

The earlier two-vertex-cover theorem [12] proves all-field actual-rank ULC whenever a minimum cover has at most two vertices on one shore. We may therefore assume

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = |B| = 3, \quad \nu(G) = 6,$$

with $A \sqcup B$ a vertex cover. Both exterior blocks have rank three, since every maximum matching avoids core edges.

For a fixed right set J , let a_J be the number of left sets that match it. Put $d_j = a_{\{j\}}$ and $p_k(\mathbf{z}) = \sum_{|J|=k} a_J z_J$.

Theorem 26.2. *For every such genuine 3 + 3 cover, the homogeneous polynomial*

$$F(s, \mathbf{z}) = 20s^3 + 10p_1(\mathbf{z})s^2 + 4p_2(\mathbf{z})s + p_3(\mathbf{z}) \quad (134)$$

is Lorentzian.

Its support is M -convex: it is the homogenized independent-set support of the rank-three truncation of the right transversal matroid. Thus it suffices to show that every first derivative has a Hessian with at most one positive eigenvalue [1, Theorem 2.25]. After $z_j = tv_j$, the degree-three Newton inequality gives $(4p_2)^2 \geq 3(10p_1)p_3$, which is (133).

Two precise companion inputs are bundled with this report. The selected-left marginal of a graph with three left vertices, homogenized to degree three, is Lorentzian [20]. [*Merge note:* Source 10's Theorem 23.1.] For a displayed $2 + q$ cover, the two-vertex lift gives a degree- $(q + 2)$ Lorentzian marginal in *all individual variables on the two-core-vertex shore*, with the opposite-shore activities fixed [12, Section 3]. [*Merge note:* Source 74's Theorem 2.1, proved in Section 5 (the retained-variable statement at its end).] We use this retained-variable statement after transposing a $3 + 2$ cover.

26.2 The homogenizing derivative

We first allow arbitrary positive left activities. Let d_i be the total activity adjacent to right vertex i and let a_{ij} be its weighted two-endpoint coefficient. Set

$$G_{ii} = d_i^2, \quad G_{ij} = d_i d_j - a_{ij} \quad (i \neq j).$$

For common-neighborhood mass s_{ij} and squared-activity sum h_{ij} , endpoint counting gives $G_{ij} = (s_{ij}^2 + h_{ij})/2$. [*Merge note:* This is source 10's overlap identity (108).]

Take one feature for each left vertex, weighted by u_ℓ^2 , and one for each unordered left pair, weighted by $u_\ell u_m$. A right vertex belongs to a feature if it meets every left vertex of that feature. Their Gram matrix K has exactly the off-diagonal entries of G , and

$$G = K + \text{diag} \left(\sum_{\ell < m: \ell, m \sim i} u_\ell u_m \right). \quad (135)$$

Split $d = \alpha + n$, according to contributions from A and L . Retain the three singleton A features; by Cauchy their Gram matrix dominates $\alpha\alpha^\top/3$. Retain all singleton and pair L features and their diagonal corrections. Applied to the transpose of B versus L , the three-vertex marginal theorem gives the bound $nn^\top/3$. Every discarded term in (135) is positive semidefinite. Hence

$$G \succeq \frac{\alpha\alpha^\top + nn^\top}{3} \succeq \frac{dd^\top}{6}. \quad (136)$$

The Hessian of $\partial_s F$ has entries $ss = 120$, $sz_i = 20d_i$ and $z_i z_j = 4a_{ij}$, with zero z_i^2 entries. Its positive s pivot leaves the negative of $4(G - dd^\top/6)$. This settles $\partial_s F$.

26.3 Exterior derivatives and rank two compression

The Hessian of $\partial_{z_j} F$ has s pivot $20d_j$, cross entries $4a_{ji}$, and off-diagonal entries a_{jik} . Its required Schur matrix is

$$Q^{(j)} = \frac{4}{5d_j} \mathbf{a}_j \mathbf{a}_j^\top - B^{(j)}, \quad (\mathbf{a}_j)_i = a_{ji}, \quad B_{ik}^{(j)} = a_{jik} \ (i \neq k), \quad B_{ii}^{(j)} = 0. \quad (137)$$

[Merge note: Lemma 26.3 is source 60's Theorem 22.1 at $q = 0$, transposed; source 05's direct Hessian proof is printed as a second route.]

Lemma 26.3. *For any matrix of rank at most two, with arbitrary positive row activities, its degree-two selected-column regular-minor marginal is Lorentzian.*

Proof. Ranks zero and one are immediate. Factor a rank-two matrix through a two-dimensional space and group its nonzero rows by projective direction, with total activities w_h . Put $W = \sum_h w_h$ and $Q = \sum_{h < k} w_h w_k > 0$. Compress parallel column classes by adding their variables, and let m be the number of distinct nonzero column directions. A column annihilates either one row direction or none. Its singleton coefficient is $d_i = W - w_i$, where the nonzero w_i are distinct members of the row-class list, and other w_i are zero. Every pair of distinct column directions has coefficient Q . The quadratic Hessian condition is

$$QI + \frac{1}{2}dd^\top - Q\mathbf{1}\mathbf{1}^\top \succeq 0.$$

The rank-one Schur criterion reduces this to $(m-1)(2Q + \sum_i d_i^2) \leq (\sum_i d_i)^2$. The difference is exactly

$$\left(W - \sum_i w_i\right)^2 + (m-1) \left(\sum_h w_h^2 - \sum_i w_i^2\right) \geq 0.$$

The support is the degree-two homogenization of a rank-two column matroid's independent sets. Restoring parallel variables preserves the result. $\square$

Consider a row-block matrix $(U; L)$ in which U has rank at most two and L is graph-generic on three columns B , with entries independent over the field of U . Let α, n be the singleton coefficient vectors of the two blocks, and let S_U, S_L be their weighted common-support Gram matrices. The mixed two-row determinant is nonzero exactly when one diagonal product is nonzero, by independence of the L entries. Consequently, with the definition of G above,

$$G = G_U + G_L + S_U \circ S_L + \text{diag}(\alpha_i n_i).$$

The last two terms are positive semidefinite. Lemma 26.3 and the three-vertex marginal give

$$G \succeq \frac{1}{2}\alpha\alpha^\top + \frac{1}{3}nn^\top \succeq \frac{1}{5}(\alpha + n)(\alpha + n)^\top. \quad (138)$$

Now fix $j \in R$. Its neighborhood lies in the three A vertices. Forcing j , dividing by its activity and by t , gives exactly E times a residual regular-minor polynomial, where $E = \sum_{a \sim j} u_a$. Here is the needed compression explicitly. [Merge note: It is the case $h = 1$ of source 60's Proposition 22.2 and Theorem 22.3, including its degree-three example; source 05 does not cite source 60.] One neighbor is deleted. Two neighbors are replaced by their kernel row of activity $u_1 u_2 / E$. Three neighbors are replaced by the three pair-kernel rows, of activities

$u_1u_2/E, u_1u_3/E, u_2u_3/E$. Each single kernel row has the endpoint profile of its original pair. Every pair has the same three-row minor profile, and its total activity is

$$\frac{u_1u_2u_3(u_1 + u_2 + u_3)}{E^2} = \frac{u_1u_2u_3}{E}.$$

Thus all support coefficients have the correct weights, including dependent images. Kernel rows are obtained by eliminating the forced column; these identities concern regular minors, not ordinary graph contraction.

The remaining and virtual A rows span a space of dimension at most two. The unchanged L block is graph-generic on B and independent of them. Its residual singleton and pair coefficients are a_{ji}/E and a_{jik}/E . Equation (138), multiplied by $E = d_j$, is precisely (137). This proves every exterior derivative, even with arbitrary positive original left activities.

26.4 Exact core profiles

We now use unit left activities and force a core column B_0 . Write $J, C_1, C_2 \subseteq A$ for the core neighborhoods of B_0, B_1, B_2 . Let N_S be the exterior-left population of nonempty B -mask S . Define

$$n = |N_L(B_0)|, \quad m_i = |N_L(B_i)|, \quad t_i = |N_L(B_0) \cap N_L(B_i)|, \quad p_i = nm_i - \binom{t_i + 1}{2}, \quad E = n + |J|.$$

The exterior rank gives $n \geq 1$. Let $h(U, V)$ count two-element A sets matching columns with supports U, V , and let $\chi(U, V, W)$ indicate matchability to all three A vertices. For every nonempty exterior-right mask S , put

$$\begin{aligned} r_S &= n|S| + h(J, S), \\ x_i &= p_i + n|C_i| + m_i|J| - t_i|J \cap C_i| + h(J, C_i), \\ y_i(S) &= p_i|S| + nh(C_i, S) + m_ih(J, S) \\ &\quad - t_i[h(J, S) + h(C_i, S) - h(J \cup C_i, S)] + \chi(J, C_i, S). \end{aligned} \tag{139}$$

These are exactly $a_{B_0R_S}, a_{B_0B_i}, a_{B_0B_iR_S}$. For y_i , partition by two, one or zero selected L rows. If the one L row meets both core columns, the remaining A pair uses either core column with R_S . Its Boolean union is $J \cup C_i$; the displayed subtraction removes the duplicate allocation.

Let q_{all} count bases of all three B columns. To compute it, add the three core-row mask populations to N_S , obtaining M_S , and ignore zero rows. Set $M = \sum_S M_S$ and $d'_i = \sum_{S \ni i} M_S$. Then

$$\begin{aligned} q_{\text{all}} &= \binom{M}{3} - \sum_i \binom{M - d'_i}{3} + \sum_i \binom{M_{\{i\}}}{3} \\ &\quad - \sum_i \binom{M_{\{i\}}}{2} \sum_{S \supseteq B \setminus \{i\}} M_S. \end{aligned} \tag{140}$$

Inclusion-exclusion first excludes a missed column. Every remaining Hall failure has just one selected row meeting a chosen pair of columns, and the other two rows meet only the third. The three failure types are disjoint.

The class-limit matrix has seven R coordinates and the two other B coordinates. Its R block and cross columns are

$$\begin{aligned} Q_R(S, T) &= \frac{4r_S r_T}{5E} - nh(S, T) - \chi(J, S, T), \\ v_i(S) &= \frac{4x_i r_S}{5E} - y_i(S). \end{aligned} \tag{141}$$

The remaining block has diagonal $4x_i^2/(5E)$ and off-diagonal $4x_1x_2/(5E) - q_{\text{all}}$. This matrix suffices for all finite class populations: if X_S is the sum of arbitrary real test coordinates in class S , the actual quadratic form is the class-limit form in $(B_1, B_2, \mathbf{X})$ plus

$$\sum_S [nh(S, S) + \chi(J, S, S)] \sum_{r: N(r)=S} z_r^2 \geq 0.$$

Absent classes set $X_S = 0$. This identity does not assume equal right activities.

26.5 The two dimensional Schur complement

For $|J| \leq 1$, all profile functions in a right mask are combinations of the six containment features 1, 2, 3, 12, 13, 23. The mask-7 row and column equal the sum of those for masks 3, 5, 6 minus those for masks 1, 2, 4, including the cross terms. We therefore retain masks 1, ..., 6. For $|J| = 2, 3$ retain all seven.

Every principal matrix containing R and only one other B_i is positive semidefinite. Indeed, delete the other core vertex. The graph has a genuine $3 + 2$ cover. Transpose and apply the full marginal statement of [12]: its selected-right marginal is Lorentzian of degree five. Differentiate in B_0 and then twice in its homogenizing variable. The quadratic Schur coefficient is $3/(4E)$, whereas (137) has $4/(5E)$. The difference is the positive rank-one matrix $\mathbf{aa}^T/(20E)$. Using arbitrarily many twins with equal test coordinates divided by their population proves the same assertion for the class-limit principal matrix.

For $J = 1, 3, 7$ in binary-mask notation, direct exact calculation gives the following determinants for the retained block $A_R = 5EQ_R$:

J	$\det A_R$
1	$6250n^4(n+1)^8$
3	$15625n^4(n+2)^7(2n^2+4n+3)$
7	$31250n^3(n+2)^2(n+3)^6(2n^2+7n+9)$.

They are positive for $n \geq 1$. The retained block is therefore positive definite. After eliminating it, both diagonal entries of the remaining 2×2 matrix are nonnegative by the preceding principal-matrix argument. Only its determinant remains to be proved nonnegative.

The case $J = \emptyset$ has a direct proof. Let q count L bases, let O count L pairs matchable both to $\{B_0, B_1\}$ and to $\{B_0, B_2\}$, and put $e_i = |C_i|$, $c = |C_1 \cap C_2|$. The Schur complement is exactly

$$D_i = \frac{p_i^2}{2n} + 2e_i p_i + n \binom{e_i}{2}, \quad L_{12} = \frac{p_1 p_2}{2n} - q + cO. \quad (142)$$

To verify this identity, use the six containment features above, put $u = (1, 1, 1, 0, 0, 0)^T$ and let f_i be the feature vector of C_i . The R block is $nF^T(I - uu^T/5)F$, the cross vector in feature coordinates is $nf_i - (p_i + ne_i)u/5$, and the feature inverse is $I + uu^T/2$. Substitution cancels the mixed core terms in (142).

Rank-three Rayleigh gives $nq \leq p_1 p_2$ [19]. Also $O \leq \min(p_1, p_2)$ and $c \leq \min(e_1, e_2)$. With $b = p_1 p_2 / (2n)$ we have $-b \leq L_{12} \leq b + cO$, while

$$D_1 D_2 - (b + cO)^2 \geq \frac{c}{n} p_1 p_2 (p_1 + p_2 - O) + c^2 (4p_1 p_2 - O^2) \geq 0.$$

Thus the empty-column Schur complement is positive semidefinite.

26.6 The complete finite certificate

For the nonempty column, permute A so J is 1, 3 or 7. Permutations stabilizing J , together with exchange of B_1, B_2 , leave 24, 24, 13 rooted core orbits respectively. The two shores are never transposed in this enumeration.

Choose any exterior L basis. Its neighborhood multiset is one of exactly 51 Hall-matchable triples among the 84 triples with repetition from seven masks. If b_S records this basis, write $N_S = b_S + X_S$ with $X_S \geq 0$. Every genuine exterior block has such a decomposition.

For each core, form the exact rational 2×2 Schur complement from (139)–(141). Clear a positive common denominator and then a positive scalar denominator in its determinant. For every basis choice, translate $N = b + X$. The complete certificate states that every coefficient of every resulting ordinary polynomial in the seven X_S is nonnegative. The exact totals are

$ J $	Core orbits	Basis cones	Nonzero coefficient entries
1	24	1224	10 025 635
2	24	1224	22 628 373
3	13	663	17 383 686
Total	61	3111	50 037 694

Every listed nonzero coefficient is a positive integer; the minimum is one. Absent monomials have coefficient zero. The cleared determinants have degree 10, 12 or 14. Their possible common denominators before squaring are

$$2n^2, \quad n(2n^2 + 4n + 3), \quad 2n(2n^2 + 7n + 9), \quad 2n(n + 2)(2n^2 + 7n + 9),$$

all strictly positive on the required domain.

The complete per-core records contain the denominator, scalar multiplier, degree, coefficient counts and the SHA-256 of each lexicographically ordered coefficient list. The source reconstructs every coefficient with exact rational arithmetic and checks its sign. An independent reconstruction uses labeled Hall tests for the endpoint coefficients, verifies each symbolic R inverse and uses a different population translation. The full replay status and commands are recorded in the package’s verification document. [*Merge note:* Shipped as `05-second-newton-VERIFICATION.md`; the scripts as `code/05-second-newton-*.py`, the per-core records as `data/05-second-newton-core_schur_*.json` and the independent replays as `data/05-second-newton-independent-*.json`.]

These finitely many identities prove every remaining determinant inequality on every nonnegative real cone $X \geq 0$, hence on all original integer populations. With both diagonal Schur entries nonnegative, each Schur matrix is positive semidefinite. Sections 2–5 [*Merge note:* Subsections 26.2–26.5 here] now settle every derivative of (134). Its M -convex support completes Theorem 26.2, and the minimum-cover reduction proves Theorem 26.1.

26.7 Scope and the next truncations

The next possible truncation is

$$F_4 = 15s^4 + 10p_1s^3 + 6p_2s^2 + 3p_3s + p_4, \quad \partial_s F_4 = 3F.$$

Its Lorentzian property would prove $9p_3^2 \geq 16p_2p_4$. Only the two-distinct-right-variable quadratic derivatives are new. This is a research direction, not a conclusion of the present theorem.

The analogous quintic

$$F_5 = 6s^5 + 5p_1s^4 + 4p_2s^3 + 3p_3s^2 + 2p_4s + p_5$$

is *not* universally Lorentzian. An explicit graph has core row masks $(2, 1, 0)$, eight exterior-left vertices of mask 7, and exterior-right types 1, 2, 3, 4, 5 with populations 1, 1, 7, 1, 1 and activities 15, 15, 1, 15, 1/10. All left activities are one. It is connected and has matching rank six. [*Merge note*: This is source 23's 25-vertex graph (Section 24.2); source 05 does not cite source 23. Its cubic below is source 23's (115).] Its all- B conditional cubic has coefficients

$$(C_0, C_1, C_2, C_3) = \left(120, \frac{26602}{5}, \frac{393176}{5}, \frac{1937208}{5}\right).$$

Differentiate F_5 in the three B variables and specialize the R variables. The resulting quadratic is $3C_0s^2 + 2C_1st + C_2t^2$, but

$$C_1^2 - 3C_0C_2 = -\frac{50396}{25}.$$

Its Hessian has positive diagonal and determinant $201584/25$, hence two positive eigenvalues. The source package directly verifies these counts. This refutes the stronger quintic approach, not scalar ULC₆. The fourth gap requires a weaker argument.

The present result is an ordinary mathematical proof with an exact finite coefficient certificate. It is not proof-assistant formalization or external refereeing, and no exhaustive priority claim is made.

27 Matching number six with common exterior neighbourhoods, one weighted shore (source 69)

Abstract of source 69, as delivered. We prove matching-rank ultra-log-concavity for one-shore activities on a rank-six family with an arbitrary 3×3 core and complete exterior neighborhoods. The proof reduces the last four Newton gaps to 8,964 explicit univariate coefficient polynomials, all nonnegative coefficientwise; the first gap has a general elementary proof. A small exact verifier and the complete integer certificate accompany the note. We also give a cover-changing reduction that handles certain adjacent exterior classes, and an explicit negative coefficient showing why the main certificate does not extend unchanged to arbitrary exterior neighborhoods. The general rank-six one-shore problem remains unresolved.

27.1 The theorem and its scope

For a bipartite graph $G = (X, Y; E)$ with activities $u_x, v_y > 0$, put

$$P_G(t) = \sum_{(I, J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^d p_k t^k.$$

Feasibility means $I \subseteq X$, $J \subseteq Y$, $|I| = |J|$, and $G[I, J]$ has a perfect matching. Each endpoint pair is counted once. Thus P_G is the weighted perfectly matchable set polynomial, not the matching polynomial; see Davis–Kohl [5]. Its degree is $d = \nu(G)$. Define

$$\Delta_k^{(d)}(P) = k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1}. \quad (143)$$

The property ULC _{d} means all these gaps are nonnegative, with no internal zero coefficients.

Theorem 27.1. *Let $X = A \sqcup L$, $Y = B \sqcup R$, where $|A| = |B| = 3$, $|L| = n \geq 3$ and $|R| = m \geq 3$. Let*

$$E = (L \times B) \cup (A \times R) \cup K, \quad K \subseteq A \times B.$$

Give every vertex in X activity one and every vertex in Y an arbitrary positive activity. Then P_G has degree six and is ULC_6 . The same conclusion holds with the two shores interchanged.

The core K may be any of the 512 labeled 3×3 bipartite graphs. The exterior-right activities need not agree. A common positive activity on the entire unit-weight shore can also be allowed, by rescaling t .

This is a partial rank-six result. Arbitrary positive two-shore fields are known to satisfy rank-normalized ULC through rank five, whereas a one-shore counterexample of rank 38 is available in companion notes [12, 16]. The present theorem excludes this entire common-exterior rank-six family from being a smaller one-shore counterexample. [*Merge note:* Its gaps two and five now also follow from sources 05 and 13 for every one-shore graph of matching number six; what is new here for this family is gaps three and four. Taking K complete and all activities one gives *preorder-root-polytopes*'s unit $H_{3,n}$, so the unit-weight assertion of its Corollary 177.2 (**hb:cor:asymptoticweight**) is a special case of Theorem 27.1.] It does not settle arbitrary $3 + 3$ covers or claim exhaustive literature priority. In particular, the ambient-shore normalization of Röhrle–Ulirsch [7] need not equal six here.

27.2 Exact endpoint counting and the deleted-core baseline

Write b_1, b_2, b_3 for the B activities, $r_1, \dots, r_m$ for the R activities, and $e_l(R)$ for their elementary symmetric polynomials. For $J \subseteq B$, let $b_J = \prod_{j \in J} b_j$. Out-of-range binomial coefficients are zero.

Lemma 27.2. *For $0 \leq k \leq 6$,*

$$p_k = \sum_{\substack{J \subseteq B, \ 0 \leq l \leq 3 \\ |J| + l = k}} T_{J,l}(n) b_J e_l(R), \quad (144)$$

where

$$T_{J,l}(n) = \sum_{\substack{I \subseteq A, \ |I| \geq l \\ \nu(K[I,J]) \geq |I| - l}} \binom{n}{|J| + l - |I|}. \quad (145)$$

Proof. Fix I, J and l selected exterior-right vertices. A support of this type has $h = |J| + l - |I|$ exterior-left vertices. A perfect matching must use l edges from I to those exterior-right vertices and h edges from the selected exterior-left vertices into J . The remaining $|I| - l$ edges are core edges. Hence the stated core matching condition is necessary. Conversely, take that many disjoint core edges, match the remaining l vertices of I to the selected exterior-right vertices, and match the remaining h vertices of J to the selected exterior-left vertices. Completeness of both exterior neighborhoods permits these choices. This proves existence without introducing a multiplicity for different realizing matchings. There are $\binom{n}{h}$ possible exterior-left endpoint sets, giving (145) and (144). $\square$

Delete all core edges and denote the resulting polynomial by Q . Then

$$Q(t) = \left(\sum_{j=0}^3 \binom{n}{j} e_j(B) t^j \right) \left(\sum_{l=0}^3 \binom{3}{l} e_l(R) t^l \right), \quad T_{J,l}^0(n) = \binom{3}{l} \binom{n}{|J|}. \quad (146)$$

Both factors are ULC_3 . For the first, the normalized coefficient is $\binom{n}{j}e_j(B)/\binom{3}{j}$, a product of two log-concave sequences by Newton's inequalities. For the second, it is $e_l(R)$, again log-concave. Their degree-three homogenizations are Lorentzian; the product is Lorentzian and therefore Q is ULC_6 [1, Example 2.26 and Corollary 2.32].

27.3 The finite nonnegative coefficient certificate

Set $z = n - 3 \geq 0$. Every right-activity variable occurs with exponent at most two in a Newton gap. A monomial is specified, up to permuting R , by its core exponent vector $v \in \{0, 1, 2\}^3$, the number ρ of exterior variables occurring once, and the number τ occurring twice. Necessarily

$$|v| + \rho + 2\tau = 2k.$$

For any coefficient table T , define

$$C_{a,b}(T; v, \rho, \tau) = \sum_{\substack{J, H \subseteq B \\ \mathbf{1}_J + \mathbf{1}_H = v}} \binom{\rho}{a - |J| - \tau} T_{J, a - |J|}(z + 3) T_{H, b - |H|}(z + 3). \quad (147)$$

A summand is zero unless both exterior indices lie in $\{0, 1, 2, 3\}$ and their sum is $\rho + 2\tau$. Formula (147) is the coefficient of the specified monomial in $p_a p_b$: every squared exterior variable lies in both supports, and the binomial factor chooses the singleton variables belonging to the first support.

Let $D_k(T) = k(6 - k)C_{k,k}(T) - (k + 1)(7 - k)C_{k-1,k+1}(T)$, with the monomial parameters suppressed. The certificate consists of the coefficient vectors in increasing powers of z of

$$36(D_k(T) - D_k(T^0)), \quad k = 2, 3, 4, 5. \quad (148)$$

They are integer vectors because T is a sum of $\binom{z+3}{h}$ for $0 \leq h \leq 3$, and

$$\begin{aligned} 6 \binom{z+3}{0} &= 6, & 6 \binom{z+3}{1} &= 18 + 6z, \\ 6 \binom{z+3}{2} &= 18 + 15z + 3z^2, & 6 \binom{z+3}{3} &= 6 + 11z + 6z^2 + z^3. \end{aligned}$$

There are 36 core orbits under separate row and column permutations. The binary mask of K is $\sum_{(i,j) \in K} 2^{3i+j}$ with $0 \leq i, j \leq 2$. The orbit representatives are

$$\begin{aligned} &0, 1, 3, 7, 9, 10, 11, 14, 15, 27, 29, 31, 63, 73, 74, 75, 78, 79, \\ &83, 84, 85, 87, 91, 93, 94, 95, 118, 119, 127, 219, 221, 223, 238, 239, 255, 511. \end{aligned}$$

The verifier constructs their full orbits and checks that they partition all 512 masks. It then enumerates $v \in \{0, 1, 2\}^3$, $0 \leq \tau \leq 3$, $0 \leq \rho \leq 6 - \tau$, subject to the displayed total-degree equation, and evaluates (148) by integer polynomial addition and multiplication.

Lemma 27.3 (Exact certificate). *Every integer in every vector (148) is nonnegative. The complete finite accounting is*

Gap k	Vectors checked	Nonzero vectors	Nonzero scalar entries
2	1,224	1,001	2,475
3	2,196	1,878	5,247
4	2,952	1,618	6,105
5	2,592	420	2,205
Total	8,964	4,917	16,032

The file `data/exact_certificate.json` lists every vector, including zeros, and `checks/verify_certificate.py` independently reconstructs each one from (145)–(148) and checks all signs and values.

This is a finite exact certificate, not sampling over activity values. For example, at the complete core, gap five and monomial type $(v, \rho, \tau) = ((2, 2, 1), 5, 0)$, the gap-increment coefficient before multiplication by 36 is

$$5(z + 2)(z + 3)^2(z + 6),$$

which is nonnegative. The complete certificate has the same coefficientwise property, without needing to factor each entry.

Proof of Theorem 27.1. For $k = 2, 3, 4, 5$, Lemma 27.3 gives coefficientwise nonnegativity of $\Delta_k^{(6)}(P) - \Delta_k^{(6)}(Q)$ in z and all right activities. Since Q is ULC_6 , these four gaps of P are nonnegative.

For the first gap, $5p_1^2 \geq 12p_2$, which is the first gap since $p_0 = 1$. [*Merge note:* This is *preorder-root-polytopes*'s Theorem 180.2 (Part XI) at $r = 6$. Source 69 proves it by the weighted Motzkin–Straus route on the edge-disjointness graph, the route printed in Section 11; that argument is not reprinted.]

Finally, $A \sqcup B$ is a six-vertex cover, and disjoint three-matchings $L \rightarrow B$ and $A \rightarrow R$ exist. Thus the degree is exactly six. Subsets of a maximum matching and positivity of all activities give positive coefficients at every degree from zero through six. $\square$

27.4 Adjacent exterior classes and a precise barrier

The following reduction applies to arbitrary exterior neighborhoods and arbitrary positive fields on both shores.

Proposition 27.4. *Suppose a graph of matching rank six has a displayed cover $A \sqcup B$ with $|A| = |B| = 3$. If some $a \in A$ has only private leaves as its exterior neighbors, then its support polynomial is ULC_6 . The analogous statement holds on the other shore.*

Proof. All those exterior neighbors have neighborhood exactly $\{a\}$. A feasible support uses at most one of them, so replace them by one leaf whose activity is their sum. This preserves every support coefficient and the matching rank. [*Merge note:* Weighted leaf compression, *preorder-root-polytopes*'s Lemma 206.1.] There is at least one such leaf, since a size-six matching forces all of A to match to exterior-right vertices. In the compressed graph, replace a in the cover by its single leaf. The resulting six-vertex minimum cover has split $2 + 4$. The rank-normalized theorem for a minimum cover meeting a shore in at most two vertices [12] gives the result. [*Merge note:* Theorem 2.1.] $\square$

For example, this includes arbitrary cores and arbitrary exterior-left neighborhoods when exterior-right neighborhoods are singletons together with the one possible pair class $\{A_1, A_2\}$. Each exterior block has matching rank and minimum-cover size three. If the exterior block between A and R has a minimum cover using at most two vertices of A , combining it with B directly gives the same reduction. Thus a genuinely unhandled case must survive both this cover test and private-leaf compression on each shore.

The coefficientwise proof of the main theorem cannot simply be applied after allowing arbitrary exterior neighborhoods. Take complete core $K_{3,3}$ and n unit exterior-left vertices

adjacent to all of B . Include exterior-right vertices r_1, r_2, r_3 with respective neighborhoods $\{A_1\}, \{A_2\}, \{A_3\}$ and variable activities. If Q again deletes the core, then

$$[b_2 b_3 r_1 r_2](\Delta_2^{(6)}(P) - \Delta_2^{(6)}(Q)) = 56 - 4n. \quad (149)$$

This is negative for $n \geq 15$. The verifier `verify_barrier.py` derives this identity exactly by enumerating left endpoint sets for the relevant right subsets. Yet this graph is ULC_6 : its right shore has six vertices, or one can use Proposition 27.4. Thus (149) obstructs a proof method, not ULC itself.

27.5 Reproducibility and remaining question

Run `python3 checks/run_all.py` [*Merge note:* shipped as `code/69-common-exterior-run_all.py`; the files `data/exact_certificate.json`, `checks/verify_certificate.py` and `verify_barrier.py` named above are shipped as `data/69-common-exterior-exact_certificate.json`, `code/69-common-exterior-verify_certificate.py` and `code/69-common-exterior-verify_barrier.py`] to regenerate the endpoint checks, compare the complete coefficient certificate, and verify (149). The core certificate verifier uses only Python's standard library, integer arithmetic and Hall's subset criterion. It also compares the support criterion with a separate injection-based core matching computation on 100,352 support types across all 512 labeled cores. The mathematical formula and the certificate were independently reconstructed before packaging.

The next unresolved step is a uniform treatment of genuine $3 + 3$ minimum-cover graphs whose exterior blocks have arbitrary neighborhood classes and survive the cover-changing reductions above. The finite positivity certificate here does not resolve that step. Neither numerical minimization nor a negative coefficient of a gap increment would by itself settle the full one-shore question.

28 Nested families and conditional barriers at matching number six, one weighted shore (source 63)

Abstract of source 63, as delivered. We extend the rank-six one-shore ULC result to a complete core with nested exterior neighborhoods. An exact finite certificate proves the result for arbitrary positive activities on the weighted shore. We then give a 20-vertex graph with a unique $3 + 3$ minimum cover whose full right marginal is not Lorentzian, ruling out that stronger approach to the general problem. A conditional theorem excludes every quadratic sector obtained by forcing four right vertices in a genuine $3 + 3$ cover. The general rank-six one-shore ULC question remains open.

28.1 A nested exterior theorem

For a positively weighted bipartite graph $G = (X, Y)$, let

$$P_G(t) = \sum_{(I, J) \text{ feasible}} u_{I \cup V} v_J t^{|I|} = \sum_{k=0}^d p_k t^k,$$

where a feasible pair has $|I| = |J|$ and $G[I, J]$ admits a perfect matching. Each endpoint pair is counted once. Its degree is the matching rank d . We use

$$\Delta_k^{(d)}(P) = k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1}$$

for the rank-normalized Newton gap. Positive support polynomials have no internal zeros, so ULC_d is equivalent to nonnegativity of these gaps.

Theorem 28.1. *Let $A = \{A_1, A_2, A_3\}$ and $B = \{B_1, B_2, B_3\}$. Take the complete core $A \times B$, together with $n \geq 3$ exterior-left vertices L , each adjacent to all of B . Let every exterior-right vertex in R have neighborhood in the chain*

$$\{A_1\} \subset \{A_1, A_2\} \subset A.$$

Assume A can be matched into R . Give all vertices of $A \cup L$ activity one and the vertices of $B \cup R$ arbitrary positive activities. Then P_G is ULC_6 .

If the three exterior class sizes are m_1, m_2, m_3 , the rank assumption is exactly $m_3 \geq 1$, $m_2 + m_3 \geq 2$, and $m_1 + m_2 + m_3 \geq 3$. The complete-core hypothesis is part of this theorem. The earlier common-exterior result allowed an arbitrary core, but that larger core scope is not claimed here. [Merge note: The earlier result is source 69 [23], Theorem 27.1; neither family contains the other. As for source 69, gaps two and five of Theorem 28.1 also follow from sources 05 and 13, and its new content is gaps three and four for this family.]

For $I \subseteq A$ and a selected exterior multiset with class counts $c = (c_1, c_2, c_3)$, define $H(c, I)$ by

$$c_1 \leq |I \cap \{A_1\}|, \quad c_1 + c_2 \leq |I \cap \{A_1, A_2\}|, \quad |c| \leq |I|. \quad (150)$$

These are precisely the nested Hall conditions for matching those right vertices into I . Fix a subset $J \subseteq B$ of size j and an actual set of exterior-right vertices with counts c . Its coefficient as a monomial in the individual right activities is

$$T_{j,c}(n) = \sum_{\substack{I \subseteq A: H(c, I) \\ 0 \leq j + |c| - |I| \leq j}} \binom{n}{j + |c| - |I|}. \quad (151)$$

Indeed, the selected exterior-left set has size $j + |c| - |I|$. Once the selected exterior-right vertices are matched, completeness of the core and of $L \times B$ completes the support exactly when the displayed bounds hold. This counts endpoint sets rather than their realizing matchings.

Delete the core edges and call the resulting polynomial Q . Its coefficient $T_{j,c}^0$ is obtained by keeping only $|I| = |c|$ in (151). The graph for Q is a disjoint union of two graphs with shores of size three. The standard transversal-matroid basis specialization makes both factors ULC_3 ; their degree-three Lorentzian homogenizations have a Lorentzian product. Thus Q is ULC_6 [1, 12].

Lemma 28.2 (Finite coefficient certificate). *For $k = 2, 3, 4, 5$, every coefficient of $\Delta_k^{(6)}(P) - \Delta_k^{(6)}(Q)$, expanded in the individual right activities and in $z = n - 3$, is nonnegative.*

Here is the complete finite reduction used by the certificate. A right-activity monomial has core exponent vector $v \in \{0, 1, 2\}^3$. For each of the three exterior classes let o_i count distinct variables occurring once and d_i count variables occurring twice. Thus

$$|v| + |o| + 2|d| = 2k, \quad |d| \leq 3, \quad |o| + 2|d| \leq 6.$$

Put $a = \#\{i : v_i = 2\}$ and $b = \#\{i : v_i = 1\}$. Distribute the singly occurring exterior variables by a vector e with $0 \leq e_i \leq o_i$, and the b core singleton variables by an integer $h \in [0, b]$. The two factors have exterior counts $d + e, d + o - e$ and core counts $a + h, a + b - h$, with multiplicity

$$\binom{b}{h} \prod_{i=1}^3 \binom{o_i}{e_i}.$$

Use (151) for the two support coefficients, and retain the distributions of first-factor degree k with multiplier $k(6-k)$ and those of degree $k-1$ with multiplier $-(k+1)(7-k)$. Subtract the corresponding T^0 product. This gives the coefficient in Lemma 28.2 with no activity specialization.

The file `data/exact_certificate.json` contains all 3,391 coefficient vectors, scaled by 36 and listed in ascending powers of z . They are integer vectors because each T is a sum of $\binom{z+3}{h}$ with $h \leq 3$. Every entry is nonnegative; the numbers of nonzero vectors at gaps 2, 3, 4, 5 are respectively 132, 495, 532, 152. The standard-library verifier reconstructs all vectors from (150)–(151), compares them to the stored certificate, and checks their signs. An independent literal matching enumeration checks 80 small weighted graphs.

Proof of Theorem 28.1. Lemma 28.2 and ULC_6 of Q prove gaps two through five. The first gap holds for every positive weighted bipartite support polynomial; here it gives $5p_1^2 \geq 12p_2$. [*Merge note:* This is *preorder-root-polytopes*'s Theorem 180.2; source 63's one-sentence Motzkin–Straus justification (edge-disjointness graph, clique number the matching rank, two-matchings at least size-two supports) is the route printed in Section 11.]

The cover $A \cup B$ has size six. The assumed three-matching $A \rightarrow R$ and a disjoint three-matching $L \rightarrow B$ attain that size. The degree is therefore six, as required. $\square$

28.2 An obstruction to the full marginal approach

With unit activities on X , define the homogeneous right marginal of degree six by

$$F_G(s, z) = \sum_{\substack{J \subseteq Y \\ |J| \leq 6}} \#\{I \subseteq X : (I, J) \text{ feasible}\} z_J s^{6-|J|}.$$

If F_G were Lorentzian, all positive one-shore specializations would be ULC_6 . The converse is not asserted, and the stronger property already fails in the irreducible $3+3$ class.

Proposition 28.3. *There is a 20-vertex graph of matching rank six with a unique minimum vertex cover, of split $3+3$, for which F_G is not Lorentzian.*

[*Merge note:* Added 1 October 2026, batch 73: three-tail obstructions of another kind, on a thirteen-vertex graph, are in source 80 (Part V, Sections 43.3.2 and 43.4).]

Proof. Use three-bit masks 1, 2, 4 for the first, second and third core vertices. The left-core neighborhoods in B are $(2, 0, 5)$. The seven exterior-left neighborhoods in B are

$$(3, 6, 5, 4, 3, 1, 4),$$

and the seven exterior-right neighborhoods in A are

$$(2, 5, 5, 5, 1, 4, 7).$$

There are no other edges. Each exterior block satisfies $|N(S)| > |S|$ for every nonempty subset of its three core vertices, so it has a three-matching. Also, omitting any nonempty set of core vertices from a cover requires more exterior vertices than were omitted. Thus $A \cup B$ is the unique minimum cover, and the matching rank is six.

Differentiate F_G in all three B variables and the variable of the first exterior-right vertex, whose neighborhood is $\{A_2\}$. This forces A_2 into the support. Of the six remaining right variables, four are then common neighbors of A_1, A_3 , and two are private neighbors, one at each. Their support coefficients are exactly

$$\text{constant } 64, \quad \text{each common singleton } 74, \quad \text{each private singleton } 42, \quad \text{each pair } 27.$$

These integers can be obtained by enumerating left subsets of sizes four, five and six and applying Hall's criterion. The accompanying verifier independently obtains them by dynamic enumeration of matchable left endpoint sets; it also enumerates all size-six covers.

Set the four common variables equal to x and both private variables equal to y . The resulting quadratic is

$$64s^2 + (296x + 84y)s + 27(6x^2 + 8xy + y^2). \quad (152)$$

At $x = t, y = 19t$, this becomes

$$64s^2 + 1892st + 14013t^2, \quad 1892^2 - 4 \cdot 64 \cdot 14013 = -7664 < 0.$$

A nonnegative bivariate quadratic is Lorentzian exactly when its middle coefficient squared is at least four times the product of its endpoint coefficients. Derivatives and nonnegative linear substitutions preserve Lorentzian polynomials [1]. Hence F_G cannot be Lorentzian. $\square$

This is not a scalar ULC_6 counterexample. A quadratic sector forced to degrees 4, 5, 6 needs the weaker ratio $12/5$, whereas the displayed sector's ratio is close to four. The result specifically rules out universal full-marginal Lorentzianity as a proof route.

28.3 Every four right vertex sector is safe

Theorem 28.4. *Let G have matching rank six and a displayed minimum cover $A \cup B$ with $|A| = |B| = 3$. Give all left vertices activity one and all right vertices arbitrary positive activities. Force four designated right vertices by scaling their activities together and taking the leading coefficient of degree four in the scaling parameter. Suppose the resulting sector is*

$$t^4(c_0 + c_1t + c_2t^2), \quad c_0c_2 > 0.$$

Divide out the fixed positive product of the four forced right activities. Then $c_1^2 \geq 3c_0c_2$. Thus its last rank-six Newton gap is strictly positive, since the required constant is $12/5$.

The theorem concerns these conditioned sectors. It does not assert ULC at arbitrary finite activities, and it does not address cubic sectors obtained by forcing only three right vertices.

28.3.1 Hall tight subfamilies

If the forced set contains S with $T = N(S)$, $|T| = |S| = h \geq 1$, and a perfect matching between T and S , every support containing S contains all of T , matched to S . Deleting both endpoint sets gives a coefficient-preserving correspondence apart from the fixed factor $u_T v_S t^h$.

The remaining graph has rank $6 - h$. Indeed, replacing all edges incident to T in a maximum matching by a matching from T to S cannot decrease size. The weighted rank-at-most-five theorem applies to the remaining graph [12]. [*Merge note:* Corollary 2.2.] Force the other $4 - h$ right vertices and take a coefficientwise limit at this fixed normalization. Its last gap gives

$$c_1^2 \geq \frac{2(6-h)}{5-h} c_0 c_2 \geq \frac{5}{2} c_0 c_2, \quad 1 \leq h \leq 4. \quad (153)$$

This includes a forced exterior vertex with a singleton neighborhood, two forced exterior vertices with a two-vertex combined neighborhood, and any three matchable exterior-right vertices.

28.3.2 Two core vertices and two exterior vertices

This case permits arbitrary positive activities on both shores. Let $J \subseteq B$ be the forced pair, and let z be the missing- B activity. Let E be the sum of the activity products of the A -pairs that match the two forced exterior-right vertices, and put $a = \prod_{i=1}^3 u_{A_i}$. Let S be the union of the core neighborhoods of the possible unmatched A rows, namely those whose complementary A -pair matches the two forced exterior vertices. Define weighted exterior-left counts

$$\begin{aligned} b &= \#\{L\text{-pairs matching } J\}, & c &= \#\{L\text{-triples matching } B\}, \\ V &= \#\{L\text{-singletons completing } J \text{ with row } S\}, & U &= \#\{L\text{-pairs completing } B \text{ with row } S\}. \end{aligned}$$

Here each count includes the product of the selected left activities. Let W be the total activity of remaining exterior-right vertices that complete the forced exterior pair to a matching of all of A . Partitioning supports by whether they select two or three vertices of A gives

$$c_0 = Eb + aV, \quad c_1 = (Ec + aU)z + abW, \quad c_2 = aczW.$$

In the three- A case the common product a makes the virtual union row exact: an endpoint support is counted once even if several unmatched rows work. Eligibility in W depends only on the exterior-right triple.

Take the rank-three transversal matroid on the exterior-left elements, the virtual row S , and a private element at the missing- B slot. After the exterior-left activity substitution, its basis polynomial in the last two variables is $c + Ux + by + Vxy$. Every rank-three matroid is Rayleigh [19, Theorem 1.1], so $bU \geq cV$. Therefore

$$c_1^2 - 4c_0c_2 = [(Ec + aU)z - abW]^2 + 4a^2zW(bU - cV) \geq 0. \quad (154)$$

This stronger bound settles the two-core/two-exterior case.

28.3.3 A useful quadratic inequality

Suppose remaining right elements have parallel-class activity totals $x_1, \dots, x_m$ and a total g of other elements, each forming an independent pair with every other distinct element. Put $w = g + \sum x_i$. Suppose their linear costs are $q - \delta_i$ on the parallel classes and q on the other elements, where $\delta_i \geq 0$ and $\sum \delta_i \leq q$. If a pair of independent right elements has coefficient $T > 0$, then

$$c_1 = qw - \sum_i \delta_i x_i, \quad c_2 \leq \frac{T}{2} \left(w^2 - \sum_i x_i^2 \right).$$

The missing term in the second formula is the nonnegative sum of squared individual activities in the generic group. Set $D = q^2 - \sum_i \delta_i^2$. Ordinary Cauchy–Schwarz, or the reverse inequality for the quadratic form $z_0^2 - \sum z_i^2$, gives

$$c_1^2 \geq D \left(w^2 - \sum_i x_i^2 \right) \geq \frac{2D}{T} c_2. \quad (155)$$

For a direct check, replace $\sum \delta_i x_i$ by $\|\delta\| \|x\|$. Subtracting $D(w^2 - \|x\|^2)$ from $(qw - \|\delta\| \|x\|)^2$ leaves $(q\|x\| - \|\delta\|w)^2$.

28.3.4 All three core vertices and a degree three exterior vertex

All left activities are now one. Let the forced exterior vertex have neighborhood A . Remove isolated exterior-left vertices and let N be the remaining number. Write T for the number

of exterior-left triples matching B ; $T \geq 1$. For each core row A_i , let $\mathcal{P}_i$ be the family of exterior-left pairs completed to B by A_i . Put

$$W = |\mathcal{P}_1 \cup \mathcal{P}_2 \cup \mathcal{P}_3|, \quad a_i = |\mathcal{P}_i \setminus (\mathcal{P}_j \cup \mathcal{P}_k)|, \quad A_0 = \sum_i a_i \leq W.$$

Let V count exterior-left singletons completed to B by some two core rows. Exact endpoint counting gives

$$C := c_0 = 3T + 3W - A_0 + V.$$

The three singleton-neighborhood classes of remaining right vertices have costs $2T + W - a_i$. Every right vertex with neighborhood size at least two has cost $q = 3T + W$. Any pair completes the forced exterior vertex to a matching of A , except two vertices in the same singleton class. Thus (155) applies with $\delta_i = T + a_i$, since $\sum \delta_i \leq q$.

We show $D \geq (3/2)CT$. The exterior-left rank-three matroid has N nonloops. A fixed basis and one distinct replacement basis for every other element show $T \geq N - 2$. Since $V \leq N$, we have $V \leq T + 2$. Expanding and using $A_0 \leq W$ yields

$$\begin{aligned} D - \frac{3}{2}CT &= \frac{3}{2}T^2 + \frac{3}{2}TW - \frac{1}{2}TA_0 + W^2 - \sum_i a_i^2 - \frac{3}{2}TV \\ &\geq T(W - 3) + W^2 - \sum_i a_i^2 \geq 0 \quad (W \geq 3). \end{aligned}$$

For the small- W boundary, represent the transversal matroid generically and let F be the span of the three core-row vectors. If $\dim F \leq 1$, then $V = 0$, and

$$D - 2CT = W^2 - \sum_i a_i^2 - 2TV \geq 0.$$

If $\dim F \geq 2$, choose a generic vector $e \in F$, avoiding all N exterior-left directions. Contraction by e gives a rank-two matroid on these N nonloops, with exactly W bases: a pair extends e exactly when at least one core row extends it. Hence $W \geq N - 1$. If $W \leq 2$, necessarily $N = 3, T = 1, W = 2$. The sole exterior-left pair not extending e spans F , which is two-dimensional. Exactly one exterior-left vector is outside F , so $V = 1$. Consequently

$$D - \frac{3}{2}CT = 7 - A_0/2 - \sum_i a_i^2 \geq 7 - 1 - 4 > 0.$$

Together with (155), this proves $c_1^2 \geq 3c_0c_2$.

28.3.5 All three core vertices and a lower degree exterior vertex

Relabel its neighborhood as $\{A_1, A_2\}$. Let U count exterior-left pairs completed by a virtual row with neighborhood $N(A_1) \cup N(A_2)$, let P count those completed by A_3 , and let W count those completed by at least one of these two rows. Put

$$O = U + P - W, \quad u = U - O, \quad p = P - O.$$

Let V count exterior-left singletons completed by those two rows. With T as before,

$$C = c_0 = 2T + U + 2P + V = 2T + 3W - 2u - p + V.$$

The remaining right vertices have two parallel classes: neighborhoods contained in $\{A_1, A_2\}$, of cost $T + P$, and neighborhood $\{A_3\}$, of cost $2T + U$. All other nonempty neighborhoods have cost $q = 3T + W$. Apply (155) with $\delta_1 = 2T + u$, $\delta_2 = T + p$. Expansion gives

$$D = 2CT + (2UP - O^2) - 2TV.$$

The previously proved sharp rank-three two-row inequality is

$$4UP - 2O^2 \geq 3TV;$$

apply it to the virtual union row and A_3 [12]. [*Merge note:* Source 74's inequality (9) (Theorem 9.2).] This inequality allows arbitrary positive exterior-left activities. It gives

$$D \geq 2CT - TV/2 \geq \frac{3}{2}CT,$$

because $C \geq V$. Again $c_1^2 \geq 3c_0c_2$ follows.

If the forced exterior vertex has degree one, remove it and its forced left neighbor. Two unit core-left rows remain. With the same two-row counts T, U, P, V, O , put $W = U + P - O$, $u = U - O$, and $p = P - O$. This time

$$C = T + U + P + V, \quad q = 2T + W, \quad \delta_1 = T + u, \quad \delta_2 = T + p.$$

The exact axis costs are $T + P$ and $T + U$, and the generic cost is q . The same expansion again gives

$$D = 2CT + (2UP - O^2) - 2TV \geq \frac{3}{2}CT.$$

Thus the constant three holds in this case as well.

Completion of Theorem 28.4. A size-six matching uses all of $A \cup B$ and no core edge. Four feasible forced right vertices contain one, two or three exterior-right vertices. Three are Hall-tight with $h = 3$, giving constant three in (153); two are handled by (154); one has degree one, two or three and is covered by the corresponding argument above. These cases exhaust the claim. $\square$

Verification. Run `python3 checks/run_all.py`. [*Merge note:* Shipped as `code/63-nested-exterior-run_all.py`; `data/exact_certificate.json` above is `data/63-nested-exterior-exact_certificate.json`.] The package uses only Python's standard library and contains the complete nested certificate, independent endpoint checks, and an exact verification of Proposition 28.3, and weighted checks of (154). The proof of Theorem 28.4 is analytic; the finite experiments are not its justification. It contains no numerical searches or internal review reports. These are computer-assisted mathematical proofs, not formal theorem-prover verification or claims of exhaustive priority.

Part IV

Eventual one-shore scaling through matching number eight

Contents of Part IV. All sources of Part IV keep every left activity equal to one, fix positive right activities, fix a minimum cover $A \sqcup B$, and multiply the activities of its *right* part B by a common factor λ ; their conclusions hold for all sufficiently large λ , with a threshold depending on the graph, the cover and the field. Forcing B gives a conditional polynomial $C(t)$ whose Newton gaps, shifted to the actual matching number, decide the leading coefficients. Source 08s (Section 29), delivered only inside archive 08, is the base: a top conditional gap for every $a + b$ cover, a two-plane incidence lemma, eventual ULC for $3 + 3$, $3 + 4$ and $3 + 5$ covers, and a reduction of rank seven to one middle gap. Source 21 (Section 30) is its matching-number-six case with further detail; sources 43, 38, 35, 32 and 29 (Sections 31–35) prove the same rank-six theorem for special cores and are printed as second routes. Source 08 (Section 36) settles the rank-seven middle gap, source 03 (Section 37) the $5 + 3$ covers, source 02 (Section 38) the $4 + 4$ top gap and source 01 (Section 39) the $4 + 4$ middle gap, so that every graph with $\nu \leq 8$ is eventually ULC_ν under this scaling.

The two scalings and the two brackets. Part I's failures at every matching number ≥ 6 (Section 6) scale the whole minimum cover, on both shores; Part IV scales its right part only, and there no failure occurs through matching number eight. The least matching number of a *persistent* failure under right-cover scaling lies in $[9, 38]$ (source 01; the upper end is source 55's $H_{8,30;32,794}$); sources 21 and 08s give $[7, 38]$ and source 08 $[8, 38]$, kept as history. The least matching number of a one-shore failure at some *finite* activity lies in $[6, 38]$; its lower end rests on Part I.

Same results in several sources. Source 21's Theorem 30.1 is the case $a = b = 3$ of source 08s's Theorem 29.1 (with source 08s's constant $75/28$ equal to source 21's refinement), its incidence lemma is the case $b = 3$ of source 08s's Lemma 29.4 (printed with proof in source 13's section, Part III), and its Corollary 30.4 is the case $b = 3$ of source 08s's Corollary 29.2. The eventual theorems of sources 43, 38, 35, 32 and 29 are cases of source 21's; their proofs are printed as second routes, with what each adds. Source 08s's weighted local equality lemma is the necessity half of source 72's contraction criterion (Part I); its appendix proof is a pointer. The every-field branches of sources 38, 35, 32 and 29 (a coloop, or a single exterior-left neighbour) are corollaries of sources 74 and 72.

Non-claims of Part IV. No source of Part IV proves ULC at every finite activity at matching numbers six to eight, or a uniform threshold over graphs or fields; source 08s proves no middle-gap bound and no counterexample (source 08 later proves the bound); source 03 and source 02 do not claim a complete rank-eight theorem; source 02's unrestricted upper incidence bound is false (its Lemma 38.2); source 08s's numerical search is discovery history, not evidence for a uniform inequality.

29 Top conditional gaps and the rank-seven reduction, one weighted shore, right-cover scaling (source 08s)

Abstract of source 08s, as delivered. We prove a uniform top conditional Newton inequality for a bipartite graph with an $a + b$ minimum cover, unit left activities and arbitrary positive right activities. Its constant is $a(b + 2)^2 / ((a - 1)(b^2 + b + 2))$. A two-plane incidence inequality in rank b supplies the geometric input; elementary extension counts are the only use of unit left activities. The result proves eventual strict ultra-log-concavity for all $3 + 4$ and $3 + 5$ covers. At rank seven it also excludes the top gap of a $4 + 3$ cover. A separate weighted equality argument handles the six-exterior-left boundary. Any eventual rank-seven failure must consequently occur at the middle conditional quartic gap, with at least nineteen nonisolated vertices. No such counterexample or uniform middle-gap bound is proved here. [Merge note: Source 08s was delivered only inside archive 08 (`Rank7_Eventual_Report/companion/structural_companion_source.zip`); it is a manuscript of its own and the base of Part IV. Its uniform middle bound, which the abstract says is not proved here, is source 08's Theorem 36.2.]

29.1 Coefficients and conclusions

For a bipartite graph with shores X, Y and positive vertex activities, define

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|}.$$

A feasible endpoint pair is counted once, irrespective of its number of perfect matchings. For matching rank d , write $P_G = \sum p_j t^j$ and

$$G_j = j(d - j)p_j^2 - (j + 1)(d - j + 1)p_{j-1}p_{j+1}. \quad (156)$$

The condition $G_j \geq 0$ at every $1 \leq j < d$, together with no internal zeros, is denoted ULC_d .

Assume

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = a, \quad |B| = b, \quad \nu(G) = a + b,$$

where $A \sqcup B$ is a vertex cover and $a, b \geq 2$. There are no L – R edges. Every maximum matching avoids core edges and gives exterior matchings of sizes b and a . Remove isolated vertices and put $n = |L| \geq b$.

Give every left vertex activity one and every right vertex an arbitrary positive activity. Force all vertices of B into the selected right endpoint set and divide out $v_B = \prod_{y \in B} v_y$. The conditional polynomial is

$$t^b C(t), \quad C(t) = \sum_{j=0}^a C_j t^j, \quad (157)$$

with all $C_j > 0$. In *common cover scaling*, all original B activities are multiplied by the same parameter $\lambda > 0$.

Theorem 29.1. *Under these hypotheses,*

$$C_{a-1}^2 \geq \frac{a(b + 2)^2}{(a - 1)(b^2 + b + 2)} C_{a-2} C_a. \quad (158)$$

No assumption on the core matrix's rank or support is needed.

[Merge note: At $a = b = 3$ the constant is $75/28$, which is source 21’s refinement (Remark after Theorem 30.1); source 21’s stated constant there is $21/8$.]

Corollary 29.2. *For $a = 3$ and $b \in \{3, 4, 5\}$, P_G is strictly ULC_{b+3} for all sufficiently large common B scalings. The threshold may depend on the graph and the fixed right field. In particular this holds for every genuine $3 + 4$ and $3 + 5$ cover.*

[Merge note: At $b = 3$ this is source 21’s Corollary 30.4, and so it contains the eventual theorems of sources 43, 38, 35, 32 and 29 for special cores (Sections 31–35). The scaling multiplies the activities of the *right* part B of the cover only; see the entry “Scalings” of Section 1.3.]

Theorem 29.3. *For a rank-seven graph and a specified minimum cover, an eventual failure under common right-cover scaling can occur only for a $4 + 3$ cover. In that case all leading gaps except the middle conditional one are strictly positive. The unresolved middle expression is*

$$5C_2^2 - 9C_1C_3. \quad (159)$$

A strictly negative value is sufficient for eventual failure; if it is zero, lower powers of λ must still be examined.

Every eventual rank-seven counterexample has $|L| \geq 7$, $|R| \geq 5$, and an R vertex with at least two A neighbors. Hence it has at least nineteen nonisolated vertices. The six-exterior-left equality boundary cannot give a counterexample.

[Merge note: The middle expression is settled by source 08 (Theorem 36.2): $C_2^2 \geq \frac{15}{8}C_1C_3 > \frac{9}{5}C_1C_3$ for every genuine $4 + 3$ cover. Hence there is no eventual rank-seven failure (source 08’s Theorem 36.1).]

The cover splits with one shore of size at most two are covered by the previously established small-cover theorem [12, Theorem 1.1 and Section 2] [Merge note: source 74’s Theorem 2.1 and Section 3]; the present argument addresses the new $3 + 4$ and $4 + 3$ cases. We use standard Lorentzian facts: matroid basis polynomials are Lorentzian, and nonnegative substitutions, directional derivatives and coefficientwise limits preserve this property. Their quadratic Hessians have at most one positive eigenvalue [1].

29.2 A two-plane incidence inequality

Lemma 29.4. *Let E be a rank- b represented configuration over an infinite field, with positive element weights, and let H be a two-dimensional subspace of its ambient row space. Let q be the weighted basis count and β the weighted count of $(b - 2)$ -subsets complementary to H . Let h count independent $(b - 1)$ -subsets P whose hyperplanes do not contain H . Write h_δ for their weights grouped by the line $\delta = \text{span}(P) \cap H$. Then*

$$h^2 - \sum_{\delta} h_{\delta}^2 \geq \frac{b}{b-1} q\beta. \quad (160)$$

Repeated and parallel elements are permitted and remain separately labeled.

[Merge note: At $b = 3$ this is source 21’s Lemma 30.2, printed with source 13’s proof as (129), and the lower bound of source 08’s Lemma 36.3; at $b = 4$ it is the lower incidence bound that sources 02 and 01 use.]

Proof. The case $\beta = 0$ is immediate. Otherwise adjoin one vector on every occurring line δ , with variable x_δ , and many generic vectors in H of total activity y , each of weight y/N . Mark

original elements by t . Passing to the coefficientwise limit $N \rightarrow \infty$ gives the Lorentzian basis polynomial

$$F = qt^b + hyt^{b-1} + \frac{\beta}{2}y^2t^{b-2} + \sum_{\delta} x_{\delta}((h - h_{\delta})t^{b-1} + \beta yt^{b-2}) + \beta t^{b-2} \sum_{\delta < \eta} x_{\delta}x_{\eta}. \quad (161)$$

Three added vectors are dependent because all lie in H . Two on distinct lines form a basis with exactly the original $(b-2)$ -sets complementary to H . A fixed added line completes precisely those original hyperplanes whose intersection line is different. These facts give every coefficient in (161). Generic additions may be made over a transcendental extension.

Apply $\partial_t^{b-2}/(b-2)!$. The resulting Hessian has entries

$$tt = b(b-1)q, \quad ty = (b-1)h, \quad t\delta = (b-1)(h - h_{\delta}),$$

$$yy = \beta, \quad y\delta = \beta, \quad \delta\delta = 0, \quad \delta\eta = \beta \quad (\delta \neq \eta).$$

Eliminating the positive yy pivot leaves line block $-\beta I$, cross entries $-(b-1)h_{\delta}$, and tt entry $b(b-1)q - (b-1)^2h^2/\beta$. Its remaining Schur complement is nonpositive, so

$$b(b-1)q - \frac{(b-1)^2}{\beta}h^2 + \frac{(b-1)^2}{\beta} \sum_{\delta} h_{\delta}^2 \leq 0.$$

This is (160). □

29.3 The exact top sectors

Represent each graph edge by an algebraically independent nonzero entry. A square minor is nonzero exactly when its endpoints are matchable, since distinct matchings yield distinct determinant monomials. Write K for the $a \times b$ core matrix, ℓ_i for the exterior-left rows and r_j for the exterior-right columns. They have exterior ranks b and a respectively.

Let q be the unweighted number of independent b -subsets of L . For every independent $(b-1)$ -subset P , choose a nonzero annihilating column v_P and introduce the virtual right vector

$$w_P = Kv_P.$$

These virtual elements remain separately labeled, including loops and parallel vectors. Form the rank- a matroid on R , the virtual elements, and the private coordinate basis E in the A space. Mark these three classes by t, z, s respectively, keeping the original positive R activities. Define the following weighted basis counts:

$$\begin{array}{ll} T : a \text{ elements of } R, & A_1 : (a-1)R + 1E, \\ U : (a-2)R + 2E, & P_B : 1 \text{ virtual} + (a-1)R, \\ P_V : 1 \text{ virtual} + (a-2)R + 1E, & D : 2 \text{ virtual} + (a-2)R. \end{array}$$

Pairs of virtual elements in D are unordered.

For an independent labeled $(a-2)$ -subset $J \subseteq R$, let β_J count $(b-2)$ -subsets $I \subseteq L$ for which the square block

$$\begin{pmatrix} K & R_J \\ L_I & 0 \end{pmatrix} \quad (162)$$

is nonsingular. Put $N = \sum_J v_J \beta_J$. Dependent J contribute zero and can be omitted.

Partitioning endpoint sets by their selected L cardinality gives exactly

$$C_{a-2} = qU + P_V + N, \quad C_{a-1} = qA_1 + P_B, \quad C_a = qT. \quad (163)$$

At the first of these degrees, the possible L sizes are $b, b-1, b-2$; at the next they are $b, b-1$; at the last only b occurs. Eliminating the chosen L rows gives the stated virtual column. Complementary private coordinates record the selected A rows. Thus the partition counts endpoint sets once, with no allocation multiplicity.

Differentiate the right basis polynomial $a-2$ times in t and divide by $(a-2)!$. Its Hessian in (t, s, z) is

$$\begin{pmatrix} a(a-1)T & (a-1)A_1 & (a-1)P_B \\ (a-1)A_1 & 2U & P_V \\ (a-1)P_B & P_V & 2D \end{pmatrix}. \quad (164)$$

The positive tt pivot and the Schur inequality on $(s, z) = (q, 1)$ imply

$$C_{a-1}^2 \geq \frac{2a}{a-1} T(q(C_{a-2} - N) + D). \quad (165)$$

29.4 Quotient geometry and the unit counts

Fix an independent $(a-2)$ -set $J \subseteq R$ and quotient the right space by $\text{span}(R_J)$. The induced core map is a $2 \times b$ matrix K_J . If its rank is below two, both β_J and the corresponding contribution D_J to D vanish.

Otherwise $H = \text{row}(K_J)$ is a two-plane in the b -dimensional left row space. Row reduction along R_J identifies (162) with the condition that L_I be complementary to H . Moreover $K_J v_P = 0$ exactly when $H \subseteq \text{span}(P)$. For a surviving P , restriction of its annihilator to H has kernel $H \cap \text{span}(P)$. Consequently two virtual images are independent in the quotient exactly when their intersection lines differ. Lemma 29.4 gives

$$2D_J = h^2 - \sum_{\delta} h_{\delta}^2 \geq \frac{b}{b-1} q\beta_J.$$

The factor two is essential: the left side counts ordered virtual pairs, while D_J counts unordered pairs. Summing over J with its product activity proves

$$D \geq \frac{b}{2(b-1)} qN. \quad (166)$$

This treats every core rank, including singular cores.

Now use unit left activities. Every complementary $(b-2)$ -set extends to an original basis, which contains at most $\binom{b}{2}$ such subsets. It also has at least two distinct one-element extensions to independent $(b-1)$ -sets. Every such extension belongs to h : a hyperplane containing both the set and H would contain their full b -dimensional sum. Each hyperplane subset contains at most $b-1$ of the original complementary sets. Therefore

$$q \geq \frac{2\beta_J}{b(b-1)}, \quad h \geq \frac{2\beta_J}{b-1}. \quad (167)$$

For fixed J , its coefficient in U is at least one because private coordinates complete it to a basis. Every surviving virtual vector is completed with J by at least one private coordinate, so its coefficient in P_V is at least h . From (163) and (167),

$$C_{a-2} \geq \left(1 + \frac{2(b+1)}{b(b-1)}\right) N = \frac{b^2 + b + 2}{b(b-1)} N. \quad (168)$$

Combining (165), (166) and (168) gives

$$\begin{aligned} C_{a-1}^2 &\geq \frac{2a}{a-1} qT \left(C_{a-2} - \frac{b-2}{2(b-1)} N \right) \\ &\geq \frac{a(b+2)^2}{(a-1)(b^2+b+2)} C_{a-2} C_a, \end{aligned}$$

proving Theorem 29.1. The incidence inequality itself allows weighted left elements; (167) is where the unit-field hypothesis enters.

29.5 Scaling and the cubic consequences

We record two standard matroid specializations to fix the normalization. For a graph with exactly b vertices on the right, take the transversal matroid matching left vertices and one private element per right vertex into those b slots. Give original left elements activity tu_i and private element j activity s/v_j , then multiply by $\prod_j v_j$. The rank- b basis polynomial becomes

$$\sum_{j=0}^b b_j t^j s^{b-j},$$

where b_j are exactly its endpoint-support coefficients. Thus it is ULC_b ; if the graph has rank b , every coefficient is positive.

For the conditional polynomial (157), instead use the extended transversal matroid in the $n+a$ left coordinates and differentiate in all B elements. Specialization gives

$$s^{n-b} \sum_{j=0}^a C_j s^{a-j} t^j, \quad (169)$$

Lorentzian of degree $d_0 = n + a - b$. In particular

$$C_1^2 \geq \frac{2d_0}{d_0-1} C_0 C_2 > 2C_0 C_2. \quad (170)$$

This ambient estimate permits arbitrary positive left activities as well. [*Merge note:* At $a = b = 3$ it is source 21's ambient lower gap (175).]

Under common B scaling, the leading coefficient of $p_j(\lambda)$ for $j \leq b$ is $b_j \lambda^j$, where b_j comes from the graph restricted to right shore B . Its rank is b by the exterior matching. For $j \geq b$, the leading term is $v_B C_{j-b} \lambda^b$. At $j < b$, the required actual-rank factor

$$\frac{(j+1)(a+b-j+1)}{j(a+b-j)}$$

is strictly smaller than the pure-shore factor $(j+1)(b-j+1)/(j(b-j))$. The leading gaps are therefore strictly positive. At $j = b$, the square has degree $2b$ and the neighboring product degree at most $2b-1$.

For $a = 3$, the two remaining required factors are

$$\frac{3(b+2)}{2(b+1)}, \quad \frac{2(b+3)}{b+2}.$$

The first is below two for $b \geq 3$, and the top theorem gives constants

$$\frac{75}{28} \ (b=3), \quad \frac{27}{11} \ (b=4), \quad \frac{147}{64} \ (b=5).$$

At $b = 4, 5$ their margins above the required factors are respectively $4/33$ and $5/448$. More generally the sign of the top margin is the sign of $b(-b^2 + 2b + 16)$, positive for $2 \leq b \leq 5$. Every gap polynomial has positive leading coefficient. Hence all are positive for sufficiently large λ , proving Corollary 29.2. This is a field-dependent eventual statement, not a uniform numerical cutoff.

29.6 The rank seven boundary

For $a = 4, b = 3$, (170) exceeds the required first conditional factor $5/3$. The top theorem gives

$$C_3^2 \geq \frac{50}{21}C_2C_4 > \frac{7}{3}C_2C_4,$$

with margin $1/21$. Only (159) remains. The degree $d_0 = n + 1$ ambient bound at index two is

$$C_2^2 \geq \frac{3n}{2(n-1)}C_1C_3. \quad (171)$$

It is strictly stronger than $9/5$ for $n \leq 5$, and equal to that constant at $n = 6$. A zero leading gap alone would not settle the eventual sign, so this boundary requires its own argument.

We use the following weighted equality implication. Its unweighted form is the local characterization of Chan and Pak [3, Theorem 1.3]; a proof of the weighted form needed here is included in the appendix.

Lemma 29.5. *Let a positively weighted matroid have rank d , with red set R and blue complement. Suppose its grouped normalized basis counts attain equality at index j , with positive neighboring counts. Then there is a common $\lambda > 0$ such that every independent $(d - 2)$ -set I containing $j - 1$ red elements has the following property: in every nonloop parallel class of M/I , red weight is λ times blue weight. In particular, a monochromatic nonloop parallel class in one such contraction forces strictness for every positive field.*

[Merge note: Lemma 29.5 is the necessity half of source 72's weighted contraction criterion, Theorem 14.3 (Part I), with the same proof; see the note in the appendix below.]

Suppose $n = 6$. Contract the three B columns in the extended transversal matroid on R and the private left coordinates. This has rank seven; color R red and all private elements blue. Its grouped polynomial is

$$C_0s^7 + C_1s^6t + C_2s^5t^2 + C_3s^4t^3 + C_4s^3t^4.$$

Choose three L rows forming an exterior basis. All four private A coordinates and the three complementary private L coordinates form a blue basis after contracting B .

If a red vertex u has at least two A neighbors, extend it to an R basis and choose a perfect matching of that basis to A . Name its assigned coordinate i , and choose another neighbor j . Choose one of the other two red basis elements assigned outside i, j , say u_h . Contract u_h , the private A coordinate at the remaining fourth position, and the three chosen private L coordinates. This independent five-set has one red and four blue elements. The images of the uncontracted coordinates A_i, A_j remain independent: the contracted red element has a nonzero generic entry at its assigned coordinate h .

The contraction and every blue vector are independent of all generic entries of u . Its two entries at i, j are independent parameters multiplying independent quotient coordinates. For any nonloop blue vector w , the determinant of u, w is a nonzero polynomial in those parameters, because its two coefficients cannot both vanish. Thus u belongs to a red-only nonloop parallel class. Lemma 29.5 proves strictness in (171) when $n = 6$.

If every R vertex instead has a singleton neighborhood, compress each nonempty A -leaf class to one right vertex of summed activity. This preserves every endpoint coefficient, since at most one vertex of such a class can be selected. Exterior rank four guarantees all four classes are nonempty. The compressed graph has exactly seven right vertices and matching rank seven. The pure-shore theorem proves ULC_7 at every field and every common scaling. [*Merge note:* The compression is *preorder-root-polytopes*'s Lemma 206.1 and the pure-shore theorem its Theorem 70.1 (Lemma 3.1 here).]

It follows that a rank-seven eventual failure requires a $4+3$ cover, $n \geq 7$ and a multi-neighbor R vertex. Also $|R| \geq 5$: otherwise the full right shore has at most seven vertices and the same pure-shore theorem applies. There must therefore be at least $(4+7) + (3+5) = 19$ nonisolated vertices. Together with the previously proved small-cover case, this proves Theorem 29.3.

29.7 Appendix: The weighted local equality implication

[*Merge note:* Source 08s's appendix proves Lemma 29.5 by exactly the argument of source 72's necessity proof for Theorem 14.3 (Section 14): equality gives $H_z = 0$ for $z = f - \lambda g$, the decomposition below propagates $H_i z = 0$ to every allowed contraction, and at rank two the kernel equation is the class-ratio condition. The argument is printed once, in source 72's section, and source 08s's text of it is not reprinted. In source 08s's notation, F is the unweighted basis polynomial, f, g carry the red and blue weights, $H = \text{Hess}(D_f^{j-1} D_g^{d-j-1} F)$ and, for $d \geq 3$, $j \geq 2$ and each red nonloop i , $H_i = \text{Hess}(D_f^{j-2} D_g^{d-j-1} \partial_i F)$; its key decomposition is]

$$H = \sum_{i \in R} w_i H_i, \quad (Hz)_i = z^T H_i f, \quad (172)$$

[*Merge note:* source 72's (21). Source 08s closes by noting that this proves Lemma 29.5 without a continuity claim about strict inequalities or an assertion that equality propagates to every index of the original matroid.]

29.8 Appendix: Exact checks and supplementary searches

The included standard-library checker [*Merge note:* shipped as `code/08-rank-seven-top-gap-*.py` with outputs `data/08-rank-seven-top-gap-*`] verifies the two-plane and unit-count inequalities on 320 represented configurations of ranks three through six. A separate endpoint-set computation agrees with the compressed profile algorithm in 400 exact coefficient comparisons and checks 100 instances of the general top-gap bound. The displayed test graphs have explicit exterior bases, so their actual ranks equal their cover sizes. These finite regressions are not used in place of the proof.

[*Merge note:* The search below is discovery history: source 08 later proved the middle bound. Its NumPy/SciPy scripts and logs are shipped as `code/08-rank-seven-top-gap-supplement-search*.py`, `data/08-rank-seven-top-gap-supplement-run_*.log` and `data/08-rank-seven-top-gap-search_summary.json`; they were not rerun at placement and are not part of any proof.] The search was narrowed to (159). It tested 2000 ordinary configurations, 1000 highly uneven unit-left clone configurations, 120 configurations with a heavy and diffuse right population in each nonsingleton class, and 600 embeddings based on a corrected rank-four transversal Rayleigh example. None produced a negative candidate. The smallest numerically optimized ratio $C_2^2 / ((9/5)C_1 C_3)$ was approximately 1.2456. This is neither a certified global minimum nor evidence sufficient for a uniform inequality.

For reproducibility, the printed presentation in Choe–Wagner [25, Proposition 5.9] was independently counted by dynamic matching and Hall's criterion on all 495 four-subsets. It has 309 bases, 69 containing e , 147 containing f and 33 containing both, giving Rayleigh

difference -54 . These correct the printed counts for that presentation and still give a negative Rayleigh example. They do not give a shifted Newton counterexample. [*Merge note:* This correction of Choe–Wagner’s printed counts is source 08s’s own statement; it was not checked against the publication in this merge. The shipped `data/08-rank-seven-top-gap-rayleigh_seed_checks.json` records that the printed counts were not reproduced.]

The search logs and exact regression data are supplementary. The persistent-failure rank bracket remains [7, 38]. [*Merge note:* Superseded within the batch: [8, 38] by source 08 and [9, 38] by source 01 (Section 39).] The unresolved mathematical target is the $4 + 3$ middle conditional gap and, at its equality points, the lower-order scaling coefficients. No exact counterexample, universal middle bound, or exhaustive priority assertion is made.

30 Rank-six conditioning for every core, one weighted shore, right-cover scaling (source 21)

Abstract of source 21, as delivered. For a rank-six bipartite graph with a displayed $3 + 3$ minimum cover, unit activities on the left shore and arbitrary positive activities on the right shore, we prove both shifted Newton inequalities for the cubic obtained by forcing the three right-cover vertices. A rank-three plane-incidence inequality, proved through a matroid basis polynomial, handles every core matrix, including singular cores. This removes the core-edge restrictions in the earlier conditional results. Consequently every such fixed graph and field is strictly ultra-log-concave after sufficiently large common scaling of the right-cover activities. Arbitrary finite-activity rank-six ultra-log-concavity remains open. [*Merge note:* The “earlier conditional results” whose core-edge restrictions this source removes are sources 43, 38, 35, 32 and 29, printed after it (Sections 31–35); source 21 does not name them.]

30.1 The conditional theorem

For a bipartite graph $G = (X, Y)$ with positive vertex activities, write

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|}.$$

Each feasible pair of endpoint sets is counted once, regardless of the number of perfect matchings. If the matching rank is d , the degree-normalized Newton gaps are

$$G_k = k(d - k)p_k^2 - (k + 1)(d - k + 1)p_{k-1}p_{k+1}. \quad (173)$$

We call P_G ULC $_d$ when these gaps are nonnegative; its positive-activity coefficients have no internal zeros.

Assume

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = |B| = 3, \quad \nu(G) = 6,$$

and $A \sqcup B$ is a vertex cover. There are no L – R edges. A maximum matching must avoid core edges and consists of exterior matchings of sizes three on both sides. In particular $n = |L| \geq 3$ after isolated vertices are removed.

Force the inclusion of all B in the selected right endpoint set, and divide by $v_B = \prod_{b \in B} v_b$. The resulting polynomial is

$$t^3 C(t), \quad C(t) = C_0 + C_1 t + C_2 t^2 + C_3 t^3.$$

All four coefficients are positive: an exterior-left basis and an exterior-right independent set of each size give the required supports.

Theorem 30.1. *If every left activity is one, then, for arbitrary positive right activities,*

$$C_1^2 \geq \frac{2n}{n-1} C_0 C_2, \quad C_2^2 \geq \frac{21}{8} C_1 C_3. \quad (174)$$

In particular the two shifted degree-six inequalities are strict:

$$8C_1^2 > 15C_0 C_2, \quad 5C_2^2 > 12C_1 C_3.$$

The first inequality in (174) also holds for arbitrary positive activities on both shores.

[Merge note: This is the case $a = b = 3$ of source 08s's Theorem 29.1 (constant $75/28 \geq 21/8$, see the Remark below) together with its ambient bound (170); source 21's proof is printed as the rank-six case. Source 21 states more than source 08s at $a = b = 3$: the first bound with the explicit constant $2n/(n-1)$, also for arbitrary left activities.]

The cubic need not be ULC₃: a connected, unique-cover example fails both degree-three inequalities [27]. [Merge note: Source 23's 25-vertex graph, Section 24.2. It fails the unshifted degree-three inequalities only; the shifted inequalities of Theorem 30.1 hold for it.] The normalization in Theorem 30.1 is therefore substantive. No hypothesis on the core's edge set, rank or connectedness is imposed.

Our structural input is established Lorentzian theory: matroid basis polynomials are Lorentzian; nonnegative linear substitutions, derivatives and coefficientwise limits preserve this property; and a quadratic Lorentzian polynomial has at most one positive Hessian eigenvalue [1, Theorems 2.10, 2.25 and 3.10]. The bivariate form is the normalized Newton inequality [1, Example 2.26].

30.2 The ambient lower gap

Introduce a private element $\bar{x}$ for each $x \in X$ and use the transversal matroid of rank $n+3$ whose elements are $Y \sqcup \bar{X}$, matched into X . Its bases are exactly

$$J \sqcup \{\bar{x} : x \notin I\}$$

for the feasible endpoint pairs (I, J) . Differentiate its basis polynomial in the three B variables. Set the private variable for x to s/u_x , and the variable for $r \in R$ to tv_r . Multiplication by $\prod_{x \in X} u_x$ gives

$$s^{n-3}(C_0 s^3 + C_1 s^2 t + C_2 s t^2 + C_3 t^3). \quad (175)$$

This is Lorentzian of degree n , even for arbitrary positive left activities. The index-one Newton inequality gives the first bound in (174). Thus

$$8C_1^2 - 15C_0 C_2 \geq \frac{n+15}{n-1} C_0 C_2 > 0. \quad (176)$$

The degree here is the ambient left count n , not the degree three of C . Nevertheless its resulting factor is stronger than the shifted factor $15/8$. [Merge note: Source 08s's (170) in general.]

30.3 A plane incidence inequality

Lemma 30.2. *Let E be a loopless rank-three vector configuration over an infinite field, with positive element activities, and let H be a plane. Let q be the weighted sum of independent triples, β the total activity outside H , and h the weighted sum of independent pairs not contained in H . For each such pair P , let $\delta(P) = \text{span}(P) \cap H$, a line, and let h_δ be the total pair activity in that line class. Then*

$$h^2 - \sum_{\delta} h_{\delta}^2 \geq \frac{3}{2} q \beta. \quad (177)$$

Proof (pointer). This is the case $b = 3$ of source 08s's Lemma 29.4. [*Merge note:* Source 13 recalls this lemma from source 21 with its proof (Section 25, display (129)); that is where it is printed. Source 21's proof is the same: with p the weighted sum of all independent pairs and g those in H , so that $h = p - g$, adjoin one vector on each line δ and N generic vectors in H of activity y/N ; the limiting basis polynomial is] Taking $N \rightarrow \infty$ gives

$$F = qt^3 + hyt^2 + \frac{\beta}{2}y^2t + \sum_{\delta} x_{\delta}((h - h_{\delta})t^2 + \beta yt) + \beta t \sum_{\delta < \eta} x_{\delta}x_{\eta}. \quad (178)$$

[*Merge note:* and the Schur complement of the $\partial_t F$ Hessian at its yy pivot gives (177), as in source 13's printed proof; not reprinted.] $\square$

Generic vectors may be taken over a transcendental field extension. Only finitely many line classes occur, and the coefficientwise limit is in a fixed polynomial space. Parallel original vectors are allowed. The lemma's weights are unrestricted; unit weights will enter later.

30.4 Virtual vectors and exact support counts

Choose a matrix with the graph's zero pattern and algebraically independent nonzero entries over a rational-function field. A square minor is nonzero exactly when its endpoint sets are matchable: distinct matchings produce distinct determinant monomials. Denote the core matrix by K , the exterior-left rows in the B coordinates by ℓ_i , and the exterior-right columns in the A coordinates by r_j . Both exterior configurations have rank three. Let p, q count independent pairs and triples of the ℓ_i .

For every independent pair P of exterior-left rows, choose a nonzero column v_P annihilated by those rows and introduce the virtual right vector

$$w_P = Kv_P.$$

It may be zero; different pairs remain separately labeled even when their virtual vectors are parallel.

Consider the rank-three matroid on the original r_j , the three private coordinate vectors e_1, e_2, e_3 , and the virtual w_P . Assign the private vectors activity s , the original r_j activity tv_j , and each virtual element activity z . Let $F_R(s, t, z)$ be its Lorentzian basis polynomial. Define:

$$\begin{aligned} U &= [s^2t]F_R, & a &= [st^2]F_R, & T &= [t^3]F_R, \\ P_V &= [stz]F_R, & P_B &= [t^2z]F_R, & Q/2 &= [tz^2]F_R. \end{aligned}$$

For each j let β_j count rows ℓ_i for which

$$\begin{pmatrix} K & r_j \\ \ell_i & 0 \end{pmatrix} \quad (179)$$

is nonsingular, and put $N = \sum_j v_j \beta_j$.

Partitioning supports by their number of selected exterior-left vertices gives

$$C_1 = qU + P_V + N, \quad C_2 = qa + P_B, \quad C_3 = qT. \quad (180)$$

Indeed, three selected independent exterior-left vertices contribute q times the original right-side basis profiles. Two selected exterior-left vertices can be eliminated; their remaining column is w_P . Complementary private coordinate vectors record the selected A -row set once. One

selected exterior-left vertex contributes to C_1 exactly through (179); it cannot contribute to C_2 or C_3 . Zero selected exterior-left vertices contribute only to C_0 . This is a partition of endpoint supports, with no allocation multiplicity.

The Hessian of $\partial_t F_R$, in coordinates (t, s, z) , is

$$\begin{pmatrix} 6T & 2a & 2P_B \\ 2a & 2U & P_V \\ 2P_B & P_V & Q \end{pmatrix}.$$

Since $T > 0$, its Schur complement at $6T$ is negative semidefinite. Evaluate it on $(s, z) = (q, 1)$ to obtain

$$C_2^2 \geq 3qT(qU + P_V) + \frac{3}{2}TQ = 3qT(C_1 - N) + \frac{3}{2}TQ. \quad (181)$$

30.5 Every core rank

Let Q_j count ordered pairs of virtual elements whose vectors together with r_j are independent. Then $Q = \sum_j v_j Q_j$. We prove

$$Q_j \geq \frac{3}{2}q\beta_j \quad (182)$$

by treating the possible ranks of K .

Invertible core. Put $a_0 = K^{-1}r_j$ and $H = \{\ell : \ell a_0 = 0\}$. The block determinant (179) is nonzero exactly when $\ell_i \notin H$, so β_j is the exterior-left count outside H . A pair contained in H has v_P proportional to a_0 , hence is zero after its virtual vector is quotiented by r_j . For other pairs, two virtual vectors are independent modulo r_j exactly when their pair spans meet H in distinct lines.

To see this last assertion directly, restriction to H identifies the dual quotient by $\text{span}(a_0)$ with the dual of H . The restriction of v_P has kernel $\text{span}(P) \cap H$. Nonzero restrictions are proportional exactly when these kernels coincide. Therefore, with the notation of Lemma 30.2,

$$Q_j = (p - g)^2 - \sum_{\delta} h_{\delta}^2. \quad (183)$$

The lemma proves (182).

Core rank two. Write $\ker K = \text{span}(a_0)$ and $\text{im } K = P_K$. If $r_j \in P_K$, then $\beta_j = Q_j = 0$. If $r_j \notin P_K$, use the same plane $H = \{\ell : \ell a_0 = 0\}$. Now (179) is nonsingular exactly when $\ell_i a_0 \neq 0$. Moreover $Kv_P, Kv_{P'}$ are independent in P_K exactly when $v_P, v_{P'}, a_0$ are independent. Since $r_j \notin P_K$, the same line classification gives (183) and (182).

Core rank at most one. Every block (179) has rank at most three, so $\beta_j = 0$. All virtual vectors lie in a subspace of dimension at most one, so $Q_j = 0$. This completes the proof of (182), including zero and parallel virtual elements.

30.6 The unit count and the higher gap

Let κ_j be the coefficient of v_j in C_1 . We claim

$$\kappa_j \geq 2\beta_j. \quad (184)$$

There is nothing to prove if $\beta_j = 0$. Otherwise the plane H in the preceding section exists. Every independent left pair not contained in H gives a virtual vector independent of r_j . At

least one private coordinate vector completes this pair to a basis. Thus $P_{V,j} \geq p - g$. Also $U_j \geq 1$.

For a loopless rank-three configuration of n unit elements,

$$q \geq n - 2, \quad p \geq 2n - 3.$$

Fix a basis: every other element extends some pair of that basis to a new triple, and is independent with at least two basis elements. These give the displayed bounds without a simplicity or coloop assumption.

If H contains $f \geq 1$ original elements, all $f(n - f)$ cross pairs are independent, so $p - g \geq f(n - f) \geq n - 1$. If $f = 0$, use $p - g = p \geq 2n - 3 \geq n - 1$. Consequently

$$\kappa_j = qU_j + P_{V,j} + \beta_j \geq (n - 2) + (n - 1) + \beta_j \geq 2\beta_j,$$

because $n \geq 3$ and $\beta_j \leq n$. This is the only step using the unit-left convention.

Summing (182) and (184) against the positive right activities gives

$$Q \geq \frac{3}{2}qN, \quad C_1 \geq 2N.$$

Insert these in (181):

$$C_2^2 \geq 3qTC_1 - \frac{3}{4}qTN \geq \frac{21}{8}qTC_1 = \frac{21}{8}C_1C_3.$$

Thus

$$C_2^2 - \frac{12}{5}C_1C_3 \geq \frac{9}{40}C_1C_3 > 0. \quad (185)$$

Together with (176), this proves Theorem 30.1.

Remark 30.3. A small refinement gives $\kappa_j \geq 7\beta_j/3$ and hence $C_2^2 \geq 75C_1C_3/28$. If $f = 0$, then $q + (p - g) \geq 3n - 5 \geq 4n/3$. If $f \geq 1$, write $\beta = n - f$ and use $q + (p - g) \geq (f + 1)\beta + f - 2 \geq 4\beta/3$: for $f = 1$, rank three forces $\beta \geq 2$; for $f \geq 2$ the inequality is immediate. Adding β proves the refinement. No optimality is claimed.

[Merge note: 75/28 is the constant of source 08s's Theorem 29.1 at $a = b = 3$.]

30.7 Eventual common scaling

Corollary 30.4. *Fix the graph and positive right activities in Theorem 30.1. Multiply the three B activities by a common λ . The full polynomial is strictly ULC_6 for every sufficiently large λ .*

[Merge note: The case $b = 3$ of source 08s's Corollary 29.2. Sources 43, 38, 35, 32 and 29 prove it for special cores (Sections 31–35); they are the “earlier conditional results” of source 21's abstract.]

Proof. Let b_j be the weighted support coefficients of the graph induced by all left vertices and B , which has rank three. For $j \leq 3$, the coefficient of λ^j in $p_j(\lambda)$ is $b_j > 0$. For $k \geq 3$, the coefficient of λ^3 in $p_k(\lambda)$ is $v_B C_{k-3}$, with $b_3 = v_B C_0$.

The smaller-shore degree-three transversal basis specialization gives $b_1^2 \geq 3b_2$ and $b_2^2 \geq 3b_1b_3$. Therefore the five leading limits are

$$\begin{aligned} \lambda^{-2}G_1 &\longrightarrow 5b_1^2 - 12b_2 > 0, \\ \lambda^{-4}G_2 &\longrightarrow 8b_2^2 - 15b_1b_3 > 0, \\ \lambda^{-6}G_3 &\longrightarrow 9b_3^2 > 0, \\ \lambda^{-6}G_4 &\longrightarrow v_B^2(8C_1^2 - 15C_0C_2) > 0, \\ \lambda^{-6}G_5 &\longrightarrow v_B^2(5C_2^2 - 12C_1C_3) > 0. \end{aligned}$$

For G_3 , the adjacent product has degree at most five. The last two limits follow from Theorem 30.1.

There is also an explicit coefficient-dominance bound. For each exact gap polynomial $G_k(\lambda) = a_d\lambda^d + \sum_{j<d} a_j\lambda^j$, with $a_d > 0$, it suffices that

$$\lambda \geq 1 + \frac{\sum_{j<d} |a_j|}{a_d}.$$

Taking the maximum of these five quantities gives a computable, field-dependent threshold. $\square$

For any chosen minimum cover of an arbitrary rank-six graph, either the cover splits $3 + 3$, or one of its shores has size at most two. The previously established cover theorem handles the latter cases for all positive activities [12]. Thus common scaling of the right part of any chosen minimum cover is eventually safe for every rank-six graph with unit left activities. This global statement is nonstrict, since balanced unions of stars can have equality. [*Merge note:* The cover theorem is Theorem 2.1; equality occurs only for balanced star forests (source 72, Theorem 14.1).]

This improves the known bracket for the minimum rank of *persistent failure under this common scaling* to [7, 38]. [*Merge note:* Superseded within the batch: [8, 38] (source 08) and [9, 38] (source 01).] The upper witness uses exactly the same convention: the full Hall graph $H_{8,30;32,794}$ has unit left activities, unit exterior-right activities, and common activity w on the right part B of its minimum cover. Its last gap is [17]

$$G_{37} = \frac{5}{6304} D^2 w^{58} (-22w^2 + 6437219280w + 11385894392295), \quad D = \binom{32}{30} \binom{794}{7}.$$

It is negative for every sufficiently large w . The minimum rank for failure at arbitrary finite one-shore fields remains bracketed by [6, 38]; the present theorem does not decide the rank-six case at finite fields. [*Merge note:* The lower end six rests on source 74 (Part I). At matching number six and finite fields, gaps one, two and five hold for every one-shore graph (Part III); gaps three and four are open.]

30.8 Verification and scope

The accompanying standard-library exact checker [*Merge note:* shipped as `code/21-every-core-*.py`, outputs `data/21-every-core-*.json`] verifies 960 unit and 960 weighted plane-incidence instances, matrix instances for all 512 Boolean core masks, and 38 full endpoint-minor enumerations. An independently written checker verifies 600 further weighted plane configurations. The checks use exact arithmetic and no numerical tolerance. These finite regressions supplement the general proof; they are not an exhaustive verification of the theorem's infinitely many exterior configurations. The matrix assignments may contain additional algebraic dependencies, which the incidence proof also permits.

This report gives an ordinary mathematical proof using established Lorentzian theory, with independent internal mathematical review and exact checks. It is not a formal-kernel proof or an externally refereed publication. No exhaustive priority claim is made. The general finite-activity rank-six one-shore question remains open.

31 Second route for the complete core: a rank-six cubic sector, one weighted shore (source 43)

Abstract of source 43, as delivered. Consider a rank-six bipartite graph with a displayed $3 + 3$ minimum vertex cover and a complete core. Both exterior neighborhood systems are

arbitrary. With unit activities on the left shore and positive activities on the right, we prove strict shifted Newton inequalities for the cubic sector obtained by forcing all three right-core vertices. A generic extension of a rank-three transversal matroid supplies three coupled Hessian inequalities; these control the two required gaps using ordinary log-concavity of the exterior-left independence counts. Consequently sufficiently large common scaling of the right-core activities gives strict rank-six ultra-log-concavity. The threshold may depend on the graph and fields.

Scope. This is a conditional-sector and eventual-scaling theorem. It does not prove ultra-log-concavity at arbitrary finite activities, and it does not yet cover a noncomplete 3×3 core. [*Merge note:* Every core is covered by source 21 and, more generally, source 08s.] The general rank-six one-shore problem remains open. The argument is an ordinary proof supported by exact checks, not a formal proof-assistant development or an external priority claim.

31.1 The conditional cubic

Let G have shores $C \sqcup L$ and $D \sqcup R$, where

$$|C| = |D| = 3, \quad \nu(G) = 6.$$

The displayed cover is $C \sqcup D$, so there are no L – R edges. Assume every edge of $C \times D$ is present. The exterior graphs $L \rightarrow D$ and $C \rightarrow R$ are arbitrary. All left activities are one, and the right activities are arbitrary positive real numbers.

The matching-support polynomial counts each feasible pair of endpoint sets once, regardless of its number of perfect matchings. Condition on selecting every vertex of D , and divide out the fixed product of their activities. The resulting polynomial has the form

$$t^3(c_0 + c_1t + c_2t^2 + c_3t^3). \quad (186)$$

Theorem 31.1. *Under these hypotheses all four c_i are positive, and*

$$8c_1^2 > 15c_0c_2, \quad 5c_2^2 > 12c_1c_3. \quad (187)$$

Thus the conditional cubic satisfies the two nontrivial shifted ULC_6 inequalities strictly.

[*Merge note:* Same theorem as source 21 (Section 30), restricted to the complete core $K_{3,3}$: its eventual statement is the case of that core in source 21’s Corollary 30.4, and its two conditional inequalities ($8C_1^2 > 15C_0C_2$, $5C_2^2 > 12C_1C_3$) follow from source 21’s Theorem 30.1, whose constants $2n/(n-1)$ and $21/8$ are larger and which needs no hypothesis on the core; both are the case $a = b = 3$ of source 08s’s Theorem 29.1 and Corollary 29.2. Source 43’s proof is printed as a second route. What it adds: the closed support partition of Lemma 31.2, an elementary lower-gap proof by six quadratic forms that uses only $p^2 \geq nq$, and an explicit higher margin.]

A matching of size six uses no core edge, so both exterior graphs have matching rank three. Delete isolated exterior vertices. Let

$$n = I_1, \quad p = I_2, \quad q = I_3$$

be the counts of independent exterior-left subsets in their rank-three transversal matroid on slots D . In particular,

$$n \geq 3, \quad p \geq n, \quad q \geq 1, \quad p^2 \geq nq. \quad (188)$$

For the pair bound, fix a basis. Every other nonloop forms an independent pair with at least two basis elements. Hence $p \geq 3 + 2(n-3) = 2n-3 \geq n$. The final inequality in (188)

is ordinary log-concavity of matroid independence counts. The stronger established Mason inequality is

$$p^2 \geq \frac{3n(n-1)}{2(n-2)}q.$$

Its normalization uses the *ground-set size* n , not the matroid rank three [1, Theorem 4.14]. [*Merge note:* Source 43's bibliography also lists [24] as an independent proof of that stronger independence-count theorem.]

For the exterior-right graph on the three slots C , use the following weighted counts:

- U, A, T are its matching-support coefficients of orders one, two, three;
- W, B count its independent right subsets of sizes one and two;
- V counts pairs consisting of a two-element subset of C and one right vertex, with at least one incidence.

The weight of a right subset is the product of its activities. Thus U counts incidences, whereas W counts nonloop right vertices only once. A singleton-neighborhood right vertex contributes twice its activity to V ; a vertex with at least two neighbors contributes three times. Genuine rank six ensures $W, T > 0$.

Lemma 31.2 (Exact support partition). *The coefficients in (186) are*

$$\begin{aligned} c_0 &= q + 3p + 3n + 1, & c_1 &= qU + pV + nW, \\ c_2 &= qA + pB, & c_3 &= qT. \end{aligned} \tag{189}$$

Proof. Suppose j exterior-right vertices are selected, together with i core-left vertices. The exterior-left set must have size $3 + j - i$ and must be independent into D . The selected right set must match into the selected core-left set. These two conditions are sufficient: after matching those exterior vertices, the complete core matches the remaining $i - j$ selected core-left vertices into the remaining slots of D .

For $j = 0$, the four core-left sizes give $q, 3p, 3n, 1$. For $j = 1$, core-left sizes one, two, three give qU, pV, nW . For $j = 2$, sizes two and three give qA, pB . For $j = 3$, only size three is possible, giving qT . Every support is counted once. $\square$

31.2 Three coupled inequalities from one matroid

Take the rank-three transversal matroid on the right exterior vertices and three private coordinate elements. The private elements form a basis. Assign activity s to each private element and t times its fixed activity to each exterior-right element.

Represent this transversal matroid by a generic three-row matrix. Add N generic elements in the full rank-three flat, each with activity z/N . One new element extends every independent original pair; two extend every original nonloop; three form a basis. Their basis polynomial specializes to

$$\begin{aligned} F_N &= s^3 + Us^2t + Ast^2 + Tt^3 + z(3s^2 + Vst + Bt^2) \\ &\quad + \frac{N-1}{2N}z^2(3s + Wt) + \frac{(N-1)(N-2)}{6N^2}z^3. \end{aligned}$$

The coefficient V here is the private/right independent-pair count: a singleton-neighborhood element is independent with two private elements, and any element with at least two neighbors is independent with all three.

Matroid basis polynomials are Lorentzian, and this property is preserved by nonnegative linear substitutions and coefficientwise limits [1]. Therefore the cubic

$$F = s^3 + Us^2t + Ast^2 + Tt^3 + z(3s^2 + Vst + Bt^2) + \frac{z^2}{2}(3s + Wt) + \frac{z^3}{6} \quad (190)$$

is Lorentzian. No real-stability or general Rayleigh assertion is needed.

Lemma 31.3 (Coupled Hessian bounds). *The following matrix is positive semidefinite:*

$$\begin{pmatrix} A^2 - 3UT & AB - \frac{3}{2}VT \\ AB - \frac{3}{2}VT & B^2 - \frac{3}{2}WT \end{pmatrix} \succeq 0. \quad (191)$$

Moreover,

$$A \leq A_* := \frac{U^2 - (V - 2U)^2/2}{3}, \quad B \leq B_* := \frac{W^2 - (V - 3W)^2/3}{2}. \quad (192)$$

Proof. Every first derivative of F is a Lorentzian quadratic, so its Hessian has at most one positive eigenvalue. In variable order (s, t, z) , the three Hessians are

$$\begin{aligned} \nabla^2 \partial_t F &= \begin{pmatrix} 2U & 2A & V \\ 2A & 6T & 2B \\ V & 2B & W \end{pmatrix}, & \nabla^2 \partial_s F &= \begin{pmatrix} 6 & 2U & 6 \\ 2U & 2A & V \\ 6 & V & 3 \end{pmatrix}, \\ \nabla^2 \partial_z F &= \begin{pmatrix} 6 & V & 3 \\ V & 2B & W \\ 3 & W & 1 \end{pmatrix}. \end{aligned}$$

For the first matrix, take the Schur complement of its positive t, t entry $6T$. That complement is negative semidefinite; multiplication by $-3T/2$ gives (191). For the second matrix, complement its s, s entry 6. The resulting determinant condition is

$$2(U^2 - 3A) - (V - 2U)^2 \geq 0.$$

For the third, complement the z, z entry 1, obtaining

$$3(W^2 - 2B) - (V - 3W)^2 \geq 0.$$

These are exactly (192). □

31.3 The two shifted Newton gaps

31.3.1 The higher gap

Set $X = A^2 - 3UT$, $Y = B^2 - \frac{3}{2}WT$, and $Z = 2AB - 3VT$. By (191),

$$Q = q^2X + qpZ + p^2Y \geq 0.$$

Using (189), an exact rearrangement gives

$$c_2^2 - \frac{12}{5}c_1c_3 = Q + \frac{3}{5}q^2UT + \frac{3}{5}qpVT + \frac{3}{2}p^2WT - \frac{12}{5}nqWT.$$

Each nonloop right vertex has at least one neighbor, and intersects at least two two-slot subsets. Hence $U \geq W$ and $V \geq 2W$. Together with (188), these give

$$c_2^2 - \frac{12}{5}c_1c_3 \geq \frac{3}{10}qWT(2q + 4p - 3n) > 0. \quad (193)$$

Indeed $p \geq n$ and all of n, q, W, T are positive.

31.3.2 The lower gap

Let x, y, z_0 be the total activities of right vertices with exactly one, two, three neighbors. Then

$$U = x + 2y + 3z_0, \quad V = 2x + 3y + 3z_0, \quad W = x + y + z_0.$$

Put $S = q + 3p + 3n + 1$ and $\ell = qU + pV + nW$. By (192),

$$8c_1^2 - 15c_0c_2 \geq 8\ell^2 - 15S(qA_* + pB_*). \quad (194)$$

The following six coefficient forms give an explicit positive certificate:

$$\begin{aligned} P_{xx} &= 8n^2 + 17np + nq + 17p^2 + 12pq - 5p + 3q^2 - 5q, \\ P_{xy} &= 16n^2 + 35np - 12nq + 51p^2 + 37pq - 15p + 12q^2 - 20q, \\ P_{xz} &= 16n^2 + 35np - 26nq + 51p^2 + 39pq - 15p + 18q^2 - 30q, \\ P_{yy} &= 16n^2 + 51np - 41nq + 99p^2 + 72pq - 15p + 29q^2 - 35q, \\ P_{yz} &= 16n^2 + 51np - 55nq + 99p^2 + 90pq - 15p + 51q^2 - 45q, \\ P_{zz} &= 16n^2 + 51np - 39nq + 99p^2 + 138pq - 15p + 99q^2 - 45q. \end{aligned}$$

Direct expansion yields

$$\begin{aligned} 2[8\ell^2 - 15S(qA_* + pB_*)] &= 2P_{xx}x^2 + 2P_{xy}xy + 2P_{xz}xz_0 \\ &\quad + P_{yy}y^2 + 2P_{yz}yz_0 + P_{zz}z_0^2. \end{aligned} \quad (195)$$

Every P is strictly positive for $n, p, q \geq 1$ and $p^2 \geq nq$. In the last five forms, absorb the negative nq term into p^2 . The remaining p^2 coefficients are, respectively,

$$39, \quad 25, \quad 58, \quad 44, \quad 60.$$

Group each negative linear p term with np and each negative linear q term with pq ; both groups remain positive for $n, p \geq 1$. The same grouping works directly for P_{xx} , whose nq term is already positive. Since $x + y + z_0 = W > 0$, at least one square term in (195) is positive. This proves the first inequality in (187) and completes Theorem 31.1.

31.4 Eventual strictness and a computable threshold

Corollary 31.4. *Fix a graph and positive right fields satisfying the theorem. Multiply all three right-core activities by a common factor λ . For every sufficiently large λ , the full matching-support polynomial is strictly ULC₆. The threshold may depend on the graph and the fixed relative fields.*

Proof. Let $a_1, a_2, a_3 > 0$ be the leading coefficients at orders one, two, three when the common factor is introduced. They are the support coefficients of the induced graph on the three right-core vertices and all left vertices. The weighted smaller-shore theorem gives

$$a_1^2 \geq 3a_2, \quad a_2^2 \geq 3a_1a_3.$$

Consequently the leading coefficients of the first two rank-six gaps are

$$5a_1^2 - 12a_2 \geq 3a_2 > 0, \quad 8a_2^2 - 15a_1a_3 \geq 9a_1a_3 > 0,$$

at powers λ^2 and λ^4 . At the third gap, the square has degree six in λ , whereas the adjacent product has degree at most five, so its leading coefficient is $9a_3^2 > 0$.

Let γ be the product of the three fixed right-core activities. At orders three through six, the leading coefficients are $\gamma c_0, \dots, \gamma c_3$. Thus the last two gap-leading coefficients are

$$\gamma^2(8c_1^2 - 15c_0c_2) > 0, \quad \gamma^2(5c_2^2 - 12c_1c_3) > 0,$$

both at degree six. Every gap is therefore eventually positive. $\square$

There is an elementary field-dependent numerical prescription. Write each gap as

$$\Delta_k(\lambda) = a_{k,d_k} \lambda^{d_k} + \sum_{j < d_k} a_{k,j} \lambda^j, \quad a_{k,d_k} > 0.$$

Then all gaps are positive whenever

$$\lambda \geq 1 + \max_{1 \leq k \leq 5} \frac{\sum_{j < d_k} |a_{k,j}|}{a_{k,d_k}}.$$

This bound is computable from the support coefficients; no uniform threshold is asserted.

31.5 Verification and the remaining frontier

The proof uses the established matroid basis theorem and ordinary independence-count log-concavity. Its new step is the simultaneous use of three Hessian consequences of the auxiliary cubic to control the forced support sector.

The source package [*Merge note:* shipped as `code/43-complete-core-verify_complete_core.py` and `data/43-complete-core-*`] includes exact symbolic verification of (195) and the Schur-complement matrix, together with 120 deterministic direct conditional-support enumerations using arbitrary exterior neighborhoods and positive integer right fields. These are supplementary checks of the identities; the population and activity quantifiers follow from the proof.

The complete-core hypothesis is used in Lemma 31.2. With a missing core edge, the left completion counts depend on which core vertices remain available, so the three numbers n, p, q no longer suffice. A flag of minor or extension profiles is a possible next route, but no noncomplete-core theorem is claimed here. The full finite-activity rank-six one-shore question remains open. [*Merge note:* See the notes at the scope paragraph and after Theorem 31.1; the finite-activity status is in Part III's introduction.]

32 Second route for matching-hole cores, one weighted shore (source 38)

Abstract of source 38, as delivered. We extend the complete-core conditional theorem to rank-six bipartite graphs whose displayed $3 + 3$ core is obtained from $K_{3,3}$ by deleting a matching of up to three edges. Exterior neighborhoods are arbitrary, left activities are one, and right activities are positive. The all-right-core sector satisfies strict shifted Newton inequalities in the remaining flag regime; the complementary cases have a $4 + 2$ minimum cover and are already ultra-log-concave at every positive field. The higher gap follows from a Lorentzian polynomial with generic coordinate-plane extensions. The lower gap has a direct certificate for the 25 independent pairs of right-neighborhood types. This proves eventual strict rank-six ultra-log-concavity under common right-core scaling.

Scope. This is an eventual-scaling and conditional-sector theorem. It does not prove ultra-log-concavity at arbitrary finite fields in the new cases, or cover cores with two missing edges in one row or column. [*Merge note:* Those cores are treated by sources 35 and 32, and every core by source 21.] The general rank-six one-shore problem remains open. The result is an ordinary mathematical proof with exact rational certificates, not a formal proof-assistant development or a claim of external priority.

32.1 The graph and the conclusion

Let G have shores $A \sqcup L$ and $B \sqcup R$, with

$$A = \{A_1, A_2, A_3\}, \quad B = \{B_1, B_2, B_3\}, \quad \nu(G) = 6.$$

There are no L – R edges. After relabeling, the missing core edges form a subset of

$$\{A_1B_1, A_2B_2, A_3B_3\}.$$

All left activities are one, and all right activities are arbitrary positive real numbers. The matching-support polynomial counts each feasible pair of endpoint sets once, irrespective of its number of matchings.

Theorem 32.1. *For every graph and field in this family, sufficiently large common scaling of the three right-core activities makes the full support polynomial strictly ULC_6 .*

More precisely, suppose each B_i has at least two exterior-left neighbors. Condition on selecting every vertex of B , divide out their fixed activity product, and write the result as

$$t^3(C_0 + C_1t + C_2t^2 + C_3t^3).$$

Then all C_i are positive and

$$8C_1^2 > 15C_0C_2, \quad 5C_2^2 > 12C_1C_3. \tag{196}$$

If some B_i has only one exterior-left neighbor, the graph already has strict ULC_6 at every positive field.

[*Merge note:* Same theorem as source 21 (Section 30), restricted to cores obtained from $K_{3,3}$ by deleting a matching (source 21 also needs no restriction on the exterior-left neighbourhoods of the B_i): its eventual statement is the case of that core in source 21’s Corollary 30.4, and its two conditional inequalities ($8C_1^2 > 15C_0C_2$, $5C_2^2 > 12C_1C_3$) follow from source 21’s Theorem 30.1, whose constants $2n/(n-1)$ and $21/8$ are larger and which needs no hypothesis on the core; both are the case $a = b = 3$ of source 08s’s Theorem 29.1 and Corollary 29.2. Source 38’s proof is printed as a second route. What it adds: the every-field branch when some B_i has a unique exterior-left neighbour, which is a corollary of Part I (source 74’s theorem for the resulting $4 + 2$ cover and source 72’s strictness), the exact subtraction formula, and a 25-pair certificate.]

Every six-matching avoids core edges, so both exterior graphs have rank three. In particular no B_i has zero exterior-left neighbors. If one has a unique such neighbor ℓ , replace B_i by ℓ in the displayed cover. This produces a minimum cover with four left and two right vertices. The established small-cover theorem [12] gives ULC at every positive field. Its strictness companion [26] applies because the nontrivial graph is connected and contains the nonstar core. We may therefore assume each B_i has at least two exterior-left neighbors.

32.2 Unit left counts and exact plane defects

Delete isolated exterior vertices. Let n, p, q be the counts of independent exterior-left subsets of sizes one, two, three in their rank-three transversal matroid on slots B .

For an actual hole $A_i B_i$, let d_i count independent exterior-left pairs whose neighborhoods both avoid B_i . For an absent hole set $d_i = 0$. Put $D = d_1 + d_2 + d_3$. The following 18 polynomials are nonnegative:

$$\begin{aligned}
g_1 &= n - 3, & g_2 &= p - 2n + 3, & g_3 &= q - n + 2, \\
g_4 &= 3q - p, & g_5 &= p - D, \\
g_{5+i} &= d_i, & g_{8+i} &= q - 2d_i & (1 \leq i \leq 3), \\
g_{11+i} &= p - d_i - 2n + 4 & (1 \leq i \leq 3), \\
g_{15} &= 2p^2 - 3nq, & g_{16} &= n(n-1) - 2p, \\
g_{17} &= 2p^2(n-2) - 3n(n-1)q, \\
g_{18} &= n(n-1)(n-2) - 6q.
\end{aligned} \tag{197}$$

Here and below the formula order fixes the certificate's generator indices.

For g_2, g_3 , fix one basis: each additional nonloop forms independent pairs with at least two basis elements and belongs to a new basis with two fixed basis elements. Every independent pair extends to a basis, whereas each basis contains three pairs, proving $p \leq 3q$. The upper bounds on p and q are immediate subset counts.

An independent pair cannot lie in two different coordinate planes, since their intersection has rank one; hence $D \leq p$. Every pair counted by d_i extends with each of the at least two exterior vertices adjacent to B_i , giving $2d_i \leq q$. For $p - d_i \geq 2n - 4$, let n_0, n_1 be the numbers of exterior-left vertices inside and outside that plane. If $d_i > 0$, both $n_0, n_1 \geq 2$, and every cross pair is independent. Thus

$$p - d_i \geq n_0 n_1 \geq 2(n_0 + n_1) - 4 = 2n - 4.$$

If $d_i = 0$, use $p \geq 2n - 3$ instead.

The two Mason constraints are consequences of the established *ground-set-normalized* independence inequality [1, Theorem 4.14]:

$$p^2 \geq \frac{3n(n-1)}{2(n-2)} q.$$

The weaker consequence $p^2 \geq 3nq/2$ is also recorded in (197). No matching-rank-three normalization of these independence counts is asserted.

32.2.1 Right profiles

For the exterior-right graph on the three slots A , let U, A_2, T be its support coefficients of orders one, two, three; let W, B_2 count independent right singles and pairs; and let V count private-coordinate/right independent pairs. A right vertex of degree one contributes twice its activity to V , and one of degree two or three contributes three times. In formulas below abbreviate A_2 by a and B_2 by b .

Let R_i be the right vertices avoiding A_i , with total activity W_i . Let u_i, v_i be the two-slot support coefficients of their restricted graph on the other two slots.

Lemma 32.2 (Exact conditional coefficients). *The conditional cubic in Theorem 32.1 has*

$$\begin{aligned} C_0 &= q + 3p + 3n + 1 - D, & C_1 &= qU + pV + nW - \sum_i d_i u_i, \\ C_2 &= qa + pb - \sum_i d_i v_i, & C_3 &= qT. \end{aligned} \quad (198)$$

Proof. Partition supports by the number of selected exterior-left vertices. With three, their set is a basis and every selected core-left vertex is used on the right exterior; this contributes q times the exterior-right support polynomial. With one, any remaining 2×2 core submatrix has a perfect matching, so the contribution is unchanged from the complete core. With none, the core itself has a perfect matching.

Only two selected exterior-left vertices can lose a support. If their pair can leave two different B slots unused, every core-left vertex can match one of those slots, since each core row misses at most one edge. Otherwise the pair lies in one coordinate plane and leaves only B_i . An actual hole there removes the profile

$$d_i(1 + u_i t + v_i t^2).$$

At exterior-right degrees zero, one, two, the losses are respectively one forbidden leftover core vertex, one right vertex avoiding it, and an independent right pair avoiding it. The plane classes are disjoint, so these subtractions do not overlap. The complete-core count [32] then gives (198). $\square$

32.3 The Lorentzian flag polynomial and the higher gap

Start with the rank-three transversal matroid on the right exterior vertices and three private coordinate elements. Give the private elements activity s and the right elements their fixed fields times t . Add diffuse generic elements in the whole rank-three flat with total activity z , and in each relevant coordinate plane with total activity f_i . Write $Z = z + \sum_i f_i$.

The coefficientwise limit of their basis polynomials is

$$F = F_{\text{base}}(s, t, Z) - \sum_i \left[f_i(s^2 + u_i s t + v_i t^2) + \frac{f_i^2}{2}(2s + W_i t) + \frac{f_i^3}{6} \right], \quad (199)$$

where

$$F_{\text{base}} = s^3 + U s^2 t + a s t^2 + T t^3 + Z(3s^2 + V s t + b t^2) + \frac{Z^2}{2}(3s + W t) + \frac{Z^3}{6}.$$

The subtractions have a direct basis interpretation. One plane point fails to extend exactly the original independent pairs spanning that plane. Two same-plane points fail to extend exactly the original elements inside it. Three same-plane points are dependent. Generic selections from distinct coordinate planes have the required independent extension profiles; their mixed terms need no correction.

This constructs F as a Lorentzian polynomial in the original nonnegative variables s, t, z, f_i , using matroid basis Lorentzianity and its closure properties [1]. For the Hessian calculation, we merely change linear coordinates to (s, t, Z, f_i) . Inertia of a quadratic form is preserved by this invertible change; no Lorentzian-cone closure under negative substitutions is being used.

The Hessian of $\partial_t F$ has t, t entry $6T > 0$ and cross-column $2(a, b, -v_i)$. On the remaining variables its quadratic form is

$$2Us^2 + 2VsZ + WZ^2 - 2s \sum_i u_i f_i - \sum_i W_i f_i^2.$$

Its Schur complement at $6T$ is negative semidefinite. Evaluate the resulting positive-semidefinite inequality at $(s, Z, f_i) = (q, p, d_i)$ to obtain

$$C_2^2 \geq 3qT(C_1 - nW) + \frac{3}{2}T \left(Wp^2 - \sum_i W_i d_i^2 \right). \quad (200)$$

For a right vertex with nonempty neighborhood S , its contribution to $\sum_i d_i u_i$ is $|S| \sum_{i \notin S} d_i$ times its activity. There are at most two indices outside S , each $d_i \leq q/2$. Hence $C_1 \geq pV + nW \geq (2p + n)W$. Also

$$\sum_i W_i d_i^2 \leq \frac{q}{2} \sum_i W_i d_i \leq \frac{1}{2} Wqp.$$

Equation (200) now yields

$$\begin{aligned} C_2^2 - \frac{12}{5}C_1C_3 &\geq WT \left(\frac{3}{2}p^2 + \frac{9}{20}pq - \frac{12}{5}nq \right) \\ &\geq WT \left(-\frac{1}{10}p^2 + \frac{9}{20}pq \right) \geq \frac{1}{20}WTp^2 > 0. \end{aligned} \quad (201)$$

The last two steps use $p^2 \geq 3nq/2$ and $p \leq 3q$. Genuine exterior rank three ensures $W, T > 0$.

32.4 The lower gap by 25 neighborhood pairs

For a right vertex with nonempty neighborhood $S \subseteq \{1, 2, 3\}$, its coefficient in C_1 is

$$\kappa(S) = q|S| + p\epsilon(S) + n - |S| \sum_{i \notin S} d_i, \quad \epsilon(S) = \begin{cases} 2, & |S| = 1, \\ 3, & |S| \geq 2. \end{cases} \quad (202)$$

The preceding profile argument gives $\kappa(S) \geq 2p + n > 0$.

For an independent pair of distinct right vertices with neighborhoods S, T , let $a(S, T)$ count the two-slot subsets into which they match. Their coefficient in C_2 is

$$\lambda(S, T) = qa(S, T) + p - \sum_{i \notin S \cup T} d_i.$$

Thus their product coefficient in $8C_1^2 - 15C_0C_2$ is

$$J(S, T) = 16\kappa(S)\kappa(T) - 15C_0\lambda(S, T). \quad (203)$$

There are 25 unordered independent pairs of nonempty neighborhood types, allowing repeated types. The three excluded pairs are repeated singleton types; their C_2 coefficient is zero and their gap coefficient is positive. The square of an individual right activity has gap coefficient $8\kappa(S)^2 > 0$.

Proposition 32.3 (Exact pair certificate). *For all 25 independent types, the accompanying rational identities have the form*

$$(n-2)J(S, T) = 1 + \sum_{\alpha} r_{\alpha} \prod_{j=1}^{18} g_j^{\alpha_j}, \quad r_{\alpha} \geq 0, \quad \alpha_j \in \mathbb{Z}_{\geq 0}. \quad (204)$$

There are 1165 nonzero terms across the complete certificate.

Exact verification. The supplied standard-library verifier reconstructs κ , λ , every target polynomial, and all 25 rational identities. It verifies nonnegative weights and exponents, exact sparse-polynomial equality, complete pair coverage, and the strict remainder one. No floating-point calculation enters this replay. The original linear-programming search served only to discover the certificate. $\square$

All $g_j \geq 0$ by Section 2 [*Merge note*: Subsection 32.2 here], and $n - 2 > 0$. Therefore every $J(S, T) > 0$. Every quadratic coefficient of the first shifted gap is positive, proving its strict positivity for arbitrary positive real right activities. Together with (201), this proves the conditional assertion of Theorem 32.1.

32.5 Eventual strictness and reproducibility

Scale the three right-core fields by a common factor λ , keeping their positive relative fields fixed. At support orders one, two, three, let the leading coefficients be $a_1, a_2, a_3 > 0$. They are the coefficients of the induced three-right-vertex support graph, so the weighted smaller-shore theorem gives

$$a_1^2 \geq 3a_2, \quad a_2^2 \geq 3a_1a_3.$$

The first two rank-six gap-leading coefficients are consequently positive:

$$5a_1^2 - 12a_2 \geq 3a_2 > 0, \quad 8a_2^2 - 15a_1a_3 \geq 9a_1a_3 > 0.$$

At the third gap the square has degree six in λ , whereas the adjacent product has degree at most five. The final two leading coefficients are positive multiples of the two expressions in (196). All five full-polynomial gaps are therefore eventually strictly positive. This completes Theorem 32.1, including its small-cover branch.

A field-dependent threshold can be obtained directly. If

$$\Delta_k(\lambda) = a_{k,d_k} \lambda^{d_k} + \sum_{j < d_k} a_{k,j} \lambda^j, \quad a_{k,d_k} > 0,$$

then every gap is positive for

$$\lambda \geq 1 + \max_{1 \leq k \leq 5} \frac{\sum_{j < d_k} |a_{k,j}|}{a_{k,d_k}}.$$

No uniform threshold is claimed.

The package [*Merge note*: shipped as `code/38-matching-hole-*.py` and `data/38-matching-hole-*.json`] includes the exact 25-case coefficient data and its independent standard-library replay. A separate endpoint enumerator checks 160 graphs, 40 for each number of holes, with arbitrary exterior neighborhoods and positive integer fields. All support identities agree; all 104 cases in the flag regime also pass the stated constraints and strict inequalities. These are supplementary regressions. The real-field and unbounded-population claims follow from the Lorentzian argument and the exact constraint-cone identities.

The next structural case has two missing edges in the same row or column. Then some 2×2 core minors vanish, and exterior-left singleton profiles also change. The present proof does not cover those additional degeneracies or settle arbitrary finite-field ULC at rank six. [*Merge note*: The next cases are sources 35 and 32; every core, source 21.]

33 Second route for cores with two complete rows, one weighted shore (source 35)

Abstract of source 35, as delivered. Consider a genuine rank-six bipartite graph with a displayed $3 + 3$ cover, two complete core rows, and an arbitrary third core row. Exterior neighborhoods are arbitrary, left activities are one, and right activities are positive. We prove eventual strict rank-six ultra-log-concavity under common right-core scaling. The new cases have an exceptional row of degree one or zero. For degree one, paired left and right Lorentzian flag polynomials control the higher conditional gap, while an exact 25-case rational certificate controls the lower gap. For degree zero, both inequalities have direct analytic proofs. The degree-two and degree-three cases follow from the preceding matching-hole and complete-core theorems.

Scope. The new general conclusion is eventual strictness, with a graph- and field-dependent threshold. ULC at rank six for arbitrary finite activities remains open. The transposed column-defect family is a separate question under the fixed unit-left convention. This is an ordinary proof with exact rational checks, not formal kernel verification or an external priority claim.

33.1 The family and its conditional sector

Let G have shores $A \sqcup L$ and $B \sqcup R$, where

$$A = \{A_1, A_2, A_3\}, \quad B = \{B_1, B_2, B_3\}, \quad \nu(G) = 6.$$

There are no L – R edges. The rows of A_2, A_3 are complete to B ; the neighborhood of A_1 inside B is arbitrary. All left activities are one and all right activities are positive real numbers.

Theorem 33.1. *Fix such a graph and its right activities. Multiplying all three right-core activities by a sufficiently large common factor makes the matching-support polynomial strictly ULC₆.*

[Merge note: Same theorem as source 21 (Section 30), restricted to cores with two complete rows: its eventual statement is the case of that core in source 21’s Corollary 30.4, and its two conditional inequalities ($8C_1^2 > 15C_0C_2$, $5C_2^2 > 12C_1C_3$) follow from source 21’s Theorem 30.1, whose constants $2n/(n-1)$ and $21/8$ are larger and which needs no hypothesis on the core; both are the case $a = b = 3$ of source 08s’s Theorem 29.1 and Corollary 29.2. Source 35’s proof is printed as a second route. What it adds: the every-field branch when the exterior-left matroid has a coloop (a corollary of Part I’s sources 74 and 72), a direct proof for an exceptional row of degree zero, and a 25-case certificate for degree one.]

Matching supports count each feasible pair of endpoint sets once. Every size-six matching avoids core edges, so both exterior graphs have rank three. Condition on selecting all of B and divide out their fixed activity product. The resulting polynomial is

$$t^3(C_0 + C_1t + C_2t^2 + C_3t^3).$$

The two nontrivial shifted inequalities are

$$8C_1^2 > 15C_0C_2, \quad 5C_2^2 > 12C_1C_3. \tag{205}$$

They supply the last two leading coefficients needed for Theorem 33.1.

The cases $\deg_B(A_1) = 3, 2$ were proved in the complete-core and matching-hole reports [32, 33]. We prove the remaining two degrees below. Isolated exterior vertices can be deleted throughout.

33.1.1 Common notation

Let n, p, q count independent exterior-left subsets of sizes one, two, three in the rank-three transversal matroid on slots B . We use

$$n \geq 3, \quad p \geq 2n - 3 \geq n, \quad q \geq n - 2 \geq 1, \quad p^2 \geq \frac{3}{2}nq. \quad (206)$$

The extension bounds follow by fixing a basis. The last inequality is a weaker consequence of the established ground-set-normalized Mason theorem [1, Theorem 4.14]:

$$p^2 \geq \frac{3n(n-1)}{2(n-2)}q. \quad (207)$$

The normalization here uses n , not the matching rank three.

For the exterior-right graph on slots A , let U, a, T be its support coefficients of orders one, two, three; let W, b count weighted independent right singles and pairs; and let V count private-coordinate/right independent pairs. A right vertex of degree one contributes twice its activity to V , and one of degree two or three contributes three times.

Let R_0 consist of right vertices avoiding A_1 . Write W_0 for its total activity and u, v for its two-slot support coefficients. Put $W_1 = W - W_0 > 0$. This positivity follows from exterior-right rank three.

33.2 The exceptional row of degree one

Relabel so that A_1 is adjacent only to B_3 inside the core. Let e count exterior-left vertices adjacent only to B_3 . Let β count exterior-left pairs whose projections to slots B_1, B_2 are independent. Define

$$d = p - \beta, \quad \mu = n - e.$$

Thus d counts independent pairs whose represented span contains the B_3 axis. Exterior rank three gives $\mu \geq 2$ and $\beta \geq 1$.

Lemma 33.2 (Exact subtraction). *The conditioned coefficients are*

$$\begin{aligned} C_0 &= q + 3p + 3n + 1 - d - 2e, \\ C_1 &= qU + pV + nW - du - eW_0, \\ C_2 &= qa + pb - dv, \quad C_3 = qT. \end{aligned} \quad (208)$$

Proof. Relative to the complete core, an independent exterior-left pair fails to permit A_1 as its leftover core vertex exactly when it cannot match to B_1, B_2 . This removes $d(1 + ut + vt^2)$. A singleton causes an additional loss exactly when it is parallel to the B_3 axis. Then the remaining core pair must be A_2, A_3 , losing two constant supports; with one exterior-right vertex, that vertex must see A_1 , losing W_0 . The additional subtraction is $e(2 + W_0t)$. These are disjoint left-cardinality cases, so no support is counted twice. $\square$

33.2.1 A left flag and the higher gap

Add diffuse generic global points and one point on the B_3 axis to the left rank-three transversal matroid. Its basis-polynomial limit is

$$\mathcal{L}(t, z, f) = qt^3 + pzt^2 + \frac{n}{2}z^2t + \frac{z^3}{6} + f\left(\beta t^2 + \mu zt + \frac{z^2}{2}\right). \quad (209)$$

It is Lorentzian by the matroid basis theorem and its closure properties [1]. The Hessian of $\partial_t \mathcal{L}$ is

$$\begin{pmatrix} 6q & 2p & 2\beta \\ 2p & n & \mu \\ 2\beta & \mu & 0 \end{pmatrix}.$$

It has at most one positive eigenvalue, and its determinant is twice

$$H = 4p\beta\mu - 2n\beta^2 - 3q\mu^2 \geq 0.$$

The exact identity

$$2\mu \left[\left(p^2 - \frac{3}{2}nq \right) - (d^2 - 2eq) \right] = H + e(2\beta^2 + \mu q)$$

therefore proves

$$d^2 - 2eq \leq p^2 - \frac{3}{2}nq. \quad (210)$$

On the right, use the global/coordinate-plane extension from [33]. In coordinates (s, t, Z, f) it is

$$\mathcal{R} = F_{\text{base}}(s, t, Z) - f(s^2 + ust + vt^2) - \frac{f^2}{2}(2s + W_0t) - \frac{f^3}{6},$$

where

$$F_{\text{base}} = s^3 + Us^2t + ast^2 + Tt^3 + Z(3s^2 + Vst + bt^2) + \frac{Z^2}{2}(3s + Wt) + \frac{Z^3}{6}.$$

The original nonnegative variables are (s, t, z, f) , with $Z = z + f$; the displayed coordinate change is used only for Hessian inertia. The Schur complement of the positive $6T$ entry of $\nabla^2 \partial_t \mathcal{R}$, evaluated at $(s, Z, f) = (q, p, d)$, gives

$$C_2^2 \geq 3qT(C_1 - nW + eW_0) + \frac{3}{2}T(Wp^2 - W_0d^2). \quad (211)$$

Since $p^2 - \frac{3}{2}nq \geq 0$ and $W_0 \leq W$, (210) implies

$$W_0(d^2 - 2eq) \leq W \left(p^2 - \frac{3}{2}nq \right).$$

Hence (211) yields

$$C_2^2 \geq 3qTC_1 - \frac{3}{4}nqWT.$$

Also $U \geq W$, $V - u \geq W$, $d \leq p$, and $\mu \geq 2$ imply

$$C_1 \geq (q + p + n - e)W \geq 2nW$$

by (206). Consequently

$$C_2^2 - \frac{12}{5}C_1C_3 \geq \frac{9}{20}nqWT > 0. \quad (212)$$

This higher-gap proof does not require the following coloop reduction.

33.2.2 Coloops and unit flag constraints

If the exterior-left matroid has a coloop ℓ , deleting it leaves an exterior graph of matching rank two. A two-vertex cover of that graph, together with ℓ and all three core-left vertices, is a minimum cover of G meeting the right shore in at most two vertices. The small-cover theorem [12] and strictness classification [26] apply. [*Merge note:* Theorem 2.1 and Theorem 14.1.] In the present positive-degree case the nontrivial graph is connected and nonstar.

We may therefore assume the exterior-left matroid is coloop-free. Then $n \geq 4$ and $2p \leq 3q$, since every independent pair has at least two basis extensions. Its projected rank-two matroid is also coloop-free: if deleting one projected element left rank one, all remaining original vectors would lie in the preimage plane, making that element a coloop before projection. Thus $\mu \geq 3$. Each projected nonloop has at least two independent partners, so double counting gives $\beta \geq \mu$.

The exact first-gap certificate uses the following 19 nonnegative polynomials, in the displayed order:

$$\begin{aligned}
g_1 &= n - 4, & g_2 &= p - 2n + 3, & g_3 &= q - n + 2, & g_4 &= 3q - 2p, \\
g_5 &= e, & g_6 &= \mu - 3, & g_7 &= d, & g_8 &= \beta - \mu, \\
g_9 &= d - e\mu, & g_{10} &= q - e\beta, & g_{11} &= q + e\beta - 2d, \\
g_{12} &= 2p^2 - 3nq, & g_{13} &= n(n-1) - e(e-1) - 2p, \\
g_{14} &= \mu(\mu-1) - 2\beta, \\
g_{15} &= 2p^2(n-2) - 3n(n-1)q, \\
g_{16} &= n(n-1)(n-2) - 6q, \\
g_{17} &= 4p\beta\mu - 2n\beta^2 - 3q\mu^2, \\
g_{18} &= 2p^2 - 3nq - 2d^2 + 4eq, \\
g_{19} &= \mu(\mu-1)(\mu-2) + 6e\beta - 6q.
\end{aligned} \tag{213}$$

The basic count bounds and Mason constraints have already been justified. Every axis/nonaxis pair is independent and counted by d , proving $d \geq e\mu$. Bases containing an axis element are exactly the $e\beta$ choices of that element and a projected independent pair. This proves $g_{10} \geq 0$ and the bound $g_{19} \geq 0$ on the remaining nonaxis triples.

Every d -pair has at least two extensions. A basis containing an axis element has exactly two d -pairs. A basis containing none has at most one: the spans of two pairs in a basis intersect in their common coordinate line. Thus

$$2d \leq q + e\beta.$$

The pair upper bound excludes the $\binom{e}{2}$ dependent axis pairs, and g_{14} is the projected pair upper bound. Equations (209)–(210) give $g_{17}, g_{18} \geq 0$. All certificate hypotheses are therefore established.

33.2.3 The first gap

For a nonempty right neighborhood $S \subseteq \{1, 2, 3\}$, put

$$\epsilon(S) = \begin{cases} 2, & |S| = 1, \\ 3, & |S| \geq 2. \end{cases}$$

Its coefficient in C_1 is

$$\kappa(S) = q|S| + p\epsilon(S) + n - \mathbf{1}_{\{1 \notin S\}}(d|S| + e) > 0.$$

For an independent pair of distinct right vertices of types S, T , its coefficient in C_2 is

$$\lambda(S, T) = q a(S, T) + p - \mathbf{1}_{\{1 \notin S \cup T\}} d,$$

where $a(S, T)$ counts its two-slot endpoint supports. Therefore the product coefficient in the first shifted gap is

$$J(S, T) = 16\kappa(S)\kappa(T) - 15C_0\lambda(S, T).$$

For all 25 unordered independent neighborhood pairs, including repeated types, the supplied exact data give

$$(n-2)J(S, T) = 1 + \sum_{\alpha} r_{\alpha} \prod_{j=1}^{19} g_j^{\alpha_j}, \quad r_{\alpha} \geq 0, \quad \alpha_j \in \mathbb{Z}_{\geq 0}. \quad (214)$$

There are 1112 nonzero rational terms. A separately written standard-library verifier reconstructs every generator, pair profile, target polynomial and rational identity, and checks complete coverage and the strict remainder one. The numerical search that found the identities is not used by this replay.

Since all $g_j \geq 0$ and $n \geq 4$, every $J(S, T) > 0$. Dependent repeated singleton types have zero C_2 coefficient, as do individual activity squares; their gap coefficients are positive. This proves the first inequality in (205) for degree one, and (212) proves the second.

33.3 The exceptional row of degree zero

Now A_1 has no core neighbors. The direct conditioned count is

$$\begin{aligned} C_0 &= q + 2p + n, & C_1 &= qU + p(V - u) + nW_1, \\ C_2 &= qa + p(b - v), & C_3 &= qT. \end{aligned} \quad (215)$$

An exterior-left triple contributes q times the full right support polynomial. A pair leaves one core-left vertex for B , which must be A_2 or A_3 , giving the plane-point profile $2 + (V - u)t + (b - v)t^2$. A singleton leaves both A_2, A_3 for B , giving $1 + W_1t$. No support with no exterior-left vertex can match all three B vertices, because the core has rank two.

The right plane Hessian evaluated at $(s, Z, f) = (q, p, p)$ gives

$$C_2^2 \geq 3qT(C_1 - nW_1) + \frac{3}{2}p^2W_1T.$$

Using (206) yields

$$C_2^2 - \frac{12}{5}C_1C_3 \geq \frac{3}{5}qTC_1 - \frac{3}{4}nqW_1T.$$

Since $U \geq W_1$, $V - u \geq 2W_1$, and $p \geq n$,

$$C_1 \geq (q + 2p + n)W_1 \geq 3nW_1.$$

The higher gap is therefore at least $\frac{21}{20}nqW_1T > 0$.

33.3.1 A direct first-gap proof

For a right neighborhood S of size k , its C_1 coefficient is

$$\kappa(S) = \begin{cases} qk + p, & 1 \notin S, \\ qk + p\epsilon(S) + n, & 1 \in S. \end{cases}$$

For an independent pair S, T , its C_2 coefficient is q if both avoid 1, and $q a(S, T) + p$ otherwise.

If both avoid 1, then $\kappa(S), \kappa(T) \geq q + p$. Their gap coefficient is at least

$$16(q + p)^2 - 15q(q + 2p + n) = q^2 + 2pq + 16p^2 - 15nq \geq q^2 + 2pq + 6p^2 > 0.$$

Otherwise take $1 \in S$. It suffices to show $\kappa(S)\kappa(T) \geq C_0\lambda(S, T)$.

1. If $S = \{1\}$, then $\kappa(S) = C_0$. If T avoids 1, $a(S, T) = |T|$ and $\kappa(T) = \lambda(S, T)$. If T contains 1, $a(S, T) = |T| - 1$ and $\kappa(T) \geq \lambda(S, T)$.
2. If $|S| \geq 2$ and T avoids 1, use $\kappa(S) \geq 2q + 3p + n$. For $|T| = 1$ and $|T| = 2$, respectively, the required comparisons follow from

$$(2q + 3p + n)(q + p) - C_0(2q + p) = p^2 - nq \geq 0,$$

$$(2q + 3p + n)(2q + p) - C_0(3q + p) = q^2 + pq + p^2 - nq \geq 0.$$

3. If both neighborhoods contain 1 and have size at least two, use

$$(2q + 3p + n)^2 - C_0(3q + p) = q^2 + 5pq + 7p^2 + nq + 5np + n^2 > 0.$$

These cases are exhaustive. Dependent pairs and individual squares again have zero C_2 contribution and positive gap coefficients. This proves the first inequality in (205). The degree-zero proof applies directly even when the graph is disconnected.

33.4 Eventual strictness and verification

Outside the already settled small-cover branch, the conditional inequalities just proved for degrees zero and one give the last two positive leading Newton gaps under common right-core scaling. At support orders one, two, three, let $a_1, a_2, a_3 > 0$ denote the leading coefficients in the scaling variable λ . They form the support sequence of the induced three-right-vertex graph, so the weighted smaller-shore theorem gives

$$a_1^2 \geq 3a_2, \quad a_2^2 \geq 3a_1a_3.$$

The first two full rank-six gaps consequently have positive leading coefficients

$$5a_1^2 - 12a_2 \geq 3a_2 > 0, \quad 8a_2^2 - 15a_1a_3 \geq 9a_1a_3 > 0.$$

At the third gap the square has scaling degree six and the adjacent product has degree at most five. Thus every full gap is eventually positive. Together with the inherited degrees two and three, this proves Theorem 33.1.

For fixed graph and fields, write each gap as $\Delta_k(\lambda) = a_{k,d_k}\lambda^{d_k} + \sum_{j < d_k} a_{k,j}\lambda^j$, with $a_{k,d_k} > 0$. A computable sufficient threshold is

$$\lambda \geq 1 + \max_{1 \leq k \leq 5} \frac{\sum_{j < d_k} |a_{k,j}|}{a_{k,d_k}}.$$

No uniform field-independent threshold is asserted.

The package [*Merge note:* shipped as `code/35-two-rows-*.py` and `data/35-two-rows-*.json`] includes the 25 exact degree-one coefficient certificates and their standard-library replay, together with 100 direct conditional-support checks for each new row degree. The left flag identities and the grouped degree-zero algebra also have exact polynomial checks. These verify the formulas; the real-field and unbounded-population conclusions follow from the proof.

The next case is a column defect, which is not obtained by transposing this theorem while keeping the same shore unit-weighted. General cores and arbitrary finite-activity rank-six ULC remain open. [*Merge note:* The column defect is source 32; general cores are covered eventually by source 21. The finite-activity status is in Part III's introduction.]

34 Second route for cores with two complete columns, one weighted shore (source 32)

Abstract of source 32, as delivered. We prove eventual strict rank-six ultra-log-concavity for bipartite graphs with a displayed $3 + 3$ minimum cover, two complete core columns, and an arbitrary third core column. Exterior neighborhoods are arbitrary; left activities are one and right activities are positive. This extends the preceding two-complete-row theorem under the same shore convention. The degree-one column uses coupled plane and axis extensions of rank-three matroids, followed by an exact 25-case rational certificate. The zero-column case reduces algebraically to the earlier complete-core inequalities. [*Merge note:* “The preceding two-complete-row theorem” is source 35 (Section 33); source 32’s bibliography does not list it.]

Scope. The scaling threshold depends on the graph and its activities. General rank-six one-shore ULC for arbitrary finite activities remains open. The argument is an ordinary mathematical proof with exact algebraic checks, not formal kernel verification or an external priority claim.

34.1 The family and exact support counts

Let the shores be $A \sqcup L$ and $B \sqcup R$, where

$$A = \{A_1, A_2, A_3\}, \quad B = \{B_1, B_2, B_3\}, \quad \nu(G) = 6.$$

There are no L – R edges. The columns B_2, B_3 are complete to A , while B_1 has arbitrary core degree. All left activities are one, and all right activities are positive real numbers.

Theorem 34.1. *For every fixed graph and set of activities as above, sufficiently large common scaling of the three right-core activities makes the matching-support polynomial strictly ULC_6 .*

[*Merge note:* Same theorem as source 21 (Section 30), restricted to cores with two complete columns: its eventual statement is the case of that core in source 21’s Corollary 30.4, and its two conditional inequalities ($8C_1^2 > 15C_0C_2$, $5C_2^2 > 12C_1C_3$) follow from source 21’s Theorem 30.1, whose constants $2n/(n-1)$ and $21/8$ are larger and which needs no hypothesis on the core; both are the case $a = b = 3$ of source 08s’s Theorem 29.1 and Corollary 29.2. Source 32’s proof is printed as a second route. What it adds: the coloop branch at every field (a corollary of sources 74 and 72), the reduction of a zero column to source 43’s inequalities, and a 25-case certificate.]

Matching supports count each feasible pair of endpoint sets once. Every size-six matching avoids core edges, so both exterior graphs have rank three. The core-column degrees three and two are the earlier complete-core and matching-hole cases [32, 33]. We prove degrees one and zero. This is not obtained by transposing the row theorem, because the unit-weighted shore is fixed.

Delete isolated exterior vertices. Let n, p, q be the numbers of independent exterior-left subsets of sizes one, two and three. Let

$$E = \{\ell \in L : \ell \not\sim B_1\}, \quad e = |E|, \quad d = I_2(E), \quad \mu = n - e, \quad h = p - d.$$

In a generic representation the E vectors lie in the B_2, B_3 coordinate plane. We use the standard bounds

$$n \geq 3, \quad p \geq 2n - 3 \geq n, \quad q \geq n - 2 \geq 1, \quad p^2 \geq \frac{3n(n-1)}{2(n-2)}q \geq \frac{3}{2}nq. \quad (216)$$

The last is the ground-set-normalized Mason inequality [1, Theorem 4.14]; its normalization uses the number n of elements, not matching rank three.

For the exterior-right graph on slots A , write U, a, T for its support coefficients of orders one, two and three, and W, b for its weighted independent-set counts of orders one and two. Let V count private-coordinate/right independent pairs: a singleton-neighborhood right vertex contributes twice its activity, and a vertex of degree two or three contributes three times. Let ω be the total activity of right vertices whose neighborhood is exactly $\{A_3\}$. Let u, v be the support coefficients after projecting right neighborhoods to $\{A_1, A_2\}$.

34.1.1 The column of degree one

Relabel so that B_1 sees only A_3 . Condition on selecting all three B vertices and divide out their fixed activity product. The resulting polynomial is

$$t^3(C_0 + C_1t + C_2t^2 + C_3t^3),$$

where

$$\begin{aligned} C_0 &= q + 3p + 3n + 1 - 2d - e, \\ C_1 &= qU + pV + nW - d(V - u) - e\omega, \\ C_2 &= qa + pb - d(b - v), \quad C_3 = qT. \end{aligned} \tag{217}$$

An independent L pair in the coordinate plane forces B_1 to use A_3 . Its allowed extension profile is $1 + ut + vt^2$, instead of $3 + Vt + bt^2$. An L singleton in that plane loses the core pair $\{A_1, A_2\}$ and the right-axis profile ωt . All other profiles are unchanged from the complete core. The empty- L support exists because this core has a perfect matching. This disjoint support partition proves (217) without matching-allocation multiplicities.

The required shifted inequalities are

$$8C_1^2 > 15C_0C_2, \quad C_2^2 > \frac{12}{5}C_1C_3. \tag{218}$$

34.2 Plane and axis extensions

Adjoin diffuse generic full-space points and diffuse generic points in the B_2, B_3 plane to the represented left matroid. Its degree-three basis-polynomial limit is

$$\mathcal{L} = qt^3 + pzt^2 + \frac{n}{2}z^2t + \frac{z^3}{6} + f\left(ht^2 + nzt + \frac{z^2}{2}\right) + \frac{f^2}{2}(\mu t + z). \tag{219}$$

It is Lorentzian by the matroid basis theorem and its nonnegative-specialization and limit closures [1]. The Hessian of $\partial_t \mathcal{L}$ is

$$\begin{pmatrix} 6q & 2p & 2h \\ 2p & n & n \\ 2h & n & \mu \end{pmatrix}.$$

Its principal (t, f) block has at most one positive eigenvalue, giving

$$h^2 \geq \frac{3}{2}q\mu. \tag{220}$$

For the other bound put $Z = z + f$. This is only a congruence for Hessian inertia, not a claim of Lorentzian closure under negative substitutions. The Hessian becomes

$$\begin{pmatrix} 6q & 2p & -2d \\ 2p & n & 0 \\ -2d & 0 & -e \end{pmatrix}.$$

When $e > 0$, eliminating the negative entry $-e$ leaves a two-by-two block of nonpositive determinant. When $e = 0$, also $d = 0$. Thus in all cases

$$nd^2 \leq e \left(p^2 - \frac{3}{2}nq \right). \quad (221)$$

Since $e \leq n$, (216)–(221) imply

$$p^2 - d^2 \geq \frac{3}{2}nq, \quad h^2 + 2eq \geq \frac{3}{2}nq. \quad (222)$$

On the right, adjoin diffuse generic full-space points and one point on the A_3 axis. The Lorentzian cubic is

$$\mathcal{F} = \mathcal{F}_{\text{base}} + f \left[s^2 + ust + vt^2 + z(2s + (W - \omega)t) + \frac{z^2}{2} \right],$$

where

$$\mathcal{F}_{\text{base}} = s^3 + Us^2t + ast^2 + Tt^3 + z(3s^2 + Vst + bt^2) + \frac{z^2}{2}(3s + Wt) + \frac{z^3}{6}.$$

The Hessian of its t derivative, in coordinates (t, s, z, f) , is

$$\begin{pmatrix} 6T & 2a & 2b & 2v \\ 2a & 2U & V & u \\ 2b & V & W & W - \omega \\ 2v & u & W - \omega & 0 \end{pmatrix}.$$

Eliminate its positive $6T$ entry. Evaluating the remaining negative-semidefinite Schur form at $(s, z, f) = (q, h, d)$ yields

$$C_2^2 \geq 3qT(C_1 - nW + e\omega) + \frac{3}{2}T \left[W(p^2 - d^2) - 2\omega dh \right]. \quad (223)$$

Since $0 \leq \omega \leq W$, the left side of

$$W(p^2 - d^2) + 2\omega(eq - dh) \geq \frac{3}{2}nqW$$

is the mixture of the two expressions in (222), with coefficients $W - \omega$ and ω . Therefore

$$C_2^2 \geq 3qTC_1 - \frac{3}{4}nqWT. \quad (224)$$

Every right activity has C_1 coefficient at least $2n$. The axis type has coefficient $q + 2h + \mu$. Here $\mu \geq 1$ and $h \geq n - 1$: if $e = 0$, use $p \geq 2n - 3$; otherwise all $e\mu$ cross-plane pairs are independent, and $e\mu \geq e + \mu - 1 = n - 1$. Thus $q + 2h + \mu \geq 2n$. Every nonaxis type has coefficient at least $q + 2p + n - d = q + p + h + n \geq 2n$. It follows that $C_1 \geq 2nW$, and (224) gives

$$C_2^2 - \frac{12}{5}C_1C_3 \geq \frac{9}{20}nqWT > 0. \quad (225)$$

34.3 The lower conditional gap

34.3.1 Coloops and valid count constraints

If the exterior-left matroid has a coloop ℓ , deleting it leaves matching rank two. A two-vertex cover of that deletion, together with ℓ and all three A vertices, gives a minimum cover of size six meeting the right shore in at most two vertices. The earlier small-cover theorem and equality classification [12, 26] [*Merge note*: Theorems 2.1 and 14.1] give strict ULC_6 at every positive field: this degree-one core makes the nontrivial graph connected and nonstar.

It remains to treat the coloop-free case. Then $n \geq 4$, $2p \leq 3q$, and $\mu \geq 2$. The last holds because deleting a sole vector outside a plane would lower rank. The first-gap certificate uses the following twenty nonnegative polynomials:

$$\begin{aligned} g_1 &= n - 4, & g_2 &= p - 2n + 3, \\ g_3 &= q - n + 2, & g_4 &= 3q - 2p, \\ g_5 &= e, & g_6 &= \mu - 2, \\ g_7 &= d, & g_8 &= q - 2d, \\ g_9 &= e(e - 1) - 2d, & g_{10} &= h - e\mu, \\ g_{11} &= e\mu + \frac{1}{2}\mu(\mu - 1) - h, & g_{12} &= h - n + 1, \\ g_{13} &= q - \mu d, & g_{14} &= 2p^2 - 3nq, \\ g_{15} &= n(n - 1) - 2p, & g_{16} &= 2p^2(n - 2) - 3n(n - 1)q, \\ g_{17} &= n(n - 1)(n - 2) - 6q, & g_{18} &= 2ep^2 - 3enq - 2nd^2, \\ g_{19} &= 2h^2 - 3q\mu. \end{aligned}$$

The remaining polynomial is

$$g_{20} = 6\mu d + 3e\mu(\mu - 1) + \mu(\mu - 1)(\mu - 2) - 6q.$$

Besides the usual extension and subset upper bounds, Mason and the two flag inequalities, the necessary incidence arguments are elementary. The h pairs consist of all $e\mu$ cross-plane pairs and independent pairs of outside elements, proving $g_{10}, g_{11} \geq 0$. Every independent E pair extends with each outside element, giving exactly μd bases of this type, hence $g_{13}, g_8 \geq 0$. The other possible bases have one E element and two outside elements, or three outside elements, proving $g_{20} \geq 0$. The bound $g_{12} \geq 0$ was proved above. In a coloop-free rank-three matroid each independent pair has at least two basis extensions; counting pair–basis incidences proves $g_4 \geq 0$.

34.3.2 Twenty-five exact coefficient identities

For a nonempty right neighborhood $S \subseteq \{1, 2, 3\}$, put $\epsilon(S) = 2$ for a singleton and 3 otherwise. Its C_1 coefficient is

$$\kappa(S) = q|S| + p\epsilon(S) + n - d(\epsilon(S) - |S \cap \{1, 2\}|) - e \mathbf{1}_{\{S=\{3\}\}}.$$

For an independent pair S, T , let $a(S, T)$ count its feasible two-slot supports, and let $\rho(S, T)$ be one when the two projected neighborhoods on $\{1, 2\}$ are independent, zero otherwise. Its C_2 coefficient is

$$\lambda(S, T) = qa(S, T) + p - d(1 - \rho(S, T)).$$

Thus the product coefficient in the first gap is

$$J(S, T) = 16\kappa(S)\kappa(T) - 15C_0\lambda(S, T).$$

For all 25 independent unordered neighborhood types, the supplied rational data give the exact identity

$$(n-2)J(S, T) = 1 + \sum_{\alpha} r_{\alpha} \prod_{j=1}^{20} g_j^{\alpha_j}, \quad r_{\alpha} \geq 0, \quad \alpha_j \in \mathbb{Z}_{\geq 0}. \quad (226)$$

There are 1134 nonzero terms. A separate standard-library verifier reconstructs every generator, neighborhood profile, target and rational identity, and checks complete coverage and strict remainder one. No numerical optimizer is used in this replay.

All g_j are nonnegative and $n \geq 4$, so every $J(S, T) > 0$. Dependent repeated-singleton types and individual activity squares have no C_2 contribution and a positive C_1 -square contribution. This proves the first inequality in (218), while (225) proves the second.

34.4 The zero-column case

If B_1 has no core neighbors, direct support counting gives

$$(C_0, C_1, C_2, C_3) = (q + 3h + 3\mu, qU + hV + \mu W, qa + hb, qT). \quad (227)$$

There is no empty- L support. A singleton in E contributes nothing; an outside singleton has profile $3 + Wt$. A pair in E contributes nothing; every other independent pair has profile $3 + Vt + bt^2$. The triple profile is unchanged.

Here $\mu, q \geq 1$ and $h \geq \mu$: for $e = 0$ use $p \geq n$, and otherwise use $h \geq e\mu$. Equation (220) gives $h^2 \geq 3q\mu/2$.

The earlier complete-core analytic inequalities [32] apply algebraically to the triple $(n, p, q) = (\mu, h, q)$. For precision, that first-gap proof expands six quadratic coefficients in the activity sums of right neighborhoods of sizes one, two and three; their positivity requires only positive integers μ, h, q and $h^2 \geq \mu q$. Its higher-gap proof additionally uses $h \geq \mu$ and gives a positive lower bound

$$\frac{3}{10}qWT(2q + 4h - 3\mu) > 0.$$

The complete-core comparison has the larger constant $q + 3h + 3\mu + 1$. Reducing that constant by one strictly increases the first gap and does not change the second. Both inequalities therefore hold for (227). This direct argument allows disconnected graphs and does not require a coloop reduction.

34.5 Eventual scaling and verification

Outside the already settled small-cover branch, the two conditional inequalities give positive leading terms in the last two full rank-six Newton gaps when all B activities are multiplied by λ . At support orders one through three, let $a_1, a_2, a_3 > 0$ be the leading coefficients in λ . The induced three-column graph gives

$$a_1^2 \geq 3a_2, \quad a_2^2 \geq 3a_1a_3.$$

The first two full gaps consequently have positive leading coefficients

$$5a_1^2 - 12a_2 \geq 3a_2 > 0, \quad 8a_2^2 - 15a_1a_3 \geq 9a_1a_3 > 0.$$

At the third gap, the square has degree six in λ and the adjacent product has degree at most five. Every gap is therefore eventually positive. The inherited core-column degrees two and three complete Theorem 34.1.

For fixed graph and fields, write each gap as $\Delta_k(\lambda) = a_{k,d_k} \lambda^{d_k} + \sum_{j < d_k} a_{k,j} \lambda^j$ with $a_{k,d_k} > 0$. A computable sufficient bound is

$$\lambda \geq 1 + \max_{1 \leq k \leq 5} \frac{\sum_{j < d_k} |a_{k,j}|}{a_{k,d_k}}.$$

No uniform threshold is claimed.

The package [*Merge note:* shipped as `code/32-two-columns-*.py` and `data/32-two-columns-*.json`] contains all 25 rational identities, their exact standard-library replay, and 100 direct support enumerations for each new column degree. The degree-one checker also verifies all twenty constraints on 66 coloop-free samples. Exact flag-algebra checks accompany them. The finite graph tests are regressions; unbounded population and positive-real-field conclusions follow from the mathematical proof and exact coefficient identities.

The next frontier consists of more general core shapes. This theorem and the corresponding row theorem together handle every core with at most two missing edges, but do not settle arbitrary finite-activity rank-six one-shore ULC. [*Merge note:* The row theorem is source 35; every core is covered by source 21.]

35 Second route for cores with at least six edges, one weighted shore (source 29)

Abstract of source 29, as delivered. Consider a genuine rank-six bipartite graph with a displayed $3 + 3$ minimum cover. If its 3×3 core has at least six edges, then, with unit left activities and arbitrary positive right activities, sufficiently large common scaling of the three right-core activities makes every actual-rank Newton inequality strict. The three remaining six-edge core types are handled by nested, transverse and two-plane matroid extensions. Their higher conditional gaps follow from a common Schur-complement criterion. Their lower gaps have 75 exact rational coefficient identities, with 5617 positive terms, checked by two independent reconstructions. An explicit orbit classification completes the stated core threshold.

Scope. The threshold depends on the graph and its fixed activities. General rank-six one-shore ULC for arbitrary finite activities remains open. This is an ordinary mathematical proof with exact algebraic verification, not formal kernel verification or an external priority claim.

35.1 The theorem and the finite frontier

Let the shores of G be $A \sqcup L$ and $B \sqcup R$, with

$$A = \{A_1, A_2, A_3\}, \quad B = \{B_1, B_2, B_3\}, \quad \nu(G) = 6,$$

and no L – R edges. The core is the graph between A and B . All left activities are one and all right activities are positive. A matching support is a feasible pair of endpoint sets, counted once regardless of the number of perfect matchings.

Theorem 35.1. *If the core has at least six edges, then for every fixed graph and set of right activities, sufficiently large common scaling of all three B activities makes the matching-support polynomial strictly ULC₆.*

[*Merge note:* Same theorem as source 21 (Section 30), restricted to cores with at least six edges: its eventual statement is the case of that core in source 21’s Corollary 30.4, and its two

conditional inequalities ($8C_1^2 > 15C_0C_2$, $5C_2^2 > 12C_1C_3$) follow from source 21's Theorem 30.1, whose constants $2n/(n-1)$ and $21/8$ are larger and which needs no hypothesis on the core; both are the case $a = b = 3$ of source 08s's Theorem 29.1 and Corollary 29.2. Source 29's proof is printed as a second route. What it adds: the exhaustive orbit census of cores with six edges, the coloop branch at every field (a corollary of sources 74 and 72), and 75 exact identities for the three remaining six-edge cores.]

Every size-six matching avoids core edges, so both exterior graphs have rank three. Let n, p, q be the exterior-left independent-set counts of orders one, two and three, after removing isolated vertices. Condition on selecting all B vertices and divide out their fixed activity product. Write the resulting sector as

$$t^3(C_0 + C_1t + C_2t^2 + C_3t^3).$$

The two conditional inequalities needed for the last leading gaps are

$$8C_1^2 > 15C_0C_2, \quad C_2^2 > \frac{12}{5}C_1C_3. \quad (228)$$

We use the established count bounds

$$n \geq 3, \quad p \geq 2n - 3 \geq n, \quad q \geq n - 2 \geq 1, \quad p^2 \geq \frac{3n(n-1)}{2(n-2)}q \geq \frac{3}{2}nq. \quad (229)$$

The last is the ground-set-normalized Mason inequality [1, Theorem 4.14]. Its normalization is the number n of elements, not matching rank three.

35.1.1 The coloop branch

If the exterior-left matroid has a coloop ℓ , deleting it leaves matching rank two. A two-vertex cover of that deletion, together with ℓ and all three A vertices, is a minimum cover of size six meeting the right shore in at most two vertices. The earlier small-cover theorem applies at every positive field [12]. [*Merge note:* Theorem 2.1; the equality classification below is Theorem 14.1.] A six-edge core on six vertices contains a cycle, so the full graph is not a union of stars. The established equality classification therefore makes every gap strict [26], even if the graph is disconnected. We may assume below that the exterior-left matroid is coloop-free.

35.1.2 Exhaustive core classification

With at least seven core edges, the missing graph has at most two edges. Zero or one missing edge is the complete-core or one-hole family. Two missing edges are disjoint, incident on one left vertex, or incident on one right vertex. The matching-hole, two-complete-row and two-complete-column theorems cover these possibilities [32, 33, 34, 35].

For exactly six core edges, the missing graph has three edges. If its maximum degree is three, it is a row star or column star. If its maximum degree is at most one, it is a matching. Otherwise a degree-two vertex exists: the third edge either extends its two-edge star to a three-edge path or is disjoint from it. The disjoint case has two orientations, according to the shore of the degree-two vertex. Thus there are exactly six orbits under independent row and column permutations:

matching, row star, column star, path, star-edge, transposed star-edge.

The first three complements are already settled. Convenient core row-neighborhood representatives for the remaining three are

$$(123, 12, 1), \quad (123, 23, 1), \quad (13, 12, 12). \quad (230)$$

Shore transposition is not used as an equivalence of weighted theorems.

35.2 A common higher-gap criterion

On the exterior right, let U, a, T be support coefficients of orders one, two and three, and let W, b be weighted independent-set counts of orders one and two. Let V count private-coordinate/right independent pairs: a right vertex of degree one contributes twice its activity, and one of degree two or three contributes three times.

For nonempty $S \subseteq \{1, 2, 3\}$, let V_S count a generic S vector, one right vertex and a private coordinate forming a basis. A right neighborhood D contributes the number of feasible two-slot supports of (S, D) . Let B_S count right pairs which, with a generic S vector, form a basis. Write W_S for the total activity of right vertices meeting S . Thus $V_{123} = V$ and $B_{123} = b$.

Adjoining generic vectors in fixed subspaces to a represented matroid preserves representability. Give a family of N generic vectors activity z/N and let $N \rightarrow \infty$. The resulting basis-polynomial limits are Lorentzian by the matroid basis theorem and closure under non-negative specialization and limits [1]. This supplies all flag polynomials used below. Invertible signed coordinate changes are used only for Hessian inertia.

Each right flag gives, after eliminating the positive $6T$ entry of the t -derivative Hessian, an inequality of the form

$$C_2^2 \geq 3qT \left(C_1 - \sum_D w_D \beta_D \right) + \frac{3}{2}T \sum_D w_D Q_D. \quad (231)$$

Here w_D is an individual right activity, β_D is its contribution from singleton exterior-left supports, and Q_D is the evaluated pair quadratic. If

$$Q_D \geq \frac{3}{2}q\beta_D, \quad \kappa(D) \geq 2\beta_D, \quad (232)$$

where $\kappa(D)$ is its C_1 coefficient, then

$$C_2^2 - \frac{12}{5}C_1C_3 \geq \frac{9}{20}qT \sum_D w_D \beta_D > 0. \quad (233)$$

In every case below $\beta_D > 0$ for nonempty D . The last strict inequality uses the genuine rank-six hypothesis.

35.3 The triangular core

Take core rows $(123, 12, 1)$. Let e count axis-1 exterior-left vectors, f count vectors in coordinate plane 12, d count independent pairs whose span contains axis 1, and g count independent pairs in plane 12. Put

$$u = p - d, \quad v = d - g, \quad \mu = n - e, \quad m = n - f.$$

Then $e \leq f$, $g \leq d$, and $m \geq 1$. The exact support count is

$$\begin{aligned} C_0 &= q + 3p - d - g + 3n - f - e + 1, \\ C_1 &= qU + uV + vV_{12} + gV_1 + (n - f)W + (f - e)W_{23} + eW_3, \\ C_2 &= qa + ub + vB_{12} + gB_1, \quad C_3 = qT. \end{aligned} \quad (234)$$

For a left pair, core row 1 always extends it; row 2 fails precisely for a plane-12 pair, and row 3 fails precisely for an axis-spanning pair. The resulting line profiles have counts u, v, g . For a left singleton, the normal supports of its allowed right extension are 123, 23, 3, with counts $n - f, f - e, e$. This is a disjoint endpoint-support partition.

The left global/plane/axis flag is

$$\begin{aligned}\mathcal{L} = & qt^3 + pzt^2 + \frac{n}{2}z^2t + \frac{z^3}{6} \\ & + y \left((p-g)t^2 + nzt + \frac{z^2}{2} \right) + \frac{y^2}{2}((n-f)t + z) \\ & + x \left(ut^2 + \mu zt + \frac{z^2}{2} \right) + xy(mt + z).\end{aligned}$$

The global/plane Hessian gives $p^2 - g^2 \geq 3nq/2$, as in [35]. The global/axis Hessian gives

$$H = 4pu\mu - 2nu^2 - 3q\mu^2 \geq 0.$$

The identity

$$2\mu \left(p^2 - d^2 - \frac{3}{2}q\mu \right) = H + 2eu^2$$

therefore yields $p^2 - d^2 \geq 3q\mu/2$. Finally the principal Hessian on (t, y, x) is

$$\begin{pmatrix} 6q & 2(p-g) & 2u \\ 2(p-g) & m & m \\ 2u & m & 0 \end{pmatrix}.$$

Its determinant is $2m[4(p-g)u - 2u^2 - 3qm] \geq 0$. Hence

$$p^2 - g^2 \geq \frac{3}{2}qn, \quad p^2 - d^2 \geq \frac{3}{2}q\mu, \quad u(u+2v) \geq \frac{3}{2}qm. \quad (235)$$

The dual right extension uses generic global points, plane-12 points and an axis-1 point. Its pair block in the t -derivative Hessian is

$$\begin{pmatrix} W & W & W_{23} \\ W & W_3 & W_3 \\ W_{23} & W_3 & 0 \end{pmatrix}.$$

Evaluate at pair variables (u, v, g) , with the private-coordinate variable equal to q . For a right neighborhood D ,

$$\beta_D = m + (f - e)\mathbf{1}_{D \cap 23 \neq \emptyset} + e\mathbf{1}_{3 \in D}.$$

The values of Q_D , according as D meets slot 3, meets 2 but not 3, or equals $\{1\}$, are

$$p^2 - g^2, \quad p^2 - d^2, \quad u(u+2v).$$

Thus (235) proves the first half of (232).

By neighborhood monotonicity, the second half reduces to the singleton coefficients

$$q + 2p - d - g + m, \quad q + 2p - d + \mu, \quad q + 2p - g + n.$$

The last two use $p \geq n$. For the first, $p - g \geq n - 1$: if $f = 0$ use $p \geq 2n - 3$; otherwise the $f(n - f)$ cross-plane pairs suffice. Also $u \geq 1$. This proves (232), and hence the strict higher gap.

35.4 A star plus a disjoint missing edge

Take core rows $(123, 23, 1)$. Let e count axis-1 vectors, f count plane-23 vectors, d count axis-spanning pairs and g count independent plane-23 pairs. Here the axis and plane are transverse:

$$e + f \leq n, \quad d + g \leq p, \quad \mu = n - e, \quad u = p - d.$$

The exact cubic is

$$\begin{aligned} C_0 &= q + 3p - d - g + 3n - 2e + 1, \\ C_1 &= qU + (p - d - g)V + dV_{12} + gV_{13} + (n - e)W + eW_3, \\ C_2 &= qa + (p - d - g)b + dB_{12} + gB_{13}, \quad C_3 = qT. \end{aligned} \tag{236}$$

The two pair defects are disjoint because a plane-23 pair cannot span axis 1. A singleton loses core-pair choices exactly when it is on axis 1, leaving only A_3 for the right extension.

The left extension uses global points z , one axis point x and diffuse plane points y . Its mixed xyt coefficient is μ ; an $xy^2/2$ term also occurs. Put $Z = z + y$. The t -derivative Hessian in coordinates (t, Z, x, y) is

$$\begin{pmatrix} 6q & 2p & 2u & -2g \\ 2p & n & \mu & 0 \\ 2u & \mu & 0 & 0 \\ -2g & 0 & 0 & -f \end{pmatrix}.$$

When $f > 0$, eliminating $-f$ replaces q in the axis Hessian by $q' = q + 2g^2/(3f)$. Its determinant bound gives

$$2\mu \left(p^2 - d^2 - g^2 - \frac{3}{2}q\mu \right) \geq 2g^2\mu \left(\frac{\mu}{f} - 1 \right) + 2eu^2 \geq 0.$$

Here $f \leq \mu$. If $f = 0$, then $g = 0$ and the axis inequality applies directly. Together with the individual axis and plane bounds,

$$p^2 - g^2 \geq \frac{3}{2}nq, \quad p^2 - d^2 \geq \frac{3}{2}q\mu, \quad p^2 - d^2 - g^2 \geq \frac{3}{2}q\mu. \tag{237}$$

On the right use generic coordinate-plane points in 12 and 13. The Schur evaluation at weights $(p - d - g, d, g)$ gives

$$Q_D = p^2 - d^2 \mathbf{1}_{3 \notin D} - g^2 \mathbf{1}_{2 \notin D}, \quad \beta_D = \mu + e \mathbf{1}_{3 \in D}.$$

The bounds (237), and $p^2 \geq 3nq/2$, prove $Q_D \geq 3q\beta_D/2$. Singleton types 1, 2, 3 have pair contributions $2p - d - g, 2p - d, 2p - g$, all at least p . Thus $\kappa(D) \geq 2\beta_D$ by $p \geq n$ and monotonicity, proving the strict higher gap.

35.5 The transposed star-edge core

Take core rows $(13, 12, 12)$. Let f_1, f_2 count left vectors in planes 13, 12 and g_1, g_2 their independent-pair counts. The pair classes are disjoint. The support count is

$$\begin{aligned} C_0 &= q + 3p - g_1 - 2g_2 + 3n - f_2 + 1, \\ C_1 &= qU + (p - g_1 - g_2)V + g_1V_{23} + g_2V_1 + (n - f_2)W + f_2W_{23}, \\ C_2 &= qa + (p - g_1 - g_2)b + g_1B_{23} + g_2B_1, \quad C_3 = qT. \end{aligned} \tag{238}$$

A plane-13 pair loses core row 1, and a plane-12 pair loses both rows 2, 3. A singleton in plane 12 loses only the core pair $\{A_2, A_3\}$.

For the left two-plane/global extension put $Z = z + y_1 + y_2$. The Hessian becomes

$$\begin{pmatrix} 6q & 2p & -2g_1 & -2g_2 \\ 2p & n & 0 & 0 \\ -2g_1 & 0 & -f_1 & 0 \\ -2g_2 & 0 & 0 & -f_2 \end{pmatrix}.$$

Eliminating negative entries gives

$$p^2 \geq \frac{3}{2}nq + n \left(\frac{g_1^2}{f_1} + \frac{g_2^2}{f_2} \right).$$

Terms with $f_i = 0$ are omitted, since then $g_i = 0$. As $f_i \leq n$,

$$p^2 - g_1^2 - g_2^2 \geq \frac{3}{2}nq, \quad (p - g_2)^2 \geq \frac{3}{2}q(n - f_2), \quad (239)$$

the latter being the individual plane bound.

The right dual extension is plane 23 plus transverse axis 1. The Schur quadratic is $p^2 - g_1^2 - g_2^2$ when D meets 23 but avoids 1, $p^2 - g_2^2$ when it meets both, and $(p - g_2)^2$ when $D = \{1\}$. The corresponding β_D values are $n, n, n - f_2$. Thus (239) proves the first criterion. The singleton coefficients prove the second using $p \geq n$ and $p - g_2 \geq n - 1$. Hence the higher gap is strict.

35.6 The exact lower-gap certificates

For each of the three cores and a right neighborhood S , let $\kappa(S)$ be the coefficient of its activity in C_1 . For an independent pair S, T , let $\lambda(S, T)$ be the coefficient of its activity product in C_2 . Both are given directly by the support formulas above: V_D uses two-slot matchability, and B_D uses three-slot matchability.

The product coefficient in $8C_1^2 - 15C_0C_2$ is

$$J(S, T) = 16\kappa(S)\kappa(T) - 15C_0\lambda(S, T).$$

For every independent unordered pair of nonempty neighborhoods, the package gives an exact identity

$$(n - 2)J(S, T) = 1 + \sum_{\alpha} r_{\alpha} \prod_j G_j^{\alpha_j}, \quad r_{\alpha} > 0, \quad \alpha_j \in \mathbb{Z}_{\geq 0}. \quad (240)$$

There are 25 pair types for each core. The generator counts and positive-term counts are

core	generators	identities	terms
(123, 12, 1)	34	25	2369
(123, 23, 1)	46	25	989
(13, 12, 12)	32	25	2259

All 75 identities, totaling 5617 terms, pass a standard-library rational reconstruction and a separately written symbolic reconstruction. The former hardcodes the generator and profile formulas and does not import the discovery code or evaluate stored expressions. No numerical optimizer is used in either replay.

All generators are nonnegative counts or established flag inequalities; the complete families and their justifications are given below. Thus (240) proves strict positivity. Dependent repeated-singleton types and individual activity squares have no C_2 contribution and a positive C_1 -square coefficient. This proves the first inequality in (228).

35.6.1 Generator families and their hypotheses

The base generators are the nonnegative quantities in (229), the pair and triple subset upper bounds, and $n - 4$, $3q - 2p$. Coloop-freeness gives $n \geq 4$ and every independent pair has at least two basis extensions, so $2p \leq 3q$.

For an axis class of size e , put $\mu = n - e$ and $u = p - d$. The projection onto the complementary coordinate plane is a coloop-free rank-two matroid: a coloop there would put all other original vectors into the preimage of a line, contradicting the original coloop-free rank three. Consequently $\mu \geq 3$ and $u \geq \mu$. The axis generators, besides nonnegative counts, are

$$\begin{aligned} d - e\mu, \quad q - eu, \quad q + eu - 2d, \\ n(n-1) - e(e-1) - 2p, \quad \mu(\mu-1) - 2u, \\ H = 4pu\mu - 2nu^2 - 3q\mu^2, \\ 2p^2 - 3nq - 2d^2 + 4eq, \quad \mu(\mu-1)(\mu-2) + 6eu - 6q. \end{aligned}$$

All axis/nonaxis pairs contribute $e\mu$ to d ; bases containing an axis element number exactly eu . A basis containing an axis element has two axis-spanning pairs, and another basis has at most one. Pair-basis incidence counting yields $2d \leq q + eu$. The subset upper bounds are immediate. The Hessian gives $H \geq 0$, and

$$2\mu \left[p^2 - \frac{3}{2}nq - d^2 + 2eq \right] = H + e(2u^2 + \mu q)$$

gives the remaining mixed inequality.

For a plane class of size f , pair count g , $m = n - f$ and $h = p - g$, the generators are

$$\begin{aligned} f, \quad m - 2, \quad g, \quad f(f-1) - 2g, \quad h - fm, \\ 2fm + m(m-1) - 2h, \quad q - mg, \quad h - n + 1, \\ 6mg + 3fm(m-1) + m(m-1)(m-2) - 6q, \\ 2fp^2 - 3fnq - 2ng^2, \quad 2h^2 - 3qm. \end{aligned}$$

Coloop-freeness gives $m \geq 2$. The fm cross-plane pairs are all independent. Bases containing an independent plane pair number exactly mg ; the remaining bases have one plane element and two outside elements, or three outside elements. These observations prove the counting inequalities. The last two are the individual plane Hessian bounds.

In the nested case, $e \leq f$, and the stronger inequalities used are

$$\begin{aligned} g &\geq e(f - e), \quad 2g \leq f(f - 1) - e(e - 1), \\ e(n - f) &\leq d - g \leq e(n - f) + \binom{n - f}{2}, \\ 2u[2(p - g) - u] &\geq 3q(n - f). \end{aligned}$$

An axis-spanning pair with one vector in the plane and one outside must have its plane vector on the axis; otherwise both vectors would lie in the plane. The other such pairs have both vectors outside. The final inequality was proved by the nested Hessian.

In the transverse axis/plane case put $k = n - e - f \geq 0$. In addition to $d + g \leq p$ and the combined bound (237), the certificate uses

$$\begin{aligned} p - g &\geq e\mu + fk, \quad q \geq eu + kg, \\ q &\leq eu + kg + f \binom{k}{2} + \binom{k}{3}, \\ 2f(n - f) + (n - f)(n - f - 1) - e(e - 1) - 2(p - g) &\geq 0. \end{aligned}$$

The first counts axis/nonaxis and plane/outside-both pairs. Bases are partitioned according to whether they contain an axis element; without one, two plane elements and one outside-both element give exactly kg . The last upper bound removes the dependent axis pairs from the outside set.

Two stronger linear bounds are also used:

$$u \geq 2\mu - 4, \quad p - g \geq 2n - 4.$$

For the first, every parallel class in the projected coloop-free rank-two matroid has size at most $\mu - 2$. If its largest class has size at least two, cross pairs alone give $2\mu - 4$; if that size is one, all pairs are independent. For the second, if $2 \leq f \leq n - 2$ use $f(n - f) \geq 2n - 4$, and if $f \leq 1$ use $g = 0$ and $p \geq 2n - 3$.

Finally the two-plane case adds $p - g_1 - g_2 \geq 0$ and (239). The separate plane pair classes are disjoint. Redundant linear generators such as $d, u, \mu, n, p, q, n - f, p - g, n - e - f$ are explicitly retained when nonnegative in the relevant geometry. This matters for the bounded-degree certificate, although it adds no hypothesis. The exact ordered generator recipes and all rational multipliers are supplied in the package.

35.7 From conditional gaps to eventual strictness

Outside the small-cover branch, scale all B activities by λ . At support orders three through six the leading λ^3 coefficients are the fixed forced-activity product times C_0, C_1, C_2, C_3 . Thus (228) makes the last two leading gaps positive.

At orders one, two and three, write the leading coefficients as $a_1, a_2, a_3 > 0$. They form the support sequence of the induced graph on the three B vertices, so

$$a_1^2 \geq 3a_2, \quad a_2^2 \geq 3a_1a_3.$$

The first two rank-six gaps have positive leading coefficients

$$5a_1^2 - 12a_2 \geq 3a_2 > 0, \quad 8a_2^2 - 15a_1a_3 \geq 9a_1a_3 > 0.$$

At the third gap the square has degree six in λ and the adjacent product has degree at most five. This proves Theorem 35.1.

Writing each gap as $\Delta_j(\lambda) = a_{j,d_j}\lambda^{d_j} + \sum_{i < d_j} a_{j,i}\lambda^i$, a computable sufficient threshold is

$$\lambda \geq 1 + \max_{1 \leq j \leq 5} \frac{\sum_{i < d_j} |a_{j,i}|}{a_{j,d_j}}.$$

The threshold is field-dependent. Nothing here establishes all-finite-activity ULC6.

The package [*Merge note:* shipped as `code/29-six-edges-*.py` and `data/29-six-edges-*.json`; the three `find*_first.json` files are `data/29-six-edges-find*_first.json`] includes 300 exact support regressions, all generator checks on their coloop-free examples, both complete rational reconstructions and an independent census of the six-edge core orbits. The finite graph tests check the formulas; the unbounded-population and positive-real-activity conclusions follow from the proof and exact identities.

36 Eventual one-shore scaling through rank seven (source 08)

Abstract of source 08, as delivered. For a bipartite graph of matching rank seven, unit left activities and arbitrary positive right activities, common scaling of the right part of any

fixed minimum cover is eventually ultra-log-concave after normalization by the actual matching rank. The new case is the middle gap of the quartic obtained by forcing a $4 + 3$ cover's three right vertices. We prove the uniform bound $C_2^2 \geq (15/8)C_1C_3$, strictly stronger than the required factor $9/5$. The proof compares the conditional quartic with a rank-four matroid specialization. A coefficientwise upper incidence inequality complements the earlier lower incidence bound, and a summation over pair annihilators controls the remaining cubic sector. This closes the last rank-seven eventual-scaling case and improves the known persistent-failure bracket to $[8, 38]$. General finite-activity rank-six and rank-seven one-shore ultra-log-concavity remain open. [*Merge note:* At matching number six, gaps one, two and five hold at every finite one-shore activity (Part III); gaps three and four, and rank seven at finite activity, stay open.]

36.1 Main theorem and normalization

For a finite bipartite graph $G = (X, Y)$, let

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, \bar{J}] \text{ perfectly matchable}}} u_I v_J t^{|I|}.$$

Every feasible endpoint pair is counted once, regardless of its number of perfect matchings. If the matching rank is d and $P_G = \sum p_j t^j$, the actual-rank Newton gaps are

$$G_j = j(d - j)p_j^2 - (j + 1)(d - j + 1)p_{j-1}p_{j+1}. \quad (241)$$

We write $P_G \in \text{ULC}_d$ if all these gaps are nonnegative and there are no internal zeros.

Fix a minimum vertex cover $A \sqcup B$, with $A \subseteq X$, $B \subseteq Y$. Put $L = X \setminus A$ and $R = Y \setminus B$. Give every left vertex activity one and every right vertex a fixed positive activity. In common right-cover scaling, replace v_y by λv_y for $y \in B$ only, with $\lambda > 0$.

Theorem 36.1. *For every graph of matching rank at most seven, every fixed minimum cover and every fixed positive right field, $P_G \in \text{ULC}_d$ for all sufficiently large λ . For rank seven, the gaps are eventually all strict when the cover splits $3 + 4$ or $4 + 3$.*

The global statement is nonstrict: the small-cover cases can include equal-weight unions of stars. The threshold depends on the graph, the cover and the fixed field. [*Merge note:* Source 01 (Theorem 39.1) extends Theorem 36.1 to matching number eight.]

The earlier structural companion [28], included unchanged with this report, proves the $3 + 4$ case and the first and last conditional gaps for a $4 + 3$ cover. [*Merge note:* Source 08s, printed as Section 29; the embedded archive is not shipped.] We settle its one remaining middle gap. Assume henceforth

$$|A| = 4, \quad |B| = 3, \quad \nu(G) = 7.$$

Exterior matchings have ranks three on $L-B$ and four on $A-R$. Forcing every B vertex into the right endpoint set and dividing out v_B gives

$$t^3 C(t), \quad C(t) = C_0 + C_1 t + C_2 t^2 + C_3 t^3 + C_4 t^4, \quad (242)$$

with every coefficient positive.

Theorem 36.2. *For every genuine $4 + 3$ cover under the stated activities,*

$$C_2^2 \geq \frac{15}{8}C_1C_3 > \frac{9}{5}C_1C_3. \quad (243)$$

In particular the conditional middle gap has the uniform strict margin

$$C_2^2 - \frac{9}{5}C_1C_3 \geq \frac{3}{40}C_1C_3 > 0.$$

No hypothesis is imposed on the core's support, rank, connectivity or number of exterior vertices.

36.2 Two bounds for plane incidence

Let $\mathcal{L}$ be a loopless rank-three vector configuration, allowing repeated and parallel labeled vectors, and fix a plane H in its ambient row space. Original elements may have arbitrary positive weights. Let q count weighted independent triples and let β be the mass outside H .

For every independent pair P whose plane is not H , its intersection $\text{span}(P) \cap H$ is a line. Let D_H be the weighted count of unordered pairs of such labeled pairs having different intersection lines. Equivalently, it counts two pair annihilators independent after restriction to H ; each virtual pair has weight equal to the product of its original element weights.

Lemma 36.3. *For every such configuration,*

$$\frac{3}{4}q\beta \leq D_H \leq q\beta. \quad (244)$$

The upper inequality holds coefficientwise in the original element weights.

[Merge note: The lower bound is source 08s's Lemma 29.4 at $b = 3$ (also source 21's Lemma 30.2); the coefficientwise upper bound is new here.]

Proof. For the lower bound, write h for the weighted count of independent pairs not spanning H , grouped by intersection-line counts h_δ . Then

$$2D_H = h^2 - \sum_{\delta} h_{\delta}^2 \geq \frac{3}{2}q\beta$$

by the two-plane incidence lemma of [28, Lemma 2.1]. Its proof is a positive-pivot Schur argument for a rank-three matroid with additional generic points in H .

For the upper bound, compare coefficients of original element weights. A term in D_H uses three or four distinct labels. A squared-label monomial $w_i^2 w_j w_k$ contributes exactly when i, j, k form a basis and i lies outside H : the planes of ij and ik meet in the line of i . This is exactly its coefficient in $q\beta$. Identical pair labels contribute zero.

Now consider four distinct labels. A loop or total rank at most two makes both coefficients zero. If a pair is parallel and total rank is three, the two remaining pair partitions are good precisely when the common parallel direction is outside H ; the $q\beta$ coefficient is the same, either zero or two.

If there is no parallel pair but one dependent triple, call it i, j, k , with the fourth vector outside their plane. The three pair partitions meet in the lines of i, j, k respectively. Thus both coefficients equal the number of those three vectors outside H .

The remaining restriction is $U_{3,4}$, where every triple is independent. The coefficient of $q\beta$ is the number of the four vectors outside H . There are at most three pair partitions. At least two vectors lie outside H ; if exactly two do, the pair of inside vectors spans H , invalidating one partition and leaving at most two. Thus $q\beta$ dominates in every case. This proves the coefficientwise inequality. $\square$

Lemma 36.4. *For every independent pair P of $\mathcal{L}$, take an annihilator v_P with activity equal to the product of its original weights. Let $F_{\mathcal{L}}$ count independent unordered triples of these labeled virtual points. Then*

$$F_{\mathcal{L}} \geq \frac{3}{4}q^2. \quad (245)$$

Proof. For fixed P , apply the lower bound in (244) with $H = \text{span}(P)$. Let D_P count unordered virtual pairs completing v_P to a basis, and let β_P be the original weight outside $\text{span}(P)$. Then $D_P \geq 3q\beta_P/4$. Summing over P gives

$$3F_{\mathcal{L}} = \sum_P w_P D_P \geq \frac{3}{4}q \sum_P w_P \beta_P = \frac{9}{4}q^2.$$

Each original basis has three pair-plus-outside decompositions. Each virtual basis has three choices of marked virtual point. Pairs sharing original labels retain their product weights on both sides. Dividing by three proves the result. $\square$

36.3 The auxiliary rank four matroid

Represent graph edges by independent indeterminates. A square minor is nonzero exactly when its endpoints admit a perfect matching. Write K for the 4×3 core, ℓ_i for the exterior-left rows and r_j for the exterior-right columns. Let $q \geq 1$ be the unit count of independent triples of the ℓ_i .

For every independent labeled L pair P , choose an annihilator v_P and introduce the virtual right point

$$w_P = K v_P.$$

Retain all virtual labels, including loops and parallel elements. Adjoin the four private A -coordinate vectors E . The resulting rank-four matroid has a Lorentzian basis polynomial [1]. Give the private elements common mark s , original R elements mark t times their actual activities, and virtual elements mark z and activity one. Denote the specialization by $F(s, t, z)$.

We name the relevant coefficients as follows:

$$\begin{aligned} F(s, t, z) = & Tt^4 + Ast^3 + Us^2t^2 + Vs^3t + s^4 \\ & + Pzt^3 + Wzst^2 + Xzs^2t + \cdots \\ & + Dz^2t^2 + E_2z^2st + F_3z^3t + \cdots. \end{aligned} \quad (246)$$

All are weighted basis counts with no factorial normalization. In particular D, E_2 use unordered pairs of virtual labels and F_3 uses unordered triples.

Partition the forced coefficients by their selected L size. Let N_2 be the C_2 sector with one L vertex and all four A vertices. Let N_1 be the C_1 sector with one L vertex and three A vertices, and let Z be its sector with no L vertices. Thus Z is the sum of activities of r for which $[K \ r]$ is nonsingular. The exact identities are

$$\begin{aligned} C_1 &= qV + X + N_1 + Z, & C_2 &= qU + W + N_2, \\ C_3 &= qA + P, & C_4 &= qT. \end{aligned} \quad (247)$$

Three selected independent L rows contribute the terms multiplied by q ; two rows yield the virtual point after elimination; one and zero rows give the indicated remaining sectors. Private complements record the selected A coordinates. This is a partition of endpoint sets, not of their matching allocations.

More explicitly, for a fixed independent pair of original R vectors, quotient the right space by their span. If the induced 2×3 core has rank two, N_2 counts the original L vectors complementary to its row plane; if it has smaller rank, the contribution vanishes. For N_1 , use instead a pair (r, e) with one original R vector and one omitted private coordinate. Write its outside count as $\beta_{r,e}$; then

$$N_1 = \sum_{r,e} v_r \beta_{r,e}. \quad (248)$$

A dependent pair or deficient quotient core has $\beta_{r,e} = 0$.

Substitute $s = qu$ and $z = u$ in F . This gives a homogeneous Lorentzian quartic $f(t, u)$ whose relevant coefficients are

$$\begin{aligned} [t^3u]f &= C_3, \\ B_2 &:= [t^2u^2]f = q^2U + qW + D = q(C_2 - N_2) + D, \\ B_1 &:= [tu^3]f = q^3V + q^2X + qE_2 + F_3. \end{aligned} \tag{249}$$

36.4 Comparing the three middle coefficients

Apply the upper bound of Lemma 36.3 in the quotient for each fixed independent R pair. It gives $D_J \leq q\beta_J$; singular quotient maps give zero on both sides. Summing with product activities yields

$$D \leq qN_2, \quad B_2 \leq qC_2. \tag{250}$$

The same quotient geometry for an independent pair (r, e) , using the lower bound instead, gives

$$E_2 \geq \frac{3}{4}qN_1. \tag{251}$$

Here E_2 counts each unordered virtual pair once for each fixed r, e , exactly as in Lemma 36.3.

For a fixed r , quotient the right space by its line. The induced 3×3 core map is invertible precisely when $[K \ r]$ is nonsingular. In that case it preserves the independence of every virtual triple; otherwise no three virtual points complete r to a basis. Hence, exactly,

$$F_3 = ZF_{\mathcal{L}} \geq \frac{3}{4}q^2Z$$

by Lemma 36.4. Together with (247), (249) and (251), this proves

$$B_1 \geq q^2 \left(C_1 - \frac{N_1 + Z}{4} \right). \tag{252}$$

It remains to bound the lower sectors by C_1 , and this is where the unit left activities are used. For a fixed independent pair (r, e) , let $h_{r,e}$ count virtual points independent of its span. If the quotient core has rank two, this is the number of independent original L pairs whose plane does not contain its row plane H . Every original element outside H extends to at least two independent pairs, and each pair contains at most two outside elements. Therefore

$$h_{r,e} \geq \beta_{r,e}.$$

If the quotient core has smaller rank, the right side is zero and the inequality is trivial. For a dependent pair (r, e) we set both quantities to zero.

Every counted (P, e) gives three independent vectors r, w_P, e , which can be completed by at least one further private coordinate e' . A basis r, w_P, e, e' has two choices of first private coordinate. Thus

$$2X \geq \sum_{r,e} v_r h_{r,e} \geq N_1. \tag{253}$$

Also $q \geq 1$ and every nonzero r is completed by at least one triple of private coordinates. The indicator contributing to Z is at most one, so $qV \geq Z$. Equation (247) now gives

$$C_1 \geq \frac{3}{2}N_1 + 2Z \geq \frac{3}{2}(N_1 + Z), \quad N_1 + Z \leq \frac{2}{3}C_1. \tag{254}$$

36.5 The middle gap and eventual scaling

The degree-four Newton inequality for f has the factor $\binom{4}{2}^2 / (\binom{4}{1}\binom{4}{3}) = 9/4$. By (249)–(254),

$$\begin{aligned} q^2 C_2^2 &\geq B_2^2 \geq \frac{9}{4} B_1 C_3 \\ &\geq \frac{9}{4} q^2 \left(C_1 - \frac{N_1 + Z}{4} \right) C_3 \geq \frac{15}{8} q^2 C_1 C_3. \end{aligned}$$

Cancel $q^2 > 0$. This proves Theorem 36.2, with margin $15/8 - 9/5 = 3/40$.

For clarity, all remaining inputs to Theorem 36.1 are in the bundled structural companion [28]. Its general top-gap theorem gives, for $4 + 3$,

$$C_3^2 \geq \frac{50}{21} C_2 C_4 > \frac{7}{3} C_2 C_4.$$

The ambient transversal-matroid specialization gives $C_1^2 > 2C_0 C_2$, stronger than the required first factor $5/3$. The new middle theorem is stronger than $9/5$. Thus all three forced quartic gaps are strict for every fixed field.

At indices below $|B|$, the leading coefficients under scaling are those of the graph restricted to right shore B . Its degree- $|B|$ matroid normalization is stronger than the actual-rank-seven normalization, so those leading gaps are strictly positive. At index $|B|$, the square has one more power of λ than the neighboring product. At the remaining indices, the strict leading terms are exactly the conditional gaps above. Hence every gap polynomial has positive leading coefficient and is positive for sufficiently large λ .

The $3 + 4$ case is already proved in [28]. If a minimum cover has one shore of size at most two, the earlier small-cover theorem gives actual-rank ULC at every positive field; its full proof is included in that companion's prerequisite archive. These exhaust all minimum-cover splits at rank seven. At rank six the only further split is $3 + 3$, also handled by the structural companion; lower ranks have a cover shore of size at most two. This proves Theorem 36.1.

For any gap polynomial with positive leading coefficient, an explicit field-dependent bound is available. If

$$G_j(\lambda) = a_D \lambda^D + \sum_{i < D} a_i \lambda^i, \quad a_D > 0,$$

then $\lambda \geq 1 + \sum_{i < D} |a_i|/a_D$ makes its leading term dominate the absolute sum of the others. Taking the maximum of these finitely many bounds gives a sufficient threshold in the new strict-cover cases.

36.6 Failure ranks and verification scope

The known upper witness for persistent one-shore failure uses the same activity convention [17]. Let $H_{8,30;32,794}$ have shores $A \sqcup L$ and $B \sqcup R$, of those block sizes, and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Every left activity and every exterior-right activity is one, and each B activity is w . Its matching rank is 38, and its last gap is

$$G_{37} = \frac{5}{6304} D^2 w^{58} (-22w^2 + 6437219280w + 11385894392295), \quad D = \binom{32}{30} \binom{794}{7}.$$

It is negative for every sufficiently large w . Together with Theorem 36.1, this places the minimum rank of persistent failure under common right-cover scaling in [8, 38]. [*Merge note:* Improved to [9, 38] by source 01.]

This conclusion is distinct from the finite-activity problem. We do not prove ULC at every positive field for general rank-six or rank-seven graphs. The existing finite one-shore-failure bracket remains [6, 38] [*Merge note*: its lower end rests on source 74, Part I]; the two-shore-weighted problem has different counterexamples and is not the subject of this theorem.

The exact checker [*Merge note*: shipped as `code/08-rank-seven-*.py` (without `top-gap`) and `data/08-rank-seven-middle_checks.json`, `data/08-rank-seven-replay.log`] uses a finite-field representation and independent endpoint-set enumeration on 120 genuine $4 + 3$ graphs. It verifies 480 sector identities, all upper-incidence bounds, all pair-annihilator lower bounds, all auxiliary quartic Newton inequalities, and all $15/8$ middle bounds. Its computations use exact integer arithmetic; finite-field assignments are compared with the graph endpoint counts before they are accepted. These regressions supplement the universal proof, rather than replacing it. The proof of the upper incidence bound is the complete coefficientwise case analysis in Lemma 36.3.

The previous structural report and its source archive are included unchanged. Its search discussion remains a historical record: the present theorem resolves the middle-gap question it left open. No numerical optimization or unsuccessful search is used in this proof. These are ordinary mathematical proofs with exact checks, not formal proof-assistant certification or external refereeing; no exhaustive priority claim is asserted.

37 Eventual scaling for five-plus-three covers, one weighted shore (source 03)

Abstract of source 03, as delivered. We prove eventual strict actual-rank ultra-log-concavity under common right-cover scaling for every bipartite graph with a minimum cover consisting of five left and three right vertices. Left activities are one; all fixed right activities are arbitrary and positive. For the conditional quintic obtained by forcing the three right-cover vertices, the new lower-middle factor is $77/48$, strictly exceeding the required $8/5$. The other two difficult factors are $7/4$, with strict inequality, and $55/24 > 16/7$. The proof combines a graph-generic additive plane-incidence correction, a small represented-matroid counting lemma, and exact comparisons with a Lorentzian auxiliary quintic. Together with earlier results, this leaves $4 + 4$ as the only unresolved minimum-cover split at rank eight. No general finite-activity theorem or global rank-eight eventual theorem is asserted. [*Merge note*: The $4 + 4$ split left open here is settled by source 02 (top gap) and source 01 (middle gap).]

37.1 Main result and exact sectors

For a finite bipartite graph $G = (X, Y)$, define its endpoint support polynomial by

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|}.$$

Each feasible endpoint pair is counted once. If the matching rank is d and $P_G = \sum p_i t^i$, its actual-rank Newton gaps are

$$G_i = i(d - i)p_i^2 - (i + 1)(d - i + 1)p_{i-1}p_{i+1}. \quad (255)$$

Nonnegative gaps and no internal zeros define ULC_d .

Fix a minimum cover $A \sqcup B$, put $L = X \setminus A$, $R = Y \setminus B$, and assume $|A| = 5$, $|B| = 3$. Thus $d = 8$, the exterior L - B matching rank is three, and the A - R rank is five. All left

activities are one. Fix arbitrary positive right activities and multiply those on B by a common parameter $\lambda > 0$.

Theorem 37.1. *For every such graph and field, all actual-rank-eight gaps are strictly positive for sufficiently large λ . The threshold may depend on the graph, cover and fixed field.*

Forcing all three B vertices and dividing out their activities gives $t^3 C(t)$, where $C(t) = \sum_{j=0}^5 C_j t^j$ and every $C_j > 0$. We prove

$$C_1^2 > 2C_0C_2, \quad C_2^2 \geq \frac{77}{48}C_1C_3, \quad C_3^2 > \frac{7}{4}C_2C_4, \quad C_4^2 \geq \frac{55}{24}C_3C_5. \quad (256)$$

The actual-rank-eight factors at these four indices are $25/16$, $8/5$, $7/4$, and $16/7$. The two numerical margins are

$$\frac{77}{48} - \frac{8}{5} = \frac{1}{240}, \quad \frac{55}{24} - \frac{16}{7} = \frac{1}{168}.$$

Use independent indeterminates for the graph edges. A square minor is nonzero exactly when the corresponding endpoints are perfectly matchable. Let K be the 5×3 core matrix and $q \geq 1$ the number of independent triples of the exterior rows $\ell \in L$. Remove isolated rows. For each independent labeled pair P of L , choose its annihilator v_P and introduce the virtual point Kv_P . Keep every label, including loops and parallel images. Adjoin the five private coordinate vectors E in the A space.

The resulting rank-five matroid has a Lorentzian basis polynomial [1]. Mark private elements by s , original R elements by t times their activities, and virtual elements by z , of activity one. Call the result $F(s, t, z)$. Its coefficients are ordinary basis counts without factorial factors. Let T_j, P_j, Q_j, R_j count bases with j original R elements and respectively zero, one, two, three virtual elements; the remaining elements are private. There are no bases with four virtual elements. Let N_j, Z_j be the forced endpoint sectors with respectively one and zero selected original L rows. Exact elimination and private complementation give

$$C_j = qT_j + P_j + N_j + Z_j, \quad (257)$$

with impossible terms zero. Three original L rows give the first term, two give the virtual term, and the other sectors remain explicitly indicated.

Put $f(t, u) = F(qu, t, u)$ and $b_j = [t^j u^{5-j}]f$. Then

$$\begin{aligned} b_1 &= q^4 T_1 + q^3 P_1 + q^2 Q_1 + q R_1, \\ b_2 &= q^3 T_2 + q^2 P_2 + q Q_2 + R_2, \\ b_3 &= q^2 T_3 + q P_3 + Q_3, \quad b_4 = C_4, \quad b_5 = C_5/q. \end{aligned} \quad (258)$$

Every quotient used below is by fixed core, private and original R data, independent of the original L edge indeterminates.

37.2 A graph generic additive plane bound

Let $\mathcal{L}$ be a loopless rank-three row configuration and H a plane. Let q count its bases, β its elements outside H , and D_H unordered pairs of pair-annihilators independent after restriction to H . The earlier rank-seven report [Merge note: source 08, Lemma 36.3 and Lemma 36.4] [29] proves, even with arbitrary positive original weights,

$$\frac{3}{4}q\beta \leq D_H \leq q\beta. \quad (259)$$

The upper inequality is coefficientwise. Its proof classifies restrictions to at most four original labels. The lower inequality is a rank-three Lorentzian Hessian inequality. Both proofs are included in the unchanged companion.

At unit weights, if $F_{\mathcal{L}}$ counts independent triples of labeled pair-annihilators, applying (259) to $H = \text{span}(P)$ and summing the marked pair gives

$$\frac{3}{4}q^2 \leq F_{\mathcal{L}} \leq q^2. \quad (260)$$

Indeed $3F_{\mathcal{L}} = \sum_P D_P$ and $\sum_P \beta_P = 3q$. No independence of H from $\mathcal{L}$ is needed in (259) or (260).

Lemma 37.2. *Suppose each permitted coordinate of each row of $\mathcal{L}$ is a separate algebraically independent indeterminate over a characteristic-zero field, and H is defined over that field. With unit original activities,*

$$4D_H \geq q(3\beta + 1). \quad (261)$$

Proof. Rescale nonzero coordinates of a normal to H to one. Up to coordinate permutation its support mask is 1, 3, or 7. For two row masks S, T , the cross-product support consists of the missing coordinate for every feasible matching of the two rows to two distinct coordinates. For four distinct row labels, the determinant of their two cross products and the fixed normal is nonzero exactly when these three supports match all three coordinates. A monomial chooses one coordinate from each original row; this uniquely fixes the two cross-product coordinates and the normal coordinate. Distinct normal terms therefore cannot cancel. For a shared row use

$$\det(a \times b, a \times c, n_H) = \pm \det(a, b, c)(a \cdot n_H).$$

Expand $4D_H - 3q\beta - q$ in products of binomial population coefficients. On a three-row mask multiset admitting a basis, its coefficient is the number of outside rows minus one. On a four-row multiset, it is $4E - 3Q$, where E counts good pair partitions and Q counts outside-label deletions leaving a basis. There are 84 triple and 210 quadruple profiles for each normal support. The complete exact matching enumeration gives respectively 138, 153, 144 nonzero coefficients for supports 1, 3, 7. The only negative coefficients are four occurrences of -4 , for support 3, on

$$(1, 2, 5, 6), \quad (1, 5, 6, 6), \quad (2, 5, 5, 6), \quad (5, 5, 6, 6).$$

The bundled standard-library checker [*Merge note:* shipped as `code/03-five-plus-three-*.py`, outputs `data/03-five-plus-three-*.json`] reconstructs every profile. All remaining binomial-population terms are nonnegative, so it suffices to handle these four row types.

Write their populations as A, B, x, y , respectively, and put

$$M = A + x, \quad N = B + y, \quad f = Ax + \binom{x}{2}, \quad g = By + \binom{y}{2}, \quad \alpha = A(A-1), \quad \gamma = B(B-1).$$

Within the two mixed coordinate planes, independent pairs share one intersection line in H , of combined mass $f + g$. Cross pairs have direction-class masses AB , then B repeated x times, A repeated y times, and 1 repeated xy times. Thus

$$q = fN + Mg, \quad \beta = M + N, \quad D_H = (f + g)MN + \frac{(MN)^2 - xy - xB^2 - yA^2 - A^2B^2}{2}.$$

If $MN = 0$, the result is immediate. Otherwise direct expansion, with $F_0 = 4D_H - q(3\beta + 1)$, gives

$$\begin{aligned} 2F_0 = & MN((M - N)^2 + M + N - 2) \\ & + \alpha N(3N - M - 3) + \gamma M(3M - N - 3) - 4\alpha\gamma. \end{aligned}$$

For fixed positive integers M, N , this is bilinear on $0 \leq \alpha \leq M(M-1)$, $0 \leq \gamma \leq N(N-1)$. Its four corner values are

$$\begin{aligned} MN((M-N)^2 + M + N - 2), & \quad MN(N-1)(M+N-1), \\ MN(M-1)(M+N-1), & \quad 0. \end{aligned}$$

All are nonnegative. This proves the remaining subsystem and hence the lemma. $\square$

The independence condition is essential: the affine hexagon $(2, 0), (1, 1), (-1, 1), (-2, 0), (-1, -1), (1, -1)$, lifted to $(x, y, 1)$, with $H = \{z = 0\}$, has $q = 20$, $\beta = 6$, $D_H = 93 < 95$. We never use (261) for a plane spanned by original L rows.

37.3 The top conditional gap

Fix an independent triple $J \subseteq R$ and quotient by its span. If the quotient core has rank below two, both its N_3 and Q_3 contributions vanish. Otherwise its row plane H is independent of all L entries, and (261) applies. Write $C_J = qU_J + P_J + \beta_J$ for this sector of C_3 .

If n is the number of nonloop original rows, elementary basis extension gives $q \geq n - 2$. Let h count independent original pairs not spanning H . Then $h \geq n - 1$: if H contains $g \geq 1$ original rows, the $g\beta$ cross pairs give $g\beta \geq n - 1$; if it contains none, basis extension gives at least $2n - 3$ independent pairs. Private coordinates span the quotient, so $U_J \geq 1$ and $P_J \geq h$. Therefore

$$C_J \geq 2n - 3 + \beta_J, \quad q\beta_J - D_J \leq \frac{q}{4}(\beta_J - 1) \leq \frac{q}{12}C_J,$$

since $\beta_J \leq n$. Deficient quotient maps satisfy the final inequality with zero loss, without any additive correction. Summing with positive activities gives

$$\frac{11}{12}qC_3 \leq b_3 \leq qC_3. \quad (262)$$

The upper bound follows from (259). Degree-five Newton at index four now yields

$$C_4^2 = b_4^2 \geq \frac{5}{2}b_3b_5 \geq \frac{55}{24}C_3C_5.$$

37.4 The strict next to top gap

Fix an independent pair $J \subseteq R$ and quotient by it, leaving dimension three. Let E be the nonloop private images, $m = |E| \geq 3$, and V their basis count. Let X, N, Z be the fixed- J sectors with two, one, zero original L rows, respectively. Thus this contribution to C_2 is $qV + X + N + Z$. Let W count private images whose further quotient core has rank two. The additive plane bound and the old adjoint bound give, after summation,

$$b_2 \geq q^2 \left(C_2 - \frac{N_2 + Z_2 - W_2}{4} \right). \quad (263)$$

Here W_2 is the activity-weighted sum of these fixed- J counts.

We prove, for each fixed J ,

$$N + Z - 2W < qV + X. \quad (264)$$

We have $q \geq n - 2$, $V \geq m - 2$, and

$$2X \geq W(n - 1),$$

because each qualifying private image has at least $n - 1$ virtual completions, and each resulting private pair is counted twice.

If the quotient core has rank at most one, $N = Z = W = 0$ and $qV > 0$. If it has rank two, let its image plane be P . Exactly the private images outside P qualify. Thus $W \geq 1$, $N = W\beta \leq Wn$, $Z = 0$, and $V \geq \max(1, W - 2)$. For $W = 1, 2$, the bound $qV + X \geq n - 2 + W(n - 1)/2$ is strictly greater than $W(n - 2)$. For $W \geq 4$, its margin is at least

$$(W/2 - 2)(n - 1) + 2 > 0.$$

For $W = 3, V \geq 2$, the margin is at least $(n + 1)/2$. If $W = 3, V = 1$, the private images are exactly three basis vectors outside P . Every nonzero virtual image in P has at least two nonzero coordinates in that basis, and hence at least two private-pair completions. Thus $X \geq 2(n - 1)$ and $qV + X \geq 3n - 4 > 3n - 6 \geq N - 2W$.

If the quotient core has rank three, $W = m$, $Z = 1$, $N \leq mn$. For $m \geq 4$ the margin in (264) is at least

$$(m/2 - 2)(n - 1) + 1 > 0.$$

For $m = 3$, the private images form a basis with matrix E , and $V = 1$. The transformed row matrix is $M = LK_J^{-1}E$. Its nonzero entries, two-minors and three-minors count N, X, q , respectively. It has rank three and no zero row. Write p for its independent-pair count; basis extension gives $p \geq 2n - 3$. If $N \leq 3n - 1$, then $q + X \geq q + p \geq 3n - 5 \geq N - 4$. If $N = 3n$, every row has full support, so the cross product of any independent pair has at least two nonzero coordinates. Hence $q + X \geq q + 2p \geq 5n - 8 \geq N - 4$. In both cases $q + X > N - 5 = N + Z - 2W$. This proves (264).

Summing gives $N_2 + Z_2 - W_2 < C_2/2$, so (263) yields

$$\frac{7}{8}q^2C_2 < b_2 \leq q^2C_2. \quad (265)$$

For the upper bound use $Q_2 \leq qN_2$ and $R_2 = F_{\mathcal{L}}Z_2 \leq q^2Z_2$. Newton at index three, together with (262), now gives

$$q^2C_3^2 \geq b_3^2 \geq 2b_2b_4 > \frac{7}{4}q^2C_2C_4.$$

37.5 A four dimensional unit counting lemma

The remaining comparison needs a quotient by just one R vector. Its private configuration spans dimension four and has at most five elements. We prove the required count independently of graph genericity.

Lemma 37.3. *Let E span dimension four and have at most five unit-weighted labeled elements. For independent v_1, v_2 , let V count E bases, X_i count E triples completing v_i , and N_{12} count E pairs completing both. Then $V + X_1 + X_2 \geq N_{12}$.*

Proof. Append a loop if E has four elements. Dualize the rank-four matroid on these five labels and the two marked labels. Its dual has rank three; both its marked pair and its E restriction have full respective ranks two and three. Basis complementation transforms the assertion into

$$q(E) \leq P(v_1) + P(v_2) + \beta,$$

where $H = \text{span}(v_1, v_2)$, β counts E outside H , and $P(v_i)$ counts pairs completing v_i .

Let f count nonloop E elements in H , and let p, Q count independent pairs and triples outside H . Then $\beta + f \leq 5$, $\beta \geq 1$, and

$$P(v_1) + P(v_2) \geq \beta f + p, \quad q(E) \leq \beta \binom{f}{2} + pf + Q.$$

For $\beta \geq 2$, use $p \leq \binom{\beta}{2}$, $Q \leq \binom{\beta}{3}$, and $3Q \leq (\beta-2)p$. The upper bounds for $q - P(v_1) - P(v_2) - \beta$, at $f = 0, 1, 2, 3$, are respectively

$$-\beta, \quad -2\beta + Q, \quad -2\beta + p + Q, \quad -\beta + 2p + Q.$$

The last three have respectively $\beta \leq 4, 3, 2$, and are all nonpositive.

For $\beta = 1$, $q(E)$ is the independent-pair count of the $f \leq 4$ inside elements. If c_i count those parallel to v_i , then $P(v_1) + P(v_2) = 2f - c_1 - c_2$. For $f \leq 3$, this is at least $f \geq \binom{f}{2}$. For $f = 4$, either $c_1 + c_2 \leq 3$, giving $P(v_1) + P(v_2) + 1 \geq 6$, or all four lie in the two marked directions, giving $q = c_1 c_2 \leq 4 < 5$. This proves the lemma. $\square$

Lemma 37.4. *Let E be as above and let v_1, v_2, v_3 be arbitrary. Let X count one marked element and three private elements forming a basis, N count two marked and two private elements, and Z count three marked and one private element. Let W count independent private pairs P for which $\text{rank}(P + \text{span}(v_1, v_2, v_3)) = 4$. Then*

$$V + X \geq N + Z - 2W. \quad (266)$$

Proof. Quotient by an independent private pair P . Remaining private images span dimension two. Let n_P count independent pairs of marked images, and m_P mixed bases with one marked and one private image. If the marked rank is two, $n_P = 1, 2, 3$ gives respectively $m_P \geq 2, 3, 4$: retain two independent private images, and in the last case note that the three marked directions are distinct. Hence $m_P - 3n_P + 6 \geq 1$. At smaller marked rank, $n_P = 0$ and $m_P \geq 0$. Summing over P gives

$$3(X - N + 2W) \geq W. \quad (267)$$

If the original marked rank is at most two, $Z = 0$ and the result follows.

Otherwise let $z = Z$ count private elements outside the marked plane. If E has five nonloop elements, its unique circuit has size $c \geq 2$ and $V = c$. Every pair is independent except possibly a parallel pair when $c = 2$. Thus

$$W = 10 - \binom{5-z}{2} - \epsilon,$$

where $\epsilon \in \{0, 1\}$ is one only for a parallel pair outside the marked plane. For $z \leq 2$, $z - V \leq 0$; for $z = 3, 4, 5$, respectively, $W \geq 8, 9, 9$ whereas $3(z - V) \leq 3, 6, 9$. Therefore $W \geq 3(z - V)$ and (267) proves the assertion.

Otherwise, after removing loops, E is a four-element basis. Here $V = 1$ and $W = 6 - \binom{4-z}{2}$. Equation (267) suffices for $z \leq 3$. For $z = 4$, write the marked vectors as columns of a 4×3 matrix K . Every three-row minor is nonzero; X, N count its nonzero entries and two-minors. We need $N - X \leq 9$. If $X \geq 9$, use $N \leq 18$. If $X \leq 8$ and a row has singleton support, its three row pairs contribute at most $X - 1$ two-minors and the other three at most nine, giving $N \leq X + 8$. Otherwise all four rows have support two. Two share a support; their independent pair gives exactly one nonzero two-minor, and the other five give at most fifteen. Thus $N \leq 16 < X + 9$. This proves (266). $\square$

Lemma 37.5. *Fix a core map from dimension three to dimension four and private E as in Lemma 37.3. For any loopless unit-weighted original L configuration of rank three, let q, V, X, N, Z, W denote the corresponding basis, one-virtual, one-original-row, no-original-row and qualifying-private-pair counts. Then*

$$qV + X \geq N + Z - 2W. \quad (268)$$

Proof. Start with three original rows forming a basis. Their pair annihilators form the dual basis, and their images are the three marked vectors of Lemma 37.4. For each private pair, its one-original-row count equals the number of independent pairs of these marked images. Thus that lemma proves the starting inequality.

Add a nonzero original row ℓ . The increment of q counts old pairs complementary to ℓ and is at least one. New virtual points are the images of $\ell \times \ell_i$. If their image has rank below two, the increment of N is zero, so the slack cannot decrease. Otherwise choose two independent new virtual points v_1, v_2 . They span that image plane. The increment of N is exactly N_{12} , the private-pair count complementary to it; the increment of X is at least $X_1 + X_2$. Lemma 37.3 shows that $qV + X - N$ cannot decrease. Finally Z, W depend only on the core and private configuration, and are unchanged. Induction proves (268). $\square$

37.6 The lower middle gap and eventual scaling

Fix one original R vector and quotient by it. The five private images satisfy Lemma 37.5. The fixed-vector sector of C_1 is $qV + X + N + Z$. Each qualifying private pair has a rank-two quotient core independent of the original L entries. Lemma 37.2 therefore bounds its two-virtual count below by $q(3\beta + 1)/4$. The three-virtual count is $F_{\mathcal{L}}Z \geq 3q^2Z/4$. No four virtual elements can form a basis. Summing with the positive original R activities gives

$$b_1 \geq q^3 \left(C_1 - \frac{N_1 + Z_1 - W_1}{4} \right).$$

Lemma 37.5 gives $N_1 + Z_1 - W_1 \leq C_1/2$, and hence

$$b_1 \geq \frac{7}{8}q^3C_1. \quad (269)$$

Newton at index two has factor two. By (265), (262), and (269),

$$q^4C_2^2 \geq b_2^2 \geq 2b_1b_3 \geq 2 \cdot \frac{7}{8} \cdot \frac{11}{12}q^4C_1C_3 = \frac{77}{48}q^4C_1C_3.$$

Cancel $q^4 > 0$ to obtain the second inequality in (256).

The remaining first inequality follows from the ambient transversal matroid after contracting B . Its degree is $n + 2$, so its first Newton factor is $2(n + 2)/(n + 1) > 2$. This proves all of (256). At indices below three, the leading scaling polynomial is the pure right-shore- B endpoint polynomial, which is ULC_3 by its rank-three basis representation. Its normalization strictly improves the actual-rank-eight one. At index three, the square term has a higher power of λ than the neighboring product. At higher indices the leading gaps are precisely the four strict conditional gaps just proved. Every gap polynomial therefore has a positive leading coefficient and is positive for sufficiently large λ . This proves Theorem 37.1.

For a gap $G(\lambda) = a_D\lambda^D + \sum_{i < D} a_i\lambda^i$ with $a_D > 0$, the explicit field-dependent choice $\lambda \geq 1 + \sum_{i < D} |a_i|/a_D$ makes the leading term dominate the others. Taking the maximum gives a sufficient threshold. No uniform graph-independent threshold is asserted.

37.7 Scope and exact verification

The earlier general top-gap theorem, included in the structural companion inside [29], already proves eventual strict scaling for $3 + 5$ covers. [*Merge note:* Source 08s, Corollary 29.2.] Covers with one shore of size at most two satisfy actual-rank ULC at every positive field by the earlier small-cover theorem, whose full proof is also bundled there. Hence $4 + 4$ is the only unresolved rank-eight cover split. The known persistent-failure bracket remains [8, 38]; this report narrows

the possible rank-eight case without proving that case. General finite-activity rank-six and higher one-shore ULC remains separate.

The additive plane proof uses the complete finite set of 882 mask profiles, followed by the analytic four-variable argument for all populations. It is not a bounded population-grid argument. The portable checker also verifies the four-variable identity and all four corner formulas by exact polynomial arithmetic. A separate exact endpoint/finite-field regression checks 64 genuine $5 + 3$ configurations, 384 endpoint-sector identities, 384 fixed- R unit inequalities, and all three new gap comparisons. Each represented endpoint coefficient is checked against direct Boolean matching before use. A 13,056-case binary configuration diagnostic checks the small four-dimensional baseline independently; no part of the proof relies on this diagnostic covering characteristic zero.

The rank-seven PDF and source archive are included unchanged, providing the earlier incidence, ambient-normalization and small-cover inputs. These are ordinary mathematical proofs with exact checks, not proof-assistant certification or external refereeing. No exhaustive priority claim is made.

38 The top conditional gap for four-plus-four covers, one weighted shore (source 02)

Abstract of source 02, as delivered. For every genuine $4 + 4$ minimum cover, unit left activities and arbitrary positive right activities, we prove a strict top conditional Newton gap with the actual-rank-eight factor $16/7$. The proof uses two finite reductions valid for arbitrarily many exterior vertices: a coefficientwise upper plane-incidence bound for graph-generic rank-four configurations, and a coupled unit-count inequality whose four-label contribution supplies a strict positive correction. All finite profiles are checked by exact standard-library programs and an independently implemented replay. An explicit six-vector real example shows that the unrestricted upper-incidence bound is false, so the graph-generic scope matters. Together with the preceding $5 + 3$ theorem, only the $4 + 4$ middle conditional comparison remains in the rank-eight eventual-scaling problem. We do not claim a complete rank-eight theorem. [*Merge note:* The remaining $4 + 4$ middle comparison is proved by source 01, which completes the rank-eight theorem.]

38.1 Graph normalization and the new top gap

Let $G = (X, Y)$ be a finite bipartite graph and let

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|}.$$

Every feasible endpoint pair is counted once. For matching rank d and $P_G = \sum p_i t^i$, actual-rank ultra-log-concavity requires

$$i(d-i)p_i^2 \geq (i+1)(d-i+1)p_{i-1}p_{i+1} \quad (270)$$

and no internal zeros.

Fix a minimum cover $A \sqcup B$ with $|A| = |B| = 4$. Put $L = X \setminus A$, $R = Y \setminus B$. The graph has matching rank eight, and both exterior matching ranks are four. All left activities are one, and right activities are arbitrary fixed positive numbers. Forcing every B vertex into the selected right endpoint set and dividing out its activity product gives

$$t^4 C(t), \quad C(t) = \sum_{j=0}^4 C_j t^j, \quad C_j > 0.$$

Let S_R be the activity-weighted number of independent exterior-right pairs in the generic A - R representation. Exterior rank four gives $S_R > 0$.

Theorem 38.1. *For every such graph and field,*

$$C_3^2 \geq \frac{16}{7}C_2C_4 + \frac{8}{21}S_R C_4 > \frac{16}{7}C_2C_4. \quad (271)$$

The factor $16/7$ is the top conditional factor required by (270) at actual rank eight. The ambient matroid first gap already exceeds the required $8/5$. The remaining conditional target is

$$C_2^2 \geq \frac{7}{4}C_1C_3, \quad (272)$$

with strictness or a separate treatment of equality needed to finish eventual scaling. [*Merge note:* Proved, with strictness, by source 01 (Section 39).] The preceding companion [30] handles $5 + 3$ covers and, together with earlier results bundled there, every other rank-eight cover split except $4 + 4$.

38.2 Incidence quantities and a real obstruction

Let $\mathcal{L}$ be a rank-four row configuration and H a two-plane in its ambient space. Let q count weighted independent four-subsets and β weighted original pairs complementary to H . For each independent labeled triple P , choose an annihilator v_P with activity equal to the product of the original activities. Let D_H count unordered pairs of these virtual labels whose restrictions to H are independent. Thus $q\beta$ and D_H are homogeneous of degree six in the original activities. At unit activities also let h_H count original independent triples whose hyperplane does not contain H .

The universal rank-four lower incidence inequality is

$$D_H \geq \frac{2}{3}q\beta,$$

from the diffuse-plane Lorentzian Hessian construction in the earlier structural work included with [30]. [*Merge note:* Source 08s's Lemma 29.4 at $b = 4$.] The reverse companion $D_H \leq q\beta$ is not valid for arbitrary represented configurations.

Lemma 38.2. *With unit activities, take the six real vectors*

$$\begin{aligned} (1, 0, -1, 0), \quad (0, 1, 3, 0), \quad (1, 0, 0, -1), \\ (1, 1, 0, 2), \quad (0, 1, 0, 0), \quad (1, 1, 0, 0), \end{aligned}$$

and $H = \text{span}(e_3, e_4)$. Then $q = 12$, $\beta = 12$, and $D_H = 145 > 144 = q\beta$.

Proof. The only dependent four-subsets are 1234, 1256, and 3456. The only original pairs not complementary to H are 13, 25, and 46. Every original triple is independent and has a nonzero annihilator restriction. The five projective restriction classes, with directions

$$(0, 1), (1, 0), (1, 1), (1, 3), (2, -1),$$

have masses 6, 6, 6, 1, 1. Therefore

$$D_H = \frac{20^2 - 3 \cdot 6^2 - 1^2 - 1^2}{2} = 145.$$

The bundled integer-determinant checker lists every triple in its class. It also verifies the squarefree six-label coefficient: $q\beta$ contributes nine and D_H ten. $\square$

The displayed vectors satisfy special algebraic dependencies. This example is not a graph-generic counterexample under the hypotheses below.

38.3 A coefficientwise graph generic upper bound

We say that $\mathcal{L}$ is graph-generic over a characteristic-zero field F when each allowed row coordinate is a separate algebraically independent indeterminate over F . Every forbidden coordinate is zero. The fixed plane H is defined over F , independently of all these indeterminates.

Theorem 38.3. *For graph-generic $\mathcal{L}$ and such an independent two-plane H ,*

$$D_H \leq q\beta \quad (273)$$

coefficientwise in the original activities. In particular it holds for arbitrary positive activities and arbitrary numbers of original rows.

Proof. First consider a degree-six monomial using repeated labels. A term of D_H is formed from two distinct independent triples; their intersection S has size one or two. If S is dependent, both coefficients vanish. If $\text{span}(S)$ meets H nontrivially, both annihilator restrictions vanish on that intersection and cannot be independent. No pair containing S can complement H , so the $q\beta$ coefficient also vanishes.

Otherwise contract $\text{span}(S)$. The image of H still has dimension two, and the two coefficients become the corresponding squarefree coefficients for rank $4 - |S|$. For one shared label the coefficientwise rank-three upper-incidence theorem in the earlier rank-seven report applies. For two shared labels this is the rank-two identity: two point annihilators restrict independently to the whole quotient exactly when the two points form a basis. The cited rank-three bound is universal for represented configurations, so the contraction does not need to preserve graph genericity.

It remains to consider six distinct labels. Their supports are nonempty masks in $\{1, \dots, 15\}$. Up to relabeling, there are $\binom{20}{6} = 38,760$ nondecreasing support multisets. Represent H by a 4×2 matrix. Its four rows form a rank-two configuration; up to coordinate permutation its loop and parallel-class patterns have exactly seven types:

$$\begin{aligned} (0, 0, 1, 2), (0, 1, 1, 2), (0, 1, 2, 3), (1, 1, 1, 2), \\ (1, 1, 2, 2), (1, 1, 2, 3), (1, 2, 3, 4). \end{aligned} \quad (274)$$

Here zero denotes a loop, and equal positive entries denote parallel rows. Let $\mathcal{B}_H$ be its set of independent coordinate pairs.

An original pair complements H exactly when it can match two coordinates whose complement belongs to $\mathcal{B}_H$. A triple annihilator can have a nonzero coordinate i exactly when its three rows can match the other three coordinates. Two disjoint triple annihilators restrict independently exactly when two such possible missing coordinates form an element of $\mathcal{B}_H$. These are exact polynomial nonvanishing tests: each monomial chooses one coordinate from every independently parametrized original row and uniquely determines the two missing coordinates. Different minors of H cannot cancel.

For each support profile compute

$$\begin{aligned} Q &= \sum_{\substack{P \subseteq [6] \\ |P|=2}} [P \text{ complements } H] [[6] \setminus P \text{ is a basis}], \\ D_6 &= \sum_{\{P, P^c\}: |P|=3} [\text{the two annihilator restrictions are independent}]. \end{aligned}$$

These are exactly the coefficients of $q\beta$ and D_H . The complete exact table verifies $Q - D_6 \geq 0$ in all 271,320 cases. Its zero-coefficient counts, in the order (274), are

$$32,570, 25,036, 20,670, 23,574, 18,932, 17,732, 16,532.$$

Every other coefficient is positive. Both the producer and a separately implemented coordinate-injection replay, with identical per-type hashes, are included. This finite classification exhausts every remaining monomial and proves (273). $\square$

38.4 A coupled unit correction

Separate worst-case bounds for D_H , q , and h_H lose too much at rank eight. The following coupled inequality supplies the required correction.

Theorem 38.4. *Under the hypotheses of Theorem 38.3, now with every original activity one,*

$$T := 7D_H + q^2 + qh_H - 6q\beta \geq q > 0. \quad (275)$$

Proof. Group every product by the union of its original labels. Contributions using seven or eight labels come only from q^2 and qh_H , and are nonnegative. No contribution uses fewer than four labels.

On four labels, a nonbasis contributes zero. A basis contributes one to q^2 , h_B to qh_H , and β_B to $q\beta$. Two triples with this union share a pair; their restrictions are independent exactly when that pair complements H . Hence D_H contributes β_B , and the total is

$$1 + h_B + \beta_B \geq 1.$$

Summing over bases already gives the lower bound q in (275). It remains to check unions of five and six labels.

For a fixed k -label profile, $k = 5, 6$, put

$$\begin{aligned} A_k &= \#\{\text{ordered pairs of bases with union } [k]\}, \\ B_k &= \#\{\text{basis and good triple with union } [k]\}, \\ Q_k &= \#\{\text{basis and complementary pair with union } [k]\}, \\ D_k &= \#\{\text{unordered triple pairs with union } [k] \text{ and independent restrictions}\}. \end{aligned}$$

A good triple is independent and its hyperplane does not contain H . The coefficient to check is

$$7D_k + A_k + B_k - 6Q_k. \quad (276)$$

For six labels, every quantity is an exact support-matching count as in the preceding proof. All 271,320 coefficients are nonnegative.

For five labels, A_5, B_5, Q_5 are exact matching counts, while we certify a lower bound on D_5 by integer determinant witnesses. The two triples have exactly one shared row. The next elementary observation ensures that these witnesses are uniform over every plane of the same type, not just the numerical representative chosen for the calculation.

Write the triple annihilators as v, w . Their restriction determinant is

$$\sum_{i < j} (v_i w_j - v_j w_i) \Delta_{ij}(H). \quad (277)$$

A cofactor coordinate v_i uses every ambient coordinate once except i . Thus every monomial in $v_i w_j$ has total coordinate multidegree

$$(2, 2, 2, 2) - e_i - e_j.$$

This identifies the unordered pair $\{i, j\}$ uniquely, even when row labels are shared. Each row-coordinate monomial in (277) therefore has coefficient equal to an integer multiple of a

single H minor. Terms using that minor may cancel, but terms using different minors cannot. Polynomial nonvanishing depends only on which H minors vanish.

Represent a nonzero row of plane class c by $(1, c)$, and a loop by $(0, 0)$. For each nondecreasing five-row support profile, substitute the allowed entries from the fixed matrix

$$\begin{pmatrix} 86 & 53 & 34 & 14 \\ 96 & 50 & 25 & 15 \\ 71 & 72 & 73 & 66 \\ 96 & 20 & 54 & 39 \\ 95 & 26 & 69 & 13 \end{pmatrix}, \quad (278)$$

setting forbidden entries to zero. A nonzero evaluated determinant certifies a nonzero polynomial. By the single-minor property, that polynomial is nonzero for every H of the same type. An accidentally zero specialization can only reduce the witnessed count $\underline{D}_5 \leq D_5$.

The exact certificate verifies

$$7\underline{D}_5 + A_5 + B_5 - 6Q_5 \geq 0$$

in all $7\binom{19}{5} = 81,396$ five-label cases. This proves (276) on five labels as well. Combined with the four-label gain and the nonnegative larger unions, it establishes (275) for arbitrarily many original rows. $\square$

The finite checks in this proof are complete coefficient reductions. They do not impose a population bound. The entries of (278) are nonvanishing witnesses after a uniformity proof, rather than random numerical evidence for the inequality.

38.5 The top conditional comparison

Represent graph edges by independent indeterminates, and let K be the 4×4 core. Let $q \geq 1$ count exterior L bases. Introduce one virtual right point Kv_P for each independent exterior L triple P , retaining all labels. Adjoin the four private A coordinates. The resulting rank-four matroid has a Lorentzian basis polynomial [1]. Mark its private, original R , and virtual elements by s, t, z , with the actual positive activities on R . Put $f(t, u) = F(qu, t, u)$ and $b_j = [t^j u^{4-j}]f$. Exact endpoint elimination gives

$$b_3 = C_3, \quad b_4 = C_4/q, \quad b_2 = qC_2 - \sum_J v_J(q\beta_J - D_J),$$

where J runs over independent exterior R pairs.

Quotient by such a fixed pair J . If the induced 2×4 core has rank two, its row space is a two-plane H independent of all original L entries. The corresponding endpoint sector is

$$C_J = qU_J + P_J + \beta_J.$$

Private images span the quotient, so $U_J \geq 1$ and every good virtual point has at least one private completion: $P_J \geq h_J$. Thus $C_J \geq q + h_J + \beta_J$. By Theorem 38.4,

$$q\beta_J - D_J \leq \frac{q}{7}(q + h_J + \beta_J - 1) \leq \frac{q}{7}(C_J - 1). \quad (279)$$

If the quotient core has rank below two, $\beta_J = D_J = 0$ while $C_J \geq q \geq 1$, so (279) remains true without applying the incidence theorem. Summing yields

$$b_2 \geq \frac{6}{7}qC_2 + \frac{q}{7}S_R.$$

Degree-four Newton at index three has factor $8/3$. Consequently

$$C_3^2 = b_3^2 \geq \frac{8}{3}b_2b_4 \geq \frac{16}{7}C_2C_4 + \frac{8}{21}S_R C_4,$$

which proves Theorem 38.1. The separate upper theorem also gives $b_2 \leq qC_2$, a potentially useful comparison for the remaining middle gap.

38.6 Status and reproducibility

The graph-generic upper proof uses 271,320 complete six-label profiles, with repeated-label cases handled by contraction. The coupled proof uses another 271,320 six-label profiles and 81,396 five-label profiles. An independent implementation reconstructs all support predicates by explicit coordinate injections, rather than the producer’s endpoint dynamic programming. For five-label witnesses it uses signed-permutation cofactors and the Cauchy–Binet expression (277), rather than the producer’s projected two-vectors. It independently evaluates 1,220,940 witness determinants. Every coefficient histogram and all 21 profile-stream hashes agree.

The independent replay also expands the universal shared-row determinant symbolically: its 432 nominal terms give 408 distinct row-coordinate monomials, of which 24 cancel and 384 survive. Every survivor uses exactly one H minor, agreeing with the multidegree proof. All checks use exact integer arithmetic and remain active under Python’s optimization flag. The complete code, data, and ordinary source are included.

Under common scaling of B , the low-index gaps have positive leading terms by the pure-shore theorem, the first conditional gap is stronger than $8/5$, and Theorem 38.1 settles the top one strictly. For $4 + 4$, only (272) and its possible equality cases remain. A zero leading middle gap would still require examining lower-order scaling coefficients; searching only for a negative leading coefficient is a sufficient discovery route, not an exhaustive failure test.

The preceding 5+3 PDF and source archive are included unchanged. They also contain the sealed rank-seven and earlier structural prerequisites. The persistent-failure bracket remains [8, 38], while the general finite-activity question is separate. [*Merge note:* Improved to [9, 38] by source 01.] These are ordinary mathematical proofs supported by exact certificates, not formal proof-assistant certification or external refereeing; no exhaustive priority claim is asserted.

39 Eventual one-shore scaling through rank eight (source 01)

Abstract of source 01, as delivered. For every bipartite graph of matching rank at most eight, unit activities on the entire left shore, and any fixed positive right field, common scaling of the right part of a fixed minimum cover is eventually ultra-log-concave after normalization by the actual matching rank. The new case is the middle conditional gap of a $4 + 4$ cover. We obtain it from an auxiliary quartic using a rank-three minor-count potential, an exact parallel-class identity, and a complete finite hyperplane-incidence certificate. Ordering the exterior rows by their parallel classes supplies the strict gain missing from separate worst-case incidence bounds. The scaling threshold depends on each fixed graph, cover, and field vector. The known persistent-failure bracket improves to [9, 38]; the finite-activity bracket remains [6, 38]. General finite-activity ultra-log-concavity at ranks six through eight is not proved.

39.1 Main theorem and the remaining cover split

For a finite bipartite graph $G = (X, Y)$, let

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|}.$$

Each feasible endpoint pair is counted once, rather than once per matching. If the matching rank is d and $P_G = \sum p_j t^j$, its actual-rank gaps are

$$G_j = j(d-j)p_j^2 - (j+1)(d-j+1)p_{j-1}p_{j+1}. \quad (280)$$

The condition $P_G \in \text{ULC}_d$ means that these gaps are nonnegative and there are no internal zeros.

Fix a minimum cover $A \sqcup B$, with $A \subseteq X$, $B \subseteq Y$, and put $L = X \setminus A$, $R = Y \setminus B$. Set every left activity to one. Fix a vector $v \in (0, \infty)^Y$, and replace v_y by λv_y for $y \in B$ only.

Theorem 39.1. *If $d \leq 8$, there is a finite threshold $\lambda_0 = \lambda_0(G, A, B, v)$ such that $P_G \in \text{ULC}_d$ for every $\lambda \geq \lambda_0$.*

The theorem is globally nonstrict: pure-shore and small-cover branches can have equality. No uniform threshold over graphs or fields is claimed.

The sealed companions [30, 31] settle all rank-eight cover splits except the middle conditional comparison for $4 + 4$. Henceforth let $|A| = |B| = 4$, and remove isolated exterior vertices. Exterior ranks are four on both $L-B$ and $A-R$. Forcing all B vertices and dividing out their activities gives

$$t^4 C(t), \quad C(t) = \sum_{j=0}^4 C_j t^j, \quad C_j > 0.$$

The required conditional factors are $8/5$, $7/4$, and $16/7$. The first is below the ambient matroid bound, and the last is strict by [31]. We settle the middle factor.

39.2 The auxiliary quartic and a sufficient unit potential

Use independent indeterminates for graph edges, so square-minor nonvanishing is exactly perfect matchability. Let $q \geq 1$ count independent four-subsets of exterior L . For each independent labeled L triple, introduce the core image of its annihilator as a virtual right point, keeping loops and parallel labels. Adjoin the four private A coordinate vectors. The resulting rank-four matroid has a Lorentzian basis polynomial [1]. Mark private, original R , and virtual elements by s, t, z , with the actual positive activities on R . Set

$$f(t, u) = F(qu, t, u), \quad b_j = [t^j u^{4-j}]f.$$

The graph-generic upper incidence theorem of [31] gives

$$b_2 \leq qC_2, \quad b_3 = C_3. \quad (281)$$

Quartic Newton therefore proves the desired strict middle gap once

$$b_1 > \frac{7}{9}q^2 C_1. \quad (282)$$

Indeed $b_2^2 \geq (9/4)b_1 b_3$ then yields $C_2^2 > (7/4)C_1 C_3$.

Fix one exterior R vector and quotient by it. The quotient has dimension three; its private images form a spanning configuration E of three or four nonloop labeled elements, all with activity one. Let V count its bases and K be the induced 3×4 core map. Write X for the one-virtual, two-private basis count, N for the two-original- L , one-private endpoint sector, and Z for the one-original- L sector. The exact fixed-vector contribution to C_1 is

$$C = qV + X + N + Z.$$

Let E_2 count two-virtual, one-private bases and F_3 virtual triples. The corresponding auxiliary contribution is

$$b = q^3V + q^2X + qE_2 + F_3.$$

The universal rank-four plane incidence bound gives $E_2 \geq (2/3)qN$. The next section proves $F_3 \geq (2/5)q^2Z$ in the graph-generic setting. Hence

$$9b - 7q^2C \geq \frac{q^2}{5}\mathcal{P}, \quad \mathcal{P} = 10qV + 10X - 5N - 17Z. \quad (283)$$

If the core rank is below three, both F_3 and Z vanish. We will prove $\mathcal{P} > 0$ for the cases requiring this comparison and handle the remaining parallel-class cases exactly.

39.3 A finite hyperplane transition bound

Let H_3 be a fixed three-plane independent of the graph-generic rank-four rows L , with normal n_H . Let Z count original elements outside H_3 . For an independent triple P with $\text{span}(P) \neq H_3$, let β_P count original pairs complementary to $\text{span}(P) \cap H_3$. With arbitrary positive original activities, define

$$S = \sum_P w_P \beta_P.$$

Lemma 39.2. *Coefficientwise in the original activities,*

$$5S \geq 9qZ. \quad (284)$$

Consequently the virtual triple count after restriction to H_3 satisfies $F_3 \geq (2/5)q^2Z$.

Proof. A term of S uses a triple and a pair. If they share two labels, the contribution vanishes. If they share one label a , the complement determinant is nonzero exactly when their four-label union is a basis and $a \notin H_3$. There are three choices of the marked triple and pair, giving coefficients three in S and one in qZ .

For five distinct labels, nonempty row supports are masks $1, \dots, 15$. Normalize the nonzero entries of n_H to one. Up to coordinate permutation its support is $1, 3, 7$, or 15 . For a triple P and its complementary pair i, j , the determinant is, up to sign,

$$\det(P, \ell_i)(n_H \cdot \ell_j) - \det(P, \ell_j)(n_H \cdot \ell_i).$$

A monomial fixes the missing coordinate a of the triple. If both pair rows choose a , the terms cancel. Otherwise one pair coordinate is a , the other is a distinct normal-supported coordinate, and exactly one term contributes. Thus a split is counted precisely when a possible missing coordinate of the triple and a distinct normal-supported coordinate can be matched by the complementary pair. This is an exact support test, independent of the nonzero values of the fixed normal.

For every nondecreasing five-row profile, let Q count outside-row deletions leaving a basis, and let S_5 count these ten triple-pair splits. The complete integer table checks $5S_5 \geq 9Q$ on all

$$4 \binom{19}{5} = 46,512$$

profiles. Two independently implemented coordinate-injection replays verify the same four profile hashes. The stronger coefficientwise assertion $S_5 \geq 2Q$ has 184 exceptions for support 3 and 120 for support 7, none for the other two supports; every exception has $(S_5, Q) = (9, 5)$. Thus the weaker constant in (284) is deliberate.

Finally mark one virtual point in a virtual basis. The universal rank-four lower plane bound, applied to $\text{span}(P) \cap H_3$, gives

$$3F_3 \geq \frac{2}{3}qS \geq \frac{6}{5}q^2Z.$$

Here the intersected plane may depend on original rows, but this lower incidence theorem is universal for represented configurations. Dividing by three proves the result. $\square$

39.4 Two elementary minor count lemmas

For a matrix M , let $m_i(M)$ count its nonzero minors of size i .

Lemma 39.3. *Every rank-three matrix satisfies*

$$m_3 + m_2 \geq m_1 - 2, \quad \Phi(M) := 2m_3 + 2m_2 - m_1 \geq 5.$$

Adding a nonzero row or column while retaining rank three increases Φ by at least three. In particular, a rank-three matrix with at least four nonzero rows or columns has $\Phi \geq 8$.

Proof. Start with an invertible 3×3 submatrix. If its adjugate has at least six nonzero entries, $m_3 = 1$, $m_2 \geq 6$, $m_1 \leq 9$ prove both bounds. If the adjugate has three, four, or five entries, permute a perfect matching onto its diagonal and scale those entries to one. It is the identity plus at most two off-diagonal entries. Its inverse has at most three, four, or six entries, respectively: two arrows give either a two-cycle or at most one extra length-two path. The bounds follow in each case.

Suppose a new row has z nonzero entries. Represent the old columns as spanning vectors x_j in dimension three and the new row as a functional ℓ . For each j with $\ell(x_j) \neq 0$, the differences

$$x_k - \frac{\ell(x_k)}{\ell(x_j)}x_j$$

span the two-dimensional kernel of ℓ . Two distinct k therefore give nonzero two-minors with column j and suitable old rows. Each new two-minor is counted at most twice over j , so at least z new two-minors appear. A nonzero added row also creates at least one three-minor, since the old row space has rank three. Hence $m_3 + m_2 - m_1$ cannot decrease, and

$$\Delta\Phi \geq 2 + z \geq 3.$$

Transpose for columns. Zero added rows or columns change none of the counts. Starting from the chosen invertible submatrix and adding all remaining rows and columns proves the assertions. $\square$

Lemma 39.4. *Every rank-three 3×4 matrix satisfies $m_2 \leq 2m_1$.*

Proof. Transpose to four rows with three coordinates. If a row is zero, use the remaining invertible 3×3 matrix: $m_1 \geq 5$ gives $m_2 \leq 9 \leq 2m_1$, while $m_1 = 3, 4$ gives $m_2 = 3, 4$.

If all four nonzero rows have support at least two, $m_1 \geq 8$. At $m_1 = 8$, two supports coincide. Their pair contributes at most one two-minor, and the other five pairs at most fifteen. At $m_1 \geq 9$, use $m_2 \leq 18$.

Suppose one row has singleton support. Its three pairs contribute at most $m_1 - 1$ two-minors, and the other three at most nine. Thus $m_2 \leq m_1 + 8$, enough for $m_1 \geq 8$. For smaller m_1 , the remaining support-size patterns give the following bounds: $(1, 1, 1)$ at $m_1 = 4$ gives $3 + 3 = 6$; $(2, 1, 1)$ at five gives $4 + 5 = 9$; $(3, 1, 1)$ and $(2, 2, 1)$ at six give $5 + 5 = 10$ and

$5 + 7 = 12$; $(3, 2, 1)$ at seven gives $6 + 7 = 13$. For $(2, 2, 2)$, if a remaining row uses the singleton coordinate, the first three pairs contribute at most five and the other three at most nine. Otherwise those three rows lie in the other coordinate plane and their mutual pairs contribute at most three. Every bound is at most $2m_1$. $\square$

An independent exhaustive check of the 330 unordered four-column support patterns, including zero columns, gives a second finite verification of Lemma 39.4. Generic support counts upper-bound the actual nonzero two-minor count of every realization with that support.

39.5 The starting potential and its increments

We work in one fixed- R quotient from Section 2 [*Merge note*: Subsection 39.2 here]. Begin with four original L rows forming a basis, so $q = 1$ and the four virtual images are the four core columns after an invertible basis change.

First use three private basis elements. If the core has rank three, X, N, Z count the nonzero entries, two-minors, and three-minors of this 3×4 matrix. Lemma 39.4 gives

$$\mathcal{P} \geq 10 - 17Z. \quad (285)$$

Here Z is the number of the four original basis rows outside the row-space hyperplane H_3 of the core.

If the core has rank two, let $W \in \{1, 2, 3\}$ count private basis elements outside its image plane, $k \leq 4$ the number of nonzero virtual images, and p their independent-pair count. Then $N = Wp$. For $W = 1$, $p \leq \binom{k}{2} \leq 3k/2$ and $X \geq k$. For $W = 2$, let k_0 count virtual points in the one private direction contained in the image plane, and k_1 the others. Then

$$X \geq k_0 + 2k_1, \quad p \leq k_0k_1 + \binom{k_1}{2}, \quad k_0 + k_1 \leq 4,$$

which implies $p \leq X$. For $W = 3$, $X \geq 2k$ and $N \leq 3\binom{k}{2} \leq 2(1 + X)$. In every case $2(1 + X) \geq N$, so $\mathcal{P} \geq 0$. At smaller core rank, $N = Z = 0$.

Adding a possible fourth nonloop private element cannot lower the starting potential. Quotient by it. The old private images span dimension two, with private-pair count $\Delta V \geq 1$. Let ΔX be their mixed basis count with the four marked virtual images, and ΔN the marked-pair count. If $\Delta N > 0$, two independent marked and two independent private images give $\Delta X \geq 2$, whereas $\Delta N \leq 6$. Thus $2\Delta V + 2\Delta X \geq \Delta N$, proving $\Delta \mathcal{P} \geq 0$. The value of Z is unchanged. Hence (285), or $\mathcal{P} \geq 0$ at core rank below three, holds for the actual private configuration.

Now add a nonzero original row ℓ to a spanning old configuration. Quotienting the old original rows by ℓ gives a rank-three configuration after omitting loops. Write q', X', N' for the increments of q, X, N .

If the core has rank three and $\ell \notin H_3$, its restriction to the annihilator plane of ℓ is invertible. In suitable coordinates the transformed old-row/private matrix M has rank three, and

$$(m_3(M), m_2(M), m_1(M)) = (q'V, X', N').$$

These are the same complementary-minor identities used for the endpoint sectors. Also $\Delta Z = 1$. Lemma 39.3 gives

$$\Delta \mathcal{P} = 5\Phi(M) - 17 \geq 8. \quad (286)$$

If at least four old rows survive as nonloops modulo ℓ , then $\Phi(M) \geq 8$ and

$$\Delta \mathcal{P} \geq 23. \quad (287)$$

For an inside row, the restricted core has rank two. More generally this is the only nontrivial increment at smaller core rank. We claim

$$2q'V + 2X' - N' \geq 1, \quad \Delta\mathcal{P} \geq 5. \quad (288)$$

At restricted rank at most one, $N' = 0$ and $q'V \geq 1$. At rank two, let $n' \geq 3$ count nonloop quotient rows, let W count private images outside the core image plane, and let $\beta \leq n'$ be their common outside-row count. Then

$$N' = W\beta, \quad q' \geq n' - 2, \quad 2X' \geq W(n' - 1).$$

The last inequality counts a good virtual point and a first private element, completes by a second private element, and counts each resulting private-pair basis at most twice. With three private elements $V = 1$; with four, $V \geq 2$. Consequently

$$2q'V + 2X' - N' \geq 2(n' - 2)V - W.$$

This is at least one except for the following boundary cases at $n' = 3$.

For three private basis elements and $W = 2$, if $\beta \leq 2$, then $X' \geq 2$ gives a margin at least two. If $\beta = 3$, the three nonzero virtual images have three distinct directions. At most one is parallel to the private direction in the image plane; the others have at least two private coordinates, so $X' \geq 5$. For $W = 3$, every nonzero virtual image has at least two coordinates. There are at least two such images, so $X' \geq 4$, giving margin at least one.

For four private elements, the only remaining case is $V = 2, W = 4$. The private configuration is a parallel pair and two coloops. No nonzero virtual image in the core plane is parallel to a private element. In a private basis it has support at least two, and the duplicated coordinate ensures at least three private-pair completions. There are at least two nonzero virtual images, so $X' \geq 6$ and the margin is at least four. This proves (288) in every case.

39.6 Parallel classes and the strict middle gap

We first record the exact case that cannot be disposed of by coarse growth bounds.

Lemma 39.5. *For an $a+b$ cover with unit A activities, suppose the exterior rank- b configuration has exactly b independent parallel classes of positive masses m_i . Let $q = \prod_i m_i$. For its conditional coefficients C_j and the corresponding auxiliary specialization with private activity q , one has*

$$b_j = q^{a-j-1} C_j \quad (0 \leq j \leq a).$$

In particular the conditional polynomial is ULC_a .

Proof. Normalize the class directions to a basis. The virtual cofactor class omitting original class i has total activity q/m_i . A selected missing-class set T has virtual weight $\prod_{i \in T} q/m_i$. Its auxiliary basis uses $a - j - |T|$ private elements, giving total weight

$$\frac{q^{a-j}}{\prod_{i \in T} m_i}.$$

The complementary original L support has weight $q/\prod_{i \in T} m_i$. Complementary subsets have exactly the same nonvanishing condition after core elimination, including loops and any further parallel identifications. Summing gives the stated coefficient identity. The auxiliary polynomial is Lorentzian; its one-variable coefficients are ULC_a . Geometric rescaling preserves this property, proving the claim. $\square$

For $4 + 4$, this gives $b_1 = q^2 C_1$, $b_2 = q C_2$ and the stronger middle factor $9/4$. Since $C_1 C_3 > 0$, it is strictly stronger than $7/4$. The lemma applies to the faithful generic graph representation; arbitrary nongeneric numerical matrices are not substituted for graph endpoint supports.

Suppose now that $n = |L| \geq 9$ and there are at least five nonloop parallel classes. If the fixed- R core has rank at most two, its starting potential is nonnegative and every added row raises it by at least five, so $\mathcal{P} > 0$.

At core rank three, put $r = \text{rank}(L \cap H_3)$. Choose a starting basis with r inside and $4 - r$ outside rows. If $r \geq 2$, (285) and the increment five give

$$\mathcal{P} \geq 10 - 17(4 - r) + 5(n - 4) > 0;$$

the worst endpoint is $r = 2, n = 9$, with value one.

If $r = 0$, all rows are outside. Choose a basis and then a row in a fifth parallel class. All four old rows survive modulo it, so its increment is at least 23. Thereafter contracting any new row kills at most one of the at least five old parallel classes, leaving at least four nonloop rows. Every later increment is also at least 23. Hence

$$\mathcal{P} \geq -58 + 23(n - 4) > 0$$

for $n \geq 7$, in particular throughout the needed range.

If $r = 1$, all inside rows form one parallel class. At least four outside classes exist. Choose one inside and three outside rows as a basis, and then an outside fourth-class row. That one increment is at least 23; all later increments are at least five. Thus

$$\mathcal{P} \geq -41 + 23 + 5(n - 5) > 0$$

at $n \geq 9$, with endpoint value two.

We have proved $\mathcal{P} > 0$ in every required fixed- R quotient, or handled the four-class case exactly. Summing (283) with the positive exterior-right activities gives (282). By (281) and quartic Newton,

$$C_2^2 \geq \frac{b_2^2}{q^2} \geq \frac{9}{4} \frac{b_1 C_3}{q^2} > \frac{7}{4} C_1 C_3.$$

Thus the middle gap is strict for every genuine $4 + 4$ cover with $n \geq 9$.

39.7 The lower population boundary and eventual scaling

The ambient transversal matroid after contracting B has degree n . Its normalized factor at conditional index two is

$$\frac{3(n-1)}{2(n-2)},$$

strictly above $7/4$ for $n \leq 7$ and equal at $n = 8$.

At $n = 8$, use the weighted local equality criterion proved in the structural companion bundled with [30] [*Merge note*: source 08s's Lemma 29.5, which is the necessity half of source 72's Theorem 14.3]: equality at index two forces compatible positive red and blue masses on every nonloop parallel class of every rank-two contraction by one red and five blue elements. In particular, a contraction retaining two independent blue elements and a red-only nonloop class forces strictness.

Choose an exterior L basis. After contracting B , the blue basis consists of the four private A coordinates and four complementary private L coordinates. If an R vertex u has two A neighbors, choose a matching extending u first, name its assigned coordinate i , and choose

another neighbor j . From the other two coordinates choose a matched red element R_h . Contract it, the private A coordinate at the other remaining position, and all four complementary private L coordinates. This independent six-set has one red and five blue elements. The surviving A_i, A_j remain independent, while u has two free edge coefficients independent of the contracted and blue data. Its quotient line is parallel to no nonloop blue element. The local criterion therefore makes the middle gap strict.

If every exterior R vertex is a singleton leaf, compress its four classes to four vertices of summed activity. The whole right shore then has eight vertices, and the pure-shore theorem gives ULC_8 at every positive field. This branch needs no strictness assertion.

For all other cases the conditional first and top gaps are already strict by [31], and the middle gap is now strict. The low-index leading terms under common B scaling are those of the pure right-shore- B support polynomial, whose degree-four normalization is stronger than the actual-rank-eight normalization. At index four, the square term has higher scaling degree than the neighboring product. At higher indices the strict leading coefficients are exactly the conditional gaps proved above. Each gap polynomial is therefore positive for sufficiently large λ .

The sealed $5 + 3$ theorem [30], its earlier $3 + 5$ input, the at-most-two-cover theorem, and the rank-seven result bundled there exhaust the other cover splits at ranks at most eight. This proves Theorem 39.1, retaining unit activities on the entire left shore throughout.

For a strict branch with gap $G(\lambda) = a_D \lambda^D + \sum_{i < D} a_i \lambda^i$, $a_D > 0$, a sufficient explicit field-dependent threshold is

$$\lambda \geq 1 + \frac{\sum_{i < D} |a_i|}{a_D}.$$

Take the maximum over its finitely many gaps. Pure-shore and small-cover branches already satisfy ULC at every positive field. No finite-activity theorem for general rank-six, rank-seven or rank-eight graphs is inferred.

39.8 Failure ranks and verification scope

The unchanged rank-38 witness in the bundled prerequisites has blocks $|A| = 8$, $|B| = 30$, $|L| = 32$, $|R| = 794$, and edges $(A \times B) \cup (L \times B) \cup (A \times R)$. All left and exterior-right activities are one, while every B activity is w . Its last gap is

$$G_{37} = \frac{5}{6304} D^2 w^{58} (-22w^2 + 6437219280w + 11385894392295), \quad D = \binom{32}{30} \binom{794}{7}.$$

It is negative for all sufficiently large w . Theorem 39.1 therefore improves the persistent common-scaling failure-rank bracket to [9, 38]. The separate finite-activity bracket stays [6, 38]. [*Merge note:* Its lower end rests on source 74 (Part I); its upper end on source 55 (Part II). The persistent-failure bracket [9, 38] is the final one of the batch; sources 21 and 08s gave [7, 38] and source 08 [8, 38].]

The new finite proof input is the complete 46,512-profile hyperplane table. Its independent coordinate-injection replay agrees with every coefficient and all four hashes. The minor-count growth lemmas and parallel-class identity are elementary proofs, not bounded-population checks; the 330 support-pattern verification supplements the rectangular lemma. All mathematical checks in the portable scripts use exact arithmetic and explicit failures, so they remain effective under optimized Python.

The two preceding PDFs and source archives are included unchanged. They supply the previously cleared graph-generic upper incidence, strict top and $5 + 3$ comparisons, Lorentzian sector construction, small-cover theorem and weighted local equality criterion. This report and

its certificates are ordinary mathematical proofs and exact computation, not proof-assistant certification or external refereeing. No exhaustive priority claim is made.

Part V

Second routes and boundaries

Contents of Part V. Part V prints four manuscripts of batch 73 of ProveIt’s intake (arrival commit `f8c3a392a`, placement commit `5e4f1263c`). They are numbered 77–80 here, continuing this report’s batch-72 numbers; their batch-73 manuscript numbers are 26, 08, 02 and 01. All four work in Part I’s setting: independent vertex activities on both shores, each pair of endpoint sets counted once, normalization by the actual matching number. They were written in parallel with batch 72: source 77 (`matching-rank-four-package.zip`) on the evening of 30 September 2026, sources 78 and 79 early on 1 October, all three before batch 72 arrived, and source 80 after its arrival (its latest file is dated less than an hour later). Source 80 alone read batch-72 material, the raw archives of sources 74, 10 and 05. None of them read this report or Part XIII of *preorder-root-polytopes*. Most of what they prove is therefore already in Parts I and III or in *preorder-root-polytopes*; such results keep their statements, and their proofs are printed as marked second routes or replaced by pointers. New in Part V:

- source 77 (Section 40): at unit activities and matching number three, a non-real-rooted support polynomial needs fifteen vertices, and the fifteen-vertex example is unique (Theorem 40.6, computer-assisted), with the pair-neighbourhood family behind it;
- source 78 (Section 41): a real-stable seed for a $2 + 3$ cover whose core is a star centred at a vertex of the two-vertex part with exactly two neighbours in the three-vertex part (the exceptional two-neighbour seed in Theorem 41.2);
- source 79 (Section 42): a two-element Lorentzian deformation of every matroid basis polynomial (Theorem 42.1), with two proofs, the first from source 80’s appendix;
- source 80 (Section 43): a positive semidefinite first-layer theorem for any number of left vertices (Theorem 43.3), which proves source 10’s three-vertex marginal theorem without its 5339-monomial certificate, and three thirteen-vertex obstructions showing that Part I’s two-tail Lorentzian lifts do not extend to three tails.

Same results in Parts I–IV and in *preorder-root-polytopes*. Source 77’s weighted rank-four theorem is source 76’s Corollary 15.2 and a case of source 74’s Corollary 2.2; its real-rootedness theorem for a $2 + 2$ cover is the univariate consequence of source 76’s Theorem 15.1, and its elementary proof (weighted paths, Hermitian Gram determinants, a positive semidefinite block matrix) is printed as a second route; its thin-cover reduction is Part XIII of *preorder-root-polytopes* (pointers); its $1 + 2$ lemma is *preorder-root-polytopes*’s Theorem 207.1 (proof printed as a second route, because the rank-three minimum uses it); its strict rank-five family is a case of source 75’s Theorem 16.1 (certificate route printed). Source 78’s theorem for $2 + 3$ cores is a case of Corollary 2.2 with source 72’s strictness; its 26 certificates are printed as a computer-assisted second route. Source 79’s graph corollaries are source 74’s Theorem 2.1 and Corollary 2.2. Source 80’s scalar counterexamples are Part XI’s Theorem 177.1 of *preorder-root-polytopes* and source 74’s Section 6 (displays kept, derivations pointers), and its three-tail cubic is source 10’s Theorem 23.1 (second route). Five further batch-73 manuscripts on the same object are duplicates and are not printed (Section A.1).

Notation of Part V. Sources 78–80 write the shores as $P \sqcup X$ and $Q \sqcup Y$: P and Q are the two parts of the displayed cover (Part I’s A and B) and X , Y the exterior sets (Part I’s L and R). Here X and Y are *not* whole shores, as they are elsewhere in this report. Source 77 writes

$G = (L, R, E)$: there L and R are the whole shores, not exterior sets. “Tail” and “head” mean left and right vertex; “actual” or “surviving” degree is the matching number after zero-activity vertices are deleted; $\Gamma(t) = \sum \gamma_k t^k$ (sources 78, 80) is p_G . Source 79’s distinguished matroid elements A, B are not cover parts, and source 80 writes N for an exterior population size in Section 43.2 but for a matrix in Theorem 43.3. The table of Section 1.3 lists these readings.

Non-claims of Part V. Source 77’s minimum concerns unit activities and matching number exactly three only; it is computer-assisted and makes no claim across matching numbers (*preorder-root-polytopes*’s theta graph has eight vertices) and no priority claim; its rank-five theorem covers block-constant activities only. Source 78’s theorem is computer-assisted, its complete-exterior hypothesis is essential, its full core can have nonreal zeros, and the two-tail/three-head role cover stays open. Source 79 makes no joint Lorentzian assertion in its population parameters, identifies no physical tail and head roles, and leaves the 3 + 3 cover outside. Source 80’s obstructions concern multivariate homogenizations whose scalar sequences remain ULC; it gives no unweighted counterexample and no minimum-vertex bound, and Theorem 43.3 does not assert full Lorentzianity for more than three tails. No source of Part V claims global priority, referee review or formalization.

40 Weighted rank four, real roots for two-by-two covers, and the rank-three minimum (source 77)

Abstract of source 77, as delivered. We prove actual-degree ultra-log-concavity for weighted matching-support polynomials of every bipartite graph with matching number at most four. Each pair of equal-size shore subsets admitting a perfect matching is counted once, with product vertex activities; arbitrary nonnegative activities are allowed on both shores. A minimum-cover reduction combines pendant compression and the credited ambient Lorentzian theorem with exact weighted-path and Hermitian determinant constructions. For covers split as two vertices on each shore, a stronger real-rootedness theorem holds; the complete core is handled by a positive-semidefinite block matrix. General rank-four support polynomials need not be real-rooted. The result answers the inspected report’s weighted rank-three and rank-four questions. Two extensions give an exact computer-assisted classification of the smallest unit-weight non-real-rooted example at matching rank three, and strict rank-five normalization for a specified complete-core family with four constant activity blocks. General rank five and the unit-weight all-rank question remain open. [*Merge note:* Source 77 is batch-73 manuscript 26, revision 2 of `matching-rank-four-package.zip` (title page “Research note prepared with OpenAI for Vladimir Reshetnikov”). Its first, twelve-page edition, batch-73 archive 27 (`matching-rank-four-package (1).zip`), contains no statement, proof or example that is not printed here and is not a source (`77-rank-four-REVISION.md`). Source 77 inspected *preorder-root-polytopes* at commit `d5e863bba`, before *preorder-root-polytopes*’s Part XIII and before this report existed. Its Theorem 40.1 is source 76’s Corollary 15.2, the case $\nu \leq 4$ of source 74’s Corollary 2.2. Its Theorem 40.2 is the univariate consequence of source 76’s Theorem 15.1, which proves real stability of the basis polynomial as well; source 77’s proof (Sections 40.3–40.5) is elementary and different, and is printed as a second route to that theorem’s univariate conclusion. New here are the family p_n of Section 40.6.1, the twin remark and Theorem 40.6.]

40.1 Support weights and the main theorem

Let $G = (L, R, E)$ be a finite simple bipartite graph. A pair (I, J) with $I \subseteq L$, $J \subseteq R$, and $|I| = |J|$ is *matchable* if $G[I, J]$ has a perfect matching. For activities $u_x, v_y \geq 0$, define

$$p_G(t) = \sum_{k \geq 0} a_k t^k, \quad a_k = \sum_{\substack{I \subseteq L, J \subseteq R \\ |I| = |J| = k \\ G[I, J] \text{ matchable}}} \prod_{x \in I} u_x \prod_{y \in J} v_y. \quad (289)$$

The empty support has weight one. A support with several perfect matchings contributes *one* product of activities, not one term for each matching. This differs from the ordinary matching generating polynomial.

Write $\nu(G)$ for the maximum matching size and $d = \deg p_G$. The degree is the matching number of the induced graph on positive-activity vertices, after isolated vertices are ignored. Indeed its largest matching supplies a positive degree- d support, and a support cannot exist above its matching number. Submatchings show that $a_0, \dots, a_d$ are all positive.

Theorem 40.1 (Actual-degree normalization through rank four). *If $\nu(G) \leq 4$, then for every choice of nonnegative two-shore vertex activities, the sequence*

$$\left(\frac{a_k}{\binom{d}{k}} \right)_{k=0}^d$$

is log-concave and has no internal zeros. Equivalently, for $1 \leq k < d$,

$$k(d-k)a_k^2 \geq (k+1)(d-k+1)a_{k-1}a_{k+1}. \quad (290)$$

For $d = 0$ or 1 the assertion has no interior inequality.

[Merge note: This is source 76's Corollary 15.2 and the case $\nu \leq 4$ of source 74's Corollary 2.2, both including nonnegative activities with the surviving degree. Source 77's proof (end of Section 40.6) is a further route.]

The structural statement used for the remaining mixed cover is stronger.

Theorem 40.2 (Real roots for a two by two cover). *If G has a vertex cover consisting of two left vertices and two right vertices, then p_G has only negative real zeros, for arbitrary nonnegative activities. Its degree may be smaller than four. In particular it satisfies (290) at its actual degree.*

[Merge note: Theorem 40.2 is the univariate conclusion of source 76's Theorem 15.1 (real stability of the complemented-row basis polynomial for a $2 + 2$ cover). Its proof here is a second route to that conclusion only; it does not give multivariate stability.]

At $d = 4$, the three inequalities are

$$3a_1^2 \geq 8a_2, \quad 4a_2^2 \geq 9a_1a_3, \quad 3a_3^2 \geq 8a_2a_4.$$

The inspected ProveIt manuscript proves the unweighted result through rank three, asks for the two remaining rank-four inequalities, and separately asks for arbitrary two-shore weights at rank three [8, questions `mr:q:rankfour`, `mr:q:weighted`, and `hb:q:rank`]. Theorem 40.1 answers these finite-rank questions, including the rank-three and rank-four parts of the last one. It does not claim all-rank weighted normalization: that same manuscript already exhibits a weighted rank-six counterexample. [Merge note: The three questions are Research questions 93 (`mr:q:rankfour`), 95 (`mr:q:weighted`) and 86 (`hb:q:rank`) of *preorder-root-polytopes*. They

were open at source 77's pin `d5e863bba`. *preorder-root-polytopes* now records them as answered: weighted rank three by its Part XIII (Corollary 206.3), and ranks four and five, with the least weighted failure rank exactly six, by source 74 in Part I. Source 77's answer is a further route, not the first one.]

The ambient Lorentzian theorem, transversal-matroid correspondence, Newton inequalities, spectral theorem, Cauchy–Binet expansion, and matching–cover theorem are established inputs or classical machinery. The paper gives the finite-rank reductions and matrix constructions in detail and makes no global priority claim. No proof-assistant formalization is asserted.

40.2 The ambient theorem and the cover reduction

We use the weighted ambient normalization exactly with its legitimate parameter.

Proposition 40.3 (Weighted smaller-shore normalization). *For positive activities on a bipartite graph with shore sizes m, n , the sequence $a_k/\binom{s}{k}$, $0 \leq k \leq s$, is log-concave after zero padding, where $s = \min(m, n)$.*

Proof (pointer). This is Theorem 70.1 of *preorder-root-polytopes* (Part V, `lor:thm:weighted ULC`), Lemma 3.1 here. [Merge note: Source 77 proves it by the private-slot transversal matroid on $L^\circ \sqcup R$ with slots L , whose bases are $(L^\circ \setminus I^\circ) \sqcup J$ for matchable (I, J) , the substitution $s/u_x, tv_y$ and the bivariate characterization of Lorentzian polynomials [1, Theorems 3.10 and 2.10, Example 2.26]. That is *preorder-root-polytopes*'s proof; source 77 credits it, and the unweighted relation-bimatroid case to Röhrle–Ulirsch [7, Theorem A and Example 2.4]. It is not reprinted.] $\square$

One cannot replace m in this homogeneous lift by $\nu(G)$ merely by cancelling a power of s . Such cancellation does not generally preserve the Lorentzian property. The reductions below arrange a genuinely smaller shore or construct a separate real-rooted polynomial.

Lemma 40.4 (Pendant compression). *Suppose $P \subseteq R$ is a nonempty set of vertices with common neighborhood $\{a\}$ for some $a \in L$. Replace P by one vertex with neighborhood $\{a\}$ and activity $\sum_{y \in P} v_y$. This preserves p_G and the matching number.*

Proof (pointer). This is weighted leaf compression, Lemma 206.1 of *preorder-root-polytopes* (Part XIII, `lc:lem:compression`). [Merge note: Source 77's proof is the same: a feasible support contains at most one vertex of P , and the supports through P for fixed (I, J_0) are all matchable or none is. It is not reprinted.] $\square$

Proposition 40.5 (Covers with a small shore part). *With positive activities, if a minimum cover of size r has at most one vertex on one shore, then p_G satisfies (290) with $d = r$.*

Proof (pointer). This is Theorem 206.2 of *preorder-root-polytopes* (Part XIII, `lc:thm:singleton`), Lemma 3.2 here. [Merge note: Source 77's proof is the same argument: transpose so that one cover part has at most one vertex, compress the pendant vertices at that vertex by Lemma 40.4, and apply Proposition 40.3 to the resulting shore of exactly r vertices. It is not reprinted.] $\square$

König's classical matching–cover theorem provides a cover of size $\nu(G)$. One standard proof starts from a maximum matching and its alternating paths from unmatched left vertices: if Z_L, Z_R are the reached vertices, then $(L \setminus Z_L) \sqcup Z_R$ meets every edge and takes one endpoint of each matching edge, hence has the matching's size. An augmenting path to an unmatched right vertex is excluded by maximality. Every cover has size at least a matching's size, giving equality.

For $d \leq 3$ every split of a minimum cover has a part of size at most one. At $d = 4$, up to exchanging shores, the splits are $0 + 4$, $1 + 3$, and $2 + 2$. The first two are settled above. We now prove Theorem 40.2, which settles the last.

[*Merge note:* Sections 40.3–40.5 are source 77’s proof of Theorem 40.2, a second route to the univariate conclusion of source 76’s Theorem 15.1. For the first three core types (no core edge, one edge, two adjacent edges) that conclusion is also *preorder-root-polytopes*’s Corollary 94.3 (Part VI, `ade:mat:cor:block-family`), as noted in Section 15.2.]

40.3 Two-center support data and path realizations

Throughout the mixed-cover proof, write the cover as $\{a_1, a_2\} \subseteq L$ and $\{b_1, b_2\} \subseteq R$. Every exterior left vertex has neighborhood among the b -centers, and every exterior right vertex among the a -centers. No edge joins two exterior vertices. Ignoring isolated vertices, there are three neighborhood types on each exterior shore. The induced graph on the four centers is called the *core*.

40.3.1 The local quadratic and rooted path model

For a component with two centers a, b of activities u, v , denote exterior activity sums of neighborhoods $\{a\}, \{b\}, \{a, b\}$ by X, Y, Z . Let

$$E = \sum_{i < j} z_i z_j = \frac{1}{2} \left(Z^2 - \sum_i z_i^2 \right), \quad 0 \leq E \leq Z^2/2,$$

where z_i are the common-neighbor activities. Then its support polynomial and its root- a deletion are

$$\begin{aligned} p(t) &= 1 + (P + Q)t + Bt^2, & q(t) &= 1 + Qt, \\ P &= u(X + Z), & Q &= v(Y + Z), \\ B &= uv(XY + XZ + YZ + E). \end{aligned} \tag{291}$$

Two selected common neighbors contribute E , not $2E$. Thus

$$PQ - B = uv(Z^2 - E) \geq 0.$$

In particular the quadratic discriminant is nonnegative: $(P+Q)^2 - 4B = (P-Q)^2 + 4(PQ-B) \geq 0$. Its nonconstant zeros are negative.

If $P > 0$, take an ordinary edge-weighted four-vertex path rooted at its first endpoint, with successive edge weights

$$P, \quad Q - B/P, \quad B/P. \tag{292}$$

All weights are nonnegative. Its matching generating polynomial is p , because the only two-edge matching uses the first and third edges. Deleting the root leaves a two-edge path of total weight Q , with polynomial q . If $P = 0$, then $B = 0$; an isolated root and one edge of weight Q give the same pair, or one may take a limit.

For any weighted path with edge weights $h_i \geq 0$, let J be the real symmetric zero-diagonal tridiagonal matrix with adjacent entries $\sqrt{h_i}$. The determinant recurrence gives

$$\det(xI - J) = \sum_k (-1)^k m_k x^{N-2k},$$

where m_k are its ordinary matching coefficients. Every eigenvalue of J is real. A nonzero zero t of $\sum m_k t^k$ would satisfy this identity at an x with $x^2 = -1/t$, so x is real and $t < 0$. This is the classical path matching/spectral mechanism. Its application to supports below is justified by explicit polynomial identities.

40.3.2 No core edges or one core edge

With no core edge, the two two-center components are disjoint and $p_G = p_1 p_2$, a product of real-rooted quadratics. With the sole core edge joining distinguished roots a and b , let $c = u_a v_b$. Then

$$p_G = p_1 p_2 + c t q_1 q_2. \quad (293)$$

To see that no support is double-counted, separate it by its shore imbalance in the two components after deleting the core edge. A support not using that edge is balanced in each component and contributes to $p_1 p_2$. A support using it has imbalance one, of opposite signs in the pieces, and after removing a, b gives one pair of root-deleted balanced supports. These cases are disjoint and exhaustive.

Replace each component by its rooted path model (292) and join the roots with weight c . This gives an eight-vertex weighted path satisfying the same identity (293). Therefore its ordinary matching polynomial equals p_G , proving real-rootedness.

40.3.3 Two adjacent core edges

After transposing if needed, let the common endpoint a of the two core edges lie in the first component, and let its neighbors be both centers of the second. Write c_a for the activity of a , and u, v, X, Y, Z, E for the second component's data. Put $N = X + Y + Z$ and

$$p_2 = 1 + At + Bt^2, \quad Q_* = \frac{uvN}{u+v}.$$

An imbalanced support contains a . In the second component it either uses one center and no exterior vertex, of total weight $u + v$, or both centers and one nonisolated exterior vertex, of total weight uvN . The latter support is counted once even if that exterior vertex can use either center. Consequently

$$p_G = p_1 p_2 + c_a(u + v)tq_1(1 + Q_*t). \quad (294)$$

Set $\theta = u/(u + v)$. Expansion gives the exact nonnegative certificate

$$Q_*(A - Q_*) - B = uv \left(\left[\theta X - (1 - \theta)Y + (\theta - \frac{1}{2})Z \right]^2 + \frac{3}{4}Z^2 - E \right) \geq 0. \quad (295)$$

Also $Q_* \leq A$, since $A \geq \min(u, v)N$ and $uv/(u + v) \leq \min(u, v)$. Thus p_2 and $1 + Q_*t$ have the rooted path realization (292), now with $P = A - Q_*$ and $Q = Q_*$. Connect this path to the first rooted path with edge weight $c_a(u + v)$. Identity (294) again proves the desired real-rootedness. Degenerate zero parameters follow by limits.

40.4 Hermitian realizations for the next two core patterns

Write the center activities as α_1, α_2 for a_1, a_2 and β_1, β_2 for b_1, b_2 . For exterior left types let their sums be L_1, L_2, Z and their common-type second elementary symmetric sum be E . On the right use R_1, R_2, W, F . Set

$$c = \sqrt{Z^2 - E}, \quad d = \sqrt{W^2 - F}, \quad Z/\sqrt{2} \leq c \leq Z, \quad W/\sqrt{2} \leq d \leq W.$$

Replace the exterior left component by one common leaf adjacent to both b -centers and two private leaves. In an ordinary weighted matching model assign the two common edges weights $\beta_1 c, \beta_2 c$, and the two private edges weights

$$\beta_1(L_1 + Z - c), \quad \beta_2(L_2 + Z - c).$$

Its matching polynomial is exactly the original two-center support polynomial: the two-edge coefficient is $\beta_1\beta_2[(L_1 + Z)(L_2 + Z) - c^2]$. The identity also holds after deleting either or both centers. Make the analogous replacement on the right using d , and restore each core edge with matching weight $\alpha_i\beta_j$. Denote the ordinary matching polynomial of the resulting surrogate by $M(t)$. We compare M to p_G explicitly; equality is not assumed.

40.4.1 Cycle interference in a Gram determinant

For a complex bipartite biadjacency matrix B , Cauchy–Binet gives

$$\det(I + tBB^*) = \sum_{k, I, J} |\det B[I, J]|^2 t^k.$$

The terms pairing a matching with itself give its ordinary weighted matching polynomial when each matrix entry has modulus the square root of that edge’s weight. Phases affect only the cross terms. If the surrogate has no two vertex-disjoint cycles, two different matchings on a fixed support differ on a single even cycle C of length $2h$. Pairing its two alternating choices gives the correction

$$2(-1)^{h-1} \Re \left(\prod_C \text{oriented amplitudes} \right) t^h M_{G'-V(C)}(t), \quad (296)$$

where G' is the surrogate and the product uses conjugates on alternate edges. The determinant signs differ by an h -cycle. Outside C the matching is identical, which gives the deleted-complement polynomial. This derives the particular cycle expansion needed here from determinant permutations. A four-cycle contributes negatively and a six-cycle positively.

The cases below have intersecting cycles, and their deleted complements are forests. A successful choice of phases therefore gives $p_G = \det(I + tBB^*)$. The matrix BB^* is positive semidefinite, so such a polynomial is a product of factors $1 + \lambda t$, $\lambda \geq 0$, and has only negative real zeros.

40.4.2 Two disjoint core edges

Choose core edges a_1b_2, a_2b_1 . The surrogate has one cycle: a six-cycle through all four centers and the two common leaves. Its deleted complement has no edges. Set $T = \alpha_1\alpha_2\beta_1\beta_2$. Then

$$M(t) - p_G(t) = TZWt^3. \quad (297)$$

For justification, separate supports by the number of core edges in a realizing matching, which is fixed by their component imbalance. Zero core edges are handled exactly by the two local models. Two core edges use all four centers and no exterior vertices and have a unique realization. With one core edge, different choices can realize the same support only when it contains all four centers and one common exterior vertex on each shore. There are exactly two such choices, and the aggregate extra weight is TZW . This proves (297) at the support level.

The cycle amplitude has modulus Tcd . Choose its phase ϕ so that

$$\cos \phi = -\frac{ZW}{2cd},$$

which is possible because $2cd \geq ZW$. Equation (296) supplies exactly $-TZWt^3$ and yields the required Gram determinant. If $ZW = 0$, the conclusion follows directly or by continuity.

40.4.3 Three core edges and two coupled phases

Take core edges a_1b_1, a_1b_2, a_2b_1 . The surrogate has two four-cycles and one six-cycle: the four-cycles pass through a_1 with the left common leaf and through b_1 with the right common leaf; the six-cycle passes through all four centers and both common leaves. Choose oriented phase coordinates θ, η for the four-cycles. The six-cycle phase is $\theta - \eta$ after choosing consistent orientations. The cycle space has dimension two, so these two phases may be assigned independently, for instance using a spanning tree and the two non-tree edges.

Put

$$x = 2c \cos \theta, \quad y = 2d \cos \eta, \quad U = \alpha_1\beta_1\beta_2, \quad V = \alpha_1\alpha_2\beta_1, \quad b = R_2 + W, \quad a = L_2 + Z.$$

The two four-cycle corrections and the six-cycle correction are respectively

$$-Uxt^2(1 + \alpha_2bt), \quad -Vyt^2(1 + \beta_2at), \quad 2Tcd \cos(\theta - \eta)t^3.$$

The deleted four-cycle complements are stars with the displayed total edge weights. Direct support counting gives

$$M(t) - p_G(t) = (UZ + VW)t^2 + T(Zb + Wa)t^3. \quad (298)$$

Here the UZ and VW terms are the duplicated supports consisting of the three centers of a four-cycle and one common neighbor. They may additionally include the remaining center and a neighbor of that center, giving TZb and TWa . At their overlap, with all centers and two common neighbors, there are three realizing one-core-edge matchings, so the two corrections correctly subtract two copies. No other imbalance-one support has more than one core choice; the imbalance-zero local models already count supports correctly, and the two-core-edge matching is unique. This proves (298) without counting matching multiplicities as support weights.

It remains to find phases solving

$$\begin{aligned} Ux + Vy &= UZ + VW, \\ 2cd \cos(\theta - \eta) &= (x - Z)b + (y - W)a. \end{aligned} \quad (299)$$

The following elementary argument supplies them. Let $A = 2c$, $B = 2d$. The line $Ux + Vy = UZ + VW$ cuts the rectangle $[-A, A] \times [-B, B]$ in a closed segment containing (Z, W) in its relative interior when the common-type sums are positive. At its endpoints a rectangle coordinate saturates. For each point on the segment the two possible values of the left side of the second equation are

$$H_{\pm}(x, y) = \frac{xy \pm \sqrt{(A^2 - x^2)(B^2 - y^2)}}{2}.$$

They are continuous and coincide at the segment endpoints. The affine target $h(x, y) = (x - Z)b + (y - W)a$ vanishes at (Z, W) . At this point the branches straddle zero, because

$$(A^2 - Z^2)(B^2 - W^2) \geq Z^2W^2.$$

Indeed, writing $c^2 = Z^2/2 + s$ and $d^2 = W^2/2 + t$ with $s, t \geq 0$, the difference is $4sW^2 + 4tZ^2 + 16st \geq 0$.

Move from (Z, W) to either endpoint. Initially $H_- - h \leq 0 \leq H_+ - h$, whereas at the endpoint both differences are equal. If that common value is positive, the lower branch crosses zero; if negative, the upper one does; if zero, the endpoint works. The intermediate value theorem gives phases solving (299). The corrections then turn M into p_G , establishing its Gram determinant and real-rootedness. Absent common types or zero activities follow by adding arbitrarily small positive activities on auxiliary vertices and passing to a coefficientwise limit. A limit of real-rooted polynomials of bounded degree with constant term one remains real-rooted; roots may escape to infinity as the degree drops.

40.5 The complete core and a positive semidefinite block matrix

The full $K_{2,2}$ core requires a different aggregate construction. We first establish a two-center inequality that supplies enough phase freedom in a 2×2 coupling matrix.

40.5.1 A side matrix and its variance margin

Use positive center activities u, v and exterior data X, Y, Z, E from (291). Put

$$U = u + v, \quad a = uv, \quad A = u(X + Z) + v(Y + Z), \quad B = a(XY + XZ + YZ + E),$$

$$\alpha = \frac{a(X + Y + Z)}{U}, \quad G = \begin{pmatrix} u(X + Z) & \sqrt{a(Z^2 - E)} \\ \sqrt{a(Z^2 - E)} & v(Y + Z) \end{pmatrix}, \quad H = \text{adj}(G).$$

Both diagonal entries are nonnegative and $\det G = B \geq 0$, so G and H are positive semidefinite, with trace A and determinant B .

Write

$$\theta = \frac{u}{U}, \quad z = \theta - \frac{1}{2}, \quad h = \frac{a}{U^2}, \quad L = Z^2 - E, \quad T = \theta X - (1 - \theta)Y + zZ.$$

Direct expansion gives the following exact certificates:

$$D := \alpha(A - \alpha) - B = a(T^2 + L - Z^2/4), \quad (300)$$

$$\Delta := A^2 - 4B = U^2[(T + zZ)^2 + 4hL], \quad (301)$$

$$2D - h\Delta = a[(T - zZ)^2 + (1 + 4z^2)(L - Z^2/2)] \geq 0. \quad (302)$$

Here $L \geq Z^2/2$, so all indicated margins are nonnegative. The first identity is also the adjacent-core certificate (295).

Let λ_1, λ_2 be the eigenvalues of H . Since $D = (\alpha - \lambda_1)(\lambda_2 - \alpha)$ after ordering them, α lies between them. If $\Delta > 0$, choose probabilities p_1, p_2 with

$$p_1 + p_2 = 1, \quad p_1\lambda_1 + p_2\lambda_2 = \alpha, \quad p_1p_2 = \frac{D}{\Delta} \geq \frac{a}{2U^2}. \quad (303)$$

If $\Delta = 0$, the matrix G is scalar. Its off-diagonal entry vanishes, so $Z^2 - E = 0$; combined with $E \leq Z^2/2$ this gives $Z = 0$. Equality of its diagonals gives $uX = vY$, whence $\alpha = A/2$ is its repeated eigenvalue. Take $p_1 = p_2 = 1/2$, which satisfies the bound in (303) because $a/U^2 \leq 1/4$.

Repeat this construction for the other exterior component. Denote its quantities by $G', H', A', B', \beta, V, b$, and eigenvalue probabilities q_1, q_2 . Thus

$$q_1q_2 \geq \frac{b}{2V^2}, \quad q_1\lambda'_1 + q_2\lambda'_2 = \beta.$$

These identities cover empty exterior populations as well as positive ones.

40.5.2 The full-core support identity

Set $p_A = 1 + At + Bt^2$, $p_B = 1 + A't + B't^2$, $c = UV$, and $d_0 = ab$. For the complete core,

$$p_G(t) = p_A(t)p_B(t) + ct(1 + \alpha t)(1 + \beta t) + d_0t^2. \quad (304)$$

This follows by component imbalance. At imbalance zero no core edge is used, giving the product. At imbalance one, a component selects either one center and no exterior vertex, or both centers and one exterior vertex. Its total polynomial is

$$U + a(X + Y + Z)t = U(1 + \alpha t).$$

Every such pair of component supports can be joined because every core edge is present, and each pair is counted once irrespective of available choices. Finally imbalance two consists solely of the four centers and contributes abt^2 once. The complete core has two perfect matchings on that support, but there is only one support.

40.5.3 An aggregate coupling with a prescribed determinant

Choose eigenbases of H, H' and regard G, G' in the same respective bases. Let C be a complex 2×2 matrix whose squared entry moduli are

$$|C_{ij}|^2 = cp_iq_j.$$

Then, for every choice of entry phases,

$$\begin{aligned} \operatorname{tr}(CC^*) &= c, & \operatorname{tr}(HCC^*) &= c\alpha, \\ \operatorname{tr}(CH'C^*) &= c\beta, & \operatorname{tr}(HCH'C^*) &= c\alpha\beta. \end{aligned} \quad (305)$$

Each identity is the corresponding sum of cp_iq_j times zero, one, or two eigenvalues; the phases cancel because both H, H' are diagonal.

Begin with nonnegative real entries and vary only the phase of C_{22} . The two terms in $\det C$ then have equal modulus $c\sqrt{p_1p_2q_1q_2}$. Therefore $|\det C|^2$ takes every value in

$$[0, 4c^2p_1p_2q_1q_2].$$

By (303) its upper endpoint is at least

$$4U^2V^2 \frac{a}{2U^2} \frac{b}{2V^2} = ab = d_0.$$

Choose the phase so that $|\det C|^2 = d_0$. This enforces the four-center support coefficient without changing the four trace moments.

40.5.4 The block determinant identity

Let $G^{1/2}, G'^{1/2}$ be positive semidefinite square roots and form

$$\mathcal{M} = \begin{pmatrix} C & G^{1/2} \\ G'^{1/2} & 0 \end{pmatrix}.$$

For $t > 0$, the lower-right block $I + tG'$ of $I + t\mathcal{M}\mathcal{M}^*$ is invertible. Its Schur complement gives

$$\det(I + t\mathcal{M}\mathcal{M}^*) = \det(I + tG') \det(I + tG + tC(I + tG')^{-1}C^*).$$

For 2×2 matrices, the determinant is quadratic in the added matrix. Expansion of the last expression yields

$$\begin{aligned} \det(I + t\mathcal{M}\mathcal{M}^*) &= \det(I + tG) \det(I + tG') \\ &\quad + t \operatorname{tr}(\operatorname{adj}(I + tG)C \operatorname{adj}(I + tG')C^*) + t^2 |\det C|^2. \end{aligned} \quad (306)$$

Both sides are polynomials, so the identity holds for every t . Since $\operatorname{adj}(I + tG) = I + tH$ and similarly on the other side, substituting (305) gives

$$\det(I + t\mathcal{M}\mathcal{M}^*) = p_{APB} + ct + c(\alpha + \beta)t^2 + c\alpha\beta t^3 + d_0 t^2 = p_G(t).$$

The matrix $\mathcal{M}\mathcal{M}^*$ is positive semidefinite. This proves real-rootedness for the full core. The construction matches the aggregate coefficient polynomial; it does not claim a faithful determinant realization of each individual support minor of the original graph.

40.6 Completion and the real-rootedness boundary

The six core types above exhaust all subsets of the four possible core edges up to permutation of the two centers on either shore and transposition: their orbit sizes are 1, 4, 4, 2, 4, 1, totaling sixteen. Sections 40.3–40.5 prove Theorem 40.2 for positive activities, and coefficientwise limits give nonnegative activities. Newton’s classical inequalities give (290) with the actual degree of each real-rooted polynomial. The constant term is one and all coefficients are nonnegative, so no nonconstant factor has a zero at zero or at a positive real number.

Proof of Theorem 40.1. Delete every zero-activity vertex. This preserves p_G and leaves a positive-activity graph with matching number exactly $d \leq 4$. Remove isolated vertices and choose a minimum cover of size d by König’s theorem. If one cover part has size at most one, Proposition 40.5 applies. The only remaining split is $2 + 2$, for which Theorem 40.2 and Newton’s inequalities apply. Positivity of every coefficient through d follows from submatchings. This proves the asserted actual-degree statement, including all zero-activity degeneracies. $\square$

This initial deletion is essential for interpreting the normalization: merely taking a limit of inequalities normalized by the original rank gives that old normalization, which can be weaker than normalization by the surviving degree. The cover reduction is applied anew after deletion.

40.6.1 Real roots can fail already at rank three

Take three left centers and n right vertices of each of the three pair-neighborhood types $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3\}$, all with unit activities. The support polynomial is

$$p_n(t) = 1 + 6nt + \frac{3n(7n-1)}{2}t^2 + (4n^3 - 3n^2)t^3.$$

Indeed each chosen left pair has two singleton exterior types of size n and one common type of size n , giving $3n^2 + \binom{n}{2}$ supports. Multiply by the three left pairs for the quadratic coefficient. A three-vertex right support is matchable to all three centers unless all its vertices belong to one pair type: every one- or two-vertex subset has at least its size in neighbors, and the full neighbor union has size three exactly in the remaining cases. Hall’s elementary condition therefore gives

$$a_3 = \binom{3n}{3} - 3\binom{n}{3} = 4n^3 - 3n^2.$$

The cubic discriminant is

$$-\frac{27n^3}{2}(n^3 - 3n^2 - 3n - 1).$$

At $n = 4$, the graph has matching number three and

$$p_4(t) = 1 + 24t + 162t^2 + 208t^3, \quad \text{disc}(p_4) = -2592.$$

It has a nonreal conjugate pair while its rank-normalized sequence 1, 8, 54, 208 is log-concave. The coefficient calculation and discriminant are verified exactly by the companion checker. Section 40.7 proves its vertex minimality within unit-weight matching rank three. Historical priority and vertex minimality across other ranks are not asserted. [*Merge note:* Neither *preorder-root-polytopes* nor Parts I–IV print a non-real-rooted support polynomial of matching number three. The graph’s three centres form its only minimum cover, on one shore, so it is ULC_3 by Lemma 3.2; it has no $2 + 2$ or $1 + 2$ cover, for which real-rootedness holds (Theorem 15.1, *preorder-root-polytopes*’s Theorem 207.1).]

For comparison, the source’s known rank-four theta graph of three internally disjoint length-three paths has polynomial $1 + 9t + 24t^2 + 16t^3 + t^4$ and discriminant -5243 [8, equation `lor:sp:eq:theta`]. Its normalized sequence is $1, 9/4, 4, 4, 1$. This prior example is credited to the source and its cited antecedent. Both examples show why real-rootedness is restricted to the mixed-cover theorem, rather than asserted for every graph of matching number at most four.

40.6.2 Why one weighted twin is not always enough

Pendant compression works because at most one member of its class can be selected. A common-neighbor class adjacent to two centers can contribute two selected vertices and requires its second elementary symmetric activity sum. Even replacing $n \geq 3$ unit-weight common twins by two independent nonnegative weights would require

$$w_1 + w_2 = n, \quad w_1 w_2 = \binom{n}{2}.$$

Their discriminant would be $n^2 - 4\binom{n}{2} = n(2 - n) < 0$. Such a replacement is impossible. The path and matrix constructions instead preserve the needed aggregate support quantities, including this second moment.

40.7 The smallest unit-weight counterexample at rank three

The example of Section 40.6.1 is minimal within its matching rank. The minimality part is computer-assisted: a structural reduction leaves a finite exhaustive integer calculation, implemented independently in two ways in the package.

Theorem 40.6 (Sharp vertex minimum at matching rank three). *Among finite simple bipartite graphs with matching number exactly three and unit vertex activities, the smallest order for which the matching-support polynomial is not real-rooted is fifteen. Up to graph isomorphism and interchange of shores, the unique fifteen-vertex example has three vertices on one shore and four vertices of each two-element neighborhood type on the other. Its polynomial is*

$$1 + 24t + 162t^2 + 208t^3,$$

with discriminant -2592 .

This is minimality within matching number three and unit activities. It is not a smallest non-real-rooted graph across all matching ranks: the credited rank-four theta graph above has eight vertices. [*Merge note:* Theorem 40.6 is computer-assisted. At the placement of batch 73 the enumeration was repeated independently (a separate Hall test and composition generator, not shipped), with the same counts, 30,699 rank-three vectors for $N \leq 11$ and 18,288 at $N = 12$, and the same unique negative discriminant. The theorem concerns real-rootedness, not ULC: the least matching number of a unit-weight ULC failure is not addressed (it lies in [6, 3450], Part II). Matching number three is the only case with content: a graph of matching number ν has at least 2ν vertices, and for $\nu \geq 4$ the theta graph with $\nu - 4$ disjoint edges added has exactly 2ν vertices and the non-real-rooted unit polynomial $(1 + 9t + 24t^2 + 16t^3 + t^4)(1 + t)^{\nu-4}$; at $\nu \leq 2$ every support polynomial is real-rooted (its degree is at most two and its discriminant is nonnegative by *preorder-root-polytopes*’s Theorem 180.2).]

40.7.1 A direct lemma for one by two covers

Lemma 40.7. *Every graph admitting a cover with one left vertex and two right vertices has a real-rooted support polynomial under arbitrary nonnegative vertex activities. The cover need not be minimum.*

[Merge note: This is *preorder-root-polytopes*'s Theorem 207.1 (Part XIII, 1c:thm:real), proved there by an interlacing argument; its conclusion also follows from *preorder-root-polytopes*'s Part VI. Source 77's rooted-path proof is printed as a second route, because the proof of Theorem 40.6 uses the lemma.]

Proof. Write the cover as $\{a\} \sqcup \{b, c\}$. After deleting the core edges, the components are a rooted star at a and a two-center component at b, c . If a has activity α and its exterior neighbor activities sum to R , the star has polynomial $p_1 = 1 + \alpha R t$ and root-deletion polynomial one. Replace this rooted star by one ordinary weighted edge of weight αR , rooted at one endpoint; it preserves both polynomials. This replacement is necessary before the joined object is called a path.

For the two-center component use (291), so $p_2 = 1 + At + Bt^2$ and its rooted deletion at b is $1 + Qt$. There are three possible core types. With no core edge the polynomial is $p_1 p_2$. With only ab , the support-imbalance argument gives

$$p_G = p_1 p_2 + \alpha u t(1 + Qt),$$

where u is the activity of b . Replace the second component by its rooted three-edge path (292), and join it to the replacement edge at the roots with weight αu . This is an ordinary weighted path with polynomial p_G .

With both core edges, the same support count as (294) gives

$$p_G = p_1 p_2 + \alpha(u + v)t(1 + Q_* t), \quad Q_* = \frac{uv(X + Y + Z)}{u + v}.$$

Certificate (295) makes $p_2, 1 + Q_* t$ a rooted-path pair. Join that path to the replacement edge with weight $\alpha(u + v)$. All three cases are real-rooted by the path spectral argument. When the denominators are positive the constructions apply directly; zero denominators and zero activities follow by continuity. $\square$

40.7.2 An exact reduction to seven populations

Delete isolated vertices from any candidate. König's theorem supplies a minimum cover of size three. Up to transposition it is split $0 + 3$ or $1 + 2$. The second case is real-rooted by Lemma 40.7. In the first, every nonisolated vertex on the cover's shore belongs to those three centers. Thus a counterexample has a shore of size exactly three, and is determined by the seven populations

$$(n_1, n_2, n_{12}, n_3, n_{13}, n_{23}, n_{123})$$

of its nonempty opposite-neighborhood types, up to permuting the centers. Write N for their sum. Fewer than fifteen vertices requires $N \leq 11$; at fifteen vertices, $N \leq 12$.

Here are the exact Hall formulas used by the independent enumeration. Index the types by nonempty subsets $S \subseteq \{1, 2, 3\}$, and set $p(t) = 1 + at + bt^2 + ct^3$. Then

$$a = \sum_S |S| n_S.$$

For each center pair i, j , let L_{ij} count opposite vertices meeting $\{i, j\}$, and I_{ij}, J_{ij} count those meeting that pair only at i or only at j . Its size-two contribution is

$$\binom{L_{ij}}{2} - \binom{I_{ij}}{2} - \binom{J_{ij}}{2};$$

summing over the three pairs gives b . For each center i , let $M_i = \sum_{S: i \notin S} n_S$. Inclusion-exclusion first counts triples whose neighborhood union is all three centers. Among those triples the only remaining Hall obstruction is two vertices of the same singleton type. Consequently

$$c = \binom{N}{3} - \sum_i \binom{M_i}{3} + \sum_i \binom{n_{\{i\}}}{3} - \sum_i \binom{n_{\{i\}}}{2} (n_{\{1,2,3\} \setminus \{i\}} + n_{\{1,2,3\}}). \quad (307)$$

The bad triples in the last sum are disjoint for different i . These formulas count supports, without a choice of realizing matching. Rank three is equivalent to $c > 0$.

For a genuine real cubic of this form, all roots are real, allowing repetitions, exactly when

$$\mathcal{D} = a^2b^2 - 4a^3c - 4b^3 - 27c^2 + 18abc \geq 0. \quad (308)$$

The independent checker enumerates every weak composition of each $N = 0, \dots, 12$ into seven populations by six separator positions among $N+6$ places, asserting that exactly $\binom{N+6}{6}$ vectors are visited at every N . It discards only $c = 0$ vectors. The second implementation instead precomputes pair and triple feasibility by enumerating bijections to selected centers, and then weights type multisets by products of integer binomial coefficients. It uses a separate recursive composition enumeration.

Opposite population	Rank-three vectors	Negative discriminants
$0 \leq N \leq 11$	30,699	0
$N = 12$	18,288	1
Total	48,987	1

Both exact computations find only the vector $(0, 0, 4, 0, 4, 4, 0)$, with coefficients $(1, 24, 162, 208)$ and discriminant -2592 . Every labeled population vector is checked, not merely an asserted list of orbit representatives. The unique negative vector is invariant under all center permutations, so it gives one graph up to isomorphism and shore exchange. This finite exhaustive calculation, combined with the structural reduction, proves Theorem 40.6. A fifteen-vertex example cannot contain an isolate, since deleting it would contradict the smaller-order exclusion.

40.8 A restricted rank-five family with four activity blocks

The next theorem settles one natural continuation of the known complete-core obstruction family. It imposes a complete $2 + 3$ core and activities constant within each of four blocks.

Theorem 40.8 (Strict rank-five normalization for a complete block family). *Let G have shores $A \sqcup X$ and $B \sqcup Y$, with*

$$|A| = 2, \quad |B| = 3, \quad |X| = n \geq 3, \quad |Y| = m \geq 2,$$

and exactly the edge blocks $A \times B$, $A \times Y$, and $X \times B$. Give each of A, B, X, Y its own constant positive vertex activity. Then $\nu(G) = 5$ and all four order-five Newton inequalities are strict:

$$k(5-k)a_k^2 > (k+1)(6-k)a_{k-1}a_{k+1} \quad (1 \leq k \leq 4).$$

[Merge note: The graph is source 75's full Hall graph $H_{2,3;n,m}$ with activities constant on each of its four blocks. Source 75's Theorem 16.1 (case $\min(a, b) = 2$) gives all four inequalities strictly for *every* positive activity assignment, and source 72's Corollary 14.2 gives strictness for every connected graph of matching number five; the nonstrict inequalities are source 74's Corollary 2.2. Theorem 40.8 is therefore a special case of Part I. Source 77's finite certificate proof is printed as a computer-assisted second route; the certificate is shipped as `data/77-rank-four-rank-five-certificate.json`.]

The exterior populations n, m are unbounded within the stated ranges. Unequal activities *within* a block, missing core edges, and general rank-five graphs are not covered.

40.8.1 Support formula and removal of exterior activities

The five core vertices form a cover, while disjoint matchings from A into Y and from B into X attain size five. Thus the matching number is five. Suppose the block activities are $U, V, \ell, r > 0$. Scaling all left activities by $1/\ell$ and all right activities by $1/r$ makes the exterior activities one and the core activities

$$u = U/\ell, \quad v = V/r.$$

Every coefficient a_k is multiplied by $(\ell r)^{-k}$, so its Newton margin is multiplied by the positive factor $(\ell r)^{-2k}$. Both the inequality and strictness are preserved.

In this normalized setting, a size- k support selecting i vertices of A and j vertices of B is feasible exactly when $i + j \geq k$. Necessity follows because its $k - i$ exterior left vertices must use selected B vertices, and its $k - j$ exterior right vertices must use selected A vertices. Under the same inequality, match the exterior vertices into the opposite core, then match the equally sized remaining core subsets using the complete core. Hence

$$a_k = \sum_{\substack{0 \leq i \leq 2, 0 \leq j \leq 3 \\ i+j \geq k}} \binom{2}{i} \binom{3}{j} \binom{n}{k-i} \binom{m}{k-j} u^i v^j, \quad 0 \leq k \leq 5, \quad (309)$$

with out-of-range binomial coefficients zero. This is a support count, not a matching count.

40.8.2 A finite positive-coefficient certificate for all parameters

Put $x = n - 3 \geq 0$ and $y = m - 2 \geq 0$. Every positive pair u, v belongs to one or both of the following branches:

	u	v	a	z
0	$(x+3)(a+z)$	$(y+2)a$	v/m	$u/n - v/m \geq 0$
1	$(x+3)a$	$(y+2)(a+z)$	u/n	$v/m - u/n \geq 0$

In both rows $a > 0$. Equality of the two activity ratios is included as $z = 0$ in either branch. For each branch substitute (309) into

$$D_k = k(5-k)a_k^2 - (k+1)(6-k)a_{k-1}a_{k+1}, \quad 1 \leq k \leq 4.$$

Using the polynomial convention $\binom{N}{q} = \prod_{h=0}^{q-1} (N-h)/q!$ for $q \geq 0$, expansion in x, y, a, z gives only positive nonzero rational coefficients. The complete coefficient lists are included as `rank-five-certificate.json`; the following table records their sizes and smallest coefficients.

Branch	Certificate statistic	D_1	D_2	D_3	D_4
0	Nonzero terms	81	625	1225	1215
0	Minimum coefficient	10	3/2	6	1/6
1	Nonzero terms	81	625	1715	1701
1	Minimum coefficient	20	15/2	1/6	1/6

The package provides an exact symbolic generator and an independent checker using sparse polynomials with rational coefficients. The latter reconstructs the coefficients by exterior selection counts p, q , with core counts $i = k - p, j = k - q$ and $p + q \leq k$, then compares all eight coefficient dictionaries to the certificate monomial by monomial. It uses neither the symbolic generator nor its polynomial implementation. Thus the supplied finite algebraic certificate establishes nonnegativity on the entire parameter domain, rather than on a numerical sample.

For strictness, at $x = y = z = 0$ both branches give

$$\begin{aligned}
D_1 &= 990a^2 + 5400a^3 + 4104a^4, \\
D_2 &= 192006a^4 + 469584a^5 + 548208a^6 + 326592a^7 + 69984a^8, \\
D_3 &= 858984a^6 + 2735424a^7 + 3100032a^8 + 1492992a^9 + 279936a^{10}, \\
D_4 &= 272160a^8 + 881280a^9 + 590976a^{10}.
\end{aligned}$$

These strictly positive monomials remain in the coefficientwise positive expansions for all $x, y, z \geq 0$ and $a > 0$. Therefore every $D_k > 0$ throughout the stated domain. Submatchings of a size-five matching give $a_0, \dots, a_5 > 0$, completing the proof of Theorem 40.8 with its exact finite certificate.

This family rules out a rank-five counterexample obtained by deleting one core vertex from the corresponding complete $3 + 3$ -core Hall construction, with unequal exterior populations and arbitrary positive activities constant on each of the four blocks. It leaves the unrestricted weighted rank-five question open. [*Merge note:* Part I rules out more: by the theorems cited above, no graph with a $2 + 3$ minimum cover fails at any positive activities, so weighted rank five holds without restriction (source 74) and the least weighted failure rank is exactly six.]

40.9 Verification and further questions

The accompanying exact checker enumerates all support subsets on 128 rationally weighted mixed-cover graphs, covering all sixteen core masks. It compares these labeled counts with a Boolean neighborhood-type kernel and with an independent formula obtained from the number of selected core edges. It checks all three rank-four Newton gaps with exact rational arithmetic. Additional symbolic checks verify the scalar margins, block determinant identity, the theta discriminant, and the pair-neighborhood cubic discriminant. The latter family is also counted directly for $1 \leq n \leq 6$. These checks corroborate the rank-four proof; their exact scopes and reproduction commands appear in the README and receipt. [*Merge note:* The delivered README is not shipped. The receipts are `data/77-rank-four-verification.json`, `data/77-rank-four-rank-three-minimum-verification.json`, `data/77-rank-four-rank-three-bijections-verification.json`, `data/77-rank-four-rank-five-verification.json` and `data/77-rank-four-rank-five-generation.json`; the commands are in this report's README.]

The polynomial identity methods are not a numerical real-root search. The phase-existence argument is proved by continuity, and the full-core determinant capacity is certified by (302). The separate rank-three minimality theorem uses a complete exact enumeration after its structural reduction. The restricted rank-five theorem uses a finite rational coefficient

certificate for an unbounded family, independently regenerated and checked. Neither result relies on a random screen. No Lean or other formal proof-assistant artifact is included.

The general theorem and restricted rank-five family leave several precise questions. Does arbitrary nonnegative two-shore weighting satisfy actual-degree normalization at matching number five? The inspected source has a rank-six weighted obstruction, so after the present theorem the least possible weighted counterexample rank is either five or six. For unit activities, is actual-rank normalization true at every matching number? The weighted counterexample does not settle that question. One may also ask which larger cover patterns admit aggregate Hermitian realizations, and whether the rank-four constructions can give useful equality or quantitative deficit classifications. [*Merge note:* Settled within this report: weighted rank five holds and the least weighted failure rank is exactly six (source 74, Corollary 2.2); unit-weight normalization fails at matching number 3450 (source 71, Theorem 17.1). Equality in the rank-four inequalities is classified by source 72’s Theorem 14.1. The shipped `77-rank-four-REVISION.md` likewise calls general weighted rank five and the unit-weight all-rank question unresolved; it is kept as delivered.]

The previously published smaller-shore inequality remains a separate, general theorem. The present proof uses that theorem in the thin-cover cases and additional support-specific structure in the mixed cover. It does not assert actual-rank ultra-log-concavity for arbitrary representable bimatroids; the source already supplies an obstruction to that broader statement at linear rank three. [*Merge note:* The obstruction at linear rank three is *preorder-root-polytopes*’s Theorem 192.1 (Part XII, `mr:thm:matrix`).]

41 Two-by-three cores with complete exteriors, and stable star cores (source 78)

Abstract of source 78, as delivered. We prove actual-degree ultra-log-concavity for every bipartite graph with an arbitrary two-by-three internal core and two complete exterior blocks, under independent nonnegative vertex activities. All four inequalities are strict at degree five. The proof combines ordinary support and moment reductions with 26 exact rational certificates covering all 64 labeled cores. Two independent solver-free checkers replay the certificates. Empty and star cores satisfy the stronger conclusion of real-rootedness, by an ordinary stable-monomer argument. The complete-exterior hypothesis is essential to the scope of the result. [*Merge note:* Source 78 is batch-73 manuscript 08, `all-two-by-three-cores-result.zip` (title page “Research manuscript prepared for Vladimir Reshetnikov”; its PDF author field reads “Research manuscript prepared with OpenAI”). It cites only source 77 and the batch-73 archive `complete-two-by-three-result.zip` (manuscript 09), which it extends: that archive’s graph is the core mask 63 here, which is source 75’s full Hall graph $H_{2,3;n,m}$, and its theorem is the case $s = 3$ of source 75’s Theorem 16.1; it is not printed. The displayed cover $P \sqcup Q$ has five vertices, so Theorem 41.1 is a case of source 74’s Corollary 2.2, and strictness at $d = 5$ is source 72’s Theorem 14.1, since such a graph is not a forest of stars. Its proof by 26 exact certificates is printed as a computer-assisted second route. Of Theorem 41.2 only the exceptional two-neighbour seed is new (see the note there).]

41.1 Statement and normalization

Let P, Q, X, Y be disjoint finite sets with $|P| = 2$, $|Q| = 3$. Let H be any subset of $P \times Q$. Our bipartite graph has shores $P \sqcup X$ and $Q \sqcup Y$ and edge set

$$H \cup (P \times Y) \cup (X \times Q). \quad (310)$$

Give each left vertex activity $u_i \geq 0$ and each right vertex activity $v_j \geq 0$. A feasible support (S, T) has S, T on the two shores, $|S| = |T|$, and a perfect matching between them. Define

$$\Gamma(t) = \sum_{k \geq 0} \gamma_k t^k, \quad \gamma_k = \sum_{\substack{(S,T) \text{ feasible} \\ |S|=|T|=k}} u^S v^T.$$

Each endpoint pair contributes once, regardless of its matching witnesses; $\gamma_0 = 1$. The actual degree d is the matching number after zero-activity vertices are deleted. Positive submatchings give no internal zeros. Rank-ultra-log-concavity means

$$k(d-k)\gamma_k^2 \geq (k+1)(d-k+1)\gamma_{k-1}\gamma_{k+1} \quad (1 \leq k < d). \quad (311)$$

Theorem 41.1. *For every graph (310), (311) holds at the actual surviving degree, with arbitrary independent nonnegative vertex activities. If $d = 5$, all four inequalities are strict.*

[Merge note: A case of Part I: Corollary 2.2 (source 74) for the inequalities, Theorem 14.1 (source 72) for strictness. Sections 41.2–41.4 below are source 78’s computer-assisted second route.]

The physical cover $P \sqcup Q$ gives $d \leq 5$. If $d \leq 4$, the previously proved arbitrary-weight bipartite rank-four theorem applies to the active graph [37, Theorem 1.1]. Its frozen proof package is included inside the dependency archive. The new argument treats degree five, where both P activities, all three Q activities, and at least three activities in X and two in Y are positive. No equality of activities within a block or bound on the exterior populations is assumed.

41.2 Boolean support reconstruction

A size- k support selecting core subsets $I \subseteq P$, $J \subseteq Q$ has exterior selection sizes $k - |I|$ and $k - |J|$. It is feasible exactly when

$$0 \leq |I| + |J| - k \leq \nu(H[I, J]). \quad (312)$$

Indeed a witnessing matching must use exactly $|I| + |J| - k$ internal edges. Conversely, any internal matching of that size extends over the selected exterior vertices because the two exterior blocks are complete. Thus

$$\gamma_k = \sum_{I \subseteq P, J \subseteq Q} \mathbf{1}_{(312)} u^I v^J e_{k-|I|}(X) e_{k-|J|}(Y), \quad (313)$$

with out-of-range elementary symmetric sums zero. This is a Boolean support identity, not a count of internal matching witnesses.

At degree five, divide all left activities by $e_1(X)$ and all right activities by $e_1(Y)$. Every coefficient is scaled by a positive k th power, preserving the Newton signs. Now $e_1(X) = e_1(Y) = 1$; write

$$M = e_2(X), \quad N = e_3(X), \quad S = e_2(Y).$$

The elementary identities

$$\begin{aligned} e_1(x)^2 - 2e_2(x) &= \sum_i x_i^2, \\ 2e_2(x)^2 - 3e_1(x)e_3(x) &= 2 \sum_{i < j} x_i^2 x_j^2 + \sum_{i < j < k} x_i x_j x_k (x_i + x_j + x_k) \end{aligned} \quad (314)$$

give the strict bounds

$$0 < 2M < 1, \quad 0 < 2S < 1, \quad 0 < \frac{3N}{2M^2} < 1. \quad (315)$$

Hence every finite degree-five instance is represented by positive V, W, Z :

$$M = \frac{V}{2(1+V)}, \quad S = \frac{W}{2(1+W)}, \quad N = \frac{V^2 Z}{6(1+V)^2(1+Z)}. \quad (316)$$

Explicitly, $V = 2M/(1 - 2M)$, $W = 2S/(1 - 2S)$, and $Z = z/(1 - z)$ for $z = 3N/(2M^2)$. Core activities remain arbitrary positive variables.

41.3 Exact middle gap certificates

Let $g_0, \dots, g_5$ denote the normalized coefficients reconstructed by (313). At degree five the middle inequalities are

$$g_2^2 \geq 2g_1g_3, \quad g_3^2 \geq 2g_2g_4.$$

The coefficient of N in g_3 is always $E = v_{Q_0}v_{Q_1}v_{Q_2}$, while g_1, g_2 do not depend on N . Thus the first gap is nonincreasing in N . Replace N by the upper bound $2M^2/3$, and then substitute M, S from (316). Multiplying the resulting boundary gap by $(1+V)^2(1+W)^2$ gives a rational-coefficient polynomial $F_{H,2}$. For the second gap use (316) exactly and multiply by $(1+V)^4(1+W)^2(1+Z)^2$, obtaining $F_{H,3}$. All denominator multipliers are positive. The variable order is

$$\mathbf{x} = (u_{P_0}, u_{P_1}, v_{Q_0}, v_{Q_1}, v_{Q_2}, V, W, Z); \quad (317)$$

$F_{H,2}$ is independent of Z .

Encode H by a six-bit mask with bit $3i + j$ representing P_iQ_j . Independent row and column permutations give 13 core orbits. Their least masks are

$$0, 1, 3, 7, 9, 10, 11, 14, 15, 27, 29, 31, 63.$$

Relabeling only permutes the independent core activities, so one pair of certificates per orbit suffices. For every representative and $k = 2, 3$ the supplied data give the exact identity

$$F_{H,k}(\mathbf{x}) = \sum_{\ell} \lambda_{\ell} \mathbf{x}^{m_{\ell}} (r_{\ell} \mathbf{x}^{a_{\ell}} - \mathbf{x}^{b_{\ell}})^2 + R_{H,k}(\mathbf{x}). \quad (318)$$

Here λ_{ℓ}, r_{ℓ} are positive rationals, all exponent vectors are nonnegative integral, and every coefficient of $R_{H,k}$ is nonnegative. Every remainder is nonzero. Since all variables in (317) are positive, (318) is strictly positive. This proves both strict middle inequalities; monotonicity transfers the first from the boundary back to the actual N .

Core mask	Orbit size	Squares for $k = 2$	Squares for $k = 3$	Positive remainder terms
0	1	24	46	708
1	6	35	46	1443
3	6	35	44	1981
7	2	38	44	2063
9	3	23	44	1846
10	6	23	44	2827
11	12	23	44	2929
14	6	23	44	3151
15	6	23	44	3151
27	3	23	44	3037
29	6	23	44	3394
31	6	23	44	3394
63	1	23	44	3394

The 26 certificates contain 915 square terms and reconstruct 33,318 positive remainder terms. Both checkers reconstruct the target polynomial, subtract every expanded square, and inspect all resulting coefficients, including any monomials absent from the target. The remainder is not trusted as prelisted data. One checker uses the core-rank criterion; the other reconstructs selected full subgraphs and applies Hall’s condition, with a distinct packed-exponent rational implementation. Both verify all 64 masks, all 13 orbits, and all target/remainder hashes. The independent implementation also verifies 384 coefficient relabelings and 52 rational substitution identities. A floating-point linear program found candidate squares, but every accepted coefficient and identity is checked exactly. No optimizer is needed for replay, and no numerical solver status is a proof input.

41.4 The outer gaps and degree drops

For the first gap, take the compatibility graph on positive-weight edges, joining two when their physical endpoints are disjoint. Its clique number is five. Assign product endpoint activities as vertex masses. Merging two nonadjacent masses toward the vertex with larger weighted neighbor sum preserves total mass and does not decrease the edge sum. After finitely many merges the support is a clique of at most five vertices, whose weighted edge sum is at most $2\gamma_1^2/5$ by Cauchy’s inequality. The initial sum counts matching witnesses. The positive complete block $P \times Y$ contains a $K_{2,2}$, giving a support with two positive witnesses. Consequently $\gamma_2 < 2\gamma_1^2/5$. [*Merge note:* This is the Motzkin–Straus argument of *preorder-root-polytopes*’s Theorem 180.2 (Part XI, `hb:thm:firstidentity`), printed in Part I as second routes of sources 74 and 72 (Sections 11 and 14).]

For the last gap, use unnormalized elementary sums

$$\begin{aligned} (A, B) &= (e_1(P), e_2(P)), & (C, D, E) &= (e_1(Q), e_2(Q), e_3(Q)), \\ (L, M, N) &= (e_1(X), e_2(X), e_3(X)), & (R, S) &= (e_1(Y), e_2(Y)). \end{aligned}$$

All denominators below are positive. For nonempty H , formula (313) gives

$$\gamma_4 = AENR + BDMS + BEMR, \quad \gamma_5 = BENS.$$

These are independent of the nonempty core mask, whereas γ_3 can only increase when the core is completed. For the complete core, set

$$x = \frac{AN}{BM}, \quad y = \frac{DS}{ER}, \quad b = \frac{B}{A^2} \leq \frac{1}{4}, \quad c = \frac{CE}{D^2} \leq \frac{1}{3}, \quad h = \frac{LN}{M^2} \leq \frac{2}{3}, \quad s = \frac{S}{R^2} \leq \frac{1}{2}.$$

The core bounds are squared-difference identities, and the exterior bounds follow from (314). The complete-core computation [39, Section 3] is

$$\begin{aligned} \frac{4\gamma_4^2 - 10\gamma_3\gamma_5}{B^2E^2M^2R^2} &= 4(x + y + 1)^2 \\ &\quad - 10(xy + xs + chy^2 + hy + hs + bsx^2) \\ &\geq (x - y)^2 + \frac{7}{4}x^2 + \frac{7}{9}y^2 + 3x + \frac{4}{3}y + \frac{2}{3} > 0. \end{aligned} \tag{319}$$

It therefore holds for every nonempty H . For $H = \emptyset$, the same normalized gap equals

$$4(x + y)^2 - 10(xy + chy^2 + bsx^2) \geq (x - y)^2 + \frac{7}{4}x^2 + \frac{7}{9}y^2 > 0.$$

Thus every degree-five gap is strict. For $d \leq 4$, use the separately approved rank-four theorem on the active graph; padded order-five inequalities are not substituted for its stronger normalization. This completes the proof of Theorem 41.1.

[*Merge note:* [39] is the batch-73 archive `complete-two-by-three-result.zip`, which is not printed. Its Section 3 is the complete-core computation (319); for the complete core the graph is source 75’s $H_{2,3;n,m}$, whose gaps are strict by Theorem 16.1.]

41.5 Stable monomers for empty and star cores

There is a stronger ordinary result for 18 of the 64 masks. For a graph on physical vertex set $V(G)$, define its signed support monomer polynomial by

$$\mu(\mathbf{z}) = \sum_{(S,T) \text{ feasible}} (-1)^{|S|} u^S v^T \prod_{a \notin S \cup T} z_a.$$

A real polynomial is real stable if it is nonzero whenever every variable lies in $\mathbb{H} = \{z : \text{Im } z > 0\}$.

Theorem 41.2. *If $\nu(H) \leq 1$, then μ is real stable for arbitrary independent nonnegative vertex activities. Hence Γ has only negative real zeros and is ULC at its actual surviving degree.*

[Merge note: By *preorder-root-polytopes*’s Corollary 94.3 (Part VI, `ade:mat:cor:block-family`) the empty core and every star core except one are already covered: a star at a vertex of P with one or three neighbours in Q , and every star at a vertex of Q , makes the graph a vertex sum of complete bipartite graphs and edges. The remaining case, a star at a vertex of P with exactly two neighbours in Q , needs the exceptional two-neighbour seed below; that seed is new.]

Seed transfer. For a fixed head side Q , adjoin a private dummy d_j for each head slot. The transversal matroid on tails and dummies has a basis for each endpoint support, with dummies encoding unused heads. If its basis polynomial $B(\mathbf{x}, \mathbf{d})$ is stable, then for positive activities

$$\mu(\mathbf{a}, \mathbf{b}) = \left(\prod_{\text{tails } i} a_i \right) \left(\prod_{\text{heads } j} v_j \right) B(x_i = -u_i/a_i, d_j = b_j/v_j)$$

is stable, since all transformed arguments stay in $\mathbb{H}$. Limits give zero activities, and the all-unused monomial excludes the zero polynomial. A complete bipartite graph has a uniform dummy matroid, with stable elementary symmetric basis polynomial. An exceptional tail with a singleton neighborhood is parallel to the corresponding private dummy and is handled by a nonnegative linear substitution in that elementary polynomial.

[Merge note: The seed transfer is also the seed construction of the batch-73 archive `bipartite-and-pendant-support-result.zip`, which concerns directed endpoint supports and is placed with the batch’s directed-support manuscripts, not in this report; the shipped `78-two-three-star-proof-notes-STAR_CORE_REAL_ROOTEDNESS.md` calls it “the seed part of the previously approved pendant-head theorem”.]

The exceptional two-neighbor seed. Suppose there are three head slots, any number of universal tails X , and one exceptional tail p adjacent to heads 1 and 2. The rank-three dummy matroid has every three-subset as a basis except $\{p, d_1, d_2\}$. Put $L_0 = \{p, d_1, d_2\}$ and $O = X \sqcup \{d_3\}$, with $|O| = m \geq 1$. Its basis polynomial is $e_3(L_0 \sqcup O) - p d_1 d_2$, symmetric separately in the two groups. Its two-block diagonal is

$$y \left(3mx^2 + 3 \binom{m}{2} xy + \binom{m}{3} y^2 \right).$$

The quadratic has nonnegative coefficients and discriminant

$$9 \binom{m}{2}^2 - 12m \binom{m}{3} = \frac{m^2(m-1)(m+7)}{4} \geq 0.$$

It factors into linear forms with nonnegative coefficients, including the cases $m = 1, 2$. The diagonal is stable, and polarization with degree bounds $(3, m)$ recovers the basis polynomial. Stability follows from Borcea–Brändén [40, Proposition 2.4]. Empty, singleton and universal exceptional neighborhoods use the elementary seeds above.

Gluing and the star cases. For two relations meeting only at one vertex, use separate unused-vertex monomers a, b . Their stable product is multiaffine in these variables, say $f = Aab + Ba + Cb + D$. The exact physical gluing map is

$$ab \mapsto z, \quad a \mapsto 1, \quad b \mapsto 1, \quad 1 \mapsto 0, \quad Az + B + C = \partial_t f(t, t)|_{t=z/2}.$$

It preserves stability by diagonal substitution, differentiation and positive scaling. It also counts supports exactly: if the shared vertex is used, the imbalance of the other selected tail/head endpoints in each piece uniquely determines which piece contains its matched edge. It cannot be used in both pieces. [*Merge note:* The gluing identity is *preorder-root-polytopes*'s Lemma 93.1 (Part VI, `ade:mat:lem:vertex-sum`), with $\Phi_G = z_v A_1 A_2 + B_1 A_2 + A_1 B_2$, and its stability is *preorder-root-polytopes*'s Theorem 93.2 (Lemma 15.3 here).]

A nonempty bipartite graph of matching number one is a star. If its center is $p \in P$, glue the complete piece on $P \sqcup Y$ to the piece on $\{p\} \sqcup X \sqcup Q$. The latter has universal tails X and the one exceptional tail just analyzed. If its center is $q \in Q$, glue the complete piece on $X \sqcup Q$ to the piece on $P \sqcup Y \sqcup \{q\}$. After interchanging shores, the latter has two head slots and only singleton or universal neighborhoods. The empty core is a disjoint product. This proves monomer stability in every case. Finally, diagonal specialization gives $\mu(x, \dots, x) = x^{|V(G)|} \Gamma(-x^{-2})$, so all zeros of Γ are negative real. Newton's inequalities use its actual surviving degree.

41.6 Verification and boundaries

The all-mask ULC theorem is computer-assisted through its 26 finite rational identities after an ordinary unbounded-population reduction. The star-core stability theorem is an ordinary proof. Independent additional checks verify all 64 Boolean core kernels, 2,304 weighted Hall cases, 216 formal gluing identities, and 714 exceptional-seed triples. These supplement the proofs and are not finite-population assumptions.

The full core need not be real-rooted: with ten vertices in each exterior block, core activities 100 and exterior activities 1, the scaled polynomial

$$\Gamma(t/100) = 1 + 650t + 39780t^2 + 143170t^3 + 53475t^4 + 5400t^5$$

has negative discriminant and one nonreal conjugate pair [39, Section 5]. Thus Theorem 41.2 is not silently extended to the matching-two cores. [*Merge note:* [39] again. Its example is the sibling, at $n = m = 10$, of source 75's rank-five example at $n = m = 9$ (Section 16); the same polynomial appears in the batch-73 archive `weighted-bipartite-rank-five-result.zip`, which is not printed.]

Missing exterior edges, arbitrary orientations and independent edge activities remain outside the conclusions. The general two-tail/three-head role-cover question is still open. No proof-assistant formalization, literature-wide priority claim or external publication is asserted. [*Merge note:* The role-cover question concerns directed endpoint supports (*preorder-root-polytopes*'s Part XII); it stays open here (Section 44). Source 78's delivered README, which is not shipped, says that general rank-five graphs remain open; source 74 settles weighted rank five (Corollary 2.2).]

42 A two-element Lorentzian deformation of every matroid basis polynomial (source 79)

Abstract of source 79, as delivered. We prove a homogeneous Lorentzian deformation associated with two distinguished elements of an arbitrary finite matroid. All original element

variables remain separate; three finite populations of nonnegative weights enter as parameters. Principal extensions supply a quadratic-line seed, while matroid union proves the required support exchange property. Contraction reduces the remaining Hessian verification to ranks zero through four. No representability or spanning hypothesis is needed. A transversal specialization retains every selected-tail variable in bipartite endpoint-support polynomials with a two-tail vertex cover. Its diagonal consequence gives actual-degree ultra-log-concavity for all independently vertex-weighted bipartite graphs of matching rank at most five. The proof is ordinary; exact finite checks are supplied only as corroboration. [*Merge note:* Source 79 is batch-73 manuscript 02, `two-element-matroid-lorentzian-result.zip` (title page “Research note”). It cites no repository source. Its graph statements are Part I’s: Corollary 42.3 is source 74’s left marginal (Theorem 2.1) with the private variables of the right cover part kept as variables of their own, and Corollary 42.4 is source 74’s Corollary 2.2. New as a statement is Theorem 42.1 for every finite matroid. Source 80’s Appendix A proves it in one page by matroid union, with the six basis cases of source 74’s lift (Section 5); that proof is printed first, in Section 42.2, and source 79’s own proof (a principal-line seed and contraction to ranks at most four) follows as a second route. Its predecessor, the batch-73 archive `weighted-bipartite-rank-five-result.zip` (manuscript 03), is not printed: it proves Theorem 2.1 and Corollary 2.2 by a third route, an explicit positive semidefinite decomposition, from the same principal-line seed, which is the limit of source 74’s generic-line model (Section 8) as its number of line elements grows.]

42.1 Statement and conventions

Let M be a finite rank- q matroid on $E \sqcup \{A, B\}$, with $A \neq B$. Loops and parallel distinguished elements are allowed. For $I \subseteq E$, write $w^I = \prod_{e \in I} w_e$. Define five Boolean basis-family polynomials:

$$\begin{aligned} U_0 &= \sum_{I \subseteq E: I \text{ a basis of } M} w^I, & U_A &= \sum_{I \subseteq E: I \cup \{A\} \text{ a basis}} w^I, \\ U_B &= \sum_{I \subseteq E: I \cup \{B\} \text{ a basis}} w^I, & V &= \sum_{I \subseteq E: I \cup \{A, B\} \text{ a basis}} w^I. \end{aligned}$$

The polynomial U counts the union of the U_A and U_B families, each old subset once. Thus U is generally not $U_A + U_B$. The degrees of (U_0, U_A, U_B, U, V) are $(q, q-1, q-1, q-1, q-2)$; negative-cardinality families are zero. In particular, U_0 may vanish when E does not span M : it always counts full rank- q bases, not bases of a lower-rank restriction.

Take three finite populations of nonnegative real weights, called private- A , private- B , and common. Let L_0, L_1, H be their sums, and let S be the second elementary symmetric sum of the common population. Put

$$\begin{aligned} R_0 &= L_0 + H, & R_1 &= L_1 + H, \\ \mu &= L_0 L_1 + H(L_0 + L_1) + S, & \bar{\mu} &= L_0 L_1 + H(L_0 + L_1) + H^2/2, \\ W &= L_1 U_A + L_0 U_B + H U. \end{aligned} \tag{320}$$

All population weights are parameters. The variables are $\mathbf{w}, a, b, z$.

Theorem 42.1. *For every such finite matroid and populations, the degree- $(q+2)$ polynomial*

$$F_M = z^2 U_0 + z(a R_0 + b R_1) U_0 + ab \mu U_0 + z^2(a U_A + b U_B) + ab z W + ab z^2 V \tag{321}$$

is Lorentzian.

The polynomial is nonzero: its z^2 coefficient is the full basis polynomial $U_0 + aU_A + bU_B + abV$. The theorem makes no joint Lorentzian assertion in the population parameters.

We use the following precise form of Lorentzian theory [1]. A nonnegative homogeneous polynomial of degree $d \geq 2$ is Lorentzian exactly when its exponent support is M -convex and every order- $(d - 2)$ partial derivative has a Hessian with at most one positive eigenvalue. Here M -convexity means symmetric unit exchange of exponent vectors. Nonnegative linear substitutions, differentiation, products, and coefficient limits preserve Lorentzianity (allowing zero). Matroid basis polynomials are Lorentzian. These are Theorems 2.10, 2.25 and 3.10 and Corollary 2.32 of [1]; nonnegative directional differentiation follows by substituting $s_i \mapsto s_i + r_i y$, differentiating in y , and setting $y = 0$.

42.2 A direct all-matroid extension of the intake lift (source 80, Appendix A)

[Merge note: Printed here, from source 80's Appendix A, as the first proof of Theorem 42.1. The intake is source 74 at commit 1512ef835, and its `full_lift.tex` is Section 5; the bundled earlier note [38] is source 79. Source 80 restates the theorem in its own letters: $h_1, \dots, h_m$ are the common weights, and W of (320) is written out in (322).]

This short proof adapts the rank-compression construction and six basis cases of the intake's `full_lift.tex` [12]. Matroid union removes its transversal-seed requirement. It supplies a shorter proof of the unrestricted two-element theorem in the bundled earlier note [38]; that release remains unchanged.

Let M be any rank- q matroid on $E \sqcup \{A, B\}$. In old variables $\mathbf{w}$, let U_0 count old full bases, U_A, U_B old subsets completed to a basis by A, B , respectively, U the Boolean union of those two families, and V the old $(q - 2)$ -subsets I for which $I \cup \{A, B\}$ is a basis. Negative-cardinality families vanish. No old-spanning hypothesis is imposed; U_0 can be zero.

For nonnegative masses L_0, L_1 and common weights $h_1, \dots, h_m$, put

$$H = \sum_j h_j, \quad S = \sum_{j < k} h_j h_k, \quad R_i = L_i + H, \quad \mu = L_0 L_1 + H(L_0 + L_1) + S.$$

The degree- $(q + 2)$ polynomial

$$\begin{aligned} F = & z^2 U_0 + z(aR_0 + bR_1)U_0 + ab\mu U_0 + z^2(aU_A + bU_B) \\ & + abz(L_1 U_A + L_0 U_B + HU) + abz^2 V \end{aligned} \quad (322)$$

is Lorentzian for every finite M , including loops and dependent distinguished elements.

To prove it, adjoin plane elements $P_1, P_2, Z_1, \dots, Z_m$. Let M_1 extend M by loops on the plane. On the same ground let M_2 be the rank-two transversal matroid with two slots: A, P_1 use only slot 1; B, P_2 use only slot 2; each Z_j uses either; old E elements are loops. Their union $K = M_1 \vee M_2$ is a matroid [41, Section 2]. Its rank is $q + 2$: any M basis together with P_1, P_2 witnesses the upper bound given by the sum of the two ranks.

A K basis partitions into full bases of the two factors. Old elements must enter M_1 and plane elements M_2 . If k distinguished and ℓ plane elements are selected, exactly $k + \ell - 2$ distinguished elements enter M_1 . A basis is counted once, not once per partition.

Assume $\mu > 0$. Keep $\mathbf{w}, a, b$, substitute

$$P_1 = L_1 z / \mu, \quad P_2 = L_0 z / \mu, \quad Z_j = h_j z / \mu,$$

and multiply the basis polynomial by μ . Every distinct plane pair fills the two auxiliary slots, and its total specialized weight is

$$(z^2 / \mu^2) [L_0 L_1 + (L_0 + L_1)H + S] = z^2 / \mu.$$

There are exactly six possible (k, ℓ) classes, with contributions

(k, ℓ)	Contribution after multiplication by μ
$(0, 2)$	$z^2 U_0$
$(1, 1)$	$z(aR_0 + bR_1)U_0$
$(1, 2)$	$z^2(aU_A + bU_B)$
$(2, 0)$	$ab\mu U_0$
$(2, 1)$	$abz(L_1 U_A + L_0 U_B + HU)$
$(2, 2)$	$abz^2 V$

For the sole ambiguous class $(2, 1)$, P_1 forces B into the second factor and A into the first, giving $L_1 U_A$; P_2 gives $L_0 U_B$. A common plane element permits either allocation, so its old subset belongs to the Boolean union U and is still one basis. This explains the common coefficient without witness multiplicity. The other classes follow immediately from the forced factor cardinalities and plane-pair sum.

The six terms sum to (322). Matroid-basis Lorentzianity and nonnegative linear substitution prove the claim. When $\mu = 0$, replace both private masses by $L_i + \varepsilon$, apply the positive case, and take a coefficient limit as $\varepsilon \downarrow 0$. The result is nonzero because the coefficient of z^2 is the full basis polynomial $U_0 + aU_A + bU_B + abV$. Thus no representation, old-spanning assumption, principal-extension limit, or low-rank Hessian induction is needed in this attributed union proof. [*Merge note:* Second proof of Theorem 42.1: source 79's own, Sections 42.3–42.5.]

42.3 Two matroid constructions

We first prove Theorem 42.1 when E spans M ; Section 42.5 removes this temporary hypothesis. No representation over a field is used.

42.3.1 A principal-line seed

Write

$$\phi(s) = s_A s_B + s_A s_U + s_B s_U + s_U^2/2, \quad J = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \phi(s) = \frac{1}{2} s^T J s.$$

Lemma 42.2. *The homogeneous degree- q polynomial*

$$\mathcal{S}_M(\mathbf{w}, s) = U_0 + s_A U_A + s_B U_B + s_U U + \phi(s) V \quad (323)$$

is Lorentzian.

Proof. Add m elements freely on the flat $\text{cl}_M\{A, B\}$. For original X and a set Y of new elements, the extension rank is

$$r_+(X \cup Y) = \min\{r_M(X \cup \{A, B\}), r_M(X) + |Y|\}. \quad (324)$$

This is the free-on-flat extension formula of Bonin–Kung [41, Lemma 3.1]. For an old $(q-1)$ -set, a single new element completes a basis exactly when A or B does. For an old $(q-2)$ -set, two new elements, or either distinguished element together with one new element, complete a basis exactly when A, B do. Three line elements are dependent, since their flat has rank at most two. The extension's basis polynomial is therefore

$$U_0 + s_A U_A + s_B U_B + \left(\sum_i h_i\right)U + \left[s_A s_B + (s_A + s_B) \sum_i h_i + \sum_{i < j} h_i h_j\right]V.$$

Set $h_i = s_U/m$. Matroid-basis Lorentzianity and nonnegative substitution apply for every m ; the coefficient of $s_U^2 V$ tends to $1/2$. Closedness proves (323), including loop and parallel cases. $\square$

42.3.2 Matroid union supplies the support

Let Y index all population members, $n = |Y|$, and introduce dummy elements $D = \{d_y : y \in Y\}$. On the common ground set $E \sqcup \{A, B\} \sqcup D$, let M_1 be M extended by loops on D . Let M_2 be the rank- n transversal matroid with slots Y : each d_y is adjacent only to y ; A, B have the private/common incidences; all elements of E are loops. Use their matroid union $K = M_1 \vee M_2$, whose independent sets are unions of independent sets in the two factors [41, Section 2].

An old basis together with all dummies shows $r(K) = q + n$. Every basis of K partitions into a full M_1 basis and a full M_2 basis: make a union decomposition disjoint, then its cardinality forces both factor ranks. Old elements belong to the first factor and dummies to the second. If r dummies are omitted, exactly r distinguished elements must be assigned to the second factor. Consequently $r \leq 2$, with the following complete cases:

- $r = 0$: the original basis polynomial $U_0 + aU_A + bU_B + abV$;
- $r = 1$, one distinguished element selected: its permitted incidence times U_0 ;
- $r = 1$, both selected: U_B for a private- A head, U_A for a private- B head, and U for a common head;
- $r = 2$: U_0 , exactly when the two omitted heads match to A, B .

A basis is counted once regardless of its possible partitions. This explains the Boolean union in the common-head case.

For positive population weights ω_y , substitute $d_y = z/\omega_y$ in the basis polynomial and multiply by $\prod_y \omega_y$. The result is

$$G = z^{n-2} F_M. \quad (325)$$

It is Lorentzian. The same assertion for G at zero weights follows by the explicit coefficient limit. Its no-omission term $z^n(U_0 + aU_A + bU_B + abV)$ is nonzero. The exponent supports of G and F_M differ by a fixed translation, which preserves symmetric exchange, even when $n < 2$ and (325) is read as a Laurent identity. Hence $\text{supp } F_M$ is M -convex. We use only support translation here, not a claim that arbitrary monomial division preserves Lorentzianity.

42.4 Contraction and the five small ranks

For an old nonloop e , restriction of each basis family to subsets containing e , followed by deletion of e , gives exactly its counterpart in M/e . This commutes with the Boolean union defining U . Therefore

$$\partial_{w_e} F_M = F_{M/e}. \quad (326)$$

For an old loop the derivative is zero. The remaining old elements span the rank- $(q-1)$ contraction.

Induct on q , using the already established M -convex support. Every order- q derivative involving an old variable follows from (326) and induction. The only remaining derivatives involve a, b, z , of respective maximum degrees 1, 1, 2. They vanish for $q \geq 5$. We check ranks zero through four explicitly. In boundary arguments, add two common heads of weight $\varepsilon > 0$ and let $\varepsilon \downarrow 0$; this makes μ, R_0, R_1 positive whenever denominators require them.

42.4.1 Rank zero

Here $F_M = z^2 + (aR_0 + bR_1)z + ab\mu$. Its Hessian in (a, b, z) is

$$\begin{pmatrix} 0 & \mu & R_0 \\ \mu & 0 & R_1 \\ R_0 & R_1 & 2 \end{pmatrix}, \quad \det = 2\mu(R_0R_1 - \mu), \quad R_0R_1 - \mu = H^2 - S \geq 0.$$

When $\mu > 0$ the leading 2×2 block has one positive and one negative eigenvalue. The nonnegative determinant then excludes a second positive eigenvalue. Limits give the boundary.

42.4.2 Rank one

Write $u = \sum_{e \in E \text{ nonloop}} w_e$, $\alpha = U_A \in \{0, 1\}$, $\beta = U_B \in \{0, 1\}$, and $W_0 = L_1\alpha + L_0\beta + H \max(\alpha, \beta)$. Then $U_0 = u$, $U = \max(\alpha, \beta)$, and $V = 0$. First work in the single variable u : this is the same construction for a rank-one matroid with one old nonloop, so its support is M -convex by Section 42.3. Afterwards substitute the nonnegative sum for u .

The u derivative is the rank-zero quadratic. The a derivative has Hessian in (u, z, b)

$$\begin{pmatrix} 0 & R_0 & \mu \\ R_0 & 2\alpha & W_0 \\ \mu & W_0 & 0 \end{pmatrix}, \quad \det = 2\mu(R_0W_0 - \alpha\mu),$$

where the Boolean identity

$$R_0W_0 - \alpha\mu = \alpha(H^2 - S) + (1 - \alpha)\beta R_0^2 + \alpha\beta R_0L_0 \geq 0$$

proves the signature as above. The b case is symmetric. The z derivative has leading (u, z) Hessian block $\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$ and Schur complement

$$\begin{pmatrix} -2R_0\alpha & -H\alpha\beta \\ -H\alpha\beta & -2R_1\beta \end{pmatrix} \preceq 0,$$

since $4R_0R_1 \geq H^2$. Sylvester inertia and limits finish rank one.

42.4.3 Rank two

The seed $\mathcal{S}_M$ is now quadratic and $V \in \{0, 1\}$. Put $m = (L_1, L_0, H)^T$, so $\phi(m) = \bar{\mu}$, $e_A^T Jm = R_0$ and $e_B^T Jm = R_1$. Direct expansion gives all nonzero remaining second derivatives:

$$\begin{aligned} \partial_z^2 F_M &= 2\mathcal{S}_M(s = (a, b, 0)), \\ \partial_a \partial_b F_M &= \mu \mathcal{S}_M(s = mz/\mu) - \frac{\bar{\mu} - \mu}{\mu} z^2 V, \\ \partial_a \partial_z F_M &= R_0 \mathcal{S}_M(s = (2e_A z + mb)/R_0) - \frac{\bar{\mu}}{R_0} b^2 V, \\ \partial_b \partial_z F_M &= R_1 \mathcal{S}_M(s = (2e_B z + ma)/R_1) - \frac{\bar{\mu}}{R_1} a^2 V. \end{aligned} \tag{327}$$

Each seed substitution is nonnegative linear. Also $\bar{\mu} - \mu = H^2/2 - S \geq 0$. Thus each subtraction removes a positive semidefinite diagonal matrix from a quadratic Hessian and cannot increase its positive index. The actual derivatives have nonnegative coefficients by (321). Perturbation removes zero denominators; repeated a or b derivatives vanish.

42.4.4 Ranks three and four

At rank three the remaining derivatives are

$$\partial_a \partial_z^2 F_M = 2(U_A + bV), \quad \partial_b \partial_z^2 F_M = 2(U_B + aV), \quad \partial_a \partial_b \partial_z F_M = W + 2zV.$$

The first is $2\partial_{s_A} \mathcal{S}_M$ at $s_A = s_U = 0, s_B = b$; the second is symmetric. For the third, the nonnegative directional derivative $D_m \mathcal{S}_M$ equals $W + (m^T J s)V$. If $m \neq 0$, choose $r \geq 0$ with

$m^T J r = 2$ and substitute $s = rz$. Such r exists because Jm has a positive coordinate. If $m = 0$, use the product $2zV$, with V a rank-one contraction basis polynomial or zero.

At rank four only $\partial_a \partial_b \partial_z^2 F_M = 2V$ remains. When A, B are independent, V is the rank-two basis polynomial of $M/\{A, B\}$; otherwise it is zero. All these quadratics have the required signature. Higher ranks have no further special-variable derivative. This completes the induction with spanning old ground.

42.5 Removing the spanning hypothesis

For each distinguished nonloop, add an old parallel copy, A' or B' ; add nothing for a distinguished loop. Parallel extension is the rank-one instance of (324). The extended matroid $\widetilde{M}$ has the same rank, and its enlarged old ground spans: every distinguished nonloop lies in the closure of its old copy, and loops lie in the closure of the empty set.

Apply the proved theorem to $\widetilde{M}$ and set the added copy variables to zero. Restriction to the original ground recovers all five full-basis families, including their Boolean union. The resulting polynomial is exactly (321); nonnegative specialization preserves Lorentzianity. It is nonzero because its z^2 coefficient is the basis polynomial of M . This proves Theorem 42.1 without any old-spanning or representability assumption.

42.6 Bipartite endpoint supports

Let a finite bipartite graph have disjoint shores $P \sqcup X$ and $Q \sqcup Y$, where $P = \{p_0, p_1\}$, $|Q| = q$, and there is no X – Y edge. A feasible endpoint support is a pair (S, T) of equally sized shore subsets such that the induced bipartite graph has a perfect matching. Each pair is counted once. Give right vertices independent nonnegative activities v_j ; keep all left selected variables x_i separate.

Corollary 42.3. *The following homogeneous degree- $(q+2)$ polynomial is Lorentzian:*

$$\mathcal{F}_G(\mathbf{x}, z, \mathbf{z}_Q) = \sum_{(S,T) \text{ feasible}} \left(\prod_{j \in T} v_j \right) \left(\prod_{i \in S} x_i \right) z^{2-|T \cap Y|} \prod_{j \in Q \setminus T} z_j. \quad (328)$$

Independent nonnegative left activities may be inserted by scaling each x_i .

[Merge note: The statement is source 74's left marginal (Theorem 2.1) with the private elements of the right cover part kept as variables z_j instead of s/v_b : the basis polynomial of the transversal matroid of Section 5 gives it by the same nonnegative substitution. The proof below derives it from Theorem 42.1 instead.]

Proof. Use the transversal matroid with slots Q , old elements X and one private dummy for each slot, and distinguished elements representing p_0, p_1 . Its rank is q , and the old dummies span. Discard isolated Y vertices; classify the others by their neighborhood in P , obtaining exactly the three populations. For positive v_j on Q , set each dummy variable to z_j/v_j , set old X variables to x_i , set $a = x_{p_0}, b = x_{p_1}$, and multiply by $\prod_{j \in Q} v_j$. The support decomposition of (321) gives (328). A selected head set with one common Y vertex uses the Boolean extension union, so multiple matching witnesses cause no overcount. Zero Q activities follow by limits in the explicit endpoint formula. The empty support has coefficient one, excluding zero output. $\square$

For clarity about normalization, scale $x_i = u_i t$, set every $z_j = z$, and write the resulting polynomial as

$$z^{q+2} \Gamma(t/z) = \sum_k \gamma_k t^k z^{q+2-k}.$$

The bivariate characterization [1, Example 2.26] gives log-concavity of $\gamma_k/\binom{q+2}{k}$. This is cover-order normalization; arbitrary degree drops do not by themselves permit replacement of $q + 2$ by the surviving degree.

Corollary 42.4. *For every independently nonnegative vertex-weighted finite bipartite graph whose surviving matching rank is $d \leq 5$, its endpoint-support sequence is ultra-log-concave normalized by $\binom{d}{k}$.*

[Merge note: This is source 74’s Corollary 2.2 (Part I), with the same reduction by König’s theorem and leaf compression; the proof below is a second route through Corollary 42.3.]

Proof. Delete zero-activity vertices and ignore isolates. By König’s matching-cover theorem [42], choose a vertex cover of size d , and reverse the shores so that at most two cover vertices are left vertices. For two, use Corollary 42.3 with $q = d - 2$. For zero, the ordinary right-dummy transversal basis construction gives a degree- d Lorentzian homogenization. For one left cover vertex p and $q = d - 1$ right cover vertices, every nonisolated outside-right vertex has the same sole neighbor p . At most one can be selected, so replace them by one head of activity equal to their total. The rank- $(q + 1)$ transversal basis polynomial with private head dummies gives the same homogenized support polynomial. An artificial zero-activity head, interpreted by the same coefficient limit, handles an empty population. Thus in every case the homogenization order is exactly d . The empty graph is immediate; the bivariate characterization proves the claim. $\square$

The stronger multivariate conclusion keeps the left selected variables, but still treats Y activities as parameters. No identification of physical tail and head roles in a general directed relation follows. The balanced three-plus-three cover case at actual rank six is also outside the present theorem.

42.7 Proof status and reproducibility

The proof uses only the cited matroid and Lorentzian facts and the displayed identities. No exhaustive or randomized finite search is a premise. The accompanying package contains four exact corroborating checkers: the producer checks 60 represented matroids, 3,505 rational Hessian inertias and 94,442 support exchanges; a separate represented-matroid checker verifies 250 literal augmentation identities and 1,240 contractions; an abstract rank-oracle checker verifies 84 principal-line identities, 840 matroid-union identities over 82,331 candidate bases, and 365 contractions, including nonrepresentable examples; and a padding checker verifies 86 distinguished pairs, with 16 initially nonspanning old sets and six distinguished-loop occurrences. Frozen proof-stage approvals, full replay instructions, and all-page PDF inspection records are included. These computations corroborate the ordinary argument and are not proof-assistant formalization or a literature-wide priority claim. [Merge note: The proof-stage approvals and visual-inspection records are shipped as `79-two-element-audits-*.md`, `data/79-two-element-audits-*.json` and `data/79-two-element-qa-visual-qa.json`; the PDF they inspect is not shipped.]

43 Three-tail obstructions and an all-rank first layer (source 80)

Abstract of source 80, as delivered. We independently reconstruct the weighted rank-five threshold and higher-rank counterexamples already recorded in a ProveIt intake manuscript. A complete three-block graph gives an explicit last-Newton-gap violation at actual degree six, and the same argument supplies every higher degree, already for vertex-weighted height-two orders.

Additional sections give a smaller graph obstruction to two higher-dimensional Lorentzian extensions whose own scalar sequences remain ultra-log-concave. Positive refinements include an all-rank first-layer Hessian inequality and full selected-tail cubic Lorentzianity for a three-vertex left shore. Exact solver-free checks accompany the examples, and the unchanged ordinary rank-five proof is bundled as a dependency. [*Merge note:* Source 80 is batch-73 manuscript 01, `sharp-weighted-rank-boundary-result.zip` (title page “Research note”), pinned at commit 1512ef835, the arrival of batch 72. “The inspected intake” [12] is source 74’s raw archive; [20] is source 10 (Section 23) and [36] source 05 (Section 26); the comparison is shipped as `80-three-tail-INTAKE_COMPARISON.md`. Source 80 never cites *preorder-root-polytopes*: its Sections 43.1–43.2 re-prove source 74’s rank threshold and Part XI’s rank-six family and are printed as statements with pointers. New are Proposition 43.2, the thirteen-vertex graph, the obstruction of Section 43.4 and Theorem 43.3; Corollary 43.4 is source 10’s theorem with a new proof.]

Relation to the inspected intake. The sharp weighted threshold, the complete-block quota construction and counterexamples in every rank at least six are explicitly present in the user-repository research manuscript [12], at the pinned commit identified in the bibliography. Sections 43.1–43.2 are an independent reconstruction and additional exact checking of those results, not a claim that they are new to that intake. Its `full_lift.tex` also proves the two-tail graph theorem by a rank-compression lift. The three-vertex cubic statement is also in [20]; here an ordinary all-rank matrix argument replaces its finite determinant certificate. The source comparison is included; no external publication or global priority is claimed.

43.1 The sharp weighted rank threshold

A bipartite endpoint support is a pair (S, T) of equally sized shore subsets admitting a perfect matching. Count each feasible pair once, regardless of its matching witnesses. With nonnegative vertex activities, let γ_k be the total product weight of size- k supports and put $\Gamma(t) = \sum_k \gamma_k t^k$. Its surviving degree d is the matching number of the positive-activity graph. Actual-degree ultra-log-concavity (ULC) means that $\gamma_k / \binom{d}{k}$ is log-concave; equivalently, with no internal zeros,

$$k(d-k)\gamma_k^2 \geq (k+1)(d-k+1)\gamma_{k-1}\gamma_{k+1} \quad (1 \leq k < d). \quad (329)$$

Theorem 43.1 (Weighted threshold from the inspected intake). *Every independently nonnegative vertex-weighted finite bipartite endpoint-support polynomial of surviving matching rank at most five is actual-degree ULC. For every integer $d \geq 6$, there is a finite bipartite graph with positive vertex activities and actual matching rank d that violates the last inequality in (329). The counterexamples may be oriented as vertex-weighted height-two strict orders.*

The threshold is stated in [12, Corollary 1.2 and `negative_ranks.tex`]. The positive assertion is also proved independently in the ordinary result [38, Corollary 5.2]; its unchanged six-page proof and frozen replayable archive are included as a dependency. The remaining assertions receive a self-contained proof below. Sharpness refers to rank, not the minimum number of vertices in a counterexample. No unweighted counterexample is asserted. [*Merge note:* This is source 74’s Corollary 2.2, whose rank-six witness is *preorder-root-polytopes*’s $H_{3,33}$ (Part XI, Theorem 177.1); [38] is source 79. The height-two orientation is *preorder-root-polytopes*’s native framework (its Part II). The proof below keeps source 80’s displays and points to the derivations already in this report and in *preorder-root-polytopes*.]

43.2 Complete-block scalar counterexamples

43.2.1 The exact support formula

Take disjoint shores $P \sqcup X$ and $Q \sqcup Y$, with sizes p, m, q, n , respectively. Include precisely all edges of $P \times Q$, $P \times Y$ and $X \times Q$. Give each core vertex in $P \cup Q$ activity $\lambda > 0$, and every exterior vertex activity 1. [*Merge note:* Source 80 derives the following count by Hall's condition: a support with i vertices of P and j of Q is feasible exactly when $i + j \geq k$. This is source 74's display (5) with $a = p$, $b = q$, core activity $w = \lambda$ (and source 77's display (309)); for equal cores it is *preorder-root-polytopes*'s Theorem 175.3 (Part XI, **hb:thm:coefficients**). The derivation is not reprinted.]

$$\gamma_k = \sum_{\substack{0 \leq i \leq p, 0 \leq j \leq q \\ i+j \geq k}} \binom{p}{i} \binom{q}{j} \lambda^{i+j} \binom{m}{k-i} \binom{n}{k-j}, \quad (330)$$

with out-of-range binomial coefficients interpreted as zero. This is an endpoint count, not a matching count.

If $m \geq q$ and $n \geq p$, the actual degree is $d = p + q$: the core is a cover of size d , and matching all of P into Y and all of Q into X supplies d edges. Positive activities preserve this degree.

43.2.2 An explicit degree-six violation

[*Merge note:* The family, the closed forms of $\gamma_4, \gamma_5, \gamma_6$ and the factorization (331) are *preorder-root-polytopes*'s Theorem 177.1 (Part XI, **hb:thm:rank6phase**) and its displays, with the same prefactor and T_N equal to Part XI's Q_n ; Part XI proves more, the exact phase boundary. Its $H_{3,33}$ with weight 26804 is source 74's rank-six witness (Section 6). The two numerical instances below are new instances of that family and nothing more.]

Set $p = q = 3$ and $m = n = N \geq 3$, and abbreviate $A = \binom{N}{2}$, $B = \binom{N}{3}$. Formula (330) gives

$$\begin{aligned} \gamma_6 &= \lambda^6 B^2, \\ \gamma_5 &= \lambda^5 (\lambda A^2 + 6AB), \\ \gamma_4 &= \lambda^4 (\lambda^2 N^2 + 6\lambda NA + 9A^2 + 6NB). \end{aligned}$$

Consequently the last degree-six gap factors exactly as

$$5\gamma_5^2 - 12\gamma_4\gamma_6 = \frac{\lambda^{10} N^4 (N-1)^2}{48} T_N(\lambda), \quad (331)$$

where

$$\begin{aligned} T_N(\lambda) &= (-N^2 + 34N - 49)\lambda^2 \\ &\quad + 12(N-1)(N-2)(N+3)\lambda \\ &\quad + 8(N-1)(N-2)^2(N+1). \end{aligned} \quad (332)$$

For integer $N \geq 3$, the leading coefficient is negative exactly when $N \geq 33$, since its positive root is $17 + 4\sqrt{15}$. Every such N therefore gives a violation for sufficiently large finite λ .

For an independently checked round-number instance, choose $N = 40$ and $\lambda = 10000$. This is an 86-vertex graph with 249 edges, a six-vertex cover and a size-six matching. To shorten the certificate, put $g_k = \gamma_k / \lambda^k$, the coefficients of $\Gamma(t/\lambda)$. They are

$$(g_0, \dots, g_6) = (1, 90240, 907219080, 1024191381360, 161879846800, 6130238400, 97614400).$$

Their last gap is the negative integer

$$5g_5^2 - 12g_4g_6 = -1722535205514240000. \quad (333)$$

A positive change of variable multiplies each gap by a positive factor, so this is an actual-degree ULC counterexample for the original activity assignment. Its earlier four gaps are positive. Within the same family, $N = 33$, $\lambda = 30000$ gives a 72-vertex example with scaled coefficients

$$(1, 270198, 8117832969, 27178589394544, 983239909344, 8380804608, 29767936)$$

and last gap -38842940605759488 . The intake already gives the same 72-vertex construction with the smaller activity 26804 [12]. No minimum-vertex assertion is made.

43.2.3 Failure at every higher rank

[Merge note: Source 80's argument here is source 74's Section 6: the same limit of the top three coefficients as the core activity grows, the same ratio $pq/((p-1)(q-1))$, the same inequality $(d+1)pq - 2d(d-1) \geq d^2 - 4d - 9 > 0$ for $d \geq 6$, and $p = 3$, $q = d - 3$. It is not reprinted.]

Finally orient all edges from the left shore to the right shore. No two nontrivial edges compose, so the relation is a height-two strict partial order. Adding reflexive pairs produces a preorder with the same disjoint-endpoint supports, since a loop cannot occur in a matching between disjoint endpoint sets. A single physical-vertex activity per vertex gives exactly the weights above; independent tail/head role activities are unnecessary. This completes the negative assertions of Theorem 43.1.

43.3 A separate multivariate obstruction and its graph lift

For the abstract analogue, let M have rank q on $E \sqcup P$, where $P = \{A_0, A_1, A_2\}$ are distinguished elements. Let Y be weighted heads with neighborhoods in P . A triple (I, J, T) , with $I \subseteq E$, $J \subseteq P$, $T \subseteq Y$, is feasible if some $K \subseteq J$ matches perfectly to T and $I \cup (J \setminus K)$ is a basis of M . Define

$$\mathcal{F}_M^{(3)}(\mathbf{w}, \mathbf{a}, z) = \sum_{(I, J, T) \text{ feasible}} w^I a^J \left(\prod_{y \in T} v_y \right) z^{3-|T|}. \quad (334)$$

Each triple contributes once, even if its allocation K is not unique. Its degree is $q + 3$. Differentiation in an old nonloop variable gives the same construction for M/e , by restricting feasible triples to those containing e and then deleting e .

We use the basic Lorentzian implications from Brändén–Huh [1, Theorems 2.10 and 2.25]: nonnegative linear specialization and differentiation preserve Lorentzianity or zero, and a nonnegative quadratic Lorentzian polynomial has at most one positive Hessian eigenvalue. A binary quadratic with positive coefficients and negative discriminant therefore cannot be Lorentzian.

43.3.1 Three parallel pairs

Let M have rank two on parallel pairs $\{u_i, A_i\}$, $i = 0, 1, 2$, with every pair from distinct classes independent. It is real-representable: assign directions $(1, 0)$, $(0, 1)$, $(1, 1)$ to the three classes. Give each A_i one private head of activity p , and add one common head of activity h .

Proposition 43.2. *The transform (334) is not Lorentzian whenever $h > 0$ and $p > 15h/4$.*

Proof. Selecting all three distinguished elements gives

$$\partial_{a_0} \partial_{a_1} \partial_{a_2} \mathcal{F}_M^{(3)} = Rz^2 + Bz(u_0 + u_1 + u_2) + C(u_0u_1 + u_0u_2 + u_1u_2), \quad (335)$$

where

$$R = 3p + h, \quad B = 2p^2 + 3ph, \quad C = p^3 + 3p^2h.$$

A single selected head leaves two distinguished basis elements, giving R . With one old element u_i , the two heads must match a distinguished pair containing i : two private pairs and all three private/common pairs qualify, giving B . With two old elements, either the three private heads or the common head with any two private heads qualify, giving C . These are endpoint counts, not allocation counts.

Set all $u_i = x$. The binary discriminant is

$$(3B)^2 - 4(3C)R = 3p^2h(15h - 4p) < 0. \quad (336)$$

Differentiation and nonnegative specialization yield the contradiction. $\square$

At $p = 4, h = 1$, the quadratic is

$$Q = 13z^2 + 44z(u_0 + u_1 + u_2) + 112(u_0u_1 + u_0u_2 + u_1u_2). \quad (337)$$

Its restriction $336x^2 + 132xz + 13z^2$ has discriminant -48 . Its Hessian on this two-dimensional subspace is $\begin{pmatrix} 672 & 132 \\ 132 & 26 \end{pmatrix}$, positive definite with determinant 48. The full four-dimensional Hessian has inertia $(2, 2, 0)$, since the two sum-zero old-variable directions have eigenvalue -112 .

The seed itself is not transversal. Two parallel nonloops in any transversal presentation must share a singleton neighborhood; otherwise the pair can be matched. Three distinct nontrivial parallel classes would force three distinct singleton slots and hence an independent triple, contradicting rank two. This observation alone would not furnish a bipartite-graph counterexample. The following lift does.

43.3.2 A thirteen-vertex graph with a full matching

Take disjoint shores

$$P \sqcup X = \{p_0, p_1, p_2\} \sqcup \{c, x_0, x_1\}, \quad Q \sqcup Y = \{q_0, q_1, q_2\} \sqcup \{y_0, y_1, y_2, h\}.$$

The only edges are

$$p_iq_i, p_iy_i, p_ih, cq_i \quad (0 \leq i \leq 2), \quad x_0q_0, x_1q_1. \quad (338)$$

There are fourteen edges. The set $P \cup Q$ is a $3 + 3$ cover. The matching

$$\{p_0y_0, p_1y_1, p_2y_2, x_0q_0, x_1q_1, cq_2\}$$

has size six, which is maximal because there are six left vertices.

For individual selected-left variables, the natural degree-six support homogenization is

$$\mathcal{F}_G(\mathbf{a}, \mathbf{z}_Q, z) = \sum_{(S,T) \text{ feasible}} \left(\prod_{j \in T} v_j \right) a^S z^{3-|T \cap Y|} \prod_{j \in Q \setminus T} z_j. \quad (339)$$

Give each y_i activity 4 and every other vertex activity 1. Then

$$\partial_{a_{p_0}} \partial_{a_{p_1}} \partial_{a_{p_2}} \partial_{a_c} \mathcal{F}_G \Big|_{a_{x_0}=a_{x_1}=0} = Q,$$

where the u_i in (337) are now the unused-head variables z_i . Thus (339) is not Lorentzian, despite actual matching rank six.

Here is an exact matroid explanation of the equality. In the rank-three transversal matroid with slots Q , put both A_i and an old dummy u_i privately at slot i , and add the old universal element c . Contract c . A same-slot pair is dependent, any different-slot pair is independent, and the contraction has rank two; it is precisely the three-parallel-pair seed. Old-element contraction commutes with the Boolean allocation definition (334). Adding the two extra graph tails x_0, x_1 makes the full matching possible and setting their variables to zero leaves the displayed obstruction. Direct endpoint enumeration independently verifies the same equality.

The scalar support polynomial at these activities is

$$\Gamma_1(t) = 1 + 23t + 195t^2 + 743t^3 + 1234t^4 + 744t^5 + 112t^6. \quad (340)$$

It is strictly actual-degree ultra-log-concave; exact gaps are recorded below. Multivariate failure here is therefore not scalar failure.

43.4 The failure survives identifying selected-tail variables

Keep the graph (338). Give p_0, p_1, p_2, c activity L , x_0, x_1 activity 1, each private head y_i activity p , and all other heads activity 1. Replace every selected-left variable by its activity times t , and every unused Q variable by x . Denote the resulting degree-six polynomial by $H(t, x, z)$.

Endpoint counting by the number of selected Y heads gives

$$[t^4]H = L^2(Ax^2 + Bxz + Cz^2), \quad (341)$$

with the local scalar coefficients

$$\begin{aligned} A &= Lp^2(3L + 2)(p + 3), \\ B &= p(6L^2p + 9L^2 + 16Lp + 32L + 3p + 6), \\ C &= 3L^2p + L^2 + 10Lp + 8L + 5p + 5. \end{aligned}$$

For the positive integer choice $L = 100, p = 8$,

$$[t^4]H/10000 = 21260800x^2 + 4688240xz + 258845z^2,$$

and its discriminant is -33412806400 . If H were Lorentzian, the nonzero quadratic obtained from $\partial_t^4 H$ at $t = 0$ would be Lorentzian. Instead its Hessian is positive definite. This disproves even the three-variable $3 + 3$ cover-order Lorentzian analogue, with one selected-tail variable and full actual degree six.

The scalar coefficients in this second example are

$$(\gamma_0, \dots, \gamma_6) = (1, 3302, 3846201, 1806662900, 262078850000, 313024000000, 70400000000).$$

For either scalar sequence, actual-degree ultra-log-concavity means log-concavity of $\gamma_k / \binom{6}{k}$. Define the integer gap numerator

$$D_k = k(6 - k)\gamma_k^2 - (k + 1)(7 - k)\gamma_{k-1}\gamma_{k+1}.$$

All five are positive in both examples:

k	Activities of (340)	$L = 100, p = 8$
1	305	8361608
2	47865	28862083622208
3	1118361	13248150548946090000
4	3890168	540999656224436000000000
5	1109184	268515910400000000000000

Both sequences satisfy scalar ULC, despite failure of their chosen higher-dimensional homogenizations. The scalar counterexamples of Section 43.2 are different graphs. [*Merge note:* These are obstructions of another kind than source 63's Proposition 28.3 (Part III), a twenty-vertex graph whose full right marginal with unit left activities is not Lorentzian. Here the thirteen-vertex graph's selected-tail homogenization and its three-variable specialization fail, so Part I's two-tail lifts (Theorem 2.1, Theorem 42.1) do not extend to three tails; the scalar sequences stay ULC₆, as Part III expects.]

43.5 An ordinary positive first-layer result for every rank

Fix $r \geq 2$ left vertices and an arbitrary finite set of heads with nonnegative activities v_y . Let R_i be the total neighbor activity of tail i . For $i \neq j$, let H_{ij} be the total common-neighbor activity, Q_{ij} the sum of its squares, and R_{ij} the sum of product activities over distinct head pairs matchable to $\{i, j\}$, counted once. Repeated-head choices and duplicate witnesses for common-head pairs give exactly

$$R_{ij} = R_i R_j - \frac{1}{2}(H_{ij}^2 + Q_{ij}). \quad (342)$$

The pair-count identity also underlies the neighbor-feature Gram argument in [36, Section 2]. The following probability/projector proof gives a general- r matrix statement by a different decomposition.

Theorem 43.3. *The symmetric $r \times r$ matrix*

$$N_{ii} = (r-1)R_i^2, \quad N_{ij} = (r-1)R_i R_j - rR_{ij} \quad (i \neq j) \quad (343)$$

is positive semidefinite for every such graph and activities.

[*Merge note:* At $r = 3$ the matrix N is $3Q$ for source 10's matrix Q (display (107), with $c_I = R_I$), so the case $r = 3$ is $Q \succeq 0$, which source 10 proves by the 5339-monomial certificate of Lemma 23.2. For $r \geq 4$ the theorem is new to the repository. The identity (342) is source 10's display (108); the Gram argument cited from source 05 is its display (135).]

Proof. First suppose all $R_i > 0$. Define a probability distribution $p_i(y) = v_y/R_i$ on the neighbors of i and zero elsewhere, and put $u_i = (\sqrt{p_i(y)})_y$. Set

$$A_{ij} = (u_i \cdot u_j)^2, \quad C_{ii} = 1, \quad C_{ij} = \sum_y p_i(y)p_j(y) \quad (i \neq j).$$

Write $J = \mathbf{1}\mathbf{1}^T$. We claim that both A and C dominate J/r in the positive-semidefinite order.

For any real vector ξ , the symmetric operator $T = \sum_i \xi_i u_i u_i^T$ acts on a space of dimension at most r , has trace $\sum_i \xi_i$, and has squared Hilbert–Schmidt norm $\xi^T A \xi$. Cauchy–Schwarz on its eigenvalues yields

$$\xi^T A \xi \geq \frac{1}{r} \left(\sum_i \xi_i \right)^2.$$

Independently sample one label from each p_i . The equality partition π of the r sampled labels has at most r blocks. Its matrix K_π satisfies

$$\xi^T K_\pi \xi = \sum_{B \in \pi} \left(\sum_{i \in B} \xi_i \right)^2 \geq \frac{1}{r} \left(\sum_i \xi_i \right)^2.$$

Since $\mathbb{E}K_\pi = C$, the second claim follows. Both arguments allow arbitrary signed test vectors.

For $D = \text{diag}(R_i)$, equation (342) now gives

$$D^{-1}ND^{-1} = \frac{r}{2}(A + C) - J = \frac{r}{2}(A - J/r) + \frac{r}{2}(C - J/r) \succeq 0. \quad (344)$$

If $R_i = 0$, the corresponding row and column of N vanish. Apply the proof to the remaining indices with the original denominator r : their span dimension and number of partition blocks are still at most r . This includes all zero-activity boundaries. $\square$

For a left subset S , let R_S sum the weights of the head subsets matchable to S , counted once. Define the selected-tail homogeneous polynomial

$$F(z, \mathbf{a}) = \sum_{S \subseteq [r]} R_S a^S z^{r-|S|}, \quad R_\emptyset = 1.$$

Then

$$\partial_z^{r-2} F = \frac{r!}{2} z^2 + (r-1)! z \sum_i R_i a_i + (r-2)! \sum_{i < j} R_{ij} a_i a_j.$$

After division by $(r-2)!$, its Hessian has upper-left entry $r(r-1) > 0$ and Schur complement

$$B - \frac{r-1}{r} R R^T = -N/r, \quad B_{ii} = 0, \quad B_{ij} = R_{ij}.$$

Thus this first-layer Hessian has exactly one positive eigenvalue. This ordinary all-rank statement is compatible with the higher-layer obstructions above and does not assert full Lorentzianity for arbitrary r .

43.5.1 The full three-tail rank-zero base

The following statement appears in [20, Theorem 1.1], whose proof uses a 5,339-term determinant certificate. Theorem 43.3 gives an ordinary replacement for that certificate.

Corollary 43.4. *For a bipartite graph whose entire left shore has three vertices, with arbitrary nonnegative head activities, the polynomial*

$$F = z^3 + z^2 \sum_i R_i a_i + z \sum_{i < j} R_{ij} a_i a_j + R_{012} a_0 a_1 a_2$$

is Lorentzian, retaining each selected-tail variable individually.

[Merge note: This is source 10's Theorem 23.1 (Part III), with s, z_i, c_I there written z, a_i, R_I here, and the proof below is a second route to it: its Rayleigh step is source 10's display (106), and Theorem 43.3 replaces the certificate Lemma 23.2 as the route to $Q \succeq 0$. The certificate itself stays a valid proof.]

Proof. After deleting zero-weight heads, matchable left subsets are the independent sets of a transversal matroid L . Take its direct sum with three free dummy elements and truncate to rank three. Identify the dummy variables with z in the basis polynomial. The resulting Lorentzian polynomial has exactly the same exponent support as F , so this support is M -convex. Only support equality is used; the coefficients need not agree. This also handles a rank drop or an empty positive graph.

Next use a rank-three transversal matroid on the heads and three private dummies d_i , with the left vertices as slots. Its basis polynomial, after substituting head activities, is

$$B(\mathbf{d}) = d_0 d_1 d_2 + \sum_i R_i \prod_{j \neq i} d_j + \sum_{i < j} R_{ij} d_k + R_{012},$$

where k is the remaining index. Wagner’s theorem that every rank-three matroid is Rayleigh [19, Theorem 1.1], applied to d_j, d_k after $d_i = 0$, yields

$$R_{ij}R_{ik} - R_iR_{012} \geq 0.$$

Zero activities are included by continuity. The Hessian of $\partial_{a_i}F$ in (z, a_j, a_k) is

$$\begin{pmatrix} 2R_i & R_{ij} & R_{ik} \\ R_{ij} & 0 & R_{012} \\ R_{ik} & R_{012} & 0 \end{pmatrix}, \quad \det = 2R_{012}(R_{ij}R_{ik} - R_iR_{012}) \geq 0.$$

If $R_{012} > 0$, its bottom block has one positive and one negative eigenvalue, so the determinant excludes a second positive eigenvalue. If $R_{012} = 0$, the arrow matrix with zero bottom block directly has at most one positive eigenvalue. Theorem 43.3 supplies the remaining derivative $\partial_z F$. The support condition and all four quadratic-derivative tests prove Lorentzianity [1, Theorem 2.25]. $\square$

This is the zero-old-rank base of the three-tail construction, not a positive-rank extension. It is therefore consistent with the positive-old-rank multivariate counterexamples above.

43.6 Scope and exact reproducibility

The reconstructed Theorem 43.1, already present in [12], gives the exact universal *weighted* matching-rank threshold: actual-degree ULC holds through five and can fail at every rank six and above. The explicit complete-block examples use positive integer vertex activities and also yield height-two order counterexamples. They establish no unweighted counterexample and no minimum-vertex bound.

The smaller thirteen-vertex examples serve a different purpose. Their scalar sequences are ULC; their saturated six-vertex left shore already puts them within the ordinary one-shore transversal-basis guarantee. They instead disprove the specified multivariate and three-variable Lorentzian extensions. Theorem 43.3 proves a first-layer condition at every rank, and Corollary 43.4 proves full Lorentzianity at the three-tail zero-old-rank base. No physical tail/head collision merge is used.

The package includes exact standard-library checkers for the new scalar and multivariate examples and the positive matrix statements, together with the unchanged rank-five proof archive and readable PDF. The scalar checker independently checks all 78 quota Hall cases, reconstructs the coefficients and verifies the exact last-gap factorization. The direct-union appendix is corroborated by 1,672 identities over 337,428 candidate bases, including 352 zero-parameter interpolation checks. The earlier graph recounts find 397 supports among 1,716 candidates; the abstract union checker finds 109 bases among 210 candidates. Matrix corroboration covers 2,800 weighted cases with 8,400 exact PSD tests, plus 6,602 three-tail cases with 46,214 principal-minor checks. The ordinary results do not depend on finite numerical sampling.

Approved proof notes, source-bound mathematical reviews, deterministic receipts and all-page PDF inspections are included. Replay uses temporary copies, compares every deterministic field, rejects Python optimization mode and verifies release immutability, including the frozen rank-five dependency. The new checkers need only the Python standard library; the older dependency uses SymPy. The source comparison credits the overlapping intake constructions; the incoming source scripts are not executed. No proof-assistant formalization, literature-wide priority claim or external publication is asserted.

Appendix A of source 80. [Merge note: Source 80’s appendix, *A direct all-matroid extension of the intake lift*, is printed in Section 42.2 as the first proof of source 79’s Theorem 42.1.]

44 Questions raised by Part V

The following are open after Part V. They are written in this merge: the first is suggested by source 80’s obstructions, the others are the sources’ own closing questions.

- *A three-tail Lorentzian homogenization* (source 80). For a cover with three vertices on one shore, the selected-tail homogenization and its three-variable specialization fail on a thirteen-vertex graph (Sections 43.3.2 and 43.4), and source 63’s right marginal fails on a twenty-vertex graph (Proposition 28.3), while the first layer (Theorem 43.3) and the zero-old-rank cubic (Corollary 43.4) hold. With arbitrary activities on both shores no Lorentzian homogenization of order $\nu(G)$ can exist for all $3 + q$ covers, since it would make *preorder-root-polytopes*’s $H_{3,33}$ ULC₆. Under which restrictions (unit or one-shore activities, restricted cores) does one exist, and which avoids these obstructions?
- *The two-tail/three-head role cover* (source 78). For directed endpoint supports with a role cover of two tails and three heads, does rank normalization hold? It concerns the directed setting of *preorder-root-polytopes*’s Part XII and is not addressed by Parts I–V.
- *Larger aggregate realizations* (source 77). Which larger cover patterns admit aggregate Hermitian realizations of p_G , and can the rank-four constructions give quantitative deficit bounds? Equality is already classified by source 72’s Theorem 14.1.

A Provenance of this report

A.1 Sources

The 27 sources of Parts I–IV arrived in ProveIt’s incoming directory in commit 1512ef835 (batch 72, 79 archives) and were placed in commit 8bb543f0f. Each is dated 1 October 2026. Their delivered PDFs, manuscripts, READMEs, checksum ledgers and the identical helper `build_local.sh` are not shipped; their code, data, audit and provenance files are, under the prefixes listed (README of this report). “Pin” is a ProveIt commit that the manuscript itself cites; sources without a pin cite their siblings by archive name only. The report was built from these 27 sources; Part V adds four batch-73 sources, listed after this table.

No.	Archive and title	Pin	Labels	Printed as
74	<code>ProveIt_Weighted_Rank_Five_and_Two_Shore_Covers.zip</code> : <i>Weighted Rank Five and Covers with a Two Vertex Shore</i> (16 pages, A4)	1b3960d8a, a866ff9a2 (archives cited)	<code>mrn:w:rf:</code>	Sections 2–13 (base)
72	<code>strictness_equality_package.zip</code> : <i>Strictness and equality for weighted matching support polynomials</i> (8 pages, Letter; “AI-assisted mathematical research”)	none	<code>mrn:w:se:</code>	Section 14

No.	Archive and title	Pin	Labels	Printed as
76	ProveIt_Weighted_Rank_Four_Stability.zip: <i>Stability of Two by Two Covers and Weighted Rank Four</i> (8 pages)	1b3960d8a, a866ff9a2	mrn:w:sb:	Section 15
75	ProveIt_Sharp_Full_Hall_Classification.zip: <i>A Sharp Weighted Classification of Full Hall Bottleneck Graphs</i> (9 pages)	4a3b6991d	mrn:w:fh:	Section 16
71	ProveIt_Unit_Weight_Rank_Normalization_Counterexample.zip: <i>Unit Weight Failure of Matching Rank Normalization</i> (5 pages)	a866ff9a2	mrn:u:uw:	Section 17
73	ProveIt_One_Shore_Counterexamples.zip: <i>One Shore Activities Can Destroy Matching Rank Ultra Log Concavity</i> (6 pages, with pendant_appendix.tex)	4a3b6991d	mrn:u:os:	Section 18
55	ProveIt_Sharp_Rank_38_Hall_Endpoint.zip: <i>A Sharp Rank 38 Boundary for the Last Hall Inequality</i> (6 pages)	none	mrn:u:ep:	Section 19
47	ProveIt_Sharp_Rank_38_Full_Hall_ULC.zip: <i>The Sharp Rank 38 ULC Boundary for Full Hall Graphs</i> (6 pages)	none	mrn:u:ag:	Section 20
51	ProveIt_Last_Hall_Gap_Separation.zip: <i>The Last Hall Gap Does Not Control Ultra Log Concavity</i> (2 pages)	none	mrn:u:sp:	Section 21
60	ProveIt_Rank_Two_Matrix_Blocks_and_Conditioning.zip: <i>Rank Two Matrix Blocks and Exterior Conditioning</i> (7 pages)	none	mrn:s:rb:	Section 22
10	ProveIt_Three_Vertex_Marginals_and_Star_Cores.zip: <i>Lorentzian Marginals on Three Vertices</i> (5 pages)	none	mrn:s:tv:	Section 23
23	ProveIt_Rooted_Hall_Sectors_and_Cubic_Obstruction.zip: <i>Rooted Lorentzian Hall Sectors and an Unrooted Cubic Obstruction</i> (5 pages)	none	mrn:s:rs:	Section 24
13	ProveIt_Rank_Six_Finite_Last_Gap.zip: <i>The Last Newton Inequality at Rank Six</i> (7 pages)	none	mrn:s:lg:	Section 25

No.	Archive and title	Pin	Labels	Printed as
05	ProveIt_Rank_Six_Second_Newton_Inequality.zip: <i>The Second Newton Inequality at Rank Six</i> (7 pages)	none	mrn:s:sn:	Section 26
69	ProveIt_Rank_Six_Common_Exterior_Theorem.zip: <i>Rank Six With Common Exterior Neighborhoods</i> (5 pages)	none	mrn:s:ce:	Section 27
63	ProveIt_Rank_Six_Nested_and_Conditional_Results.zip: <i>Nested Rank Six Families and Conditional Barriers</i> (7 pages, with conditioning.tex)	none	mrn:s:ne:	Section 28
08s	inside ProveIt_Eventual_One_Shore_Scaling_Through_Rank_Seven.zip: Rank7_Eventual_Report/companion/structural_companion_source.zip, <i>Top Conditional Gaps and the Rank Seven Reduction</i> (8 pages)	none	mrn:e:tg:	Section 29
21	ProveIt_Rank_Six_Conditioning_for_Every_Core.zip: <i>Rank Six Conditioning for Every Core</i> (6 pages)	none	mrn:e:ec:	Section 30
43	ProveIt_Rank_Six_Complete_Core_Cubic.zip: <i>A Rank Six Cubic Sector with a Complete Core</i> (5 pages)	none	mrn:e:cc:	Section 31
38	ProveIt_Rank_Six_Matching_Hole_Cores.zip: <i>Rank Six Cubic Sectors for Matching Hole Cores</i> (6 pages)	none	mrn:e:mh:	Section 32
35	ProveIt_Rank_Six_Two_Complete_Rows.zip: <i>Rank Six Cores with Two Complete Rows</i> (7 pages)	none	mrn:e:tr:	Section 33
32	ProveIt_Rank_Six_Two_Complete_Columns.zip: <i>Rank Six Cores with Two Complete Columns</i> (6 pages)	none	mrn:e:tc:	Section 34
29	ProveIt_Rank_Six_At_Least_Six_Core_Edges.zip: <i>Rank Six Cores with at Least Six Edges</i> (8 pages)	none	mrn:e:se:	Section 35
08	ProveIt_Eventual_One_Shore_Scaling_Through_Rank_Seven.zip: <i>Eventual One Shore Scaling Through Rank Seven</i> (6 pages)	none	mrn:e:r7:	Section 36

No.	Archive and title	Pin	Labels	Printed as
03	ProveIt_Eventual_Five_Plus_Three_Covers.zip: <i>Eventual Scaling for Five Plus Three Covers</i> (8 pages)	none	mrn:e:fp:	Section 37
02	ProveIt_Four_Plus_Four_Top_Gap.zip: <i>The Top Conditional Gap for Four Plus Four Covers</i> (6 pages)	none	mrn:e:ff:	Section 38
01	ProveIt_Eventual_One_Shore_Scaling_Through_Rank_Eight.zip: <i>Eventual One Shore Scaling Through Rank Eight</i> (8 pages)	none	mrn:e:r8:	Section 39

The full commit hashes are 1b3960d8acdabd983147b7c609bd9dfb2d67a0a0 (the batch-70 archive of the manuscript now printed as Part XII of *preorder-root-polytopes*), a866ff9a2 (the batch-68 archive of the Hall-bottleneck manuscript, now Part XI) and 4a3b6991d0cc12cc09b884e22728e2128a54c6e3 (the commit that wrote Part XI). Source 72 cites source 74 by title, and its shipped 72-strictness-SOURCES.md records a hash of source 74's delivered PDF; that file is kept as delivered.

Batch 73 (Part V). Four further sources arrived in commit f8c3a392a (batch 73, cluster C1 of fifteen archives) and were placed in commit 5e4f1263c. They are numbered 77–80 here, continuing the batch-72 numbers in order of dependency; their batch-73 manuscript numbers are 26, 08, 02 and 01, and number 78 here has nothing to do with batch 72's retired archive 78. Each is dated 1 October 2026. Their PDFs, manuscripts, delivery READMEs and checksum ledgers are not shipped; their code, data, audit and provenance files are, under the prefixes 77-rank-four-, 78-two-three-star-, 79-two-element- and 80-three-tail- (README of this report). The report is now built from 31 sources.

No.	Batch-73 manuscript, archive and title	Pin	Labels	Printed as
77	26, matching-rank-four-package.zip: <i>Weighted Matching Support Rank Normalization Through Rank Four</i> (revision 2, 16 pages, Letter)	d5e863bba (preorder-root-polytopes), article blob 2b952dfcf	mrn:v:rf4:	Section 40
78	08, all-two-by-three-cores-result.zip: <i>Weighted matching supports with a two plus three core</i> (6 pages, A4)	none	mrn:v:st:	Section 41
79	02, two-element-matroid-lorentzian-result.zip: <i>A two-element Lorentzian deformation of every matroid basis polynomial</i> (6 pages, A4)	none	mrn:v:te:	Section 42

No.	Batch-73 manuscript, archive and title	Pin	Labels	Printed as
80	01, sharp-weighted-rank-boundary-result.zip: <i>Weighted matching-rank boundaries: independent verification and Lorentzian obstructions</i> (9 pages, A4)	1512ef835 (archives of sources 74, 10, 05)	mrn:v:tt:	Section 43; its Appendix A in Section 42.2

The full hashes are d5e863bba6b40738af292b45005a0cf0495903e4 (the batch-70 write of *preorder-root-polytopes*’s Part XII; source 77 also records the article blob 2b952dfcf443cdb014878c4bc0cd6adc1f8ef7e5 and the line numbers it read, all exact there) and 1512ef8356d10fef99e7caaa2753620f0c900c83 (the arrival of batch 72). Sources 78 and 79 cite only their batch-73 siblings, by archive name. Source 77’s shipped 77-rank-four-SOURCE.txt also records a SHA-256 for the inspected article that the pinned blob does not reproduce; the README of this report gives the details. Source 80’s shipped 80-three-tail-INTAKE_COMPARISON.md compares it with the raw batch-72 archives of sources 74 and 10 at 1512ef835; it does not compare with this report or with *preorder-root-polytopes*, and so misses that source 80’s scalar family is Part XI’s.

Batch-73 archives that are not sources. Five further archives of the cluster concern the same object and are duplicates; they are not staged and survive only in commit f8c3a392a:

- **matching-rank-four-package (1).zip** (manuscript 27) is the first, twelve-page edition of source 77; every result of it is in source 77.
- **complete-two-by-three-result.zip** (manuscript 09), *Independent weights on a complete two plus three matching core*, proves source 75’s Theorem 16.1 at $s = 3$ (its graph is $H_{2,3,n,m}$) and is the core mask 63 of source 78, which supersedes it; source 78 cites its last-gap computation and nonreal example.
- **weighted-bipartite-rank-five-result.zip** (manuscript 03), *Weighted bipartite matching supports through rank five*, proves source 74’s Theorem 2.1 and Corollary 2.2 by a third route, an explicit positive semidefinite decomposition; source 79 generalizes its construction.
- **two-pair-exterior-result.zip** (manuscript 05), *Weighted support inequalities for overlapping pair neighborhoods*, proves a case of source 74’s theorem (a $2 + 3$ cover), without the strictness that source 72 gives.
- **twin-core-lorentzian-result.zip** (manuscript 04), *Lorentzian polynomials for twin core matching supports*, proves cases of source 74’s theorem; its private-head theorem is *preorder-root-polytopes*’s leaf compression with its ambient theorem (Lemma 206.1, Theorem 70.1).

A sixth retired archive of the cluster, **common-path-rank-package (1).zip**, belongs to the report **constrained-crossover-closure**, which also receives the cluster’s two crossover-rank manuscripts. Three archives of the cluster on directed endpoint supports (**physical-cover-three-result.zip**, **three-core-rayleigh-result.zip**, **bipartite-and-pendant-support-result.zip**) are placed with the batch’s directed-support manuscripts, not here.

A.2 What each source contributed

- **74** (base): the $2 + q$ transversal lift (Theorem 2.1), weighted normalization through matching number five and the least weighted failure rank six (Corollary 2.2), the cover obstruction (Corollary 5.1), the forced-sector and scaling refinements, the negative family at every rank ≥ 6 , the Ehrhart corollary, the generic-line matroid model, the sharp all-core constants $2q/(q - 1)$ and three, the star-core transfer proofs, a second route to the first-gap equality case, and quantitative strictness thresholds.
- **72**: equality in any one gap only for balanced star forests, and strict ULC_r otherwise, under the cover hypothesis of source 74 (Theorem 14.1); a weighted Chan–Pak contraction criterion (Theorem 14.3); a second route to the first-gap equality case.
- **76**: real stability of the basis polynomial, hence real-rootedness of p_G , for every graph with a vertex cover of two vertices on each shore (Theorem 15.1); a second route to weighted rank four; the sharp stability criterion for complete $2 \times s$ mixed cores (Theorem 15.5) and its 17-vertex boundary example.
- **75**: the full Hall classification (Theorem 16.1), with a second route to its positive half, strictness, real-rootedness for a singleton core, explicit asymmetric failures, and a strictly ultra-log-concave rank-five example with nonreal zeros.
- **71**: a connected all-unit graph with 6916 vertices and matching number 3450 that is not ULC_{3450} (Theorem 17.1); a deficiency-one example and infinitely many such; failure of the two Davis–Kohl h^* -polynomials.
- **73**: one-shore counterexamples of matching number 50 (125 vertices) and 38; the endpoint quadratic and a construction criterion; the weighted pendant-arm identity (which contains source 71’s) and a rank-310 transfer example.
- **55**: for one common right-core activity on full Hall graphs, the least matching number with a failing last gap is exactly 38, with the two shapes and minimal populations (864 vertices).
- **47**: in the same family no intermediate gap fails through matching number 38, so 38 is the least failure rank of full ULC.
- **51**: a rank-49 full Hall graph whose last gap holds for every activity while the penultimate gap fails.
- **60**: Lorentzian lift for matrices with a rank-two exterior column block; exact Hall-slack conditioning (slack zero, one, two); forcing $a - 2$ exterior-right vertices of a genuine $a + q$ cover leaves a ULC_{q+2} factor; eventual strictness with an explicit threshold.
- **10**: the degree-three left marginal of any graph with three left vertices is Lorentzian (a 5339-monomial determinant certificate); star cores on the three-vertex shore give ULC_{q+3} for all activities on both shores, hence all five rank-six inequalities for both star orientations.
- **23**: rooted sector marginals are Lorentzian; the 25-vertex graph whose all- B conditional cubic fails both unshifted Newton inequalities although the graph is strictly ULC_6 for every common core activity.
- **13**: the last rank-six inequality for every one-shore graph and every finite activity (858 weighted squares, ten profiles).

- **05**: the second rank-six inequality likewise, through a Lorentzian cubic truncation (a certificate of 50,037,694 coefficients); the quintic route is refuted.
- **69**: ULC_6 for every 3×3 core with complete exterior neighbourhoods (8,964 certificate vectors); a private-leaf reduction; a barrier coefficient $56 - 4n$.
- **63**: ULC_6 for the complete core with nested exterior neighbourhoods; a 20-vertex graph whose full right marginal is not Lorentzian; forcing four right vertices gives $c_1^2 \geq 3c_0c_2$.
- **08s**: the top conditional gap for every $a+b$ cover with constant $a(b+2)^2/((a-1)(b^2+b+2))$; the two-plane incidence lemma in rank b ; eventual strict ULC for $3+3$, $3+4$, $3+5$ covers; the reduction of rank seven to the $4+3$ middle gap; a weighted local-equality lemma; the bracket $[7, 38]$ (superseded).
- **21**: for every $3+3$ core, $(n-1)C_1^2 \geq 2nC_0C_2$ (also with arbitrary left activities) and $C_2^2 \geq \frac{21}{8}C_1C_3$ (refined to $75/28$); eventual strict ULC_6 ; the global nonstrict rank-six statement.
- **43, 38, 35, 32, 29**: second routes to source 21's theorem for complete, matching-hole, two-complete-row, two-complete-column and ≥ 6 -edge cores, with the partition formula and elementary proof of 43, the orbit census of 29, the every-field corollaries of 38, 35, 32, 29, and their certificates.
- **08**: the $4+3$ middle gap $C_2^2 \geq \frac{15}{8}C_1C_3$; eventual ULC_ν through matching number seven; $[8, 38]$.
- **03**: eventual strict ULC_8 for every $5+3$ cover (factors $77/48$, $7/4$, $55/24$).
- **02**: the $4+4$ top gap $16/7$ with a coupled correction; the unrestricted upper incidence bound is false (six real vectors).
- **01**: the $4+4$ middle gap; eventual ULC_ν through matching number eight; the persistent-failure bracket $[9, 38]$.
- **77** (batch 73): at unit activities and matching number three, a nonreal zero needs fifteen vertices, with a unique extremal graph (Theorem 40.6, computer-assisted), the pair-neighbourhood family p_n and its discriminant, and the twin remark; second routes to source 76's $2+2$ real-rootedness (univariate), to *preorder-root-polytopes*'s $1+2$ real-rootedness, and to weighted rank four; a certificate for one block-constant rank-five family.
- **78** (batch 73): a computer-assisted second route for 2×3 cores with complete exteriors (26 exact certificates); real stability for empty and star cores, new at the exceptional two-neighbour seed.
- **79** (batch 73): the two-element Lorentzian deformation of every matroid basis polynomial (Theorem 42.1), by a principal-line seed and contraction; second routes to source 74's graph theorem with every left variable kept and to weighted rank five.
- **80** (batch 73): the all-rank first-layer theorem (Theorem 43.3) and with it a certificate-free proof of source 10's three-vertex theorem; an abstract three-element obstruction and two thirteen-vertex graph obstructions to three-tail Lorentzian lifts; the union proof of source 79's theorem; two further numerical instances of *preorder-root-polytopes*'s rank-six family.

A.3 Where the merge chose

- *Placement and base.* The batch’s manuscripts continue Parts XI–XIII of *preorder-root-polytopes* and answer several of its research questions, but they form a self-contained programme of 26 archives; they were placed as a new report in **log-concavity-and-unimodality**, with source 74 as the base because it proves the most general theorem and the others cite it. *preorder-root-polytopes* is cited, not reprinted, and receives dated pointers to this report.
- *Source 74’s layout.* Its eight section files were inlined into **article.tex** and removed; its appendices are ordinary sections of Part I. Its two short lemmas that restate *preorder-root-polytopes* (Lemmas 3.1 and 3.2) keep their statements and receive pointer proofs. Table 3 was added from its recorded verifier output.
- *Re-proofs of preorder-root-polytopes.* Sources 74 and 72 re-prove Part XI’s first-gap inequality and its equality case by the Motzkin–Straus route: both proofs are printed, marked as second routes. Source 74’s rank-six witness is Part XI’s $H_{3,33}$ and its (4, 4) shape Part XI’s $H_{4,11}$: printed with notes. The leaf compression of sources 72, 74 and 76 is Part XIII’s; source 76’s inversion identity, gluing lemma, expected-determinant lemma and $K_{2,k}$ stability are Part VI’s: proofs of those are pointers.
- *Same theorem in the batch.* Source 76’s weighted rank four and the positive half of source 75’s classification are cases of source 74’s theorem; both are printed in full as second routes, because their proofs (real stability; a comparison polynomial) differ from source 74’s and give more (real-rootedness, strictness).
- *Stale sentences.* Statements of sources 75 and 76 that the first weighted failure rank is five or six, or that rank five is open, are kept with notes: source 74 settles them.
- *Part II.* Source 71’s pendant lemma is printed once, in source 73’s weighted form (source 71’s statement is kept with a pointer proof). Source 47’s first-gap proof is a pointer to Part XI of *preorder-root-polytopes* (its strictness step is kept); its recall of source 55’s endpoint theorem is kept with pointers to source 55, which is printed before it. Source 73’s rank-310 example is kept as a remark; it is one-shore, not unit, weighted. Statements of sources 55, 47 and 51 that call rank-six one-shore ULC open or that leave earlier gaps open are kept with notes.
- *Part III.* Source 60’s restatement of Part XII’s rank-three matrix is a pointer. The 25-vertex graph is printed once, in source 23; source 05’s description of it is kept with a note. Source 13’s incidence lemma, which is also source 21’s (Part IV) and a case of source 08s’s, is printed with source 13’s proof, where it is first used. The first-gap arguments of sources 69 and 63 are pointers to *preorder-root-polytopes*’s Theorem 180.2. Source 05’s direct proof of its rank-two lemma is kept as a second route of source 60’s theorem. Sentences of sources 10, 13, 23, 60, 63 and 69 that call three middle gaps, or the whole rank-six one-shore problem, open are kept with notes giving the status above.
- *Part IV.* Source 08s, delivered only inside archive 08, is printed as a source of its own and is the base of Part IV, because it proves the general $a + b$ theorem of which source 21’s is the $3 + 3$ case. Source 21 is printed in full as that case; its incidence-lemma proof is a pointer to source 13’s printed proof. The five special-core sources are printed in full as second routes of source 21’s theorem (the union rule; the alternative of shipping only their provenance was not taken). Source 08s’s appendix proof of its weighted equality lemma is a pointer to source 72’s proof, and its one equation label **eq:kernel** is kept on the displayed decomposition. The brackets [7, 38] and [8, 38] are kept as history with

notes; [9, 38] is final. Source 08s’s claim that Choe–Wagner’s printed counts are wrong is labelled as its own statement.

- *Part V (batch 73)*. The four kept batch-73 manuscripts were added as one Part after Part IV, not inserted into Parts I and III: Parts I–IV were written the same day, and *preorder-root-polytopes*’s reciprocal notes cite their section and theorem numbers, which Part V leaves unchanged. Parts I and III receive five dated forward notes instead. Four Parts of a page or two of new mathematics each were not made. The order (77, 78, 79, 80) is that of dependency.
- *Re-proofs in Part V*. Statements that repeat Parts I–IV or *preorder-root-polytopes* are kept with notes. Proofs are replaced by pointers where they are the repository’s proofs (source 77’s ambient theorem, leaf compression and thin covers; source 80’s derivation of the quota count and its higher-rank limit argument), and printed as marked second routes where they differ: source 77’s path and Gram-determinant proof of $2 + 2$ real-rootedness, its rooted-path proof of the $1 + 2$ lemma (used by the rank-three minimum) and its rank-five certificate; source 78’s 26 certificates; source 79’s proofs of its graph corollaries; source 80’s proof of the three-tail cubic. Source 80’s scalar counterexample displays and its two numerical instances are kept, with the attribution to Part XI of *preorder-root-polytopes* that source 80 omits.
- *One theorem, two proofs*. Source 79’s theorem is printed once, in source 79’s section; source 80’s one-page matroid-union proof (its Appendix A, which credits source 74’s lift) is printed first, as the shorter proof, and source 79’s own proof second.
- *Source 80’s three-tail cubic*. It is printed in Part V as a second route to source 10’s theorem, with a dated note in source 10’s section, not inside that section.
- *Stale sentences in Part V*. Source 77’s closing statement that the least weighted failure rank is five or six, the same sentence in its shipped `77-rank-four-REVISION.md`, source 77’s claim to answer three research questions of *preorder-root-polytopes*, and source 78’s delivered README on rank five are kept with notes giving the current status.

References

- [1] P. Brändén and J. Huh, *Lorentzian polynomials*, *Annals of Mathematics* **192** (2020), 821–891. <https://annals.math.princeton.edu/2020/192-3/p04>, arXiv:1902.03719. [Merge note: Places used, as the sources name them: Theorem 2.10, Example 2.26, Corollary 2.32, Proposition 3.1, Theorem 3.4, Corollary 3.5 and Theorem 3.10 (source 74); Theorem 3.10, Corollary 2.11 and Theorem 2.10 (source 72); cited without a theorem number by sources 75 and 76. Sources 13 and 21 give the Annals preprint link; source 21 uses Theorems 2.10, 2.25, 3.10 and Example 2.26. Sources 35, 38 and 43 also cite its Theorem 4.14 on ground-set-normalized independence counts. Part V: Theorems 2.10 and 3.10 and Example 2.26 (source 77); Theorems 2.10, 2.25 and 3.10, Corollary 2.32 and Example 2.26 (source 79); Theorems 2.10 and 2.25 (source 80).]
- [2] J. Borcea and P. Brändén, *The Lee–Yang and Pólya–Schur programs. II. Theory of stable polynomials and applications*, *Comm. Pure Appl. Math.* **62** (2009), 1595–1631. arXiv:0809.3087.
- [3] S. H. Chan and I. Pak, *Equality cases of the Stanley–Yan log-concave matroid inequality*, arXiv:2407.19608v2 (2026). <https://arxiv.org/abs/2407.19608>. [Merge note: Theorems 1.3 and 1.11 (source 72). Theorem 1.3 (source 08s).]
- [4] Y.-B. Choe, J. G. Oxley, A. D. Sokal and D. G. Wagner, *Homogeneous multivariate polynomials with the half-plane property*, *Advances in Applied Mathematics* **32** (2004), 88–187. arXiv:math/0202034.

- [5] R. Davis and F. Kohl, *Perfectly Matchable Set Polynomials and h^* -polynomials for Stable Set Polytopes of Complements of Graphs*, Discrete Mathematics **347**(7) (2024), 114018. arXiv:2207.14759. [Merge note: Theorem 3.10 and Lemma 3.11 state the geometric identities used (source 74).]
- [6] T. S. Motzkin and E. G. Straus, *Maxima for Graphs and a New Proof of a Theorem of Turán*, Canadian Journal of Mathematics **17** (1965), 533–540, Theorem 1. doi:10.4153/CJM-1965-053-6.
- [7] F. Röhrle and M. Ulirsch, *Logarithmic concavity of bimatroids*, Annals of Combinatorics **30** (2026), 501–523, published online 14 August 2025. arXiv:2402.15317v3. [Merge note: Theorem A and Example 2.4 (sources 74, 75, 76 and 77).]
- [8] *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, ProveIt research report, SetTheory/Cardinals/docs/reports/enumerative-combinatorics/preorder-root-polytopes (“preorder-root-polytopes” in this report), in thirteen Parts. Theorem numbers cited here are those of its `article.pdf` at the placement commit `8bb543f0f`. [Merge note: Source 76 cites its Part VI (independent-column expectations, complete bipartite stability and block locality), inspected 1 October 2026. Source 77 cites it at commit `d5e863bba` (article blob `2b952dfcf`), before Part XIII, by label: `lor:thm:weightedULC` (Theorem 70.1), `lor:sp:eq:theta`, `mr:q:rankfour`, `mr:q:weighted` and `hb:q:rank` (Research questions 93, 95 and 86), with source lines at that commit.]
- [9] *Hall Bottlenecks and the Limits of Rank Normalization*, research manuscript prepared for Vladimir Reshetnikov with ChatGPT, 30 September 2026; now Part XI of *preorder-root-polytopes*. Cited as the incoming archive `ProveIt_Hall_Bottlenecks.zip` at commit `a866ff9a2` by sources 74 (“The rank-six witness is restated and verified here”) and 76, and as Part XI of *preorder-root-polytopes* pinned at commit `4a3b6991d0cc12cc09b884e22728e2128a54c6e3` by source 75. Source 71 cites the archive at `a866ff9a2`, its section “Further research questions”, question “Unit weights”, and notes that it “gives two-shore-weighted counterexamples and explicitly leaves the unit-weight question open”; source 73 cites Part XI at `4a3b6991d`.
- [10] *Rank-Three Ultra-Log-Concavity and Exact Kernels for Matching Supports*, research manuscript prepared for Vladimir Reshetnikov with ChatGPT, 30 September 2026; now Part XII of *preorder-root-polytopes*. Cited by sources 74 and 76 as the incoming archive `ProveIt_Matching_Rank_Research.zip` at commit `1b3960d8acdabd983147b7c609bd9dfb2d67a0a0`; source 76 refers to its question “Finish rank four” (Research question 93 of *preorder-root-polytopes*).
- [11] *Leaf Compression and Weighted Rank Three for Matching Support Polynomials*, research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026, delivered as `ProveIt_Weighted_Rank_Three_Leaf_Compression.zip` in batch 71 and again (byte-identical) as archive 78 of batch 72; now Part XIII of *preorder-root-polytopes*. Source 74: “The compression proof is repeated in” Lemma 3.2 (here a pointer to Part XIII).
- [12] Batch-72 source 74, *Weighted Rank Five and Covers with a Two Vertex Shore*, research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026; printed here as Sections 2–13. Its Theorem 1.1 is Theorem 2.1 here, and its Sections 2 and 3 are Sections 3 and 4. Source 72 cites it as an “AI-assisted ProveIt research report” containing “the rank- $(q + 2)$ transversal lift and weighted rank-five theorem; the lift’s exact interface is recalled in” Section 14.3. Later sources cite it as *Weighted Rank Five and Two Shore Covers*. Source 80 (Part V) cites the archive at commit `1512ef835`, its Corollary 1.2 (Corollary 2.2 here) and its files `negative_ranks.tex` and `full_lift.tex` (Sections 6 and 5).
- [13] Batch-72 source 75, *A Sharp Weighted Classification of Full Hall Bottleneck Graphs*, printed here as Section 16. Source 74 cites it as *Sharp Full Hall Classification*, a companion report: “Its general negative-shape construction is reproduced in” Section 6.
- [14] Batch-72 source 76, *Stability of Two by Two Covers and Weighted Rank Four*, printed here as Section 15. Source 74: “The present rank-five ULC proof is self-contained and does not import its stability constructions.”
- [15] P. Alexandersson and A. Jal, *Rook matroids and log-concavity of P -Eulerian polynomials*, version 3 (30 June 2026), Corollary 4.1. <https://arxiv.org/html/2410.00127v3>.

- [16] Batch-72 source 73, *One Shore Activities Can Destroy Matching Rank Ultra Log Concavity*, printed here as Section 18. Source 55 cites its Sections 2 and 3, here the subsections of Section 18 from Lemma 18.3 to Proposition 18.5. Source 69 cites it for “the rank-38 one-shore counterexample”.
- [17] Batch-72 source 55, *A Sharp Rank 38 Boundary for the Last Hall Inequality*, printed here as Section 19. Source 21 cites it under the variant title *The Sharp Rank 38 Boundary for the Last Hall Gap*; source 08 embeds an exact copy of its archive.
- [18] Batch-72 source 51, *The Last Hall Gap Does Not Control Ultra Log Concavity*, printed here as Section 21. Source 47 cites it under the wrong title *The Last Hall Gap Does Not Control All Earlier Gaps*.
- [19] D. G. Wagner, *Rank-three matroids are Rayleigh*, Electronic Journal of Combinatorics **12** (2005), N8, Theorem 1.1. arXiv:math/0403216, doi:10.37236/1975. [Merge note: Sources 05 and 10 cite the arXiv preprint (2004) as *Rank three matroids are Rayleigh*; sources 13 and 63 the journal version; source 80 both.]
- [20] Batch-72 source 10, *Lorentzian Marginals on Three Vertices*, printed here as Section 23. Source 05 embeds an exact copy of its archive as a prerequisite. Source 80 cites that nested copy at commit 1512ef835; its Theorem 1.1 is Theorem 23.1 here.
- [21] Batch-72 source 60, *Rank Two Matrix Blocks and Exterior Conditioning*, printed here as Section 22. Source 23: “Exact weighted Hall-slack compression and the rank-two-block marginal theorem”.
- [22] Batch-72 source 21, *Rank Six Conditioning for Every Core*, printed here as Section 30. Source 13 recalls its incidence lemma with proof.
- [23] Batch-72 source 69, *Rank Six With Common Exterior Neighborhoods*, printed here as Section 27. Source 63 lists it in its bibliography as a companion report without citing it in the text; its “earlier common-exterior result” is this source.
- [24] N. Anari, K. Liu, S. Oveis Gharan and C. Vinzant, *Log-concave polynomials III: Mason’s ultra-log-concavity conjecture for independent sets of matroids*. <https://arxiv.org/abs/1811.01600>. [Merge note: Listed by source 43 as an independent proof of the stronger independence-count theorem.]
- [25] Y.-B. Choe and D. G. Wagner, *Rayleigh matroids*, Combinatorics, Probability and Computing **15** (2006), 765–781, Proposition 5.9 of the preprint. <https://arxiv.org/abs/math/0307096>.
- [26] Batch-72 source 72, *Strictness and equality for weighted matching support polynomials*, printed here as Section 14. Sources 29, 32, 35 and 38 cite it as “research note prepared with OpenAI” (source 72’s own title page reads “AI-assisted mathematical research”), for the “every-gap equality classification in the small-cover class”.
- [27] Batch-72 source 23, *Rooted Lorentzian Hall Sectors and an Unrooted Cubic Obstruction*, printed here as Section 24.
- [28] Batch-72 source 08s, *Top Conditional Gaps and the Rank Seven Reduction*, delivered only inside archive 08 as its “structural companion”; printed here as Section 29.
- [29] Batch-72 source 08, *Eventual One Shore Scaling Through Rank Seven*, printed here as Section 36. Source 03 embeds an exact copy of its archive.
- [30] Batch-72 source 03, *Eventual Scaling for Five Plus Three Covers*, printed here as Section 37. Sources 02 and 01 embed exact copies of its archive.
- [31] Batch-72 source 02, *The Top Conditional Gap for Four Plus Four Covers*, printed here as Section 38. Source 01 embeds an exact copy of its archive.
- [32] Batch-72 source 43, *A Rank Six Cubic Sector with a Complete Core*, printed here as Section 31.
- [33] Batch-72 source 38, *Rank Six Cubic Sectors for Matching Hole Cores*, printed here as Section 32.
- [34] Batch-72 source 35, *Rank Six Cores with Two Complete Rows*, printed here as Section 33.
- [35] Batch-72 source 32, *Rank Six Cores with Two Complete Columns*, printed here as Section 34.

- [36] Batch-72 source 05, *The Second Newton Inequality at Rank Six*, printed here as Section 26. Source 80 cites its Section 2 for a neighbour-feature Gram argument, display (135) here.
- [37] Batch-73 source 77, *Weighted Matching Support Rank Normalization Through Rank Four*, research manuscript prepared with OpenAI for Vladimir Reshetnikov, revised 1 October 2026; printed here as Section 40. Source 78 cites its Theorem 1.1, here Theorem 40.1, and embeds an exact copy of its PDF, and of its archive inside the embedded copy of `complete-two-by-three-result.zip` (neither is shipped).
- [38] Batch-73 source 79, *A two-element Lorentzian deformation of every matroid basis polynomial*, research note, 1 October 2026; printed here as Section 42. Source 80 cites its Corollary 5.2, here Corollary 42.4, and embeds an exact copy of its archive and its PDF, the latter under the misleading name `dependencies/rank-five-proof.pdf` (not shipped).
- [39] *Independent weights on a complete two plus three matching core*, research manuscript prepared for Vladimir Reshetnikov, 1 October 2026; batch-73 archive `complete-two-by-three-result.zip` at commit `f8c3a392a`, not a source of this report (Section A.1). Source 78 cites its Sections 3 and 5 and embeds an exact copy of it (not shipped).
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